Government of Western Australia
School Curriculum and Standards Authority

Western Australian Curriculum

Mathematics

Scope and sequence | For implementation in 2026

Acknowledgement of Country

Kaya. The School Curriculum and Standards Authority (the Authority) acknowledges that our offices are on Whadjuk Noongar boodjar and that we deliver our services on the country of many traditional custodians and language groups throughout Western Australia. The Authority acknowledges the traditional custodians throughout Western Australia and their continuing connection to land, waters and community. We offer our respect to Elders past and present.

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- installing maths support add-ons to your chosen screen reader if problems with maths equation readability persist
- exporting the curriculum PDF from our website, to remove distractions such as website navigation.

General Capabilities

Critical and creative thinking

The Mathematics curriculum fosters critical and creative thinking as students analyse problems, explore strategies and justify solutions. A core part of the curriculum is engaging students in reasoning and thinking about solutions to problems and the strategies needed to find these solutions. They are encouraged to question assumptions and compare computational strategies. In statistical investigations, algebraic reasoning and modelling tasks, students develop and test ideas, simplify complex problems and consider alternative approaches to deepen their mathematical understanding.

Digital literacy

In the Mathematics curriculum, students develop digital literacy as they engage with a range of technologies to investigate, model and communicate mathematical ideas. Using digital tools, students enhance their ability to perform computations, construct and interpret graphs, conduct probability simulations and analyse data. Technologies such as spreadsheets, dynamic geometry software, computer algebra systems, statistical software and graphing applications support students in exploring mathematics, including spatial reasoning, algebraic manipulation and statistical modelling. Digital tools allow students to represent and investigate complex mathematical relationships, solve real-world problems, and deepen their understanding of core concepts.

Ethical understanding

The Mathematics curriculum provides opportunities to explore, develop and apply ethical understanding by analysing data and statistics, identifying distortions or misleading representations, and evaluating the fairness of comparisons. Students also interrogate financial claims and sources, fostering critical awareness of how mathematical information can be used ethically and responsibly in real-world contexts.

Literacy

In the Mathematics curriculum, literacy is developed through learning to communicate using appropriate mathematical language, notation and symbols. Students build a specialised vocabulary including technical terms and everyday words with specific mathematical meanings. They apply their literacy skills to interpret and create texts related to mathematics and statistics. Literacy supports students in understanding contexts, posing and answering questions, discussing strategies and justifying their reasoning and solutions.

Numeracy

Mathematics plays a central role in the development of numeracy. The Mathematics curriculum is designed to equip students with the knowledge and skills to apply mathematical thinking across other learning areas and in real-world contexts. Financial mathematics, health and well-being are important contexts for the application of number, algebra, measurement and probability. Concepts in measurement and geometry provide meaningful opportunities for students to engage with design, construction and spatial reasoning. In an increasingly data-rich world, students use statistics and probability to interpret information, critically evaluate data and make informed decisions in situations involving uncertainty.

Personal and social capability

The Mathematics curriculum enhances the development of students' personal and social capabilities by providing opportunities for initiative-taking, decision-making, communicating their processes and findings, and working independently and collaboratively in the mathematics classroom.

Year 3

Year level descriptions

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Mathematics provides opportunities for students to develop a sound grasp of numeric conventions. Concrete materials continue to assist students to make sense of mathematical concepts as they develop the ability to think in more abstract terms.

Students engage in a range of approaches to learning through the proficiencies of understanding, fluency, problem-solving and reasoning. These reinforce the significance of working mathematically with the content and describe how the content is explored or developed.

In Year 3, students recognise the place value pattern in numbers to at least four digits, explore different ways to partition numbers and the use of 'greater than' and 'less than' symbols. This knowledge is applied to a range of calculation strategies and supports the modelling of relevant real-world situations. Students start using formal units of measurement and are introduced to the concept of angles, identifying mathematical applications in familiar contexts. They recognise the likelihood of outcomes for repeated chance experiments and explore different ways to collect and represent data.

Achievement standard

By the end of the year:

Students demonstrate the behaviours of the proficiencies of understanding, fluency, problem-solving and reasoning in conjunction with year level content in routine situations. They select from and use content and mathematically model situations to solve real-world problems in familiar contexts.

Students order numbers to at least four digits. They represent and partition numbers, recall addition and subtraction facts to 20, recognise the relationship between addition and subtraction, and use these to add and subtract two- and three-digit numbers. Students recall multiplication facts of 2, 3, 4, 5 and 10 and represent multiplication and division with arrays. They create increasing and decreasing additive patterns. Students represent the unit fractions $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$ and $\frac{1}{10}$ and recognise equivalent money values.

Students make and classify three-dimensional objects according to key features. They interpret simple maps and identify and compare angles in everyday situations. Students measure and order length and capacity in metric units and compare the mass of objects to common benchmark weights. They tell the time in minutes using analogue and digital clocks, and describe duration in hours, minutes and seconds.

Students identify possible outcomes of everyday events and repeated chance experiments. They collect categorical or discrete numerical data through observation or surveys, and represent and interpret data in dot plots, tables and column graphs.

Content descriptions

Number and algebra

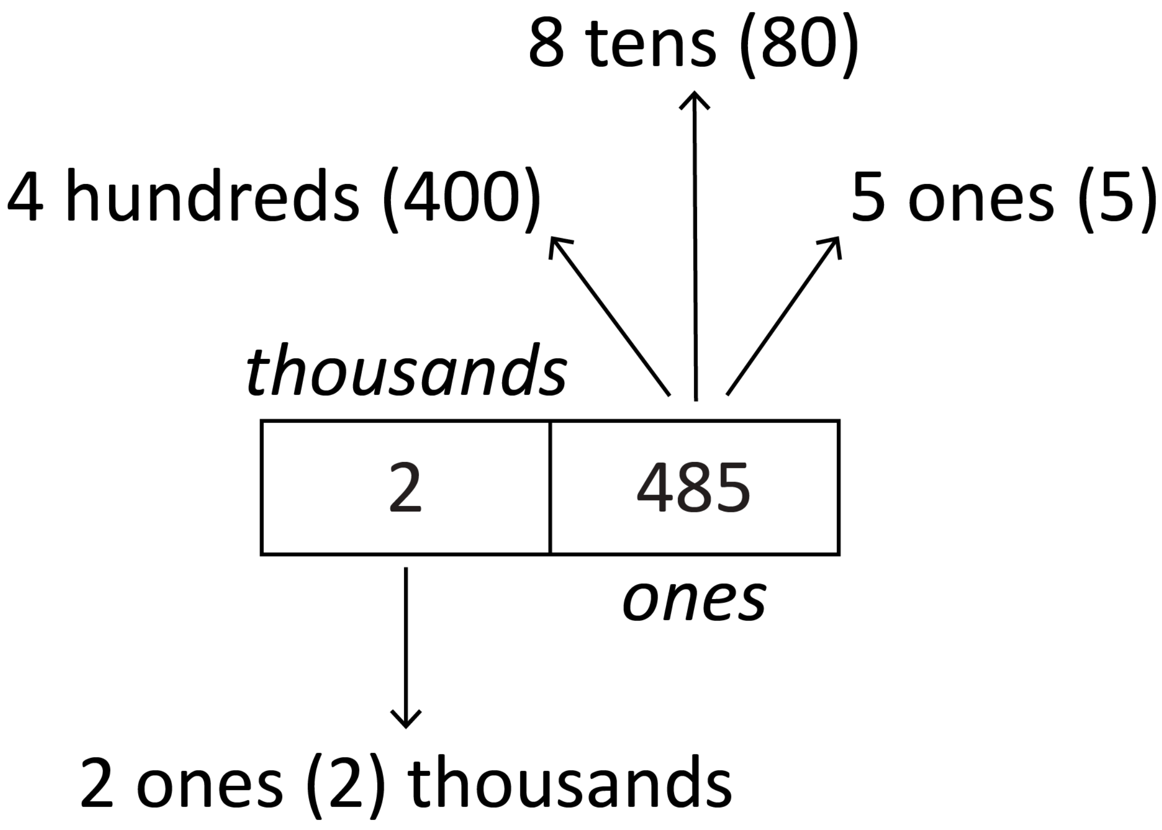
Understanding number

WA3MNAUN1

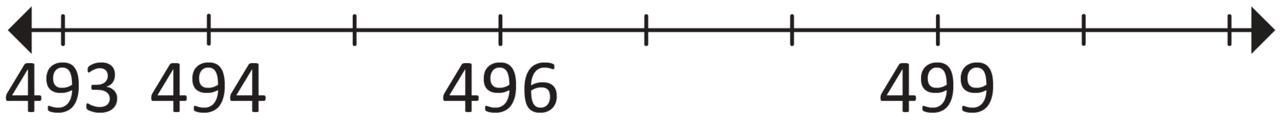
Read, write and order numbers to at least four digits, including on a number line. Recognise the repetition of the 0–999 sequence of digits

For example:

- using diagrams to support reading large numbers based on understanding of place value



- arranging numbers in the hundreds in ascending and descending order
- comparing different hundreds charts, such as 1301–1400 and 1901–2000, to identify patterns
- identifying missing numbers on the markings for a given number line



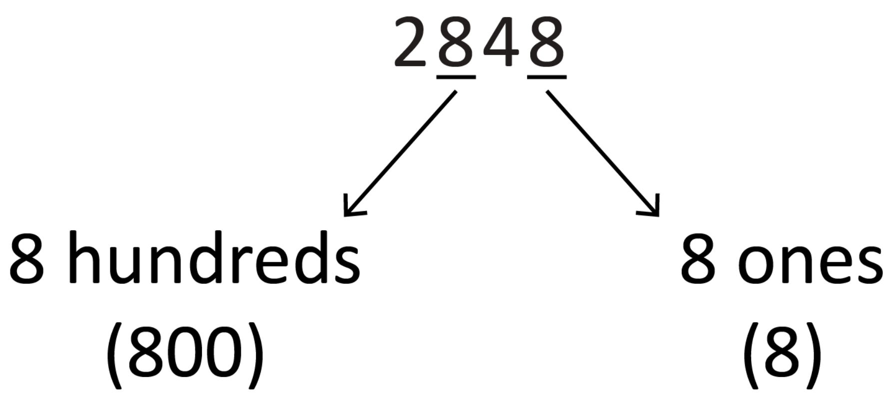
General Capabilities: Literacy, Numeracy

WA3MNAUN2

Represent and partition numbers up to four digits, including groups of 10 (tens), 10 groups of 10 (hundreds) and beyond, using concrete materials and number sentences. Recognise that the value of a digit is determined by its place in a number

For example:

- bundling materials (e.g. pipe-cleaners or popsticks) into groups of 10, 10 groups of 10 to form 100 and 10 groups of 100 to form 1000, connecting bundles to numerals
- using expanded notation to see partitions of a number, including using a calculator, such as $328 = 300 + 20 + 8$
- exploring non-standard partitions, such as $328 = 200 + 125 + 3$ or 328 as 3 hundreds and 28 ones
- recognising the different place values of a digit, such as 8 in 2848



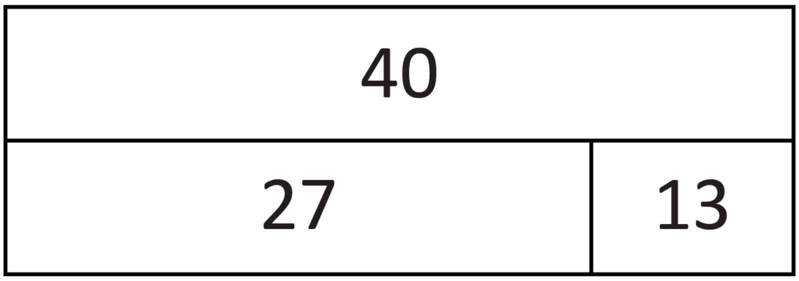
General Capabilities: Literacy, Numeracy, Critical and creative thinking

WA3MNAUN3

Represent and explain the relationship between addition and subtraction using part-part-whole models and number sentences

For example:

- using part-part-whole models to demonstrate how addition and subtraction are inverse operations



$$40 = 27 + 13$$
$$40 - 27 = 13$$
$$40 - 13 = 27$$

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA3MNAUN4

Recall addition and subtraction facts to 20

For example:

- recalling combinations of two numbers that add up to 20 and related subtraction facts
- using related number facts, such as $14 + 6 = 20$, so $20 - 4 = 16$ and $20 - 14 = 6$

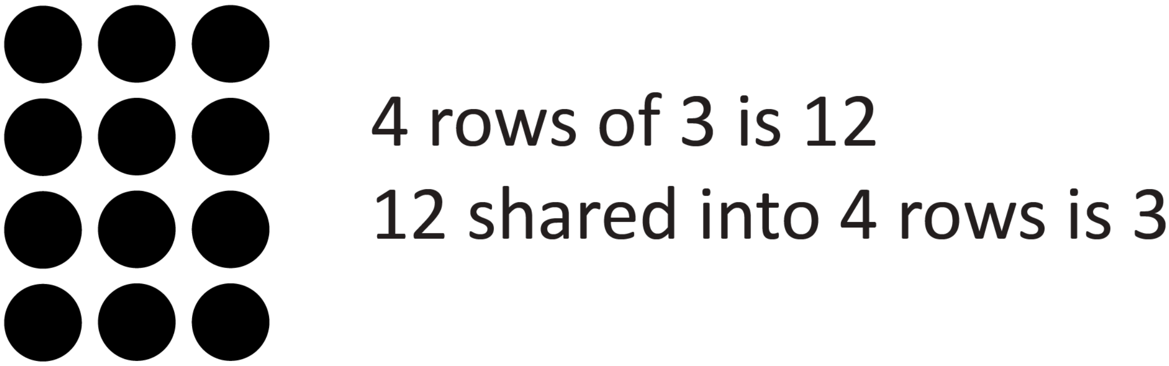
General Capabilities: Numeracy

WA3MNAUN5

Explore and represent the relationship between multiplication and division using diagrams, arrays and number sentences

For example:

- understanding that an array is a rectangular arrangement of objects used to model multiplicative situations, where rows represent the first factor and columns represent the second factor
- arranging and rearranging a collection of 12 into arrays, exploring the connection between multiplication and division, such as



- investigating the result of multiplying and dividing by one and multiplying by zero
- relating doubling to multiplication facts for two and four, recognising that doubling is multiplying by two and halving is dividing by two

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA3MNAUN6

Recall multiplication facts of 2, 3, 4, 5 and 10, and related division facts

For example:

- recognising and using the symbols for multiplied by (×) and divided by (÷)
- using the commutative property of multiplication to extend multiplication facts, such as recognising that if $10 \times 3 = 30$ then $3 \times 10 = 30$

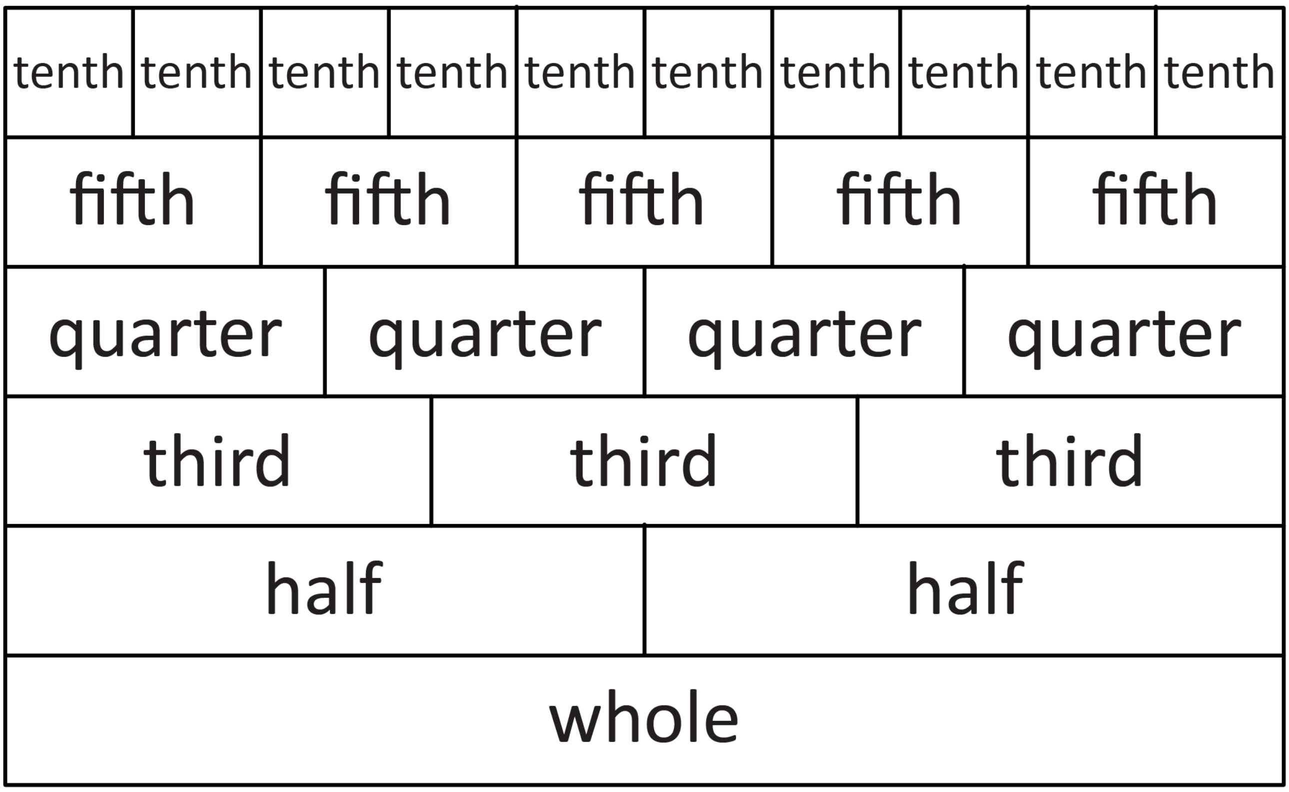
General Capabilities: Numeracy

WA3MNAUN7

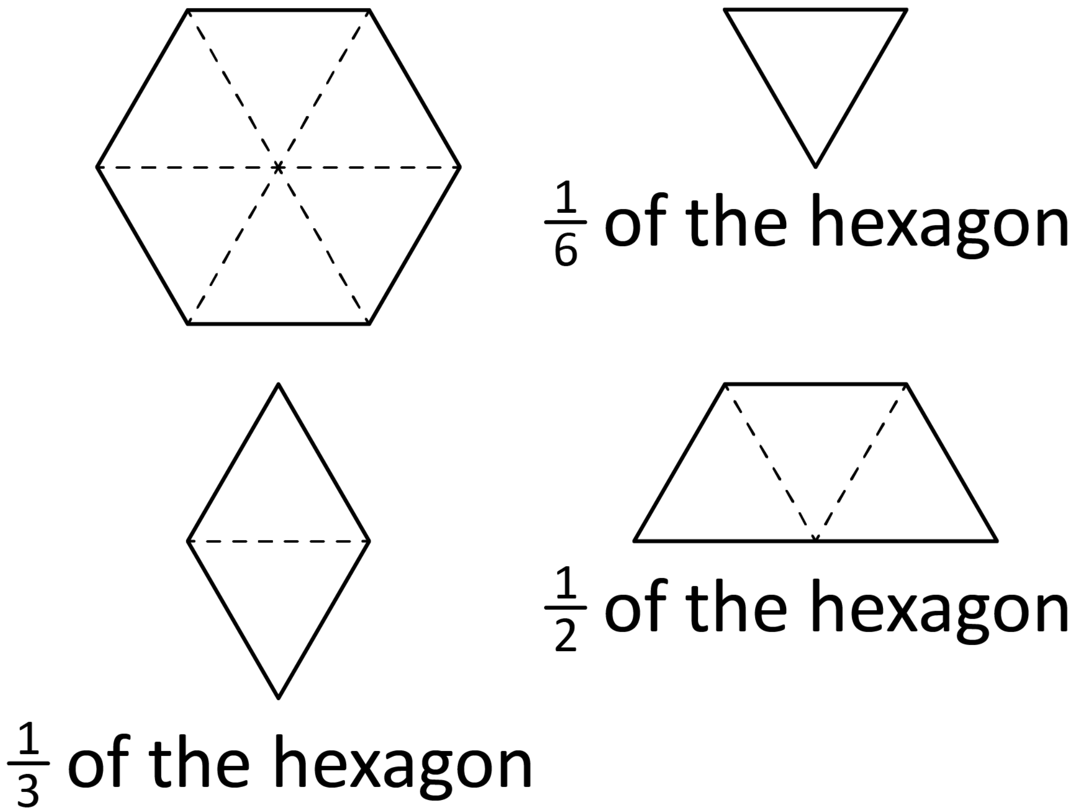
Recognise, represent and describe unit fractions $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$ and $\frac{1}{10}$. Combine unit fractions with the same denominator to create a complete whole

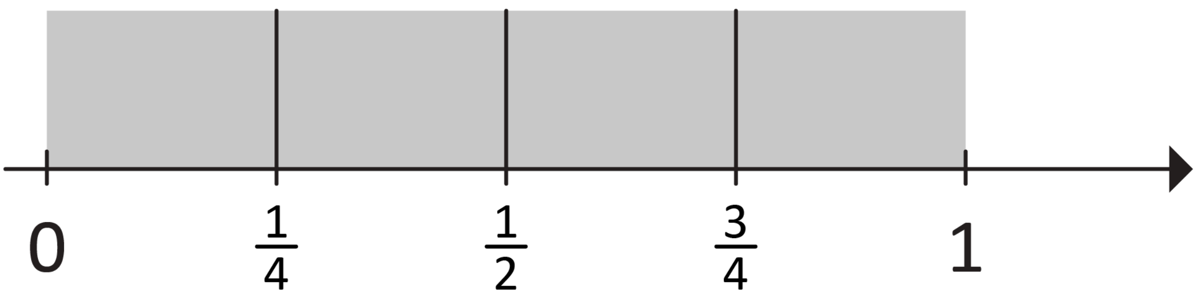
For example:

- describing fractions using words and numerical representations, such as one-third as one part out of three, $\frac{1}{3}$ or $1 \div 3$
- exploring fraction walls

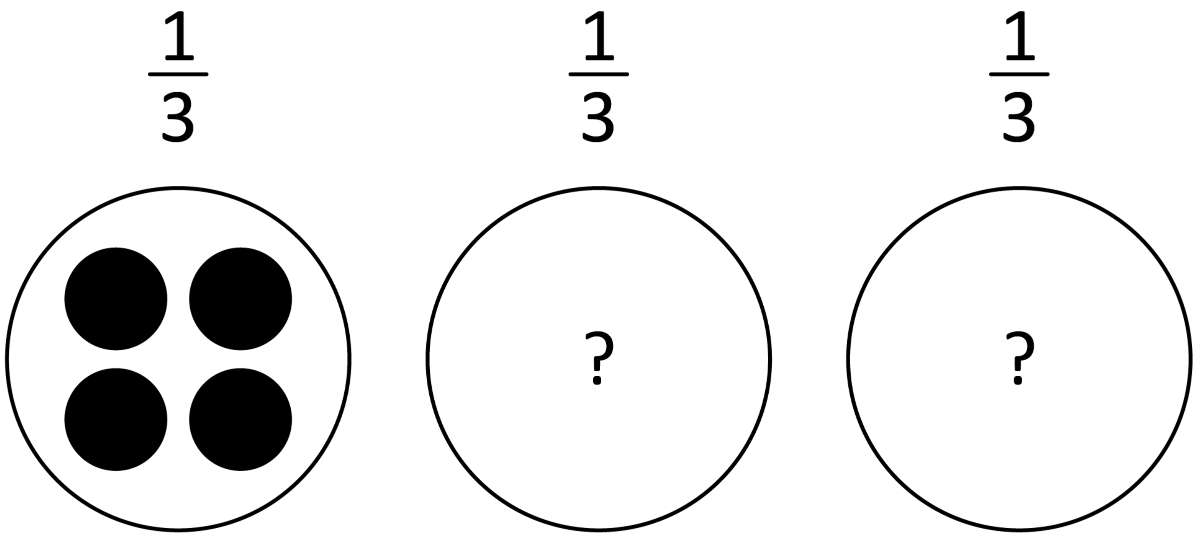


- representing fractions using shapes, objects, paper folding, fraction strips or number lines





- finding the quantity of a whole collection, given the number of objects in one-third



- sharing collections, such as popsticks or counters, between three, four and five people and connecting division with fractions, such as sharing between three people gives $\frac{1}{3}$ of the collection to each and sharing between five people gives $\frac{1}{5}$ of the collection to each

General Capabilities: Numeracy

Understanding equalities and inequalities

WA3MNAUE1

Explore and use the greater than, less than and equality symbols to compare two whole numbers or statements involving addition and subtraction

For example:

- recognising and using the symbols for greater than (>) and less than (<)
- using the correct inequality symbol to compare two given numbers, such as $5 \square 2$
- using equality and inequality symbols to compare numbers and statements, such as

  $7 + 3 \square 11$

  $12 + 8 \square 1 + 19$

  $8 + 4 \square 7 + 6$

  $20 - 12 + 2 \square 10 - 3$

General Capabilities: Numeracy, Critical and creative thinking

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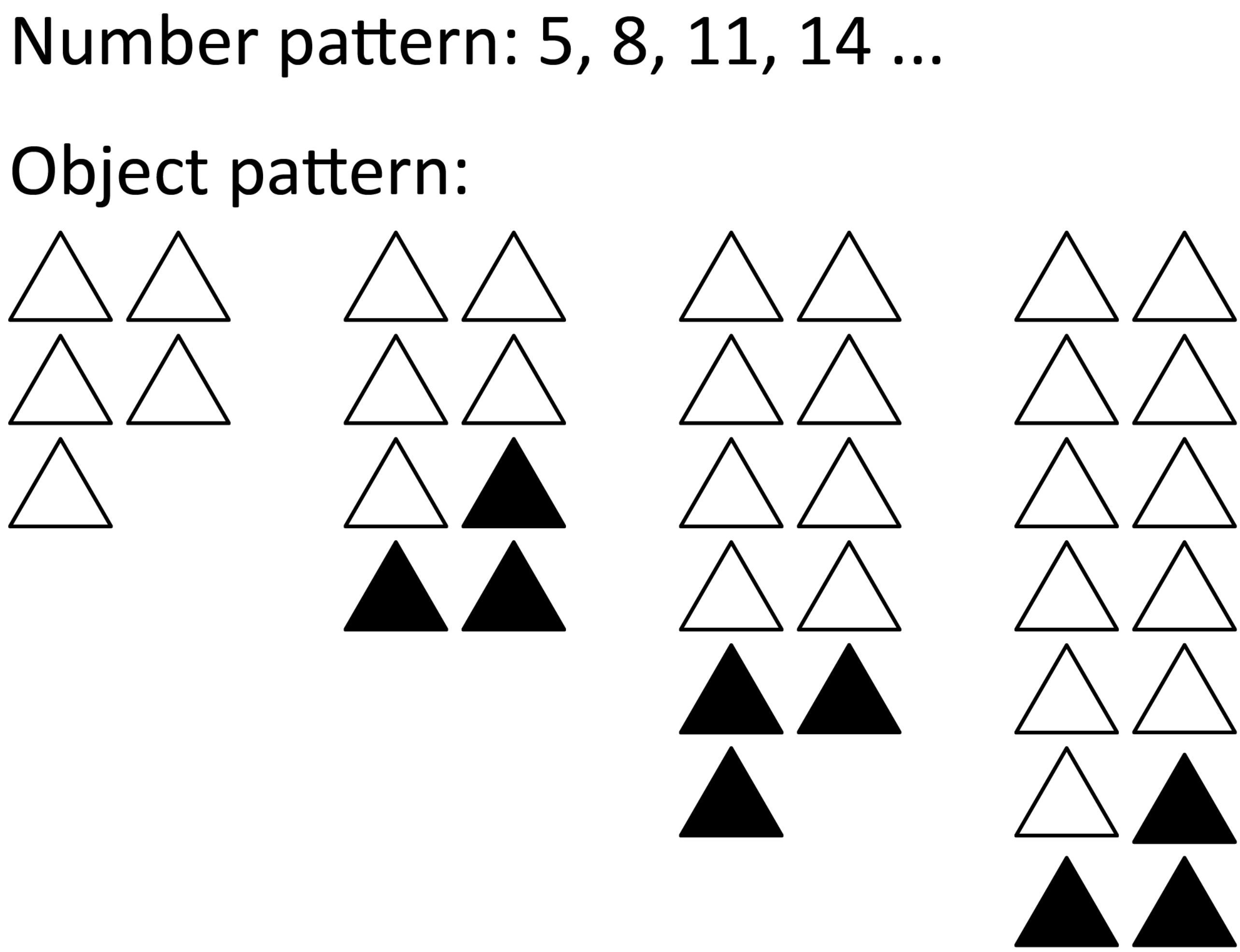
Patterns and relationships

WA3MNAP1

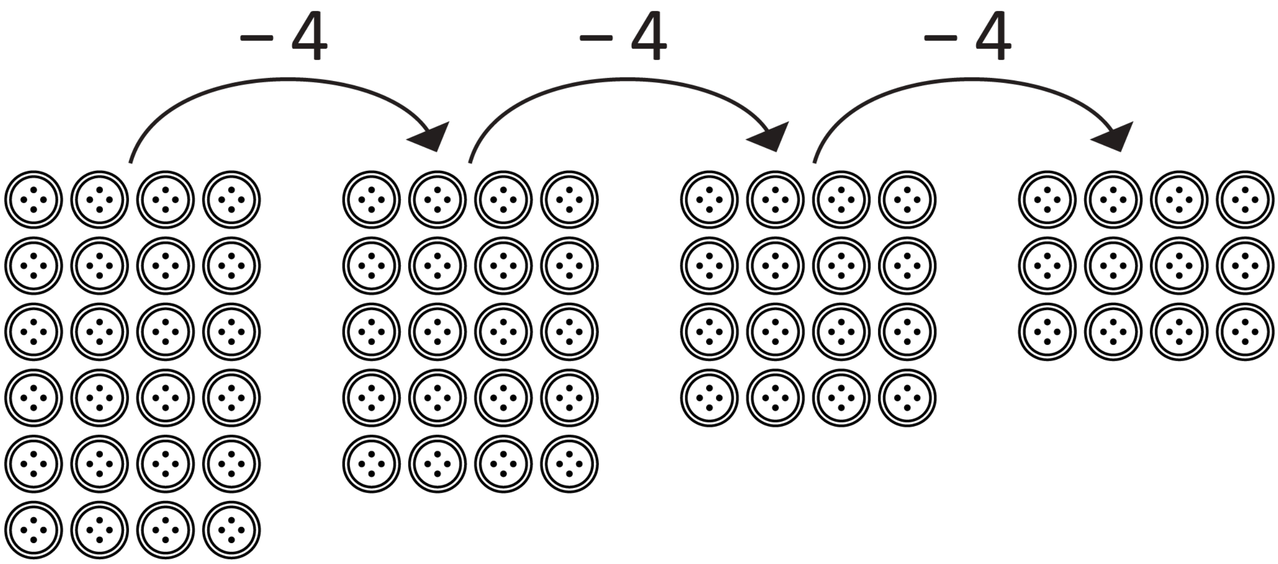
Create and represent increasing or decreasing additive patterns from any starting point, using concrete materials and numbers, and describe rules to represent the pattern

For example:

- using concrete materials to represent an increasing additive pattern and describing it, recognising the importance of the starting point, such as ‘starting at five and adding three more each time’



- using concrete materials to represent and describe a decreasing additive pattern, such as



- creating and describing a variety of number patterns that increase or decrease by a constant amount, such as 20, 25, 30 … or 16, 12, 8 …

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Calculating with number

WA3MNAC1

Add and subtract two- and three-digit numbers using a range of strategies

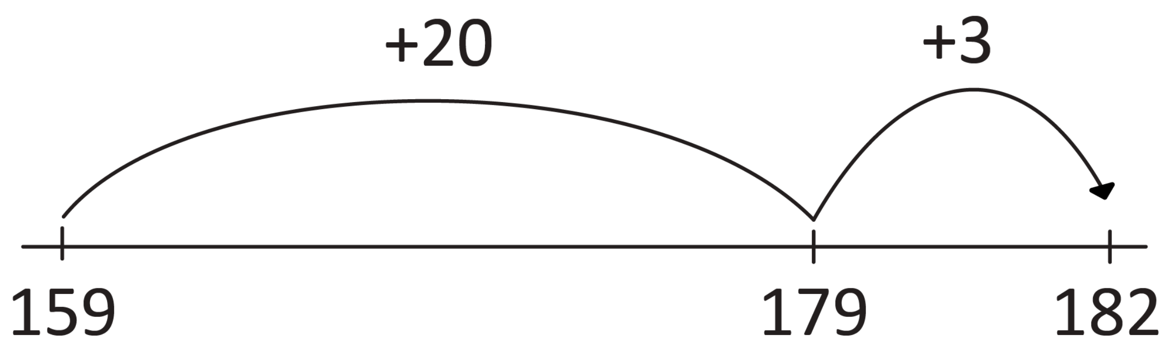
For example:

- using and applying a range of strategies, such as partitioning, rearranging, number lines, bar models and part-part-whole models
  - rearranging and partitioning to facilitate calculations, such as

    $37 + 28$ (subtract 3 from 28 and add to 37)

    $40 + 25$ (using place value partitioning)

    $60 + 5 = 65$
  - solving subtraction problems efficiently by adding or subtracting a constant amount to both numbers, such as 534 − 395 adding 5 to both numbers to make $539 - 400 = 139$
  - using a number line, such as for $159 + 23$

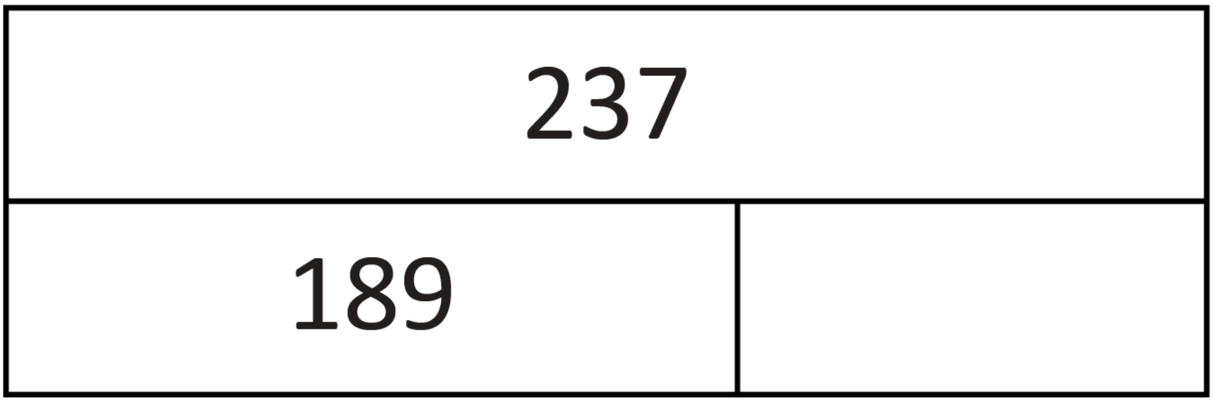


- using a part-part-whole model, such as for $237 - \square = 189$

  and using the inverse relationship to solve for change unknown in $189 + \square = 237$

  using the inverse of addition (subtraction) to solve

  $237 - 189 = \square$



General Capabilities: Numeracy

WA3MNAC2

Explore additive estimation strategies to evaluate the reasonableness of a calculation in familiar contexts

For example:

- identifying situations where estimating is useful because a high level of accuracy is not necessary
- estimating by rounding numbers to the nearest ten or hundred, such as rounding $\$219 + \$385$ to $\$200 + \$400 = \$600$
- estimating the results of a calculation to check the reasonableness of calculator results

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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Financial mathematics

WA3MNAF1

Investigate financial transactions, recognising equivalent values and change

> **For example:**
>
> - identifying different payment methods and situations where they are commonly used, such as cards for public transport and gold coins for donations
> - developing an awareness of the value of money relative to everyday items, using catalogues
> - representing equivalent values using different denominations, such as 70 cents can be made up of three 20-cent coins and a 10-cent coin, or two 20-cent coins and three 10-cent coins
> - representing transactions involving change, such as determining the change from a $50 note used to purchase a $24 shirt and $8 socks for a sport team uniform

General Capabilities: Numeracy, Critical and creative thinking

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Modelling with number

WA3MNAM1

Identify and represent a range of real-world addition and subtraction situations with part-part-whole models, and multiplication and division situations with arrays. Write number sentences to reach a solution and interpret in context

> **For example:**
>
> - modelling additive situations, such as ‘I had 75 tomatoes and then picked some more. Now I have 138. How many did I pick?’ Choosing to use a part-part-whole model and writing number sentences, explaining how each number is related to the situation. Using strategies, such as the relationship between addition and subtraction, to solve, interpreting the answer and the additional tomatoes picked
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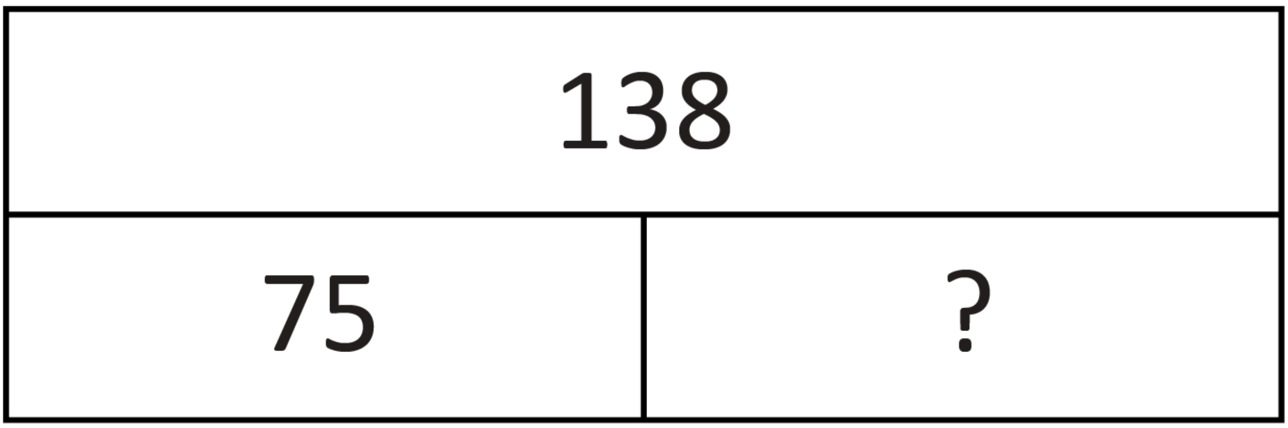
> 
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>
> - modelling multiplicative situations, such as sharing 48 horses into 2, 4, 6 or 8 paddocks using arrays, choosing to use multiplication and/or division, writing number sentences (e.g. $2 \times \boxed{} = 48$) and explaining how each number is connected to the situation (e.g. 2 is the number of paddocks, 48 is the total number of horses and the missing number will be the number of horses in each of the two paddocks)

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Measurement and geometry

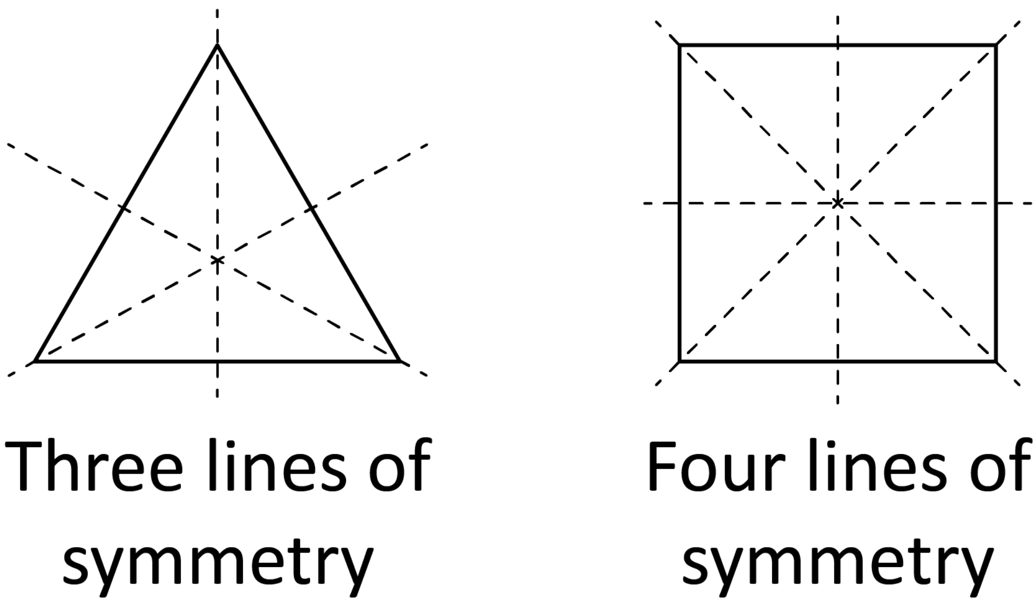
Two-dimensional space and structures

WA3MMGTW1

Explore one-step slides (translations) and flips (reflections) of familiar two-dimensional shapes, make connections to line symmetry and describe the movement of the shape

For example:

- sliding or flipping 2D shapes to identify that translations and reflections do not change the size or features
- identifying lines of symmetry on given shapes, such as triangles and squares, using paper-folding, mirrors or drawings



General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA3MMGTW2

Estimate, measure and order lengths in uniform units, including millimetres, centimetres and metres

For example:

- creating a metric unit measuring tape by drawing 1 cm markings on a long paper strip using a centimetre cube or tile and using it to estimate, measure and order the length of stationery items
- recognising the need for formal units smaller and larger than the centimetre to measure length and using the abbreviations mm, cm, m and km
- investigating the length of one metre using everyday examples, such as a desk or arm span, and exploring benchmarks for centimetres and millimetres

General Capabilities: Literacy, Numeracy

WA3MMGTW3

Compare the areas of two shapes indirectly, using uniform informal units, without gaps and overlaps

> **For example:**
>
> - selecting from a range of uniform informal units, such as tiles, counters or blocks, to cover shapes
> - using uniform informal units to compare areas, recognising why the same unit is used repeatedly
> - recognising the relationship between the size of the informal units used and the number of units needed to fill a shape, choosing units that can best cover a shape with no gaps or overlaps

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA3MMGTW4

Identify angles as measures of turn between two lines that intersect and directly compare angle sizes in everyday situations

> **For example:**
>
> - identifying angles formed by two arms that open and close in everyday situations, such as the arms of a clock, a partially open book, or an open pair of scissors, recognising that a full arm turn forms a circle
> - superimposing two angles, aligning one arm and the vertex, and checking the other arm to determine which angle is greater

General Capabilities: Numeracy, Critical and creative thinking

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WA3MMGTW5

Create and interpret simple maps to show positions and pathways, considering the relative position of key features

> **For example:**
>
> - imagining a 'bird's-eye view' of locations to create maps without scales, such as representing the layout of a classroom or bedroom
> - sketching a mud map of the school, identifying the main buildings and orientating to the real world to determine directions to travel

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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Three-dimensional space and structures

WA3MMGTH1

Visualise and make models of three-dimensional objects. Compare and classify objects according to the key features of faces, edges and vertices

> **For example:**
>
> - visualising and constructing models of 3D objects using clay, sticks, card or digital tools
> - comparing 3D objects, identifying similarities and differences, such as a pyramid has triangular faces and a cube does not
> - classifying a collection of 3D objects, including cubes, cylinders, cones, spheres, rectangular and triangular prisms and pyramids, using geometric language to describe features

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA3MMGTH2

Measure and order capacity in uniform units, including millilitres. Estimate larger capacities using a litre container

> **For example:**
>
> - creating a calibrated measuring instrument by repeatedly pouring a quantity of water, such as 50 mL, into a container, and drawing and labelling a line at each level
> - recognising that one-litre containers can be a variety of shapes, such as milk cartons and ice-cream containers

General Capabilities: Numeracy

Non-spatial measurement

WA3MMGN1

Measure mass to compare objects to everyday items using kilograms and grams

> **For example:**
>
> - using balance scales to compare the weight of everyday items to common benchmark weights including 100 g, 250 g, $\frac{1}{2}$ kg and 1 kg
> - identifying familiar items that weigh about one kilogram, such as a one kilogram bag of flour or a litre of milk
> - identifying familiar items that are measured in grams, such as chocolate or spices

General Capabilities: Literacy, Numeracy

WA3MMGN2

Tell the time in minutes using analogue and digital clocks. Describe duration in hours, minutes and seconds and identify the relationship between them

> **For example:**

- recognising that the space before the minute hand (in a clockwise direction) indicates the minutes elapsed since the hour past, and that the remaining space on the clock shows how many minutes until the next hour
- developing a sense of how long one second and one minute is, using a timer to determine the duration of simple tasks
- using time to describe the duration of events, such as the three-second rule in netball, 20 minutes for recess and a school day being about 6 hours long

General Capabilities: Literacy, Numeracy, Digital literacy

Probability and statistics

Probability

WA3MPSP1

Describe familiar events using the language of chance. Identify and list possible outcomes of everyday chance events

> **For example:**
>
> - describing events as being likely, unlikely, 50-50, possible, certain or impossible, such as
>
> Christmas happening in December (certain)
> Getting heads when you flip a coin (50-50)
> Snow in Australia in Summer (possible, but not likely)
> Having pizza for dinner tonight (possible)
> Seeing a dog on the way home from school (likely)
> Finding a unicorn in the backyard (impossible)
>
> - identifying all possible outcomes for events, such as choosing two pegs from a bag containing three red and three blue pegs

General Capabilities: Literacy, Numeracy

WA3MPSP2

Identify the likelihood of outcomes for planned, equally likely, repeated chance experiments. Conduct the experiments and recognise variation in the results

> **For example:**
>
> - rolling a dice, tossing a coin or choosing a card for a set number of trials and discussing the likelihood of outcomes using the language of chance
> - conducting experiments and recording results using tallies, pictures or objects
> - comparing results with others and with expected outcomes, discussing how results vary

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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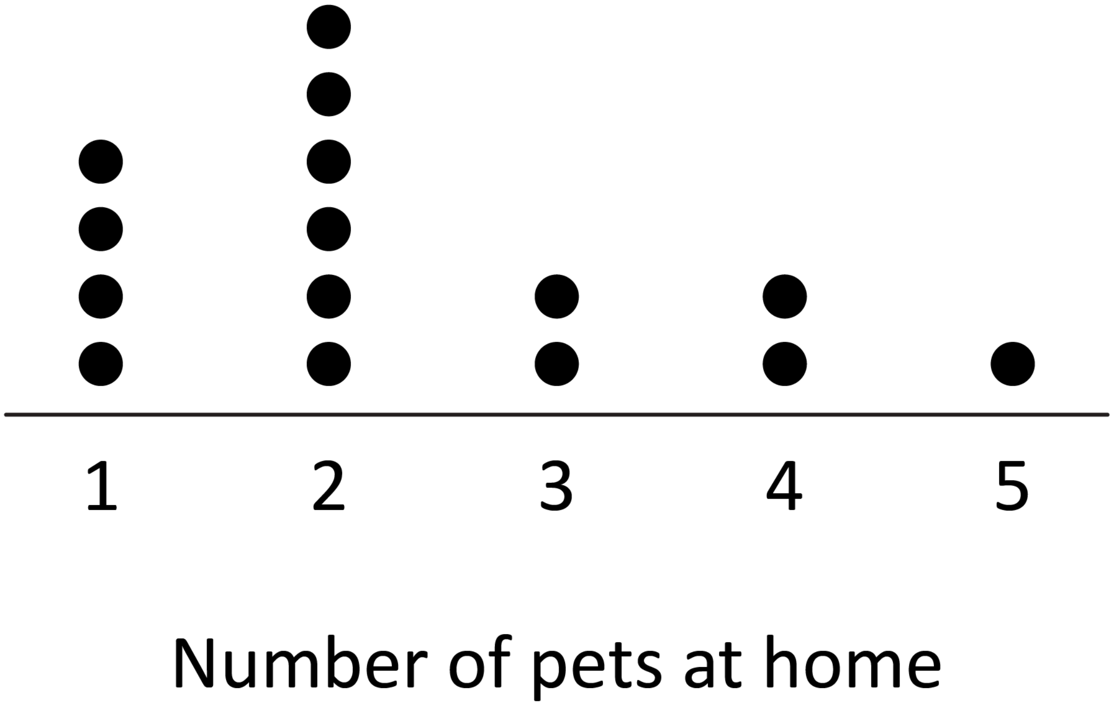
Statistics

WA3MPSS1

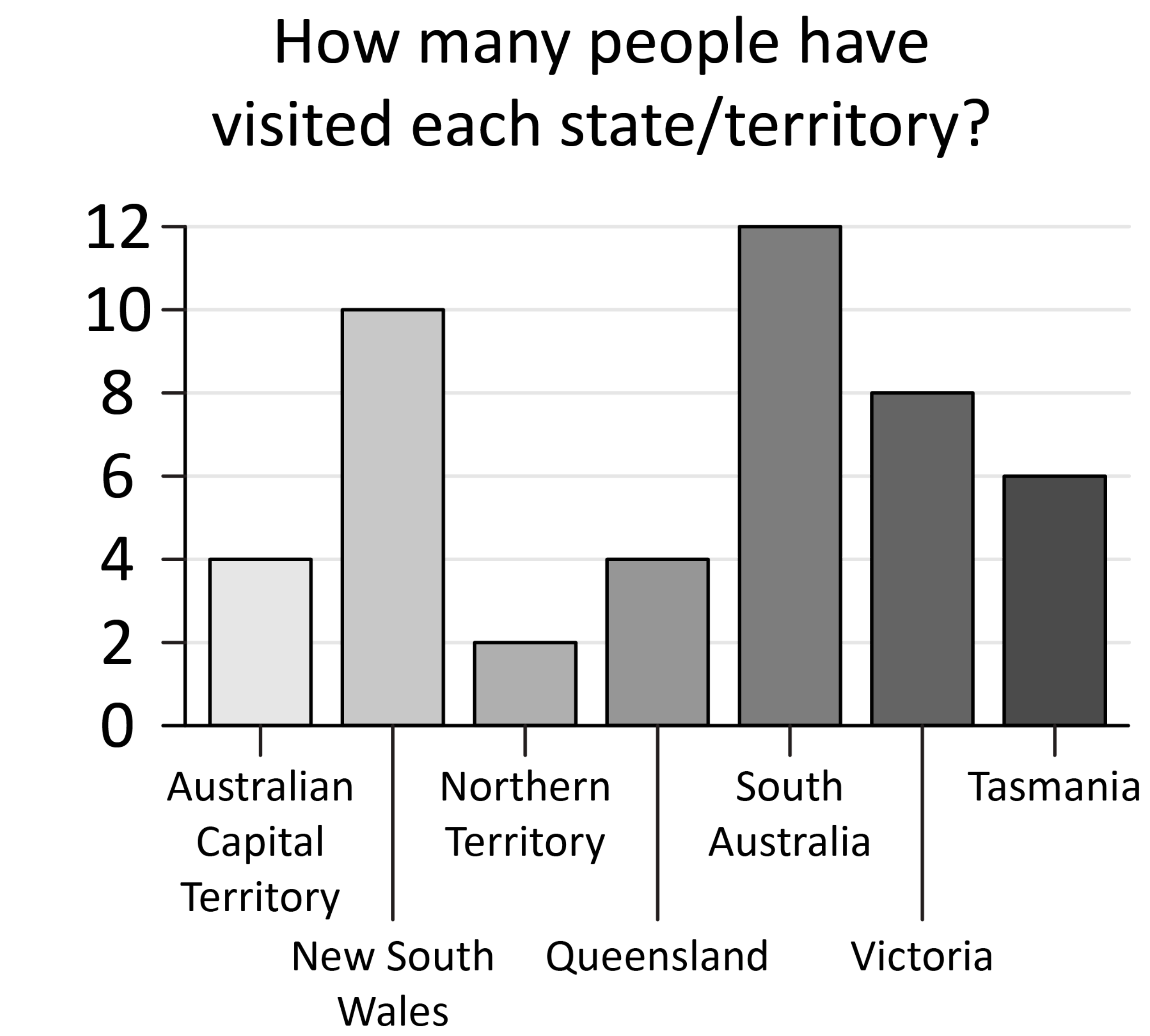
Describe and interpret real-life data represented in dot plots and column graphs with scale intervals of one

> **For example:**

- using the dot plot to answer questions, such as ‘How many students have two pets?’, ‘What was the least common number of pets?’



- identifying features of column graphs, such as labels, gaps, categories and titles, and answering questions relating to the information in the graph



General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA3MPSS2

In a real-world context, explore questions of interest by collecting categorical or discrete numerical data through observation or surveys. Organise and represent data in dot plots, tables and column graphs and interpret to answer a question

For example:

- using a suitable method to collect data (observation or survey) to explore a question, such as 'How do most Year 3 students at our school travel to school?' Creating a list or table to organise the data and constructing an appropriate display, including with the use of digital tools, to answer the initial question and interpret the data, such as 'Most students travel by car, as there are limited bus services to our school'

> - using conventions, such as equal spaces on axes, naming and labelling axes and choosing appropriate titles for dot plots and column graphs

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

Year 4

Year level descriptions

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Mathematics provides opportunities for students to develop a sound grasp of numeric conventions. Concrete materials continue to assist students to make sense of mathematical concepts as they develop the ability to think in more abstract terms.

Students engage in a range of approaches to learning through the proficiencies of understanding, fluency, problem-solving and reasoning. These reinforce the significance of working mathematically with the content and describe how the content is explored or developed.

In Year 4, students extend their knowledge of the number system to at least six digits and decimal numbers up to two decimal places. They apply this understanding when trialling strategies to calculate efficiently and model relevant real-world situations. Students continue to use formal units of measurement and reason to convert between units of time, expanding their understanding of practical applications of Mathematics in the world around them. Students identify when chance events are not affected by previous events and predict the likelihood of outcomes of repeated chance experiments. They collect, organise and represent data, checking for accuracy and consistency.

Achievement standard

By the end of the year:

Students demonstrate the behaviours of the proficiencies of understanding, fluency, problem-solving and reasoning in conjunction with year level content in routine situations. They select from and use content and mathematically model situations to solve real-world problems in familiar contexts.

Students add and subtract numbers up to four digits, multiply two-digit numbers by one- and two-digit numbers and divide whole numbers by one-digit numbers, where there is no remainder, using efficient strategies, including knowledge of odd and even numbers and recall of multiplication facts up to 10 × 10. They describe rules to represent an increasing multiplicative pattern. Students represent common equivalent fractions and make connections between fraction and decimal representation.

Students use scaled instruments to measure and compare perimeter, capacity and mass, and use arrays to compare the area of rectangles in informal units. They compare angles relative to a right angle. Students convert between units of time and determine duration.

Students order the likelihood of chance events and identify when events are not affected by previous events. They conduct repeated chance experiments and describe variation in results. Students collect categorical or discrete numerical data, checking for accuracy and consistency. They represent data in many-to-one pictographs and column graphs, and communicate findings in terms of the context.

Content descriptions

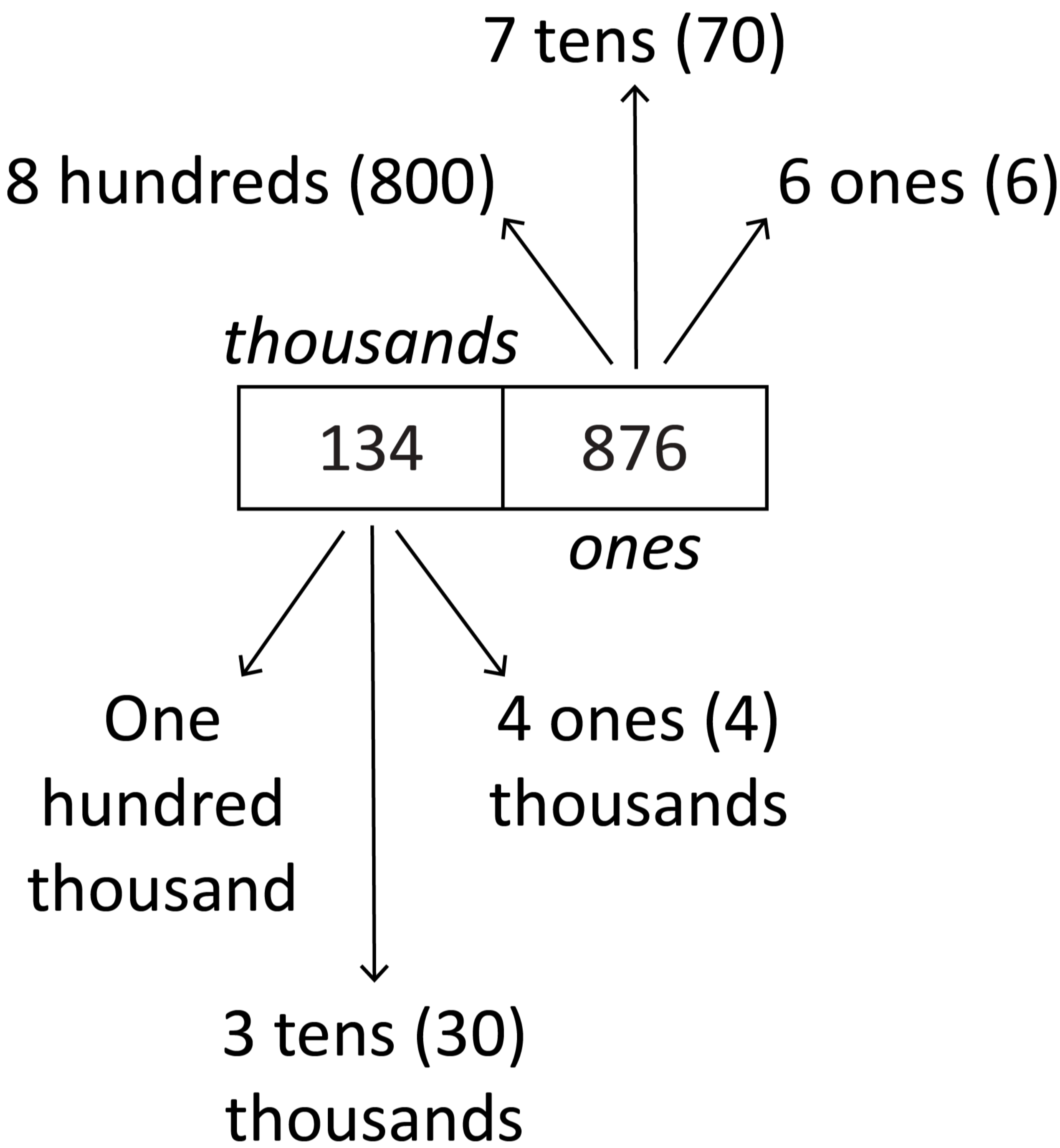
Number and algebra

Understanding number

WA4MNAUN1

Read, write and order numbers to at least six digits. Recognise the significance of the final digit to determine odd and even numbers

> **For example:**
>
> - using diagrams to support reading of numbers to at least six digits, recognising that the sequence of ones, tens and hundreds is repeated in the thousands



- arranging numbers in the thousands in ascending and descending order
- identifying that all whole numbers that end in the digits 0, 2, 4, 6 and 8 are even and that whole numbers ending in 1, 3, 5, 7 and 9 are odd

General Capabilities: Literacy, Numeracy

WA4MNAUN2

Read and write decimal numbers up to two decimal places

> **For example:**
>
> - reading decimal numbers correctly, such as 0.25 as 'zero point two five'
> - express decimals as both tenths and hundredths, such as 0.25 is 2 tenths and 5 hundredths or 25 hundredths

General Capabilities: Literacy, Numeracy

WA4MNAUN3

Represent numbers up to five digits using place value and non-standard partitions with equations. Recognise the multiplicative (10 times as many) place value relationship between adjacent places from right to left

> **For example:**
>
> - partitioning numbers using place value, such as
>
> $1276 = 1000 + 200 + 70 + 6,$
>   and non-standard partitions, such as
>
> $3733 = 3500 + 200 + 28 + 5$ or 3733 as 37 hundreds and 33 ones
>
> - recognising the place value relationships
>   - $1(\text{ones}) \times 10 = 10(\text{tens})$
>   - $10(\text{tens}) \times 10 = 100(\text{hundreds})$
>   - $100(\text{hundreds}) \times 10 = 1000(\text{thousands})$
>   - $1000(\text{thousands}) \times 10 = 10\,000\ (\text{ten thousands})$
> - identifying how rearranging digits changes the size of a number, such as forming the second largest number by rearranging the digits within 4321

General Capabilities: Numeracy, Critical and creative thinking

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WA4MNAUN4

Represent and explain the relationship between one whole being shared equally among 10 as 0.1 or $\frac{1}{10}$ and being shared equally among 100 as 0.01 or $\frac{1}{100}$ using concrete materials

> **For example:**

- dividing a long strip of paper into 10 equal parts labelled as 0.1 and then dividing the last tenth into ten equal parts to demonstrate 0.01 (hundredths of the whole)
- using a calculator to perform the same process, such as $1 \div 10 = 0.1$ (recognising that 1 is 10 times as many as 0.1) and $0.1 \div 10 = 0.01$ (recognising that 0.1 is 10 times as many as 0.01)

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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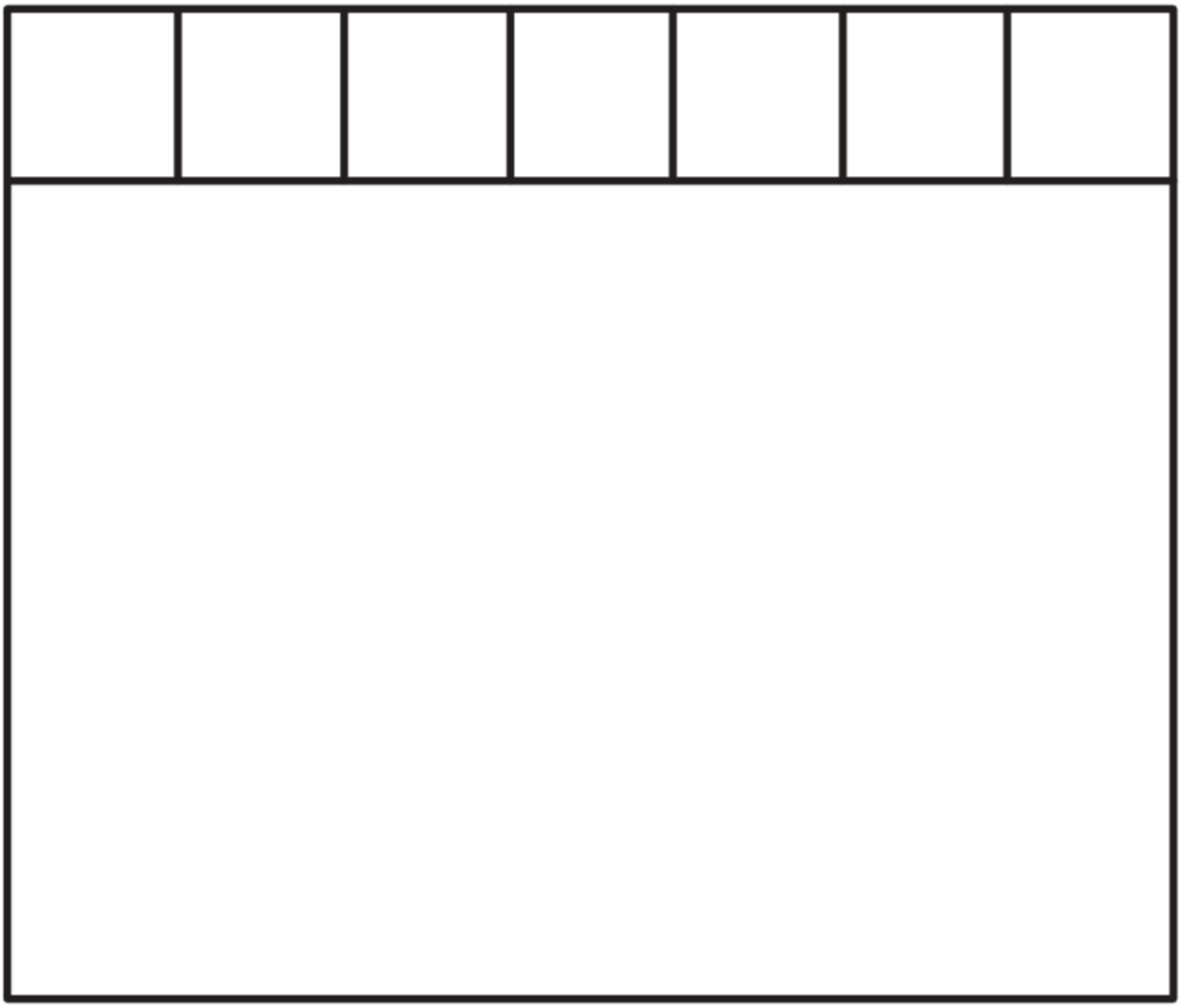
WA4MNAUN5

Represent and explain the relationship between multiplication and division using arrays and equations

For example:

- using knowledge of arrays to explain how many rows of seven are needed to make 42



$$? \times 7 = 42$$

$$42 \div 7 = ?$$

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA4MNAUN6

Recall multiplication facts up to 10×10, and related division facts

For example:

- using arrays and grid paper to represent and explain patterns in the multiplication and division facts related to 10×10

- using knowledge of doubles and near doubles to establish the multiplication facts, such as using doubles for 8×6 where

  $2 \times 6 = 12$ doubles to

  $4 \times 6 = 24$ doubles to

  $8 \times 6 = 48$

- recognising the distributive property of multiplication, such as when finding 6×7, knowing that 7 is made up of 2 and 5, so $6 \times 2 = 12$ and $6 \times 5 = 30$, and therefore $12 + 30 = 42$

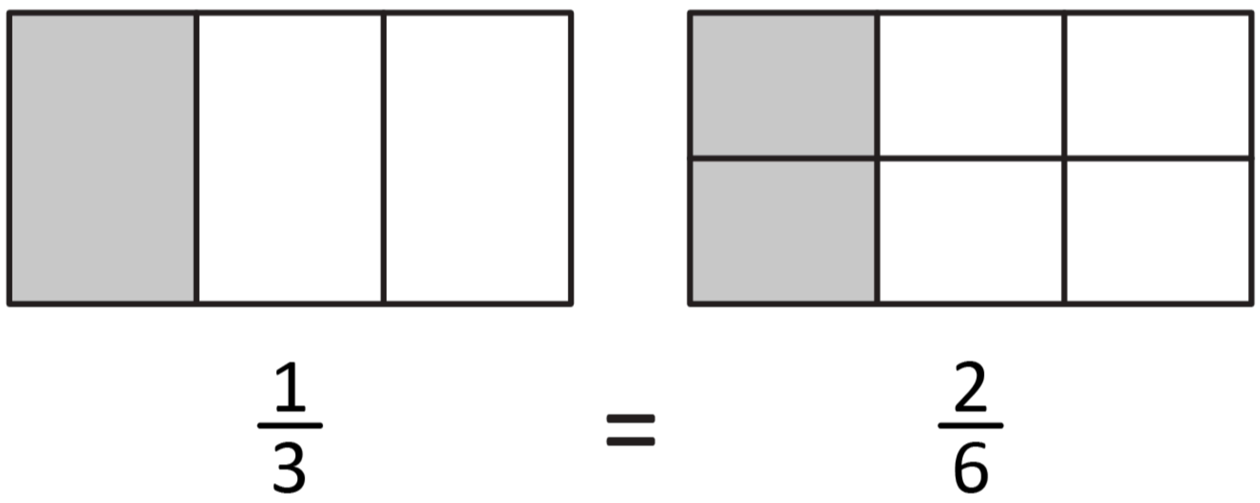
General Capabilities: Numeracy

WA4MNAUN7

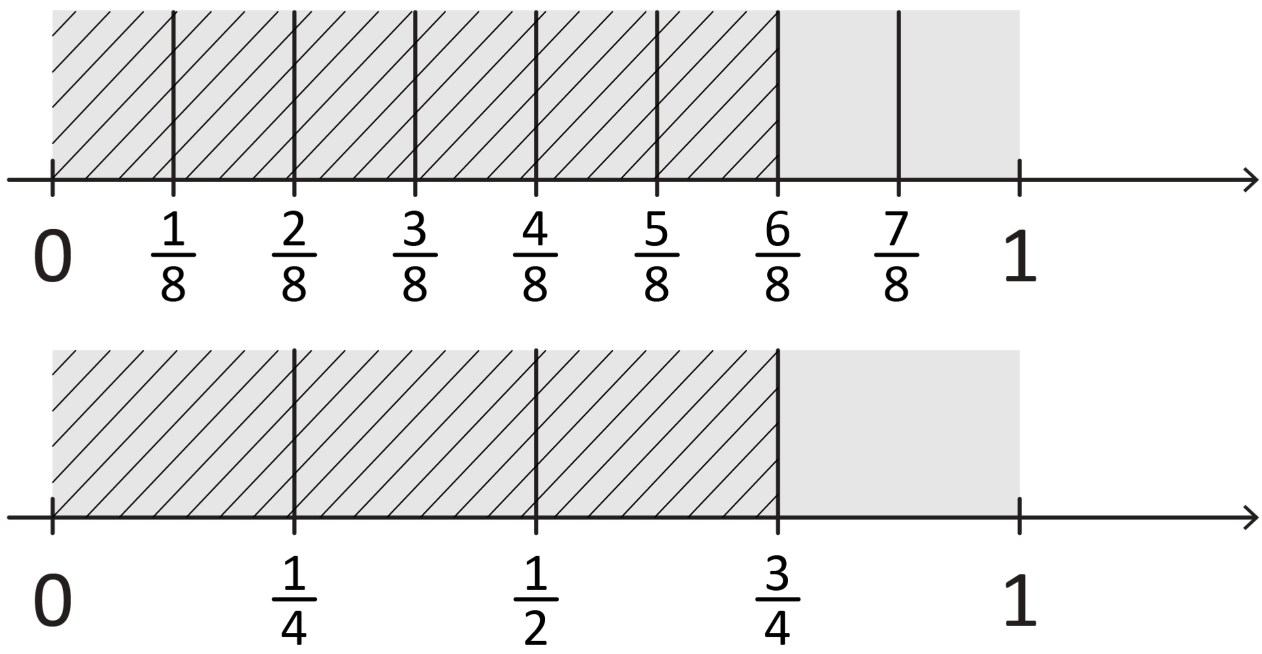
Explore and represent common equivalent fractions and make connections to their decimal representation

For example:

- representing a line, shape, object or collection using equivalent fractions



- demonstrating equivalent fractions as lengths and using paper folding



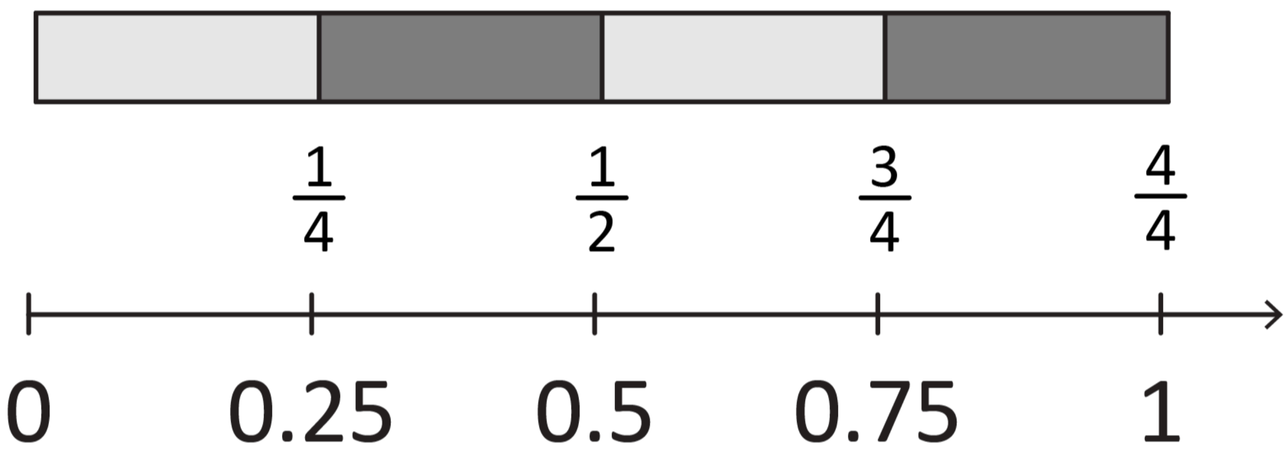
- building familiarity with common equivalent fractions, such as

$\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$

$\frac{1}{5} = \frac{2}{10}$ or $\frac{2}{5} = \frac{4}{10}$

$\frac{3}{4} = \frac{6}{8}$

- connecting fractions and decimals by aligning number lines to show equivalence



General Capabilities: Numeracy

Understanding equalities and inequalities

WA4MNAUE1

Decide if statements of equality and inequality involving the four operations are true, and explain reasoning

For example:

- classifying statements as true (T) or false (F)

$17 + 0 = 0$ (F)

$52 + 17 < 70$ (T)

$60 - 10 > 60 \div 10$ (T)

$16 \div 1 > 16 \times 1$ (F)

- reasoning to identify, without calculating, that $6 \times 12 < 6 \times 14$ because multiplying 6 by a larger number will result in a larger answer
- reasoning to insert the correct value to make the statement true when there is an unknown quantity, such as $12 + \square = 20$

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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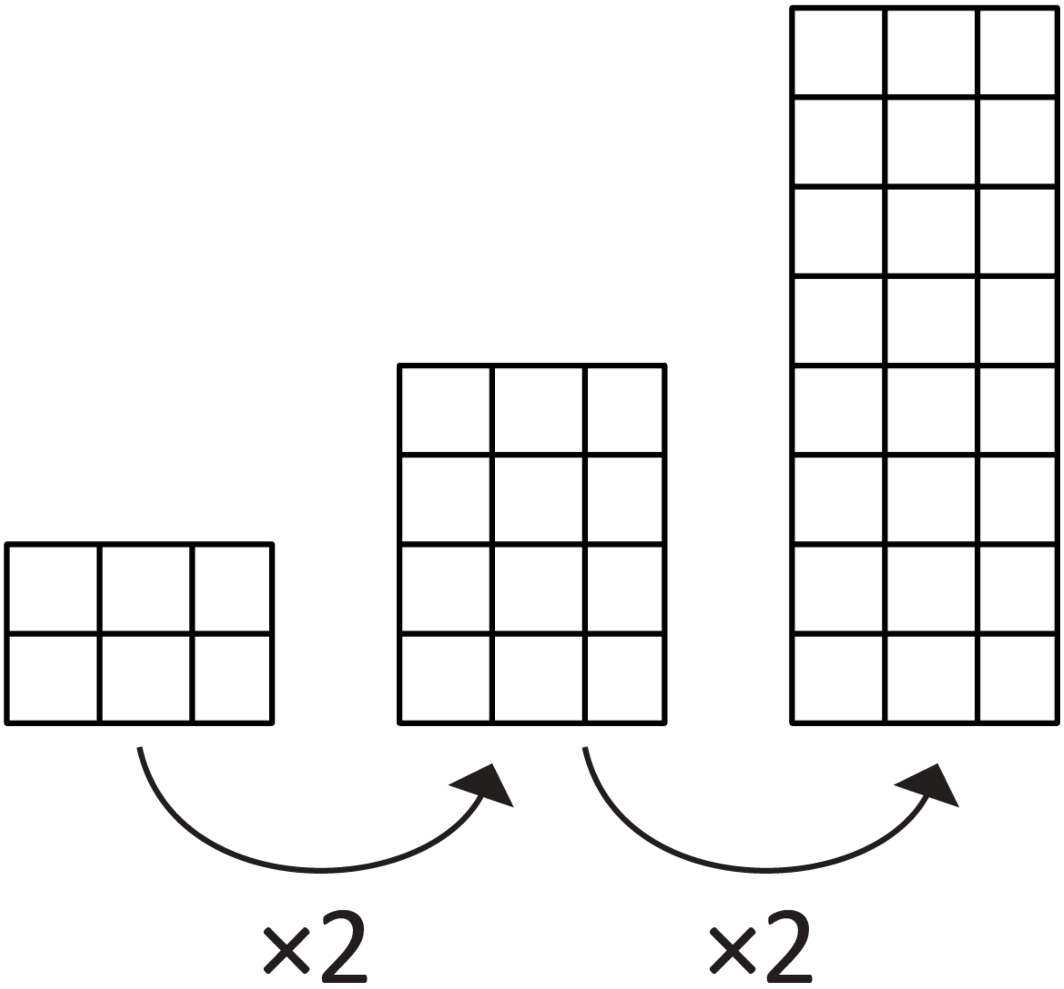
Patterns and relationships

WA4MNAP1

Create and represent increasing multiplicative patterns, using concrete materials and numbers, and describe rules to represent the pattern

For example:

- using concrete materials to represent an increasing multiplicative pattern and describing it, such as ‘starting at six and multiplying by two each time’



- creating and describing a variety of multiplicative number patterns, such as 2, 10, 50, 250 … or 4, 12, 36, 108 …

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Calculating with number

WA4MNAC1

Add and subtract whole numbers up to four digits using flexible and efficient strategies

For example:

- using and applying a range of strategies, such as Bridging the decades (385 + 47 → 385 plus 15 = 400 plus 32 is 432)

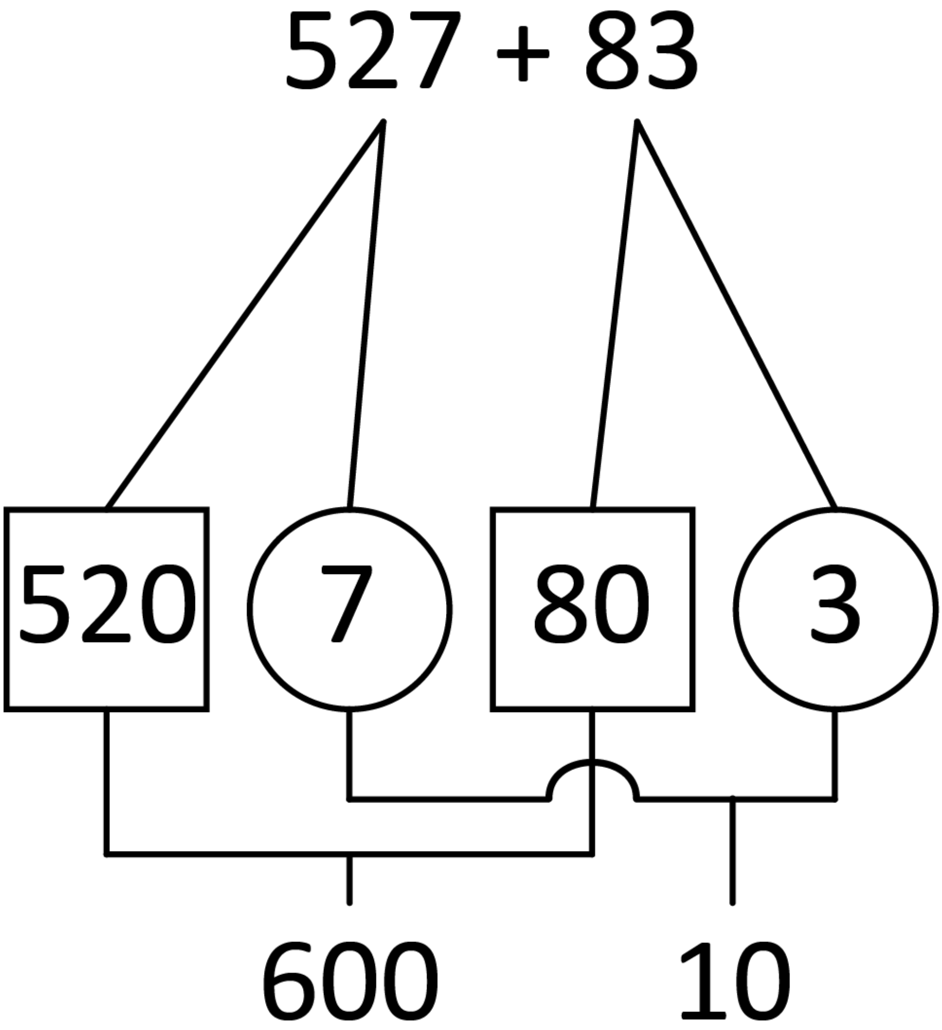
  Quantity values (36 + 27 → 36 plus 20 = 56 plus 4 is 60 and 3 is 63)

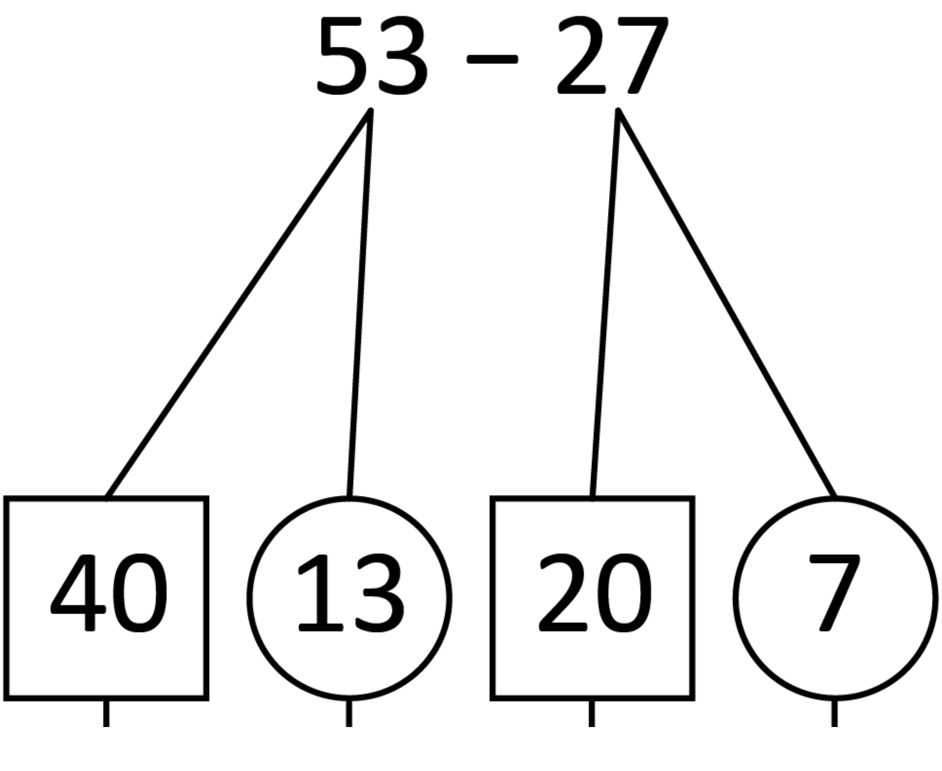
  Compensation (28 + 35 is the same as 30 + 35 = 65, subtract 2 to obtain 63) (125 − 78 → 125 − 80 = 45, add 2 to obtain 47)

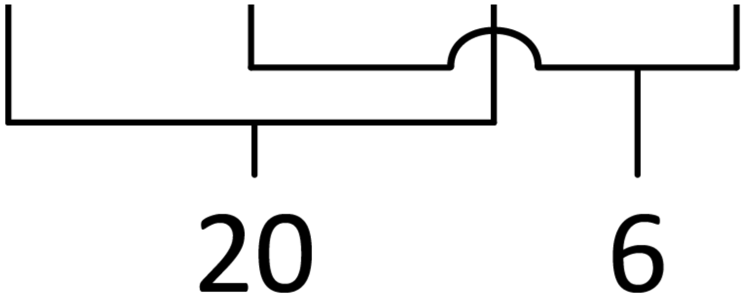
  Levelling (264 + 198 is the same as 262 + 200)

  Constant difference (125 − 78 is the same as 127 − 80)

- setting out standard and non-standard partitions to facilitate addition and subtraction


527 + 83
520
7
80
3
600
10

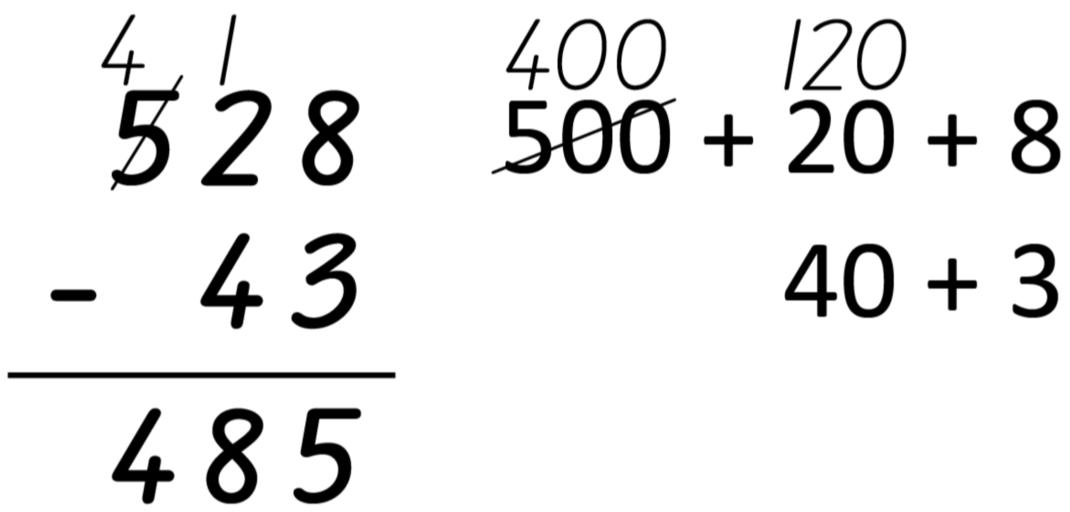

53 − 27
40
13
20
7



- using algorithms, showing understanding to record addition and subtraction calculations

286	
+437	
13	sum of ones
110	sum of tens
+600	sum of hundreds
723	

- recognising how hundreds are exchanged in subtraction algorithms requiring regrouping



- recognising when mental strategies would be more efficient than a vertical algorithm for subtraction, such as 900 − 3 or 500 − 498
- using the inverse relationship between addition and subtraction to find missing numbers in calculations, such as

$\square - 15 = 19$

$83 = 55 + \square$

$18 + 5 = \square + 16$

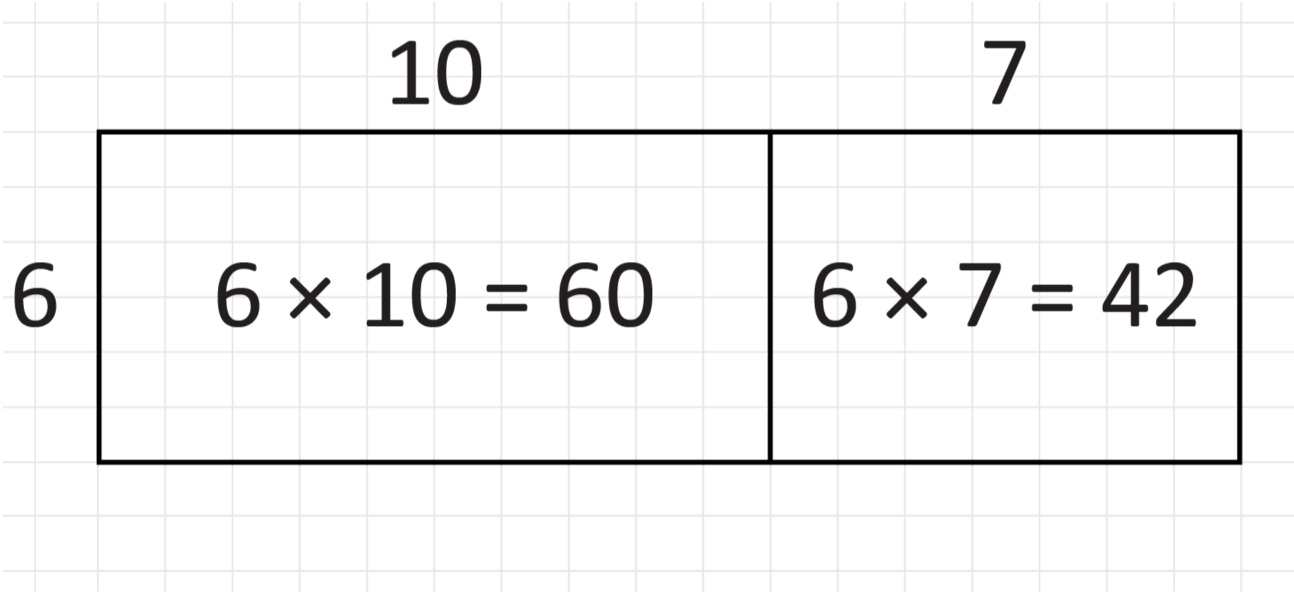
General Capabilities: Numeracy

WA4MNAC2

Multiply two-digit numbers by one- and two-digit numbers, and divide whole numbers by one-digit numbers, where there is no remainder, using flexible and efficient strategies

For example:

- using a variety of formats to partition numbers into ones and tens and represent operations in arrays to support calculations, such as 6×17



$$6 \times 17 = 60 + 42 = 102$$

or 14×16

	10	+	6
10	$10 \times 10 = 100$		$10 \times 6 = 60$
+			
4	$4 \times 10 = 40$		$4 \times 6 = 24$

$$\begin{aligned} 14 \times 16 &= 100 + 60 + 40 + 24 \\ &= 100 + 100 + 24 \\ &= 224 \end{aligned}$$

- using knowledge of multiples to partition as appropriate to divide, such as $85 \div 5 = 17$

$85 \div 5 =$		
$50 \div 5$ $\downarrow$	$35 \div 5$ $\downarrow$	
10	7	$10 + 7 = 17$

- doubling for multiplying by four and eight, such as 8×19

  $2 \times 19 = 38$ doubles to

  $4 \times 19 = 76$ doubles to

  $8 \times 19 = 152$

  and halving for dividing by four and eight, such as $104 \div 8$

  $104 \div 2 = 52$ halves to

  $104 \div 4 = 26$ halves to

  $104 \div 8 = 13$

- doubling and halving to make multiplication easier, such as 45×14 is the same as $90 \times 7 = 630$

General Capabilities: Numeracy, Critical and creative thinking

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WA4MNAC3

Explore a range of additive estimation strategies for different situations, including using knowledge of odd and even numbers

For example:

- using front-end estimation where a high level of accuracy is not required, such as estimating $764 + 239$ as $700 + 200$

- rounding to the nearest dollar to estimate cost, such as $\$16.67 + \$4.12 + \$0.97 + \$0.46$ rounding to $\$17 + \$4 + \$1 + \$0 = \$22$

- overestimating and underestimating for a range, such as for $687 + 294$, the result will be between

  $700 + 300 = 1000$ and
  $680 + 290 = 970$

- checking results of calculations using knowledge of odd and even numbers, such as the sum of two even numbers or two odd numbers must be even

General Capabilities: Numeracy, Critical and creative thinking

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Financial mathematics

WA4MNAF1

Investigate and represent saving and spending, recognising that limited amounts of money are available

> **For example:**
>
> - investigating saving patterns using different weekly amounts and various time periods, such as 4, 8 or 12 weeks
> - comparing how total savings can change when weekly savings amounts differ, such as ‘If $5 is saved each week for eight weeks, how much do you save? What if you save $7 a week for the same time?’
> - exploring purchasing choices when money is saved rather than spent immediately, recognising that spending decisions can be influenced by external factors, such as friends, media, events and advertising

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Modelling with number

WA4MNAM1

Identify and represent real-world additive and multiplicative situations with diagrams and equations to reach a solution. Interpret and communicate findings in context

> **For example:**
>
> - modelling situations, such as buying five scooters for $96 each, evaluating the situation and identifying that it is multiplicative. Representing the situation using a bar model and equation, explaining that five is the number of scooters and $96 is the cost of each scooter. Using efficient strategies to solve, such as rounding $96 to $100, multiplying 5×100 and subtracting 5×4, communicating that the answer is the total cost of buying five scooters
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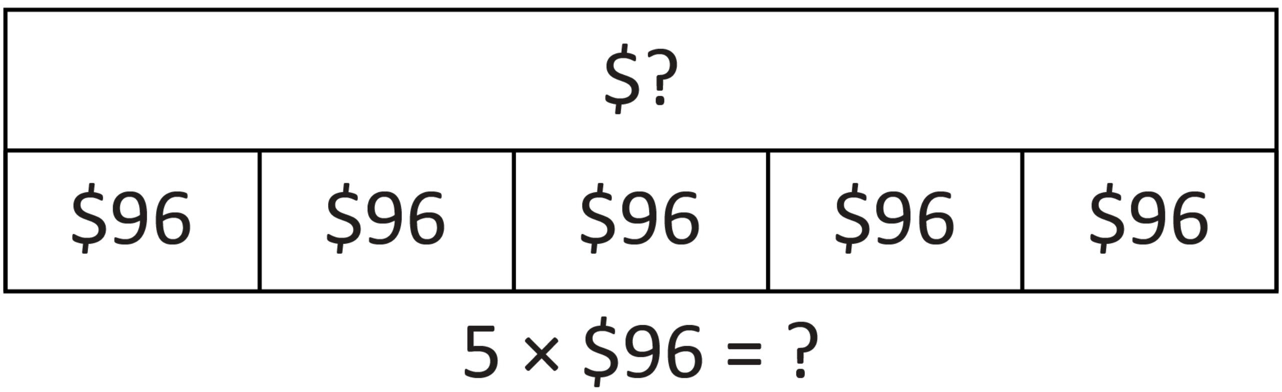
> 
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General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Measurement and geometry

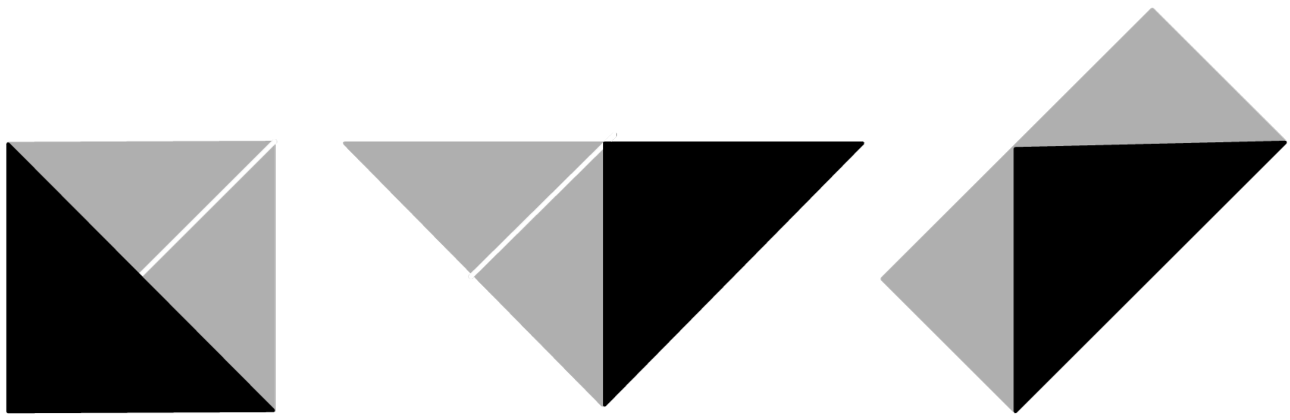
Two-dimensional space and structures

WA4MMGTW1

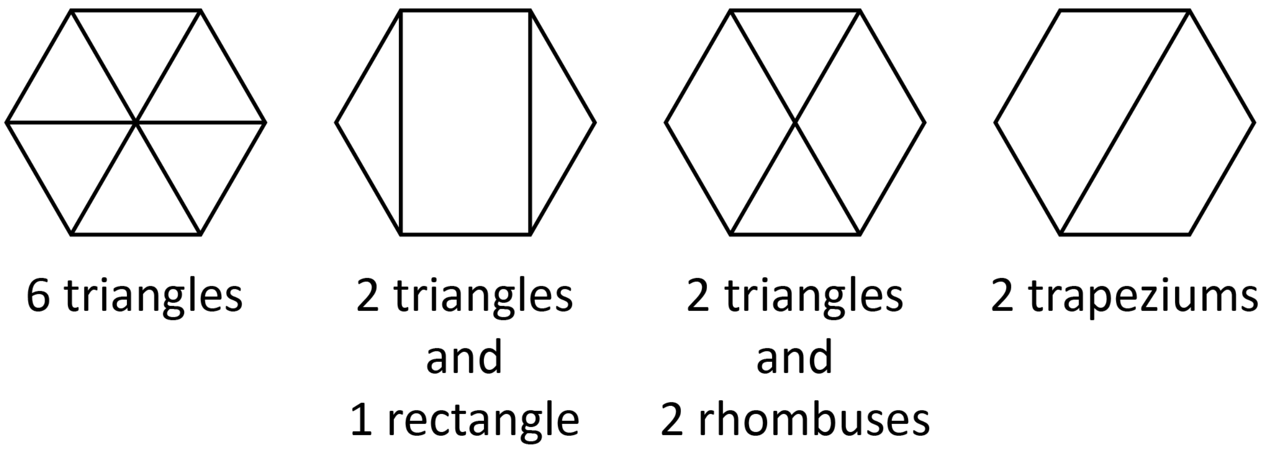
Explore, visualise, describe and create two-dimensional shapes that result from combining or splitting familiar shapes

For example:

- combining familiar 2D shapes, including quadrilaterals and other polygons to form common shapes



- splitting a given shape, such as a hexagon, into two or more common shapes



General Capabilities: Numeracy, Critical and creative thinking

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WA4MMGTW2

Estimate, measure and compare the perimeter of two-dimensional shapes, using scaled instruments and appropriate informal or formal units

For example:

- recognising that perimeter is the sum of the lengths that form the boundary of a shape or enclosed space
- selecting and using appropriate scaled instruments, such as a tape measure, ruler or trundle wheel, to measure around the boundary of a shape, e.g. a painting or a garden bed
- using a piece of string to measure the perimeter of irregular shapes, including those that have curved sides

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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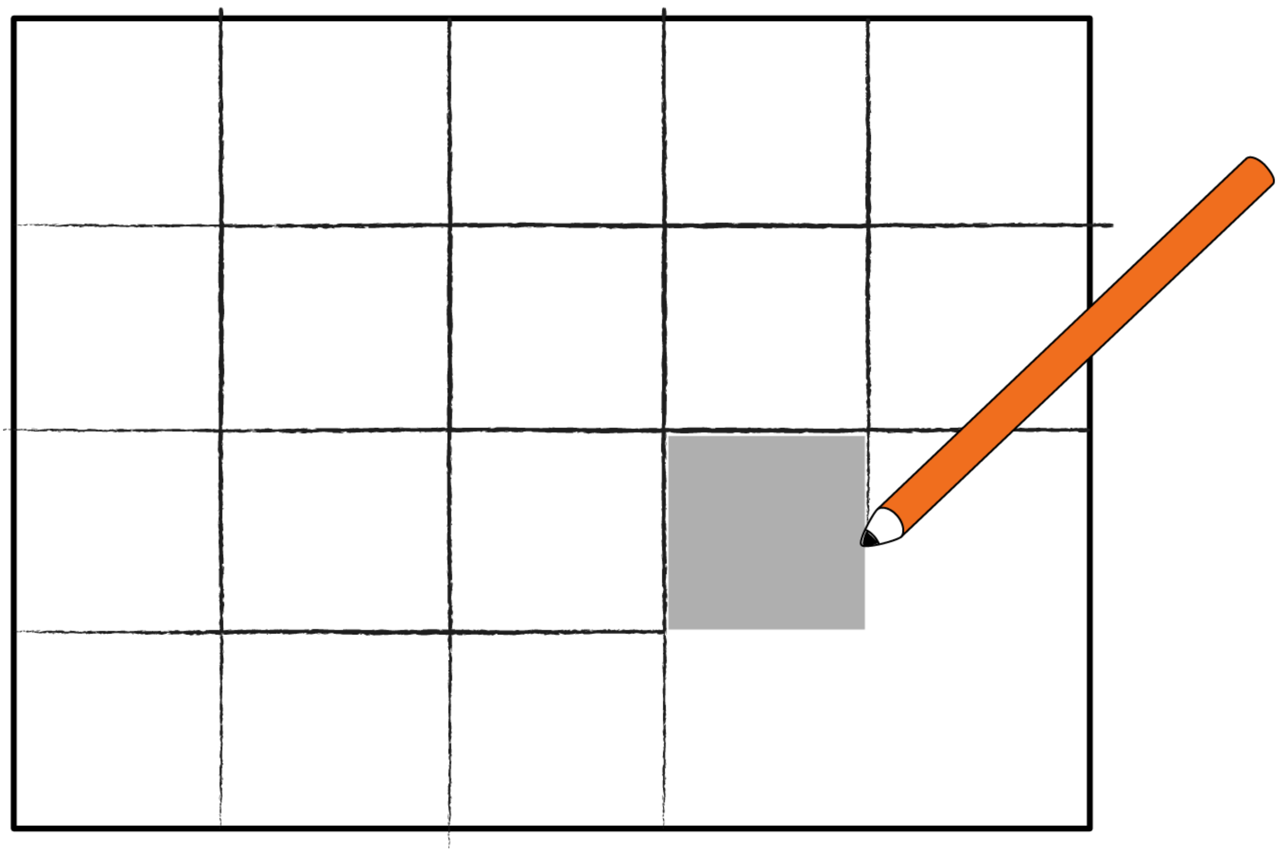
WA4MMGTW3

Estimate, measure and compare the areas of rectangles using uniform informal square units in arrays

> **For example:**
>
> - choosing from a range of uniform informal square units, such as tiles, blocks, sticky notes or paper cut-outs, tracing around this unit, creating an array to estimate, measure and compare the area of rectangles
>
> 

General Capabilities: Numeracy, Critical and creative thinking

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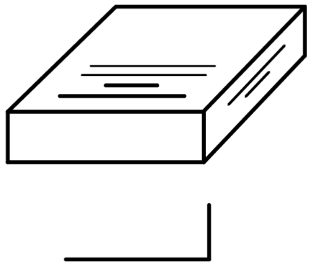
WA4MMGTW4

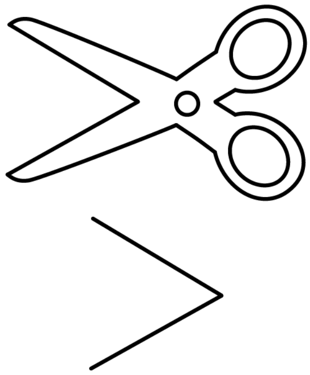
Indirectly compare angles and identify as being equal to, greater than or less than a right angle

> **For example:**

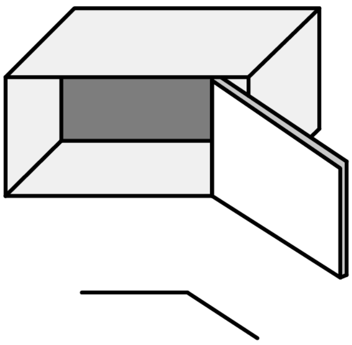
- identifying angles in the environment and comparing to a right angle, including by using an informal tool where required, such as a pipe-cleaner or bent straw
- recognising that the length of the arms does not affect the size of the angle



Corner of a book - equal to a right angle



Open blades of scissors - less than a right angle



Wide open door - greater than a right angle

General Capabilities: Numeracy, Critical and creative thinking

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WA4MMGTW5

Create or interpret grid maps, describe positions and pathways, and explore scales and legends

For example:

- locating positions on grid maps by coordinating horizontal and vertical references

- exploring scales, such as the length of one square grid equalling 10 metres
- using maps to move through pathways and describing a journey between two locations on a grid map
- using digital tools to create a grid map of a familiar area, such as the local park

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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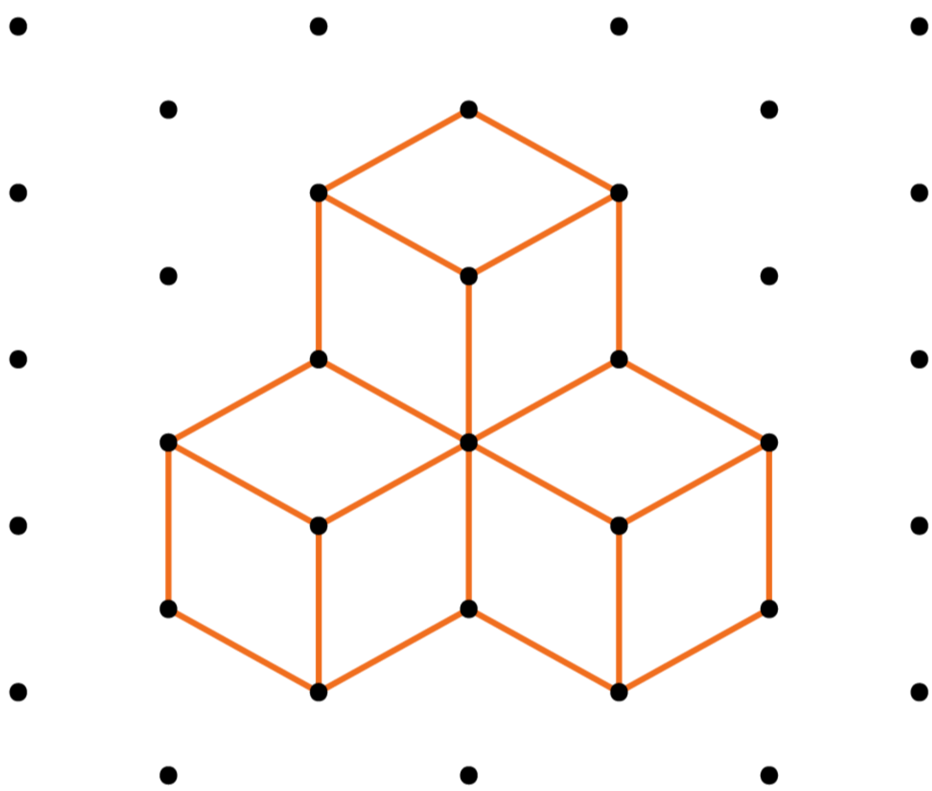
Three-dimensional space and structures

WA4MMGTH1

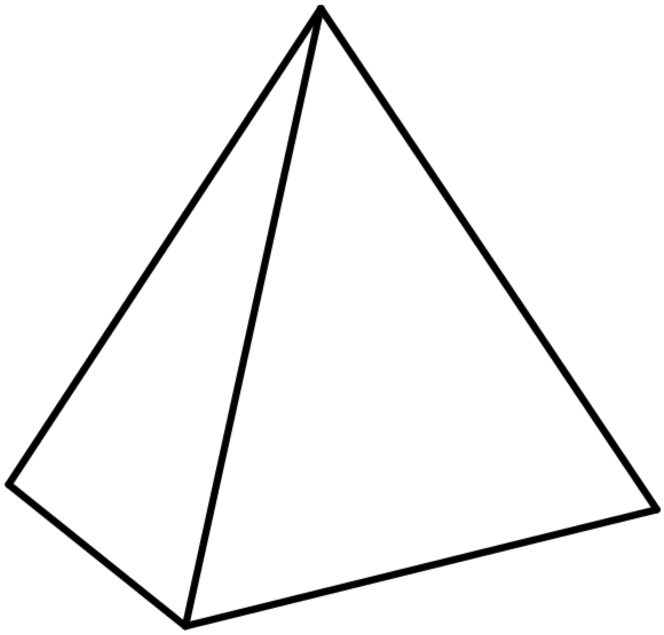
Connect three-dimensional objects to their two-dimensional representations and visualise and describe key features that cannot be seen

For example:

- connecting images or drawings to 3D objects, including prisms, pyramids and cylinders
- constructing models of 3D objects, based on 2D sketches, using cubes



- visualising features of 3D objects that cannot be seen, such as recognising the sketch is a square-based pyramid with a square base, four triangular faces, eight edges and five vertices



General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA4MMGTH2

Estimate, measure and compare capacity in litres and millilitres using scaled instruments

> **For example:**
>
> - relating benchmark capacities to everyday containers, such as a 250 mL juice container or a 600 mL water bottle
> - recording capacities using the abbreviation for litres (L) and millilitres (mL)
> - comparing capacity measurements, such as recognising that a 1500 mL container holds more than a 1 L container

General Capabilities: Literacy, Numeracy

WA4MMGTH3

Explore and compare volume, and recognise that objects with different shapes can have the same volume

> **For example:**
>
> - identifying volume as the space an object occupies and differentiating it from other measurement attributes
> - recognising that objects made with the same number of identical cubes will have the same volume
> - creating different objects using the same amount of playdough, observing that they all have the same volume even though they are shaped differently

General Capabilities:

Non-spatial measurement

WA4MMGN1

Estimate and measure mass in kilograms and grams using analogue and digital scales

> **For example:**
>
> - interpreting evenly spaced markings on analogue scales where not all values are labelled
> - estimating and weighing the same item using both analogue and digital scales and comparing the readings
> - creating a calibrated scale, and using it to measure the mass of everyday items in grams and kilograms
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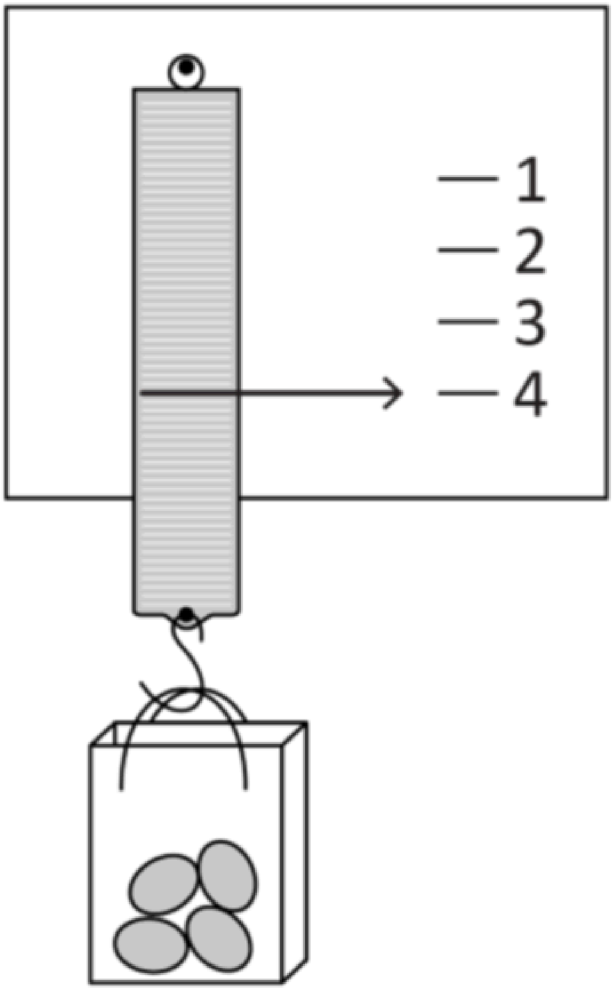
> 
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>
> - recording masses using the abbreviations for grams (g) and kilograms (kg)

General Capabilities: Numeracy, Digital literacy

WA4MMGN2

Convert between units of time, tell the time on digital and analogue clocks using 'am' and 'pm' notation and determine duration

> **For example:**

- using the multiplicative relationship between hours, minutes and seconds to convert between units, such as recognising that an hour is 60 times longer than a minute
- comparing the duration of two familiar events with durations provided in different time units, such as a movie length of 95 minutes and a bus journey of 2 hours and 10 minutes
- relating analogue notation to digital notation for time, such as 20 to 9 in the morning is the same time as 8.40 am

General Capabilities: Literacy, Numeracy, Digital literacy

Modelling with measurement and geometry

WA4MMGM1

In real-world situations involving two-dimensional shapes, three-dimensional objects, grid maps, determining length, capacity or mass in metric units or converting between units of time, mathematically represent the problem to reach a solution. Interpret and communicate findings in the context of the situation

For example:

- modelling situations involving time, such as determining the winner of a summer reading contest where some students have entered their data in minutes and some in hours. Representing the problem using a diagram, such as a table, explaining how information in the table relates to the context. Using efficient strategies to convert between the units of time to determine a winner, interpreting the winner to be the student with the longest duration of reading time
- modelling situations involving units of measurement, such as exploring how much water is needed to fill a bathtub and comparing it to how much water is used during a five-minute shower

General Capabilities: Literacy, Numeracy, Critical and creative thinking

Probability and statistics

Probability

WA4MPSP1

Order the likelihood of everyday chance events. Identify when events are not affected by previous events

> **For example:**
>
> - using lists of familiar events and ordering them from 'least likely' to 'most likely' to occur, such as seeing a rainbow during recess, finding a pencil in your school bag, having lunch at school on a regular day or winning a national raffle, and discussing why the order of some events might be different for different students
> - identifying that obtaining heads when tossing a coin does not affect the chance of obtaining heads on the next toss
> - clarifying misconceptions about events that are not affected by previous events, such as if it rains today, it is more likely to rain tomorrow

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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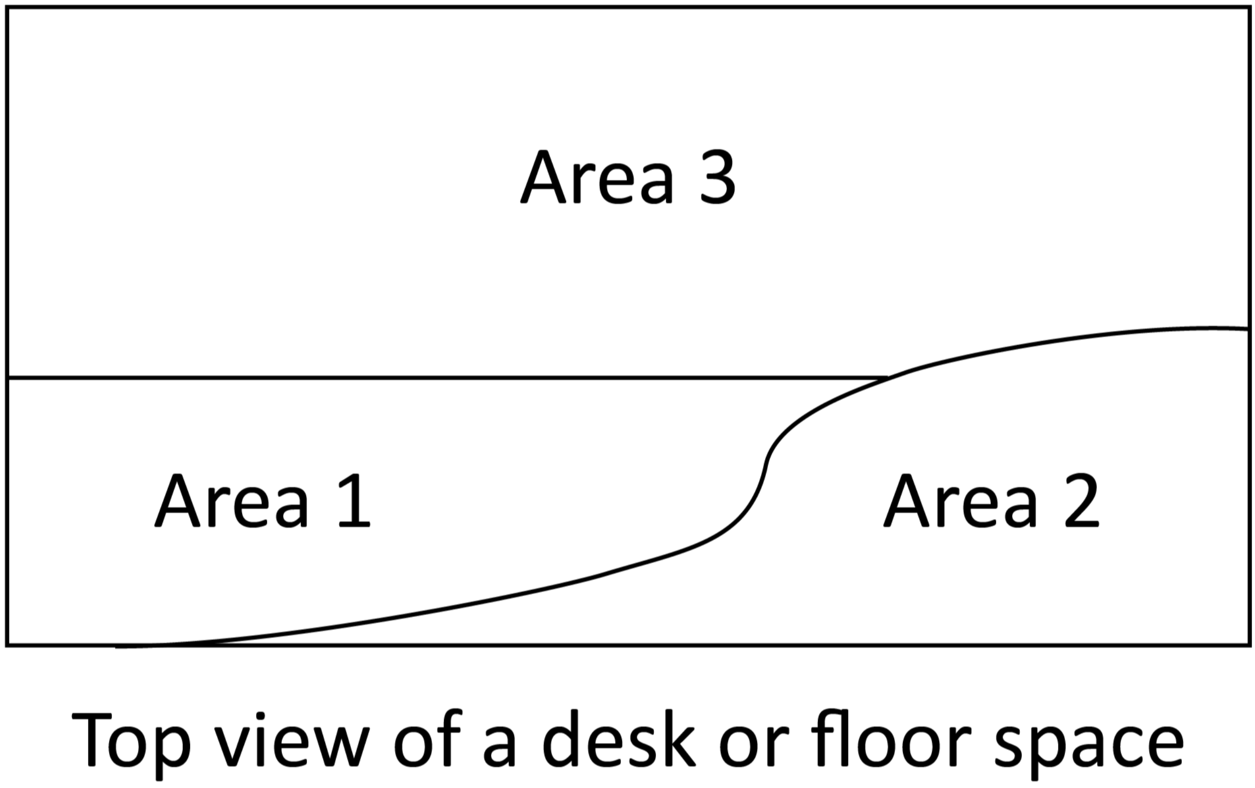
WA4MPSP2

Predict the likelihood of outcomes of unequally likely, repeated chance experiments. Conduct the experiments, describe variation and compare to the prediction

> **For example:**
>
> - naming blue as the colour most likely and red as the least likely to be drawn when choosing a block from a bag containing 7 red, 13 blue and 10 yellow blocks
> - predicting that a button being tossed will land on Area 3 most often, conducting the experiment 25 times, recording the results and describing differences between predicted and observed results



General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

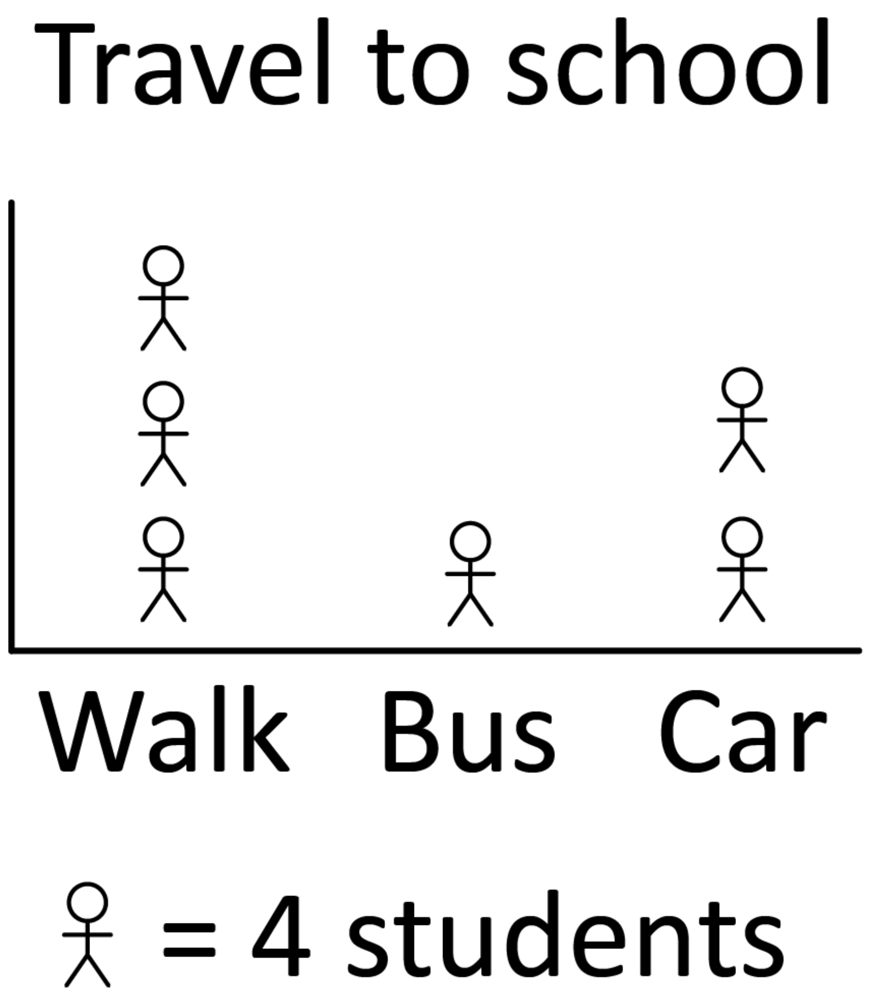
Statistics

WA4MPSS1

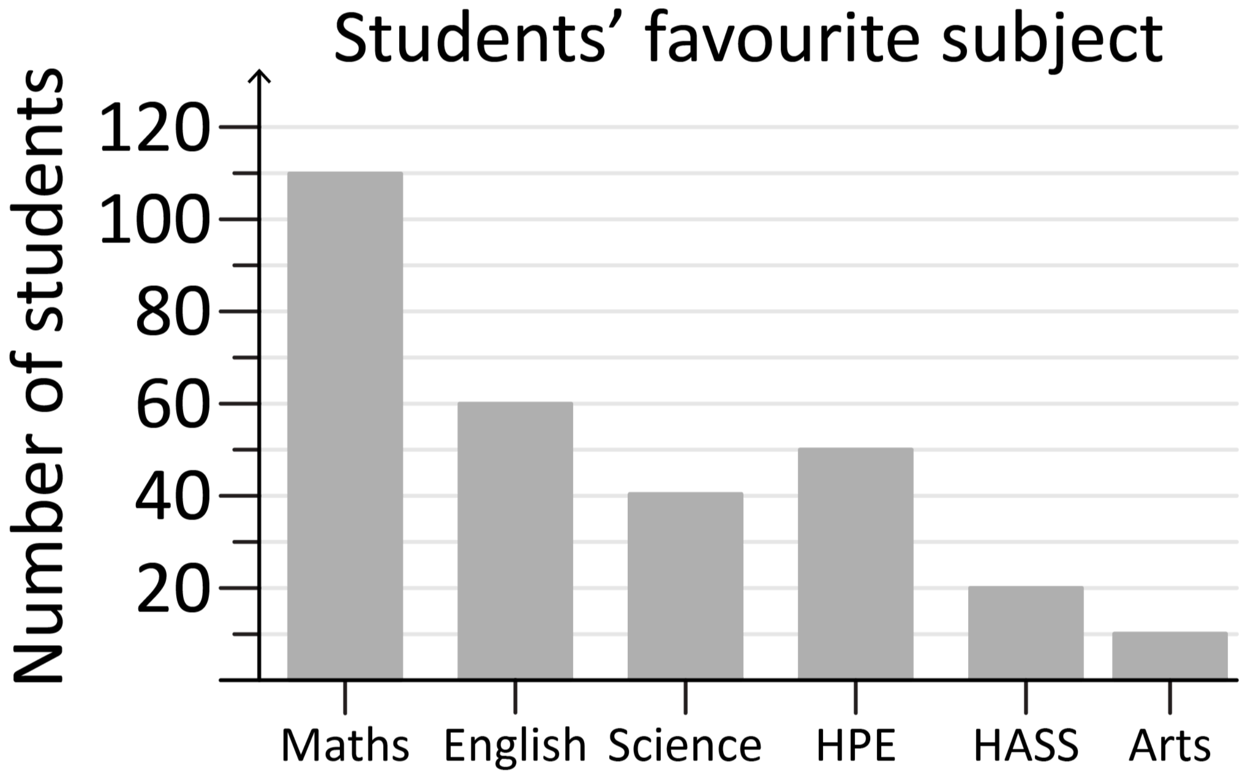
Describe and interpret real-life data represented in many-to-one pictographs and column graphs

For example:

- discussing advantages of many-to-one representation, such as it is easier to display larger numbers visually, and challenges, such as identifying 'how many' are represented by each symbol or reading values for the columns associated with many-to-one representations



- answering questions relating to the information in many-to-one graphs, such as 'How many students liked Health and Physical Education (HPE)?' 'How many more students liked Mathematics more than English?' 'How many times as many students liked Humanities and Social Sciences (HASS) more than the Arts?'



General Capabilities: Literacy, Numeracy, Digital literacy

WA4MPSS2

In a real-world context, pose questions and collect categorical or discrete numerical data, checking for accuracy and consistency. Organise and represent data in pictographs and column graphs, and interpret the data to communicate findings in terms of the context

For example:

- drafting suitable questions, response categories and recording methods to collect data on topics, such as 'What is the most popular playground game?'
- recording responses accurately, including unexpected responses
- creating data displays, including with the use of digital tools, considering appropriate titles, labels, scales and categories for the axes
- interpreting the data displays created, summarising the information, responding to the initial question and discussing any unexpected results

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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Year 5

Year level descriptions

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Mathematics provides opportunities for students to develop a sound grasp of numeric conventions. Concrete materials continue to assist students to make sense of mathematical concepts as they develop the ability to think in more abstract terms.

Students engage in a range of approaches to learning through the proficiencies of understanding, fluency, problem-solving and reasoning. These reinforce the significance of working mathematically with the content and describe how the content is explored or developed.

In Year 5, students start to generalise place value patterns in seven-digit numbers and beyond, and explore representations of factors and multiples. They apply this understanding when choosing from a range of strategies to calculate efficiently, and model additive and multiplicative problems relevant to their real world. Students choose appropriate metric units in practical situations, connect three-dimensional objects with their nets and convert between 12- and 24-hour time systems. They explore the difference between chance events with equally likely and not equally likely outcomes and discuss variation in results across trials of repeated chance experiments. Students are introduced to line graphs showing continuous data, and they make decisions about organising and representing data they have collected.

Achievement standard

By the end of the year:

Students demonstrate the behaviours of the proficiencies of understanding, fluency, problem-solving and reasoning in conjunction with year level content in routine situations. They select from and use content and mathematically model situations to solve real-world problems in familiar contexts.

Students add and subtract whole numbers, multiply larger whole numbers by one- and two-digit numbers and divide whole numbers by one-digit numbers, including those with remainders, using efficient strategies. They identify factors and multiples of whole numbers. Students order decimal numbers, locate unit fractions on number lines and recognise that 100% represents a complete whole. They add and subtract fractions with the same denominator. Students follow a rule to create patterns and create simple budgets.

Students choose and use appropriate metric units to measure length, capacity and mass. They identify the dimensions of metric square and cubic units, and use them to compare areas and volumes. Students connect three-dimensional objects to their nets, construct angles in degrees and identify types of angles. They use grid coordinates and convert between 12- and 24-hour time systems to determine duration.

Students make comparisons between events with equally likely and not equally likely outcomes and conduct repeated chance experiments with equally likely outcomes, representing results as fractions. They pose questions to collect categorical or discrete numerical data and make appropriate choices to represent the data. Students describe and interpret data represented in line graphs.

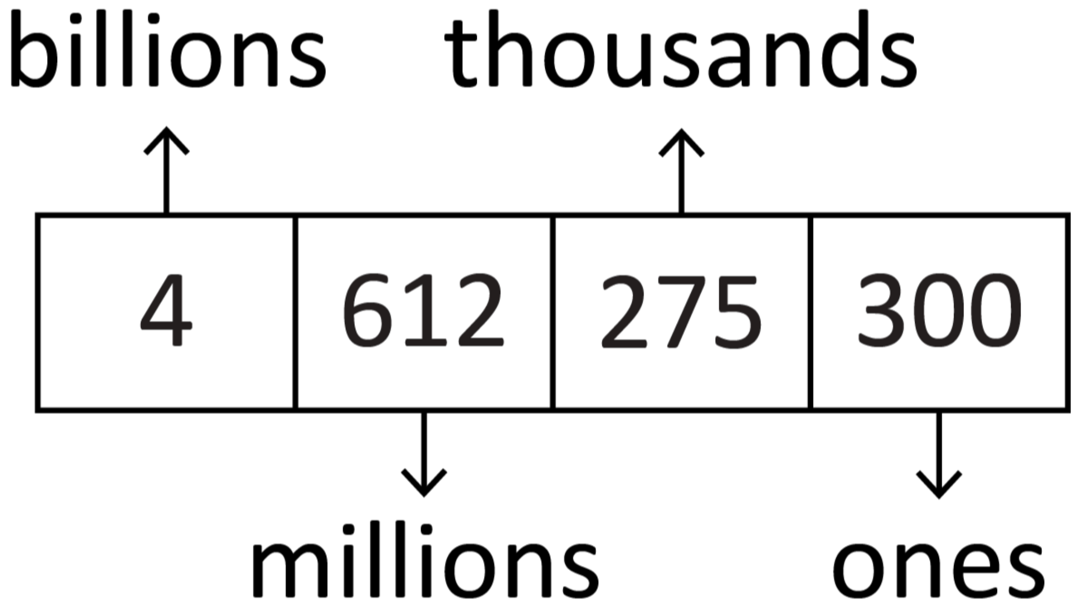
Content descriptions

Number and algebra

Understanding number

WA5MNAUN1

Read, write and order seven-digit numbers and beyond

> **For example:**
>
> - using diagrams to partition numbers in groups of three digits, associate them to the place value (i.e. billions, millions etc.) and support reading
>
> 
>
>
> - arranging numbers in the millions in ascending and descending order

General Capabilities: Literacy, Numeracy

WA5MNAUN2

Read, write, compare and order decimal numbers, including on a number line

> **For example:**
>
> - reading decimal numbers correctly, such 1.436 as 'one point four three six'
> - comparing decimal numbers, such as $0.25 < 2.5$, $2.46 > 2$ and $1.45 < 1.54$
> - comparing and ordering decimal numbers, such as 0.5, 0.75, 0.25, 0.001, and positioning them on a number line
> - interpreting zero digit/s at the end of a decimal, such as recognising that 0.170 and 0.17 have the same value

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA5MNAUN3

Represent and partition numbers up to seven digits. Use the multiplicative place value relationship between adjacent places to explain the value of a digit

> **For example:**
>
> - partitioning numbers using place value, such as
>
> $3\ 487\ 621 = 3\ 000\ 000 + 400\ 000 + 80\ 000 + 7000 + 600 + 20 + 1,$
>
>   and non-standard partitions, such as
>
> $645\ 522 = 500\ 000 + 100\ 000 + 45\ 000 + 522$
>
> - using the multiplicative place value relationship, explaining the different values of the fours in 5446 i.e. the 4 in the hundreds place is 10 times as many as the 4 in the tens place
> - identifying the place value relationships
>
> $10\ 000(\text{ten thousands}) \div 10 = 1000(\text{thousands})$
>
> $1000(\text{thousands}) \div 10 = 100(\text{hundreds})$
>
> $100(\text{hundreds}) \div 10 = 10(\text{tens})$
>
> $10(\text{tens}) \div 10 = 1(\text{ones})$

General Capabilities: Literacy, Numeracy

WA5MNAUN4

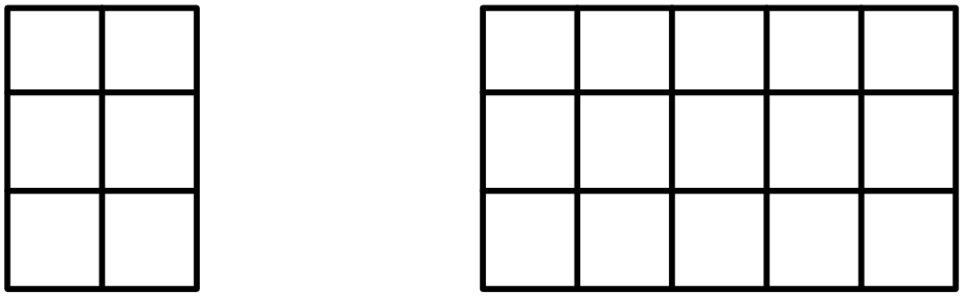
Represent and partition decimal numbers. Use the multiplicative place value relationship between adjacent places to explain the value of a digit

> **For example:**
>
> - using place value to partition decimals, such as $5.37 = 5 + 0.3 + 0.07$ and 5.429 is 5 ones and 429 thousandths
> - comparing the value of digits by determining numbers that are 10 times the original decimal number, such as 5.0 is 10 times 0.5
> - exploring multiplying and dividing by 10 using digital tools and explaining place value relationships, such as
>
> $492\ 731 \div 10 = 49\ 273.1$
>
> $49\ 273.1 \div 10 = 4927.31$
>
> $4927.31 \div 10 = 492.731$

General Capabilities: Literacy, Numeracy

WA5MNAUN5

Explore, identify and represent factors and multiples of whole numbers in arrays and explain reasoning

> **For example:**
>
> - using blocks or grids to form different rectangles to list all possible factors for that number, such as 12 can form the following rectangles: 1×12, 2×6 and 3×4
> - demonstrating that all multiples can be formed by combining or regrouping, such as multiples of seven can be formed by combining a multiple of two with the corresponding multiple of five
>
> 
>
> $$3 \times 2 \quad + \quad 3 \times 5$$
>
> - identifying lowest common multiples and highest common factors of pairs or triples of whole numbers, such as the lowest common multiple of six and nine is 18 and the highest common factor is three and the lowest common multiple of three, four and five is 60 and the highest common factor is one
> - creating a sequence of steps based on multiplication and division facts to determine if a number is a multiple or a factor of another number, and representing this in diagrams or flow charts, such as a yes/no flow chart

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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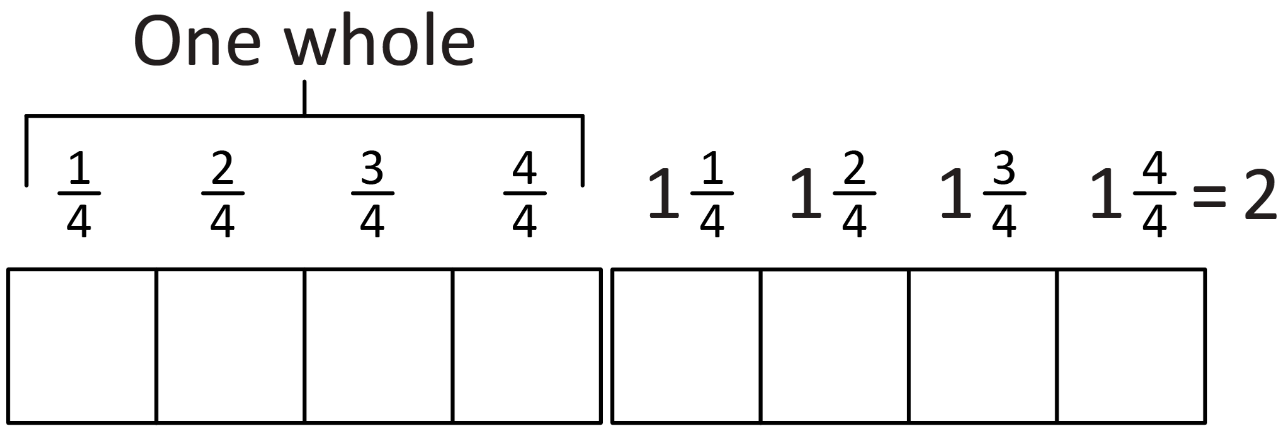
WA5MNAUN6

Count by unit fractions, locate and represent on number lines and extend to mixed numerals

> **For example:**
>
> - cutting objects, such as oranges, into sixths and counting by sixths to find the total
> - counting forwards and backwards by unit fractions

- describing and representing quantities that are more than one whole



- moving flexibly between mixed numerals and improper fractions, such as seeing that $1\,\frac{1}{4}$ is equivalent to $\frac{5}{4}$

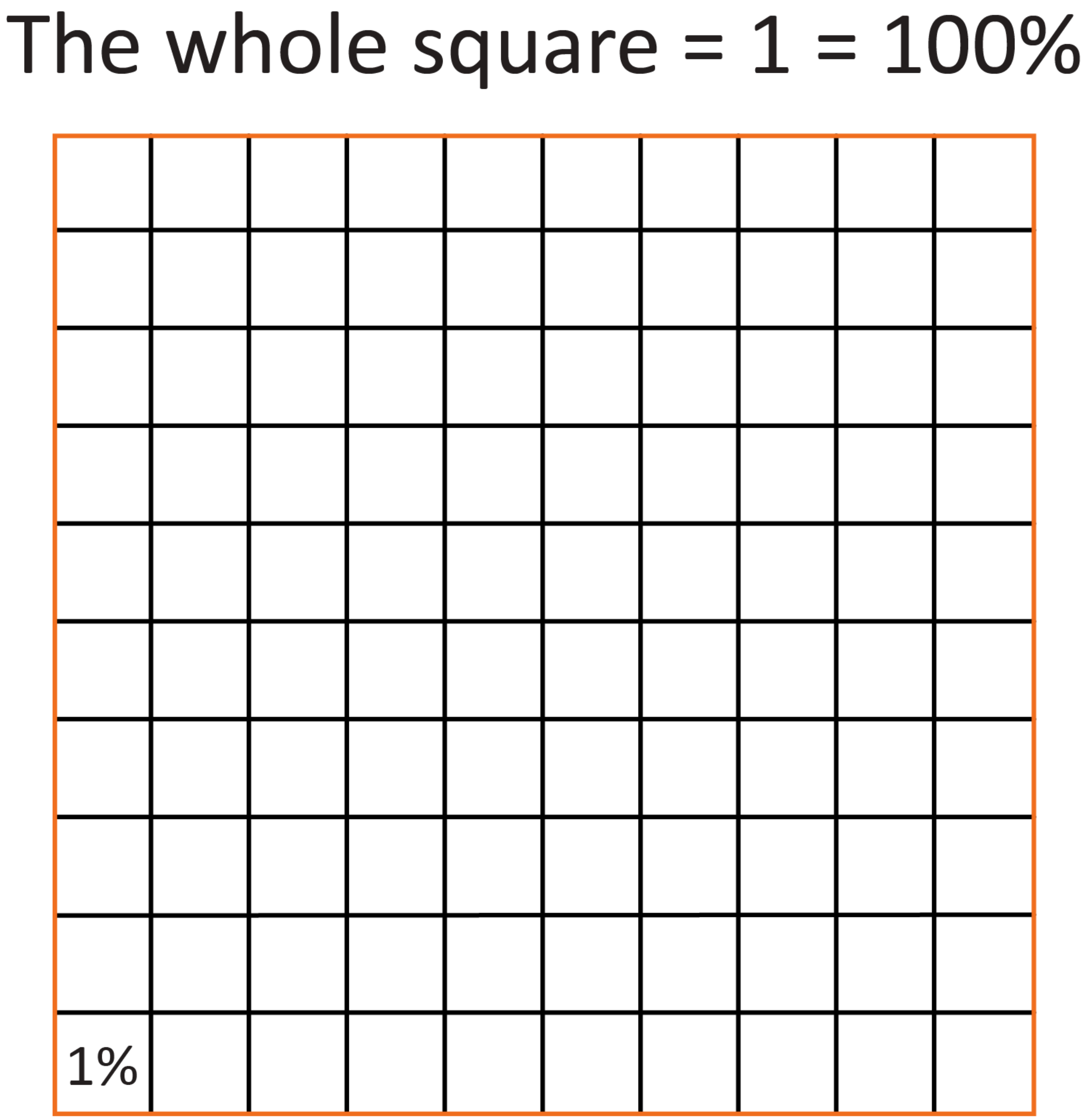
General Capabilities: Numeracy

WA5MNAUN7

Identify the use of percentages in everyday situations and recognise that 100% represents a complete whole, which is equal to one

For example:

- using grid paper to represent a whole



- identifying the use of percentages in familiar contexts, such as battery levels on laptops or tablets, sport statistics or supermarket discounts
- recognising percentages that represent more or less than a whole, such as 120% is more than a whole and 75% is less than a whole

General Capabilities: Numeracy

Understanding equalities and inequalities

WA5MNAUE1

Complete and check statements of equality and inequality involving the four operations, and explain reasoning

For example:

- completing statements, such as $28 \times 5 \div \square = 28$

  $2 \times 3 \times 4 \times 5 = 4 \times 3 \times \square$

- checking statements, such as $2 \times 18 = 4 \times 9$, to identify if they are true or false, explaining that doubling one number and halving the other will give the same result

- reasoning to insert the correct symbol to make true statements without calculating, such as $15 + 87 \ \square \ 16 + 86$

- reasoning to make the statement true when there is an unknown quantity, such as $35 \times \square = 70$

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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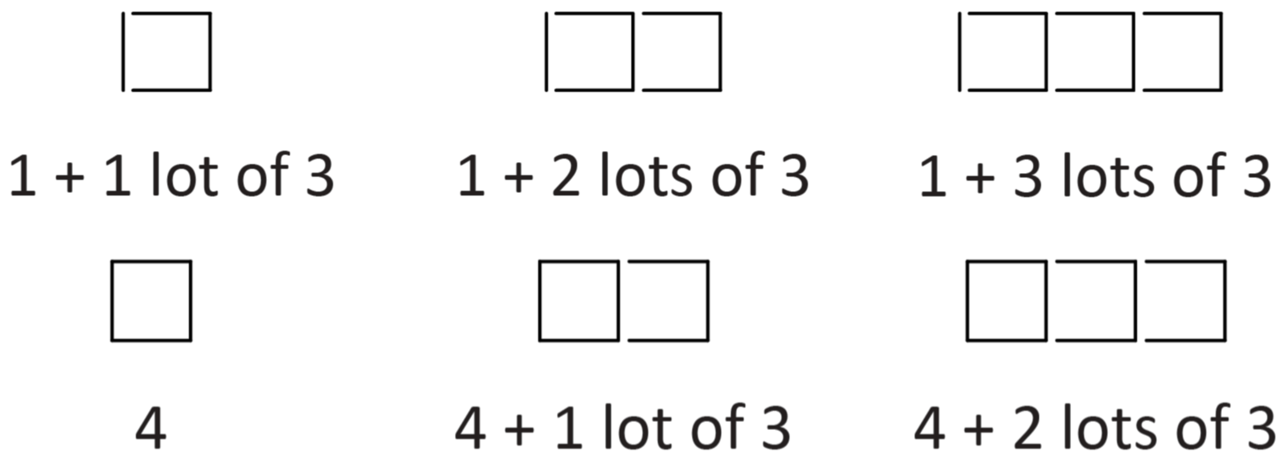
Patterns and relationships

WA5MNAP1

Follow rules to create increasing or decreasing additive and multiplicative patterns using concrete materials and numbers. Explore ways to predict unknown values

For example:

- following a rule, such as 'starting at two, double and add one, and repeat', drawing or creating the pattern and producing the sequence 2, 5, 11, 23 ...
- recognising that the same pattern can be created using different rules and predicting that the number of toothpicks needed for eight boxes will be 1 plus 8 lots of 3 or 4 plus 7 lots of 3, such as



- entering an additive and multiplicative formula, such as = (A1*3 – 5), into a spreadsheet and following the rule to generate a sequence of numbers, starting from 3, to determine the 15th term
- predicting if the next number in a pattern will be odd or even

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Calculating with number

WA5MNAC1

Add and subtract any whole numbers using flexible and efficient strategies

For example:

- applying known strategies, such as

  Levelling ($4988 + 955 \rightarrow 4990 + 953 \rightarrow 5000 + 943$)

  Addition for subtraction (5009 − 3997 add 3 is 4000 plus 1009)

  Constant difference (5009 − 3997 → 5012 − 4000)

  Bridging (4988 + 955 = 4988 plus 2 plus 10 plus 943)

  Place value (42 000 + 5123 + 246 = 40 000 + 7000 + 300 + 60 + 9)

- using algorithms to record addition and subtraction calculations

$$\begin{array}{cccccc} & & 1 & 1 & & \\ & 2 & 1 & 3 & 8 & 4 \\ + & & 3 & 7 & 9 & 1 \\ \hline & 2 & 5 & 1 & 7 & 5 \end{array} \qquad \begin{array}{cccccc} & 2 & & 1 & 16 & \\ & \cancel{3} & {}^{1}0 & 2 & \cancel{7} & {}^{1}3 \\ - & & 1 & 0 & 9 & 6 \\ \hline & 2 & 9 & 1 & 7 & 7 \end{array}$$

- identifying efficient and inefficient multidigit subtraction strategies, such as solving 9000 − 7 mentally is efficient and solving using an algorithm is inefficient

General Capabilities: Numeracy, Critical and creative thinking

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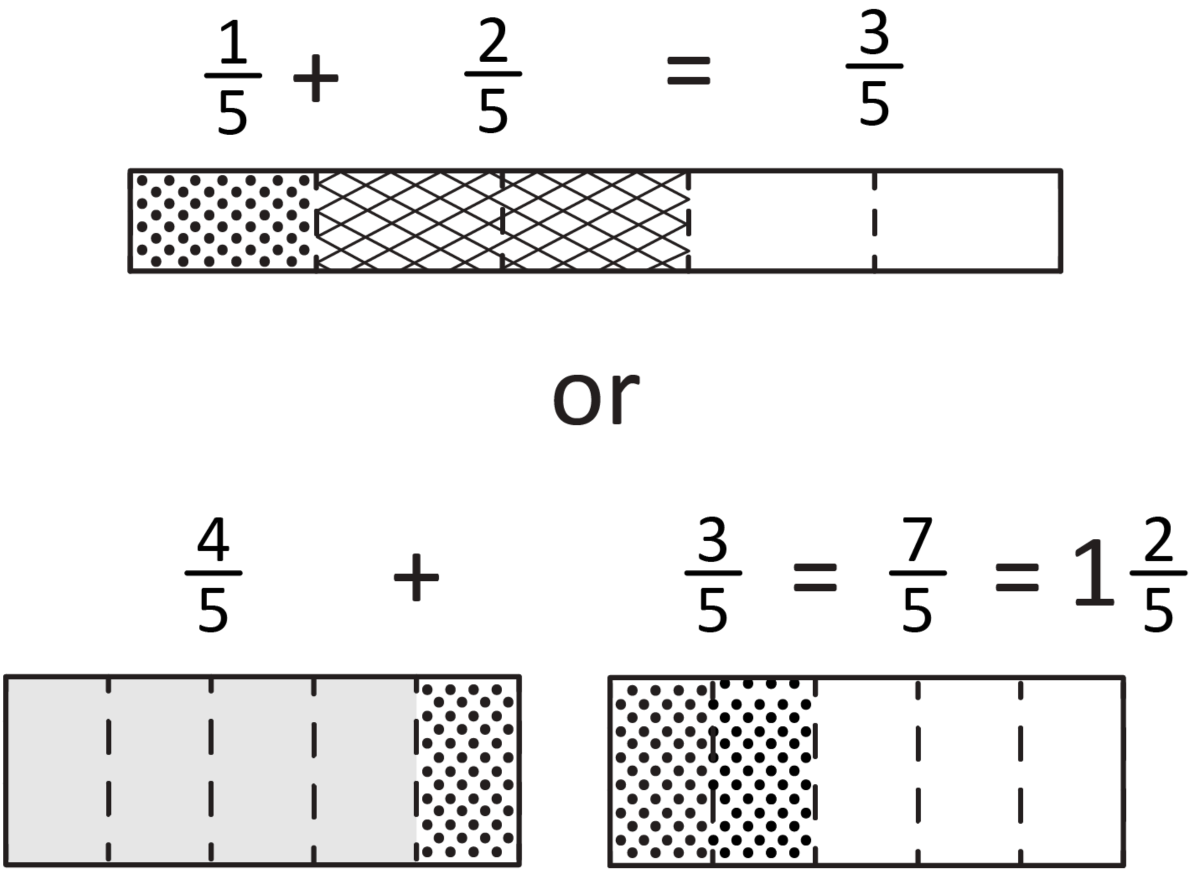
WA5MNAC2

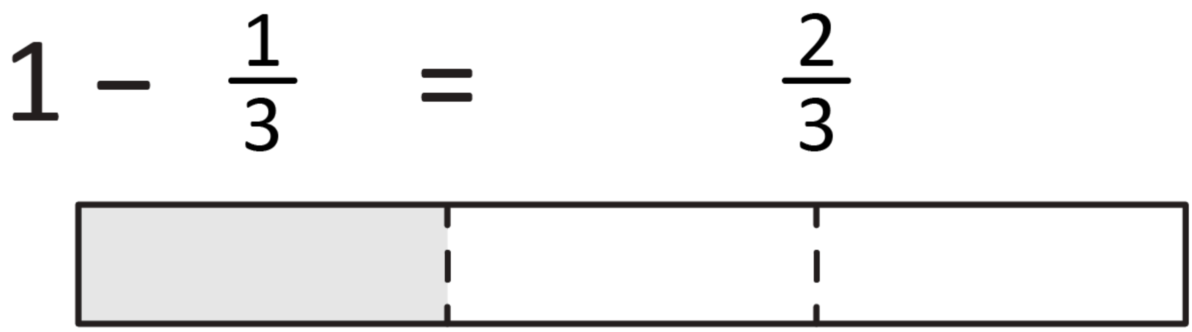
Add and subtract fractions with the same denominator using flexible and efficient strategies

For example:

- using concrete materials or diagrams



- using diagrams, objects and mental strategies to subtract a unit fraction from any whole number, including one



General Capabilities: Numeracy, Critical and creative thinking

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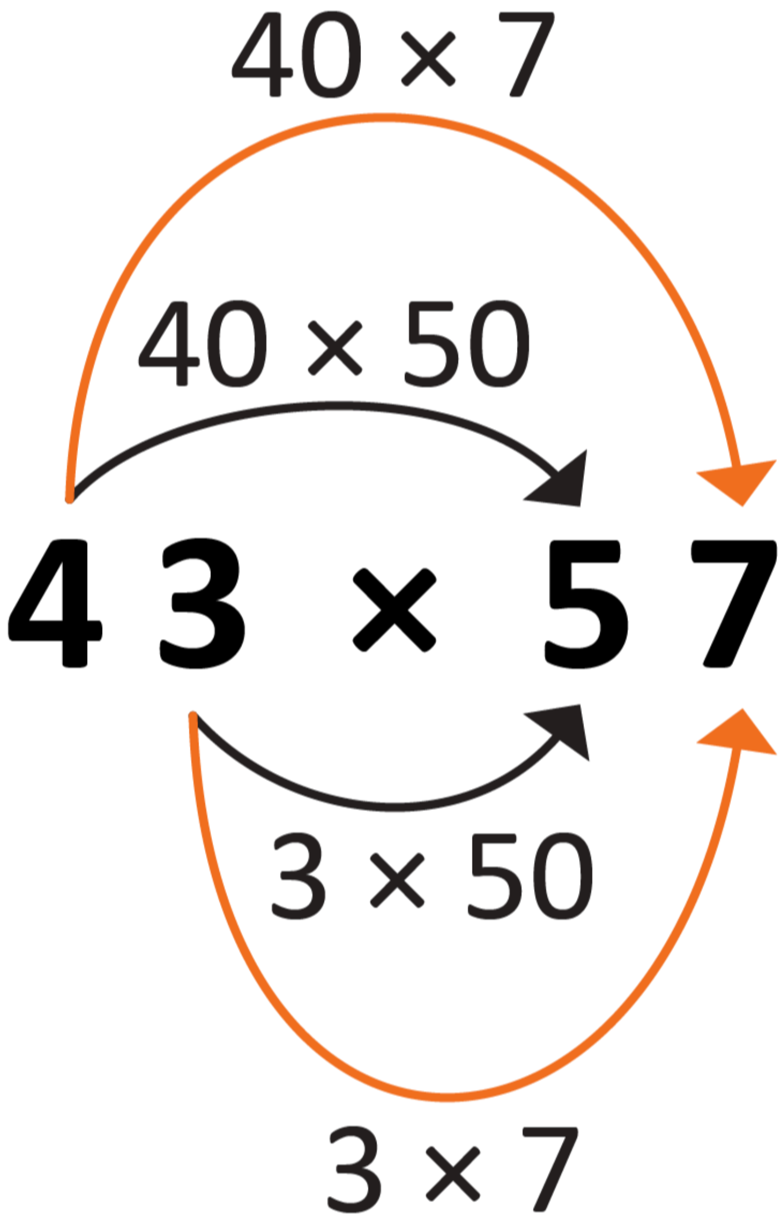
WA5MNAC3

Multiply larger whole numbers by one- and two-digit numbers and divide whole numbers by one-digit numbers, including those with remainders, using flexible and efficient strategies

For example:

- utilising factors to multiply, such as 36×25 can be rewritten as $9 \times 4 \times 25$, which is 9×100

- making division easier by multiplying or dividing by the same amount, such as $28 \div 4$ is $14 \div 2$
- partitioning numbers to simplify multiplication, such as



- using multiplication facts to facilitate division, such as solving $27 \div 6$ by using $4 \times 6 = 24$ and $5 \times 6 = 30$ to predict the result must be between 4 and 5, with a remainder $(3 = 30 - 27)$

General Capabilities: Numeracy, Critical and creative thinking

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WA5MNAC4

Explore multiplicative estimation strategies and their appropriateness in different situations

For example:

- recognising the effect of rounding both numbers up, both numbers down or one number up and one number down, explaining which estimation is the best approximation and why

- estimating how many pages are in a book series if there are five books in the series and each book (in the series) has about 120 pages, using doubling and halving to solve i.e.

  $10 \times 60 = 600$ pages

- estimating if $30 will be enough to buy six mugs at $4.95 each by rounding to $5

General Capabilities: Numeracy, Critical and creative thinking

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Financial mathematics

WA5MNAF1

Identify features of budgets and create a simple budget, comparing prices where possible

> **For example:**
>
> - using a context, such as a school trip or class party, to create a simple budget
> - distinguishing between total cost and cost per person
> - comparing prices and recognising that a product/service can be overpriced
> - developing an awareness that advertising can influence purchasing decisions

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Modelling with number

WA5MNAM1

Identify and represent a range of real-world additive and multiplicative situations with equations, using diagrams where needed. Interpret and communicate findings in context

> **For example:**
>
> - modelling situations, such as determining how many containers Amy collected for 'Containers for Change', knowing their total was 12 times as many as Ling, who collected 25. Determining the situation to be multiplicative and representing it using a diagram and equation, demonstrating how the representation is connected to the situation, such as the diagram shows that 12×25 is 12 times as many as 25. Using efficient strategies to solve, such as halving 12 and doubling 25, communicating that the answer is the number of containers Amy collected, which is 12 times as many as Ling

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Measurement and geometry

Two-dimensional space and structures

WA5MMGTW1

Explore line and rotational symmetry in two-dimensional shapes

> **For example:**
>
> - identifying line symmetry in the environment, artworks and patterns
> - identifying the rotational symmetry of shapes by tracing and rotating them to determine how many times they match their original position in a full rotation, such as a rectangle aligns with itself twice during one complete turn

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA5MMGTW2

Choose and use appropriate metric units and part units to estimate and measure lengths

> **For example:**
>
> - choosing and using an appropriate unit, such as centimetres when measuring a length of fabric
> - recognising that the choice of an appropriate unit may depend upon the need for accuracy rather than simply the size of the object, for instance a bridge may be measured in metres to estimate length, but in millimetres for engineering work
> - recording measurements using both whole units and whole and part units, such as 56 millimetres or 5.6 centimetres

General Capabilities: Numeracy, Critical and creative thinking

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WA5MMGTW3

Describe and test a sequence of steps to determine the perimeter of rectangles

> **For example:**
>
> - creating a range of rectangles representing paddocks on grid paper and establishing different methods of working out the length of boundary fences
> - exploring efficient ways to calculate the perimeters of rectangles, such as adding the length and width together and doubling the result

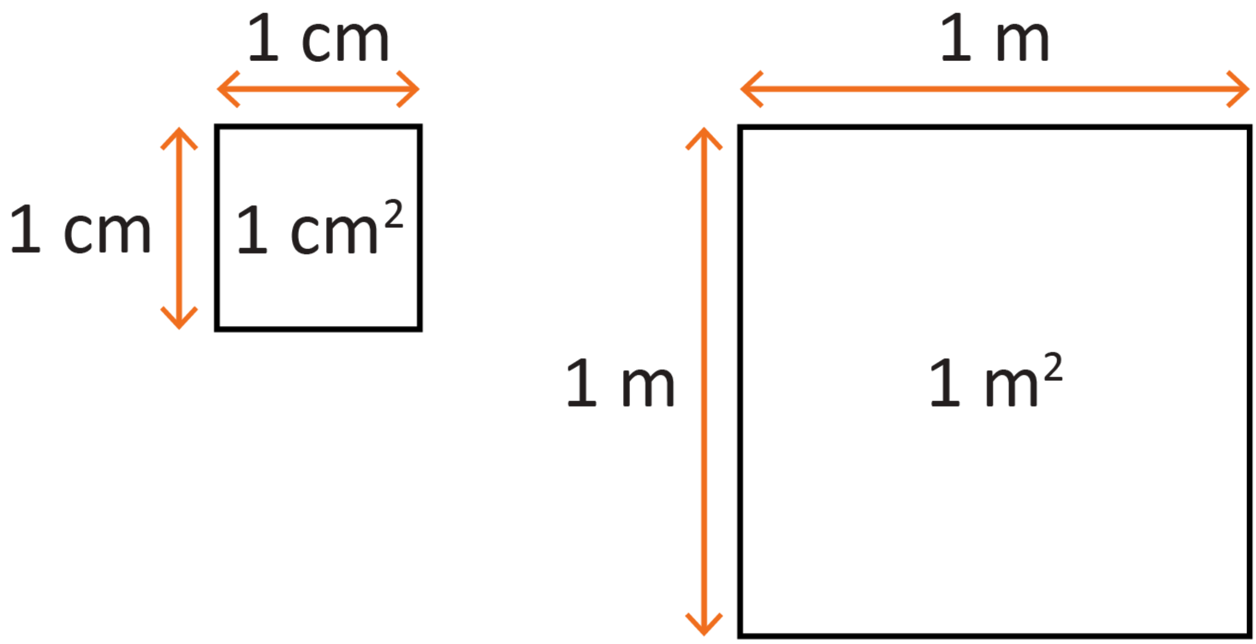
General Capabilities: Literacy, Numeracy

WA5MMGTW4

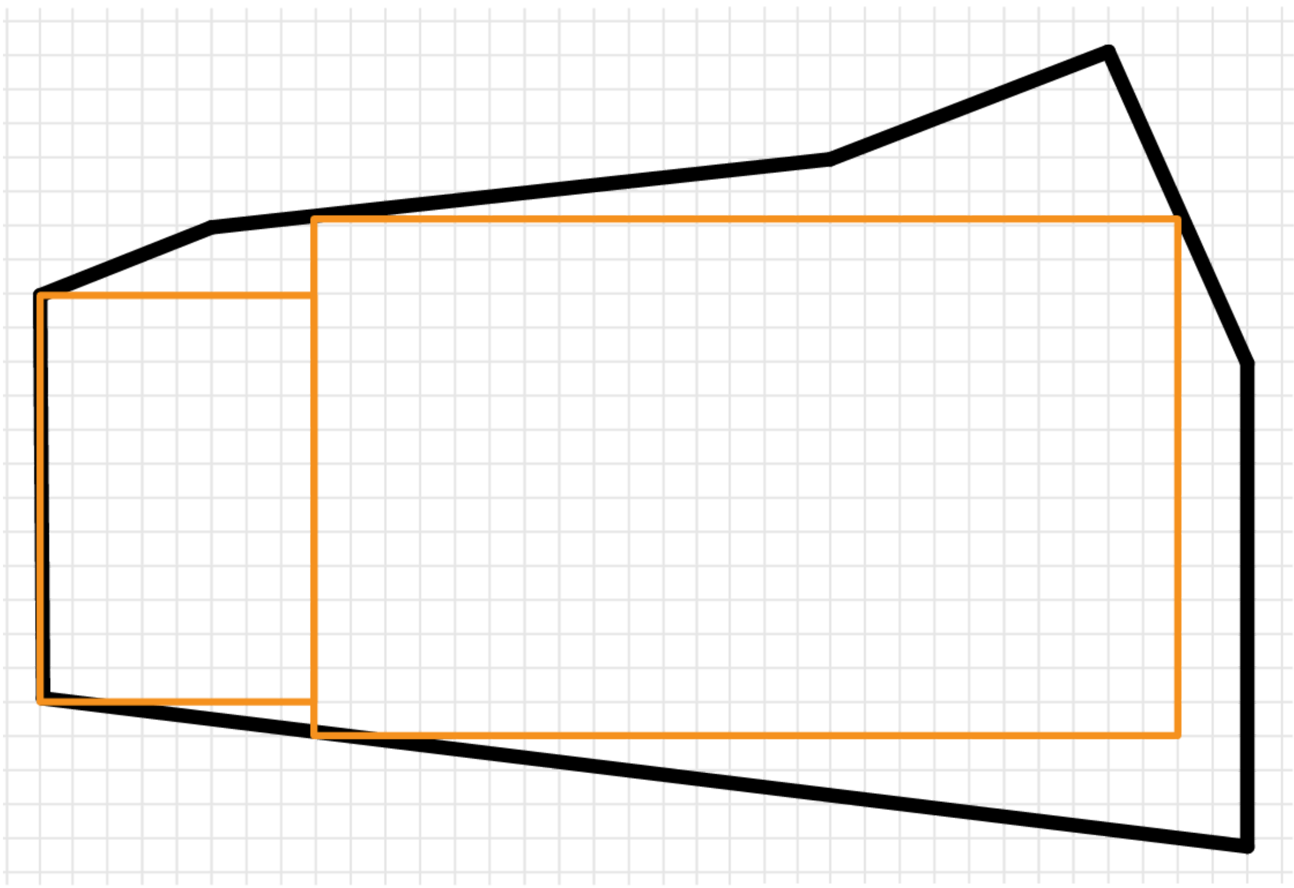
Identify dimensions of a metric square unit. Estimate, measure and compare areas using metric square units

> **For example:**

- recognising the dimensions of a metric square unit, recording the area using the abbreviations mm^2, cm^2 and m^2 and using correct terminology, such as 2 cm^2 being read as 'two square centimetres'



- recognising that a unit of area can be cut and rearranged and still be the same area, such as a square metre not needing to be 'square' – it may be 50 cm by 2 m or 25 cm by 4 m
- exploring the use of arrays within irregular shapes to approximate the area, discussing the parts that fall outside of the arrays



General Capabilities: Numeracy, Critical and creative thinking

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WA5MMGTW5

Estimate, measure and construct angles in degrees using a protractor. Classify acute, right, obtuse, reflex and straight angles

For example:

- estimating and describing the size of angles using known angles as benchmarks
- exploring the scale on a protractor, recognising that protractors usually have two sets of numbers, one for measuring angles in an anticlockwise direction and one for clockwise
- developing and following a sequence of steps to accurately measure angles using a protractor
- recording angle measurements using the symbol for degrees (°)

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA5MMGTW6

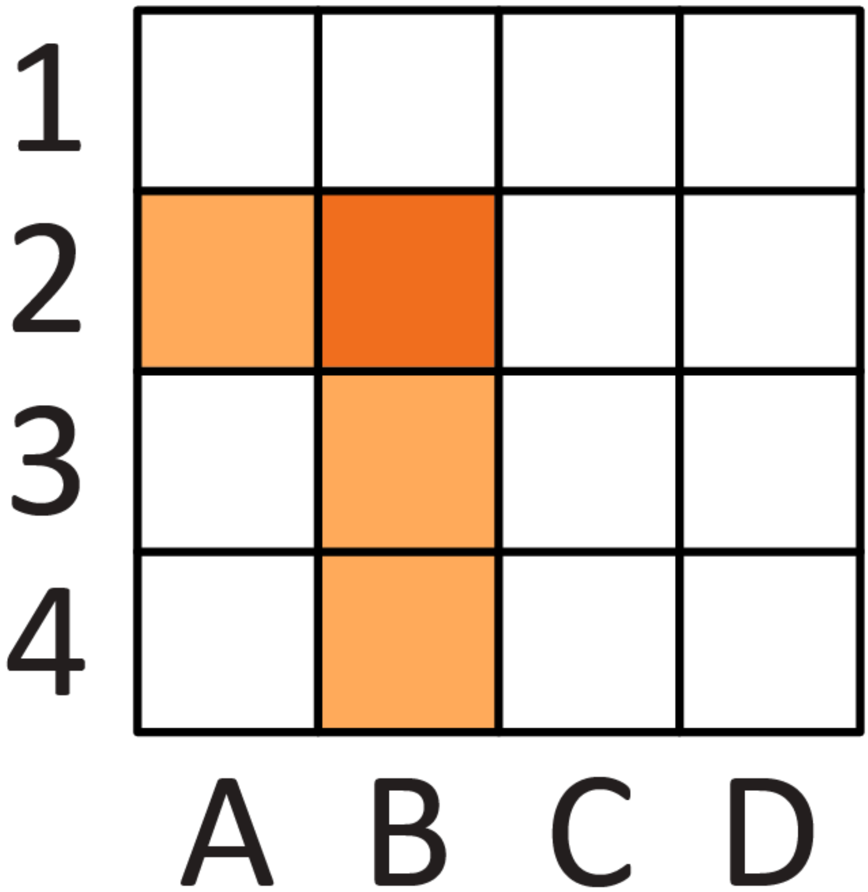
Use directional language, grid references and grid coordinates to describe positions and pathways

For example:

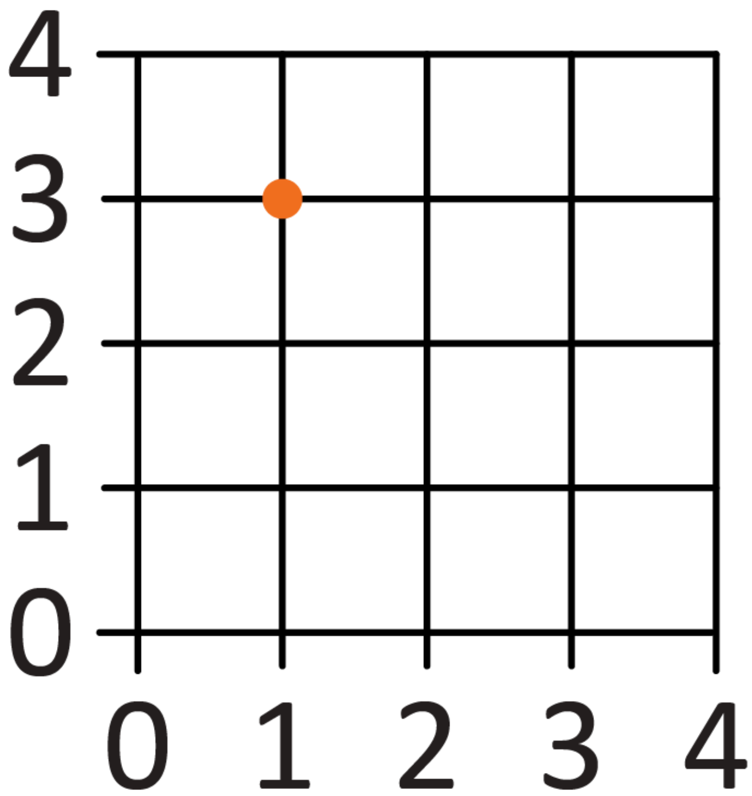
- using a given grid map and compass directions (N, S, E, W), planning, describing and showing a route from one location to another

- recognising the difference between grid references and grid coordinates.

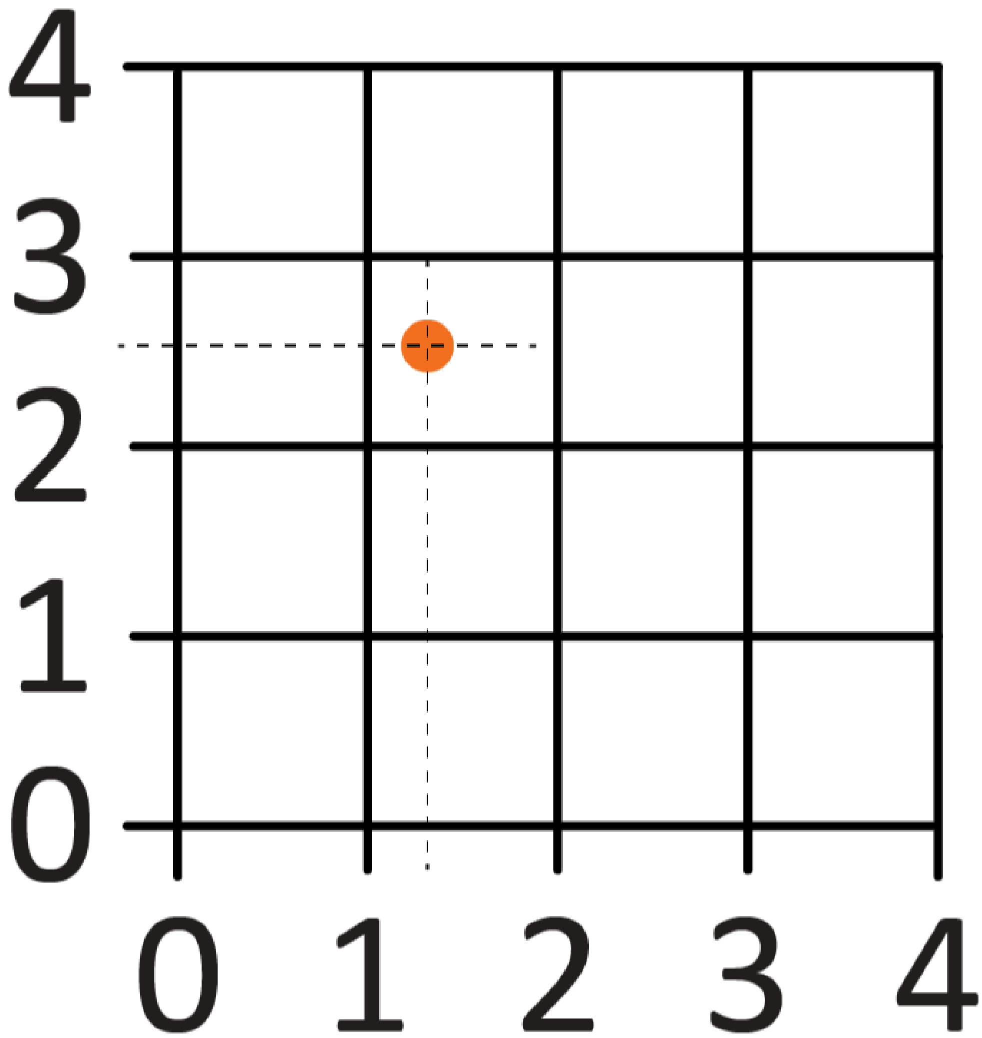
  A grid reference identifies a region by labelling the spaces e.g. B2 shows the darker shaded region in that square.



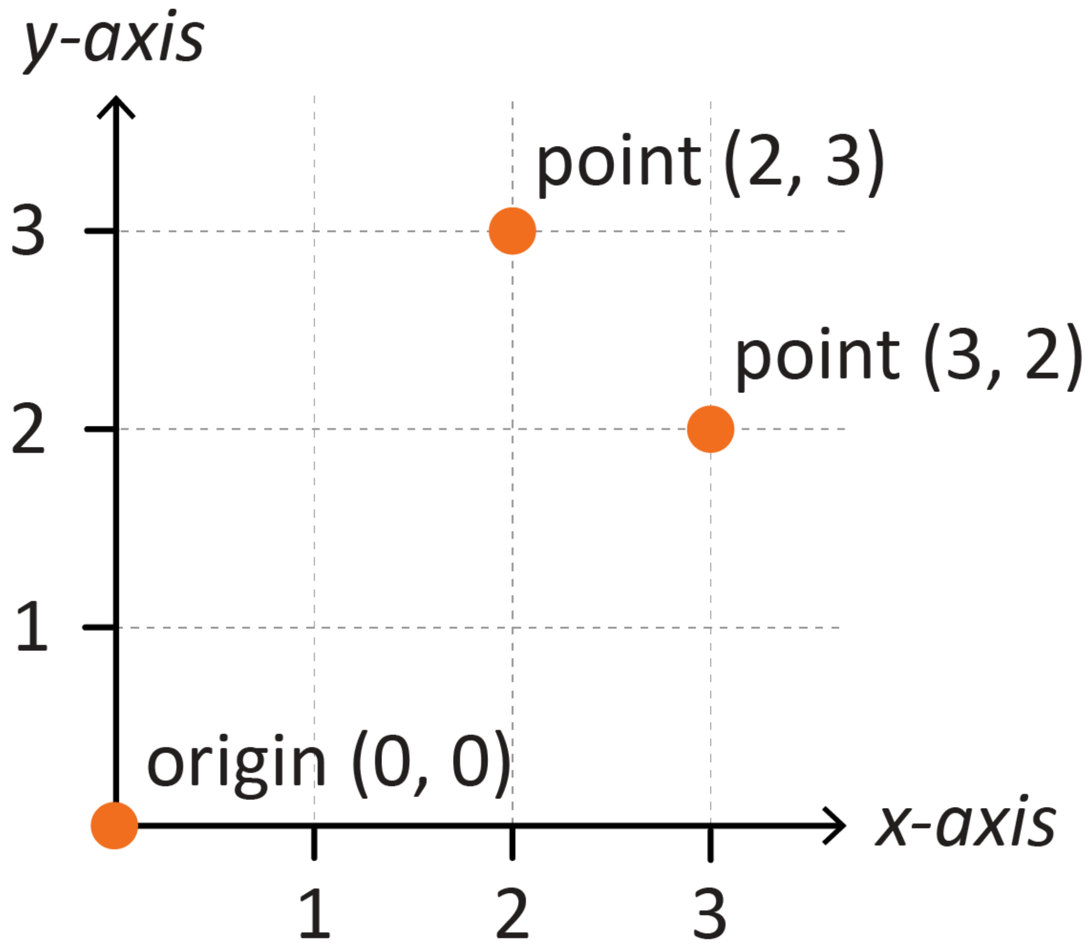
  A grid coordinate uses numbers to identify a point where two numbered lines intersect e.g. (1, 3).



Because a grid coordinate uses numbers, it can identify any point within the spaces by using decimal fractions e.g. (1.25, 2.5).



- identifying grid coordinates on the number plane in the first quadrant, describing the horizontal position first, followed by the vertical position. Recognising (2, 3) is a different location to (3, 2)



General Capabilities: Literacy, Numeracy, Digital literacy

Three-dimensional space and structures

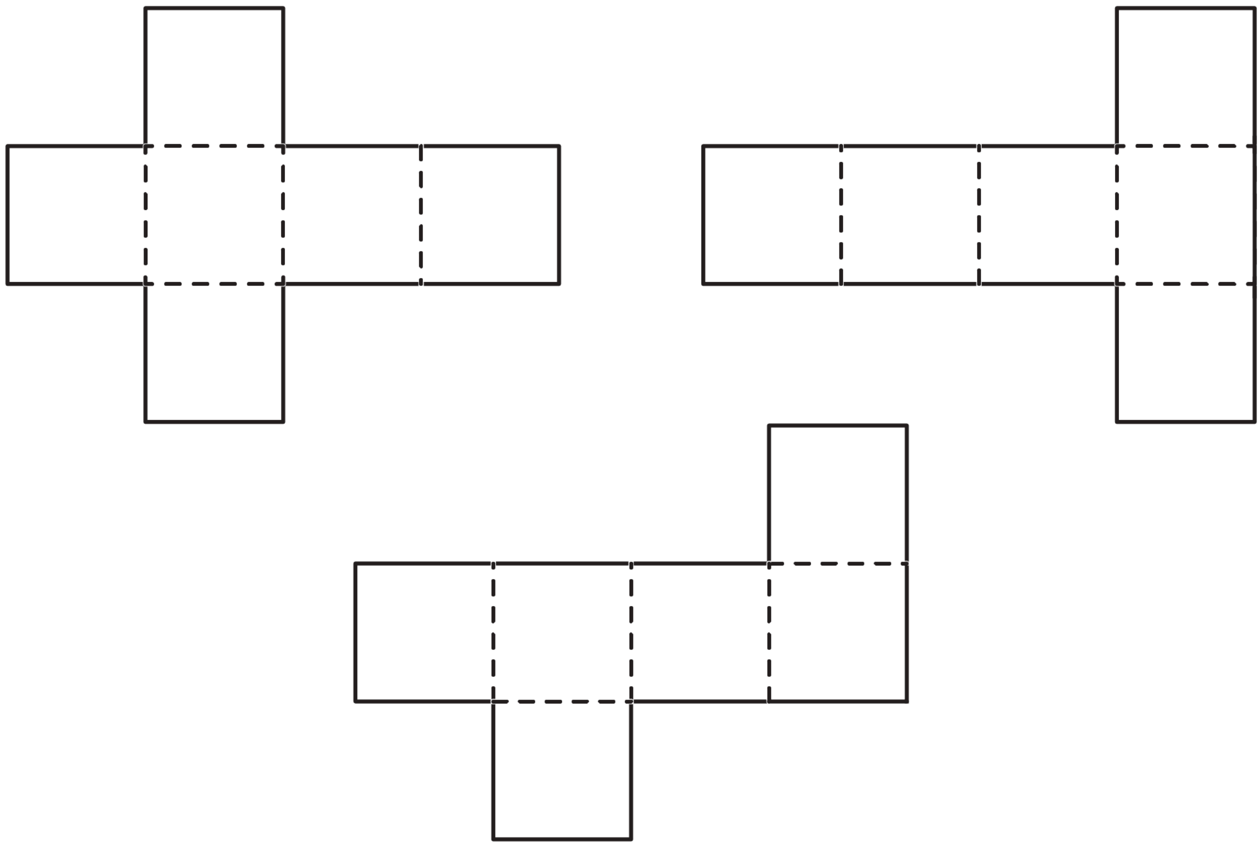
WA5MMGTH1

Visualise and connect three-dimensional objects to their nets and build objects from their nets

For example:

- deconstructing packages to see the nets of different 3D objects, including prisms, pyramids and cylinders

- investigating the variety of nets that can be used to create a particular prism



- examining a diagram to determine whether it is the net of a 3D object or not
- constructing 3D objects from provided nets and from sketching and testing nets, considering the number, shape and placement of faces

General Capabilities: Numeracy, Critical and creative thinking

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WA5MMGTH2

Choose appropriate units to estimate and measure capacity

For example:

- selecting and using appropriate units, such as millilitres for a drinking glass and litres for a bucket
- reading calibrated measuring instruments with evenly spaced markings where not all values are labelled, such as markings for 1.0, 1.2, 1.4, 1.6, 1.8, 2.0, but only 1 and 2 are labelled
- recognising and interpreting decimal notation for capacities, such as 8.7 L is the same as 8 litres and 700 mL

General Capabilities: Numeracy, Critical and creative thinking

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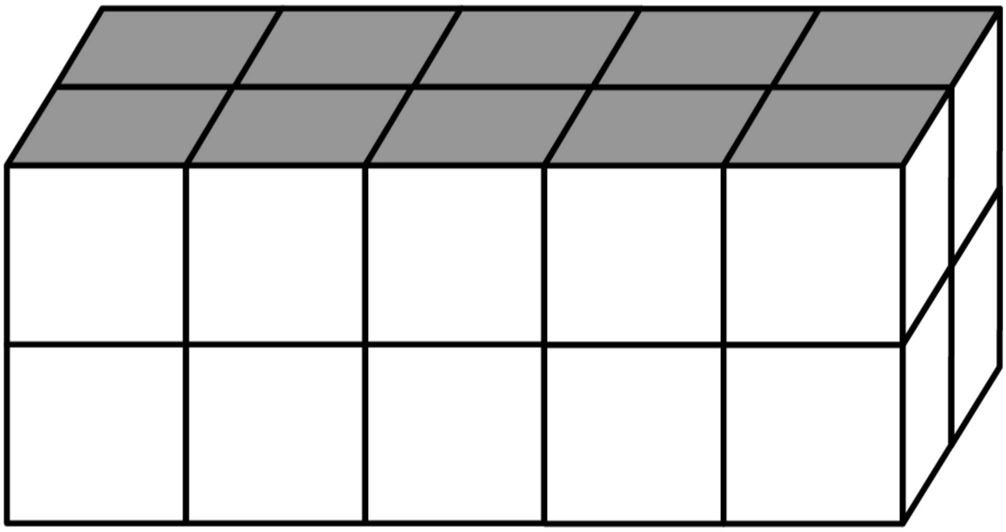
WA5MMGTH3

Identify the dimensions of a metric cubic unit. Construct and compare rectangular prisms using cubes and determine their volume

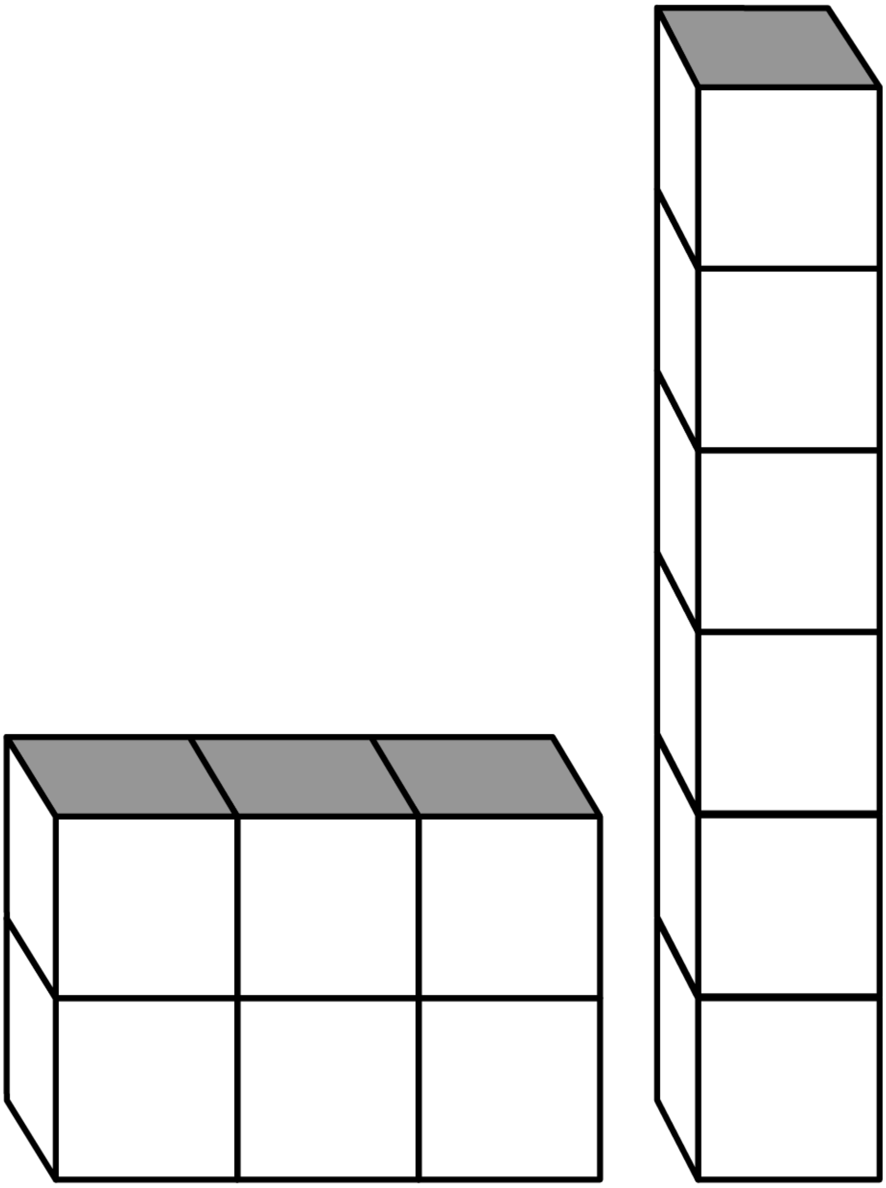
For example:

- constructing and comparing rectangular prisms using cubic-centimetre blocks and describing the volumes in terms of layers, such as two layers of 10 cubic-centimetre blocks



- recording volumes using the abbreviation for cubic centimetres (cm^3) and cubic metres (m^3)

- exploring different rectangular prisms that can be made from the same number of cubes



General Capabilities: Numeracy, Critical and creative thinking

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Non-spatial measurement

WA5MMGN1

Choose appropriate units to estimate, measure and compare mass

For example:

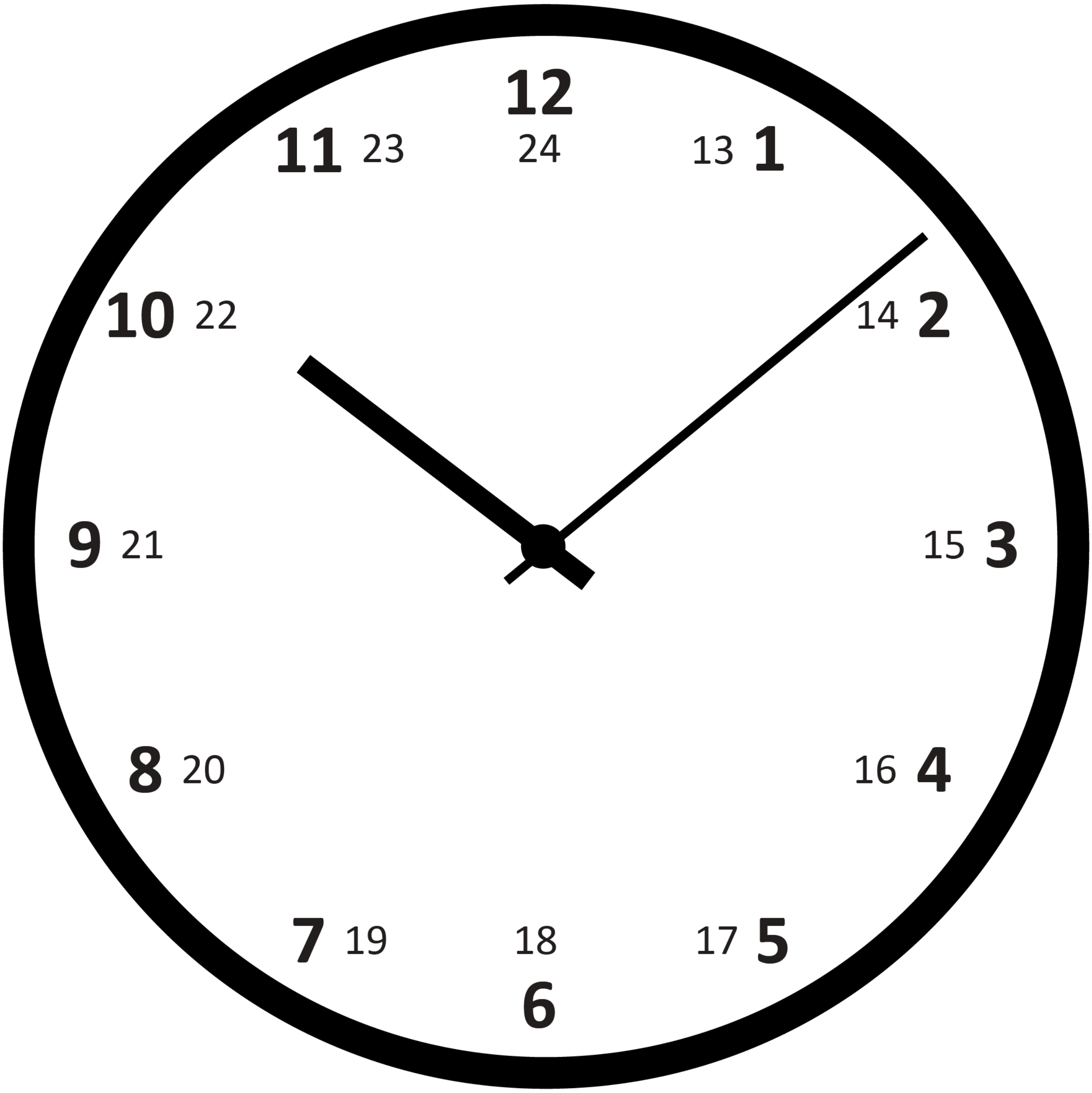
- identifying the appropriate unit and device to measure mass, such as digital scales to measure a school bag (kg) and kitchen scales to measure the weight of ingredients (g)
- comparing readings on digital scales, applying knowledge of decimal numbers, such as recognising that 3.25 kg is heavier than 3.025 kg

- recognising the equivalence of whole number and decimal representations of measurements of mass, such as 3 kg 250 g is the same as 3.25 kg

General Capabilities: Numeracy

WA5MMGN2

Explore, describe and convert between 12- and 24-hour time systems and use to determine duration

> **For example:**
>
> - using analogue clocks to make connections between 12- and 24-hour time systems
>
> 
>
>
> - applying strategies to convert between time systems, understanding that the same time can be described in different ways, such as to convert times from 12.00 pm to 11.59 pm into 24-hour time, 12 is added to the hour
>   1.00 pm → 1300 in 24-hour time
>   7.45 pm → 1945 in 24-hour time

- determining duration of events when the starting and finishing times are provided in different systems, such as the duration of a school excursion scheduled to start at 9.30 am and conclude at 1445

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Modelling with measurement and geometry

WA5MMGM1

In real-world situations involving transformation of two-dimensional shapes, nets, grid reference systems, determining length, area, capacity, volume or mass in metric units or converting between 12- and 24-hour time, mathematically represent the problem to reach a solution. Interpret and communicate findings in the context of the situation

> **For example:**
>
> - modelling situations involving metric units, such as planning items to pack for a school camp, given a list of required items and a weight and size limit for student bags. Representing the problem with diagrams and equations, determining the weights of given items, using estimation strategies where appropriate and determining the total weight. Interpreting the findings, such as the weight could be reduced if a plastic water bottle is chosen rather than a metal one
> - modelling situations, such as
>   - considering if the classroom can fit a new table by creating a grid map of the current configuration of the classroom, to an approximate scale. Using translations and grid references to analyse and solve the problem
>   - creating a pathway utilising a map of a local community, ensuring all points of interest are explored and identifying the total distance covered

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Probability and statistics

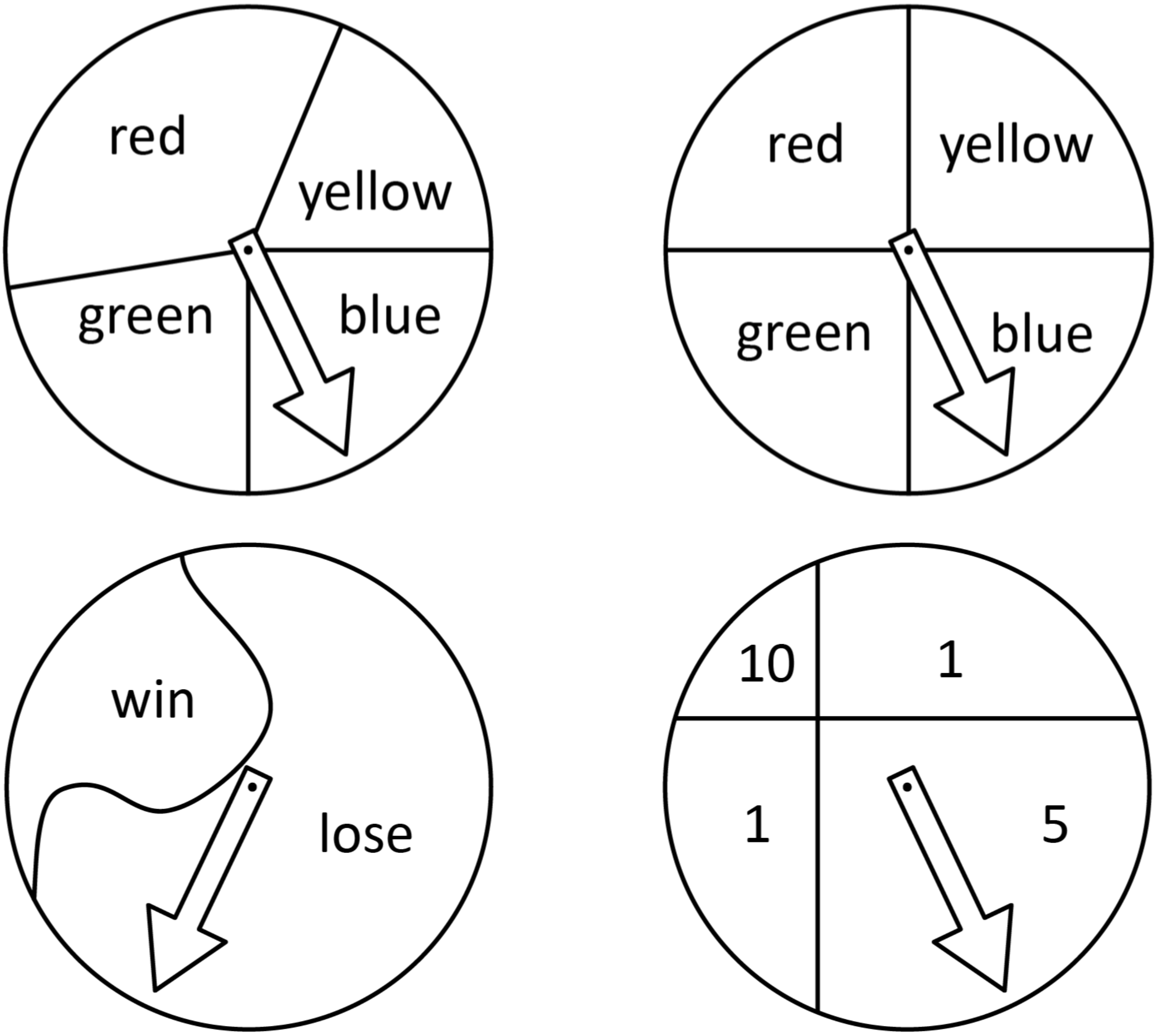
Probability

WA5MPSP1

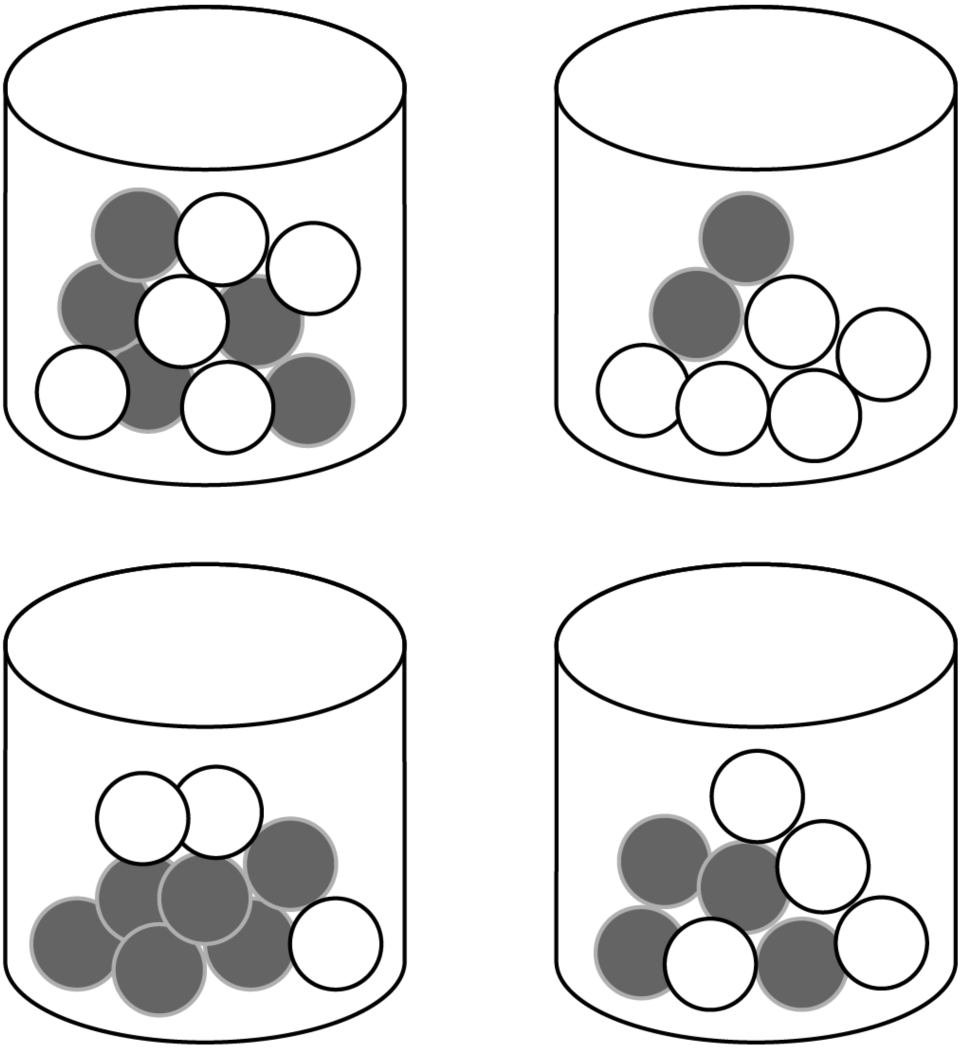
Compare a range of everyday chance events, grouping into those with outcomes that are equally likely or not equally likely

For example:

- comparing differently divided spinners and identifying if they will produce equally or unequally likely outcomes for each section



- comparing jars filled with two colours of balls and identifying those with equally likely outcomes and with unequally likely outcomes for each colour



General Capabilities: Literacy, Numeracy, Critical and creative thinking

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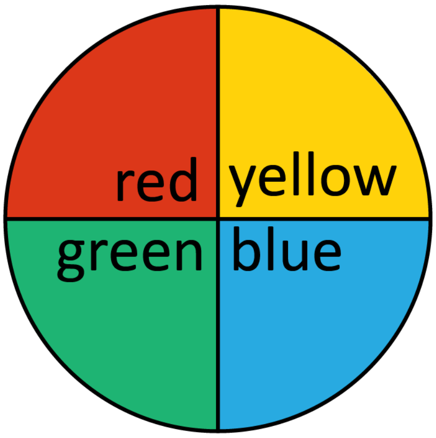
WA5MPSP2

Conduct repeated chance experiments with equally likely outcomes, including with the use of digital tools. Represent results as fractions, compare with others and discuss variation

> **For example:**
>
> - recording results for 20 trials as fractions, comparing and discussing differences between the results for each group



	Group results (20 trials)			
	Group 1	Group 2	Group 3	Group 4
red	$\frac{4}{20}$	$\frac{3}{20}$	$\frac{1}{20}$	$\frac{6}{20}$
blue	$\frac{6}{20}$	$\frac{5}{20}$	$\frac{6}{20}$	$\frac{4}{20}$
green	$\frac{5}{20}$	$\frac{5}{20}$	$\frac{7}{20}$	$\frac{6}{20}$
yellow	$\frac{5}{20}$	$\frac{7}{20}$	$\frac{6}{20}$	$\frac{4}{20}$

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

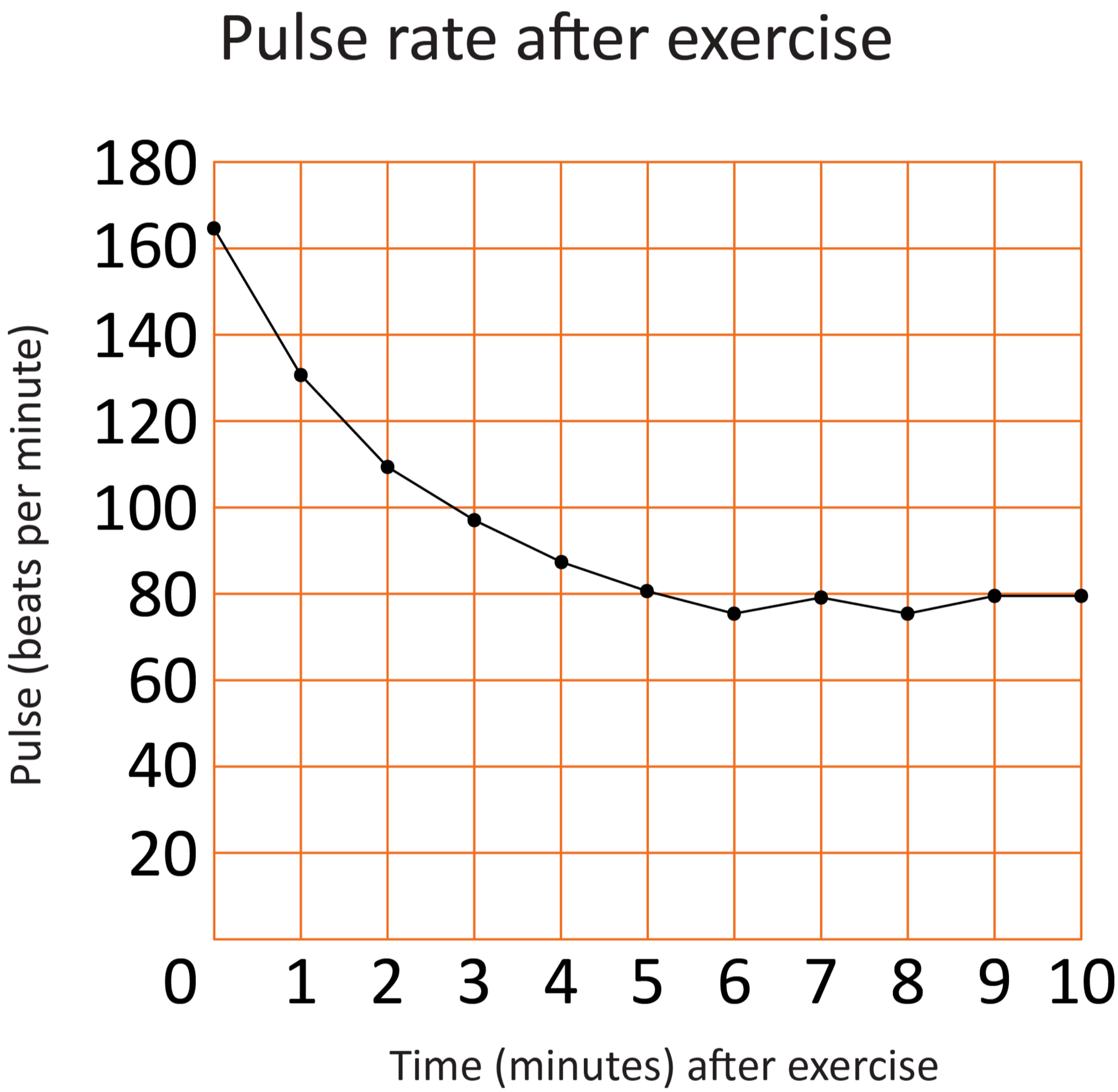
Statistics

WA5MPSS1

Describe and interpret line graphs that show how real-life continuous data changes over time

For example:

- identifying features of line graphs, such as labels, scale and title, and answering questions relating to the information in the graph



General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA5MPSS2

In a real-world context, pose and refine questions, and collect categorical or discrete numerical data. Organise and make choices to represent data. Interpret and communicate findings in terms of the context, and reflect on variation and accuracy

For example:

- formulating and refining questions, such as refining 'How many plastic bottles do people in the classroom have in their home bins at the end of a week?' to 'How many plastic bottles do people in the classroom have in their recycling bins at home?'

- collecting data using numbers to indicate rating, such as 1 = doesn't like and 5 = likes a lot, to collect data to answer, 'How did students at my school enjoy the food choices at the end-of-year event?'
- choosing and creating data displays, such as dot plots, many-to-one column graphs and pictographs, including with the use of digital tools
- discussing if the data provides the information necessary to answer the question and identifying if the type of questions, place, time or people asked could vary the results

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

Year 6

Year level descriptions

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Mathematics provides opportunities for students to develop a sound grasp of numeric conventions. Concrete materials continue to assist students to make sense of mathematical concepts as they develop the ability to think in more abstract terms.

Students engage in a range of approaches to learning through the proficiencies of understanding, fluency, problem-solving and reasoning. These reinforce the significance of working mathematically with the content and describe how the content is explored or developed.

In Year 6, students consolidate their understanding of the number system, expanding their repertoire of numbers to include integers, square, prime and composite numbers. They apply this understanding to calculate and model real-world problems efficiently and interpret and communicate findings. Students explore the Cartesian plane, describe a sequence of steps to determine the area of rectangles and volume of rectangular prisms and convert between units of measurement, connecting metric units to the decimal system. They represent probabilities numerically and conduct repeated chance experiments and simulations, comparing expected and observed frequencies. Students interpret data displays, including side-by-side column graphs, using mode, range and shape, and they describe how features of displays may influence an audience.

Achievement standard

By the end of the year:

Students demonstrate the behaviours of the proficiencies of understanding, fluency, problem-solving and reasoning in conjunction with year level content in routine situations. As part of this, they select from and use year level content along with the modelling process, in which they analyse the situation mathematically, represent the problem and interpret and communicate their findings, to solve straightforward, real-world problems in familiar contexts across all strands.

Students identify square, prime and composite numbers. They complete calculations with whole numbers using any of the four operations. Students order integers and common fractions on a number line. They add and subtract fractions with related denominators. Students add and subtract decimals to two decimal places, multiply decimals by whole numbers and multiply and divide decimals by powers of 10. They determine a familiar fraction, decimal or percentage of a whole number and make connections between them. Students describe rules that relate each element of a pattern to its position.

Students convert between units of length, mass and capacity, connecting metric units to the decimal system. They describe steps to determine the area of rectangles and the volume of rectangular prisms. Students construct three-dimensional objects, including prisms and pyramids. They determine unknown angles, explaining reasoning. Students describe translations, reflections or rotations of two-dimensional shapes and locate points in any one of the four quadrants on the Cartesian plane. They use timetables to determine duration.

Students order chance events on a scale from 0 to 1. They conduct repeated chance experiments and simulations, comparing expected and observed relative frequencies for an increasing number of trials. Students interpret a range of data displays, including using mode, range and shape, and describe how the features of displays may influence an audience. They choose the most appropriate way to collect and represent data, including in line graphs and side-by-side column graphs.

Content descriptions

Number and algebra

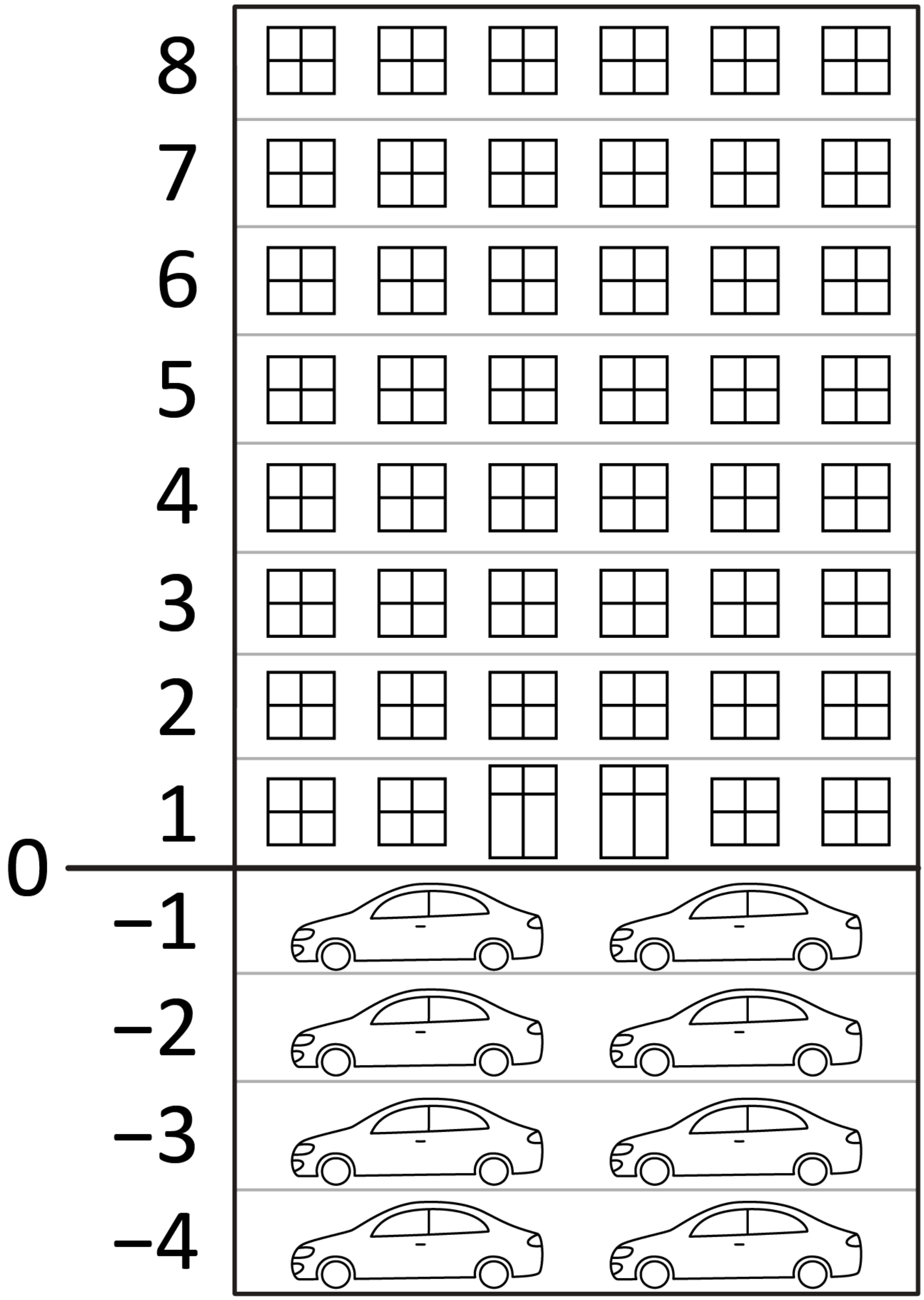
Understanding number

WA6MNAUN1

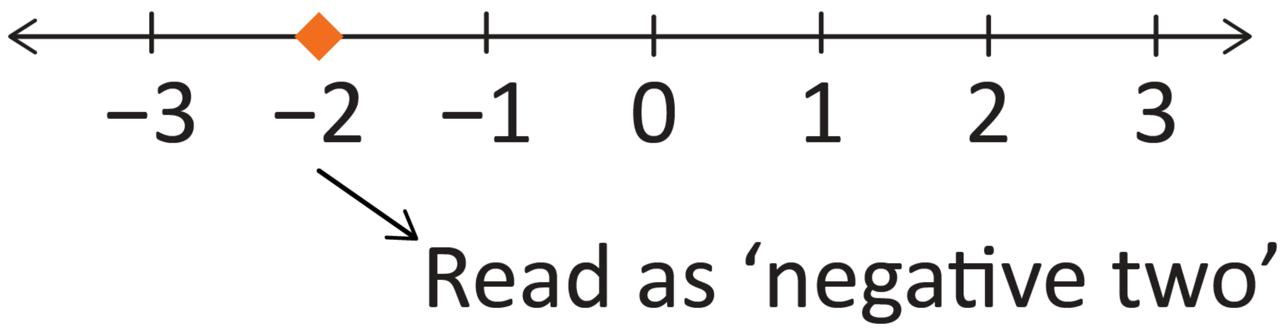
Investigate the use of positive and negative integers to represent everyday situations. Read, write and order integers on a number line

For example:

- identifying negative integers in familiar contexts, such as temperatures, sea levels, bank balances and buildings with underground parking



- interpreting the marking of increments on the number line on both sides of the zero mark



General Capabilities: Numeracy, Critical and creative thinking

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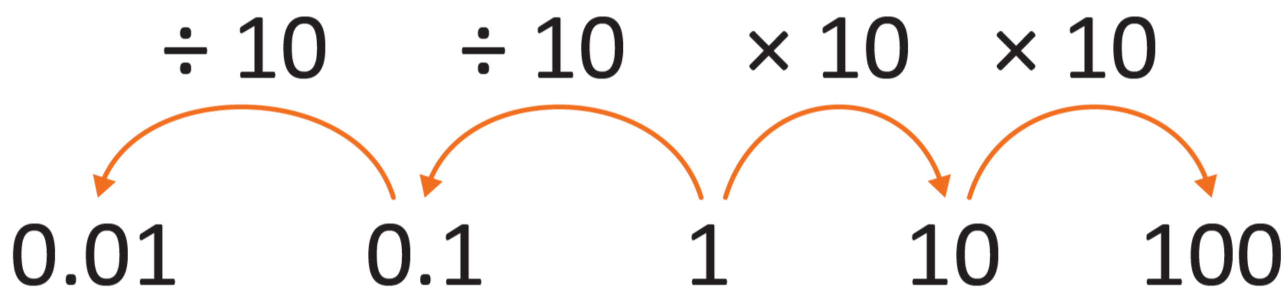
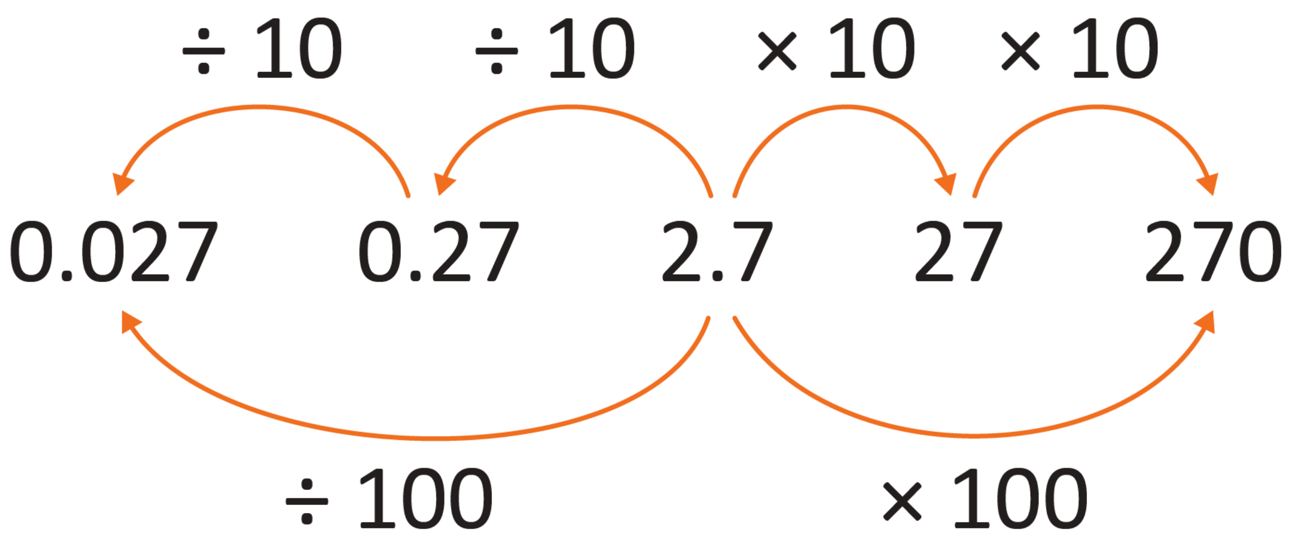
WA6MNAUN2

Represent and explain the multiplicative place value relationship between places in any number, including decimals

For example:

- using calculators to explore multiplying and dividing whole numbers and decimals by powers of 10, recording results and noticing patterns





- explaining the multiplicative place value relationship, such as
    - 62 is 10 times as many as 6.2 because 60 is 10 times as many as 6 and 2 is 10 times as many as 0.2
    - 20.8 is 100 times as many as 0.208 because 20 is 100 times as many as 0.2 and 0.8 is 100 times as many as 0.008

General Capabilities: Numeracy, Digital literacy

WA6MNAUN3

Explore, identify and represent square, prime and composite numbers in arrays and explain reasoning

For example:

- using visual representations to explore and reason about square numbers

Multiplication Facts

10	10	20	30	40	50	60	70	80	90	100
9	9	18							81	90
8	8	16						64		80
7	7	14					49			70
6	6	12				36				60
5	5	10	15		25					50
4	4	8	12	16	20					40
3	3	6	9	12	15					30
2	2	4	6	8	10					20
1	1	2	3	4	5	6	7	8	9	10
×	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

- using tiles, blocks or visual representations to form arrays to explore and explain what makes a number prime or composite
- recognising that a prime number, which has exactly two distinct factors, itself and one, has only one row when represented as an array

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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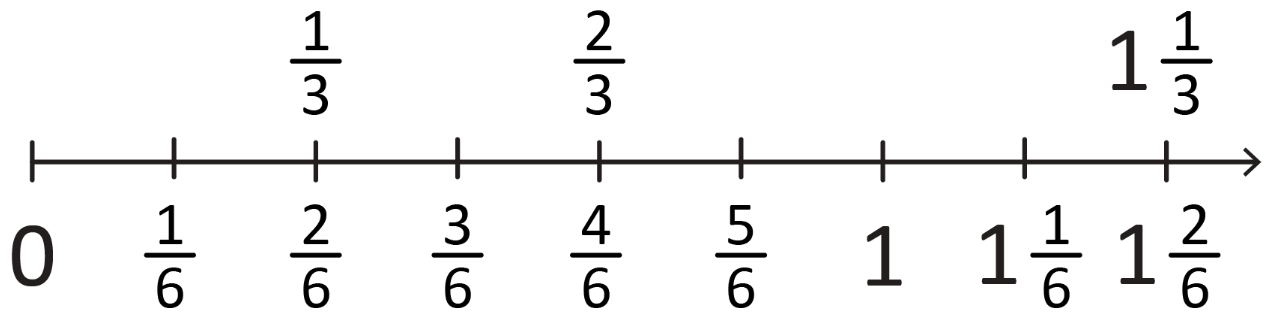
WA6MNAUN4

Order common fractions with the same and related denominators, including mixed numerals, using diagrams and number lines

For example:

- using a range of representations, such as number lines, paper strips, shapes and objects, to assist in ordering common fractions, such as $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{1}{6}$, and demonstrating that $\frac{1}{3}$ is equivalent to $\frac{2}{6}$ when ordering fractions on a number line



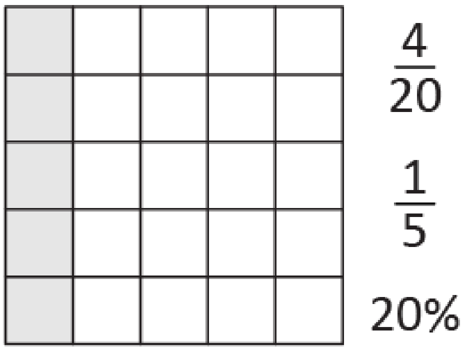
General Capabilities: Numeracy

WA6MNAUN5

Connect commonly used percentages, including 10%, 25% and 50%, to fractions and decimals, including on a number line

For example:

- recognising the percentage symbol (%) is used to show a ratio out of 100
- contextualising percentages, such as comparing the advantages of being offered a 10%, 25% or 50% discount, in connection to fractions, such as '50% off' is half price
- using representations to identify fractions and the corresponding percentages



- using visual aids to explore percentages as parts of a whole, exploring wholes that are different sizes

General Capabilities: Numeracy

Understanding equalities and inequalities

WA6MNAUE1

Complete, check and construct statements of equality and inequality involving the four operations, including the use of brackets and order of operations, and explain reasoning

For example:

- checking statements to say if they are true (T) or false (F), such as

  $\frac{1}{2} \times 0 = \frac{1}{2}(\mathrm{F})$

  $10 \times 0.2 < 10 \div 2(\mathrm{T})$

- completing statements, such as

  $(3 + 5) \times 2 \ \square \ 23 + 5 \times 2$

  and explaining reasoning using order of operations

- explaining where to put brackets in a number sentence to make the statement true, such as

  $2 + 4 \times 7 = 42$

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Patterns and relationships

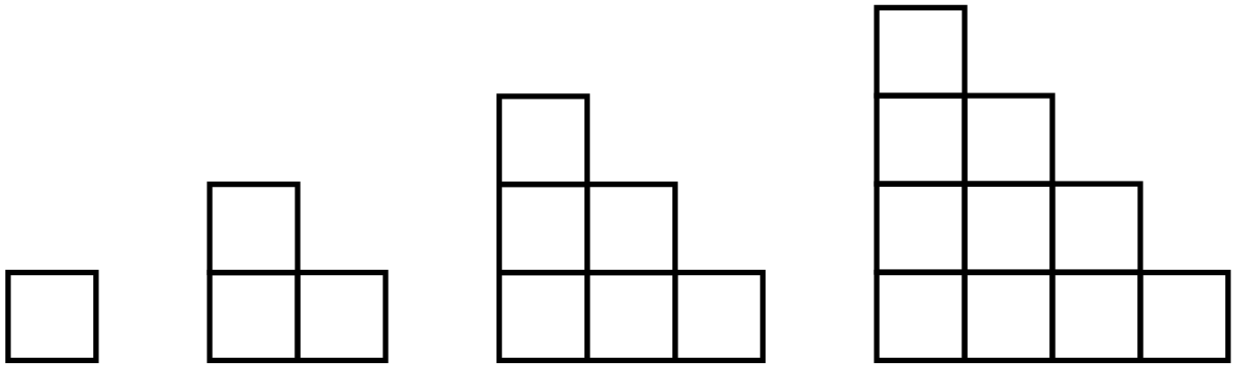
WA6MNAP1

Create and represent increasing or decreasing patterns using concrete materials and numbers. Use words to generalise rules that relate each element of a pattern to its position

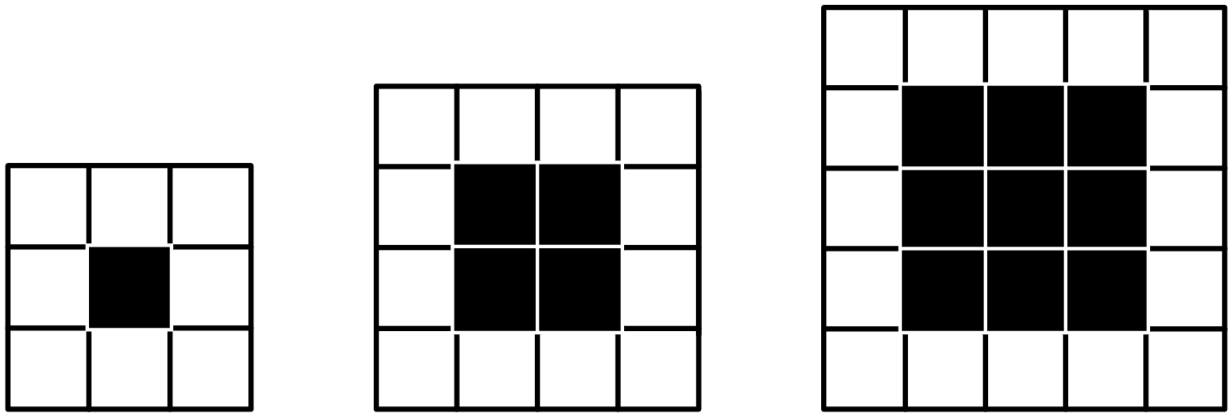
For example:

- using toothpicks, counters, blocks or drawings to create patterns, investigating and making predictions, such as the number of squares in the sixth diagram being 21 as

  $6 + 5 + 4 + 3 + 2 + 1 = 21$



- recognising that one pattern can generate many rules, such as a square with side lengths of 10 having $8 \times 8 = 64$ black squares, $10 + 10 + 8 + 8$ or $4 \times 10 - 4 = 36$ white squares and $10 \times 10 = 100$ squares altogether



- considering sequences, such as 1, 3, 5 …, and recognising the rule, which in this case is double the position number and subtract one, then using the rule to predict the 200th odd number

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Calculating with number

WA6MNAC1

Choose and use flexible and efficient strategies to calculate with whole numbers, involving any of the four operations, and explore the use of the order of operations

For example:

- selecting and using efficient strategies to multiply whole numbers, such as 5000×90 mentally but 4974×87 using a calculator

- applying the order of operations to solve equations, such as $5 + (2 \times 3) = 5 + 6 = 11$
- representing remainders appropriately, as whole numbers, fractions or decimals

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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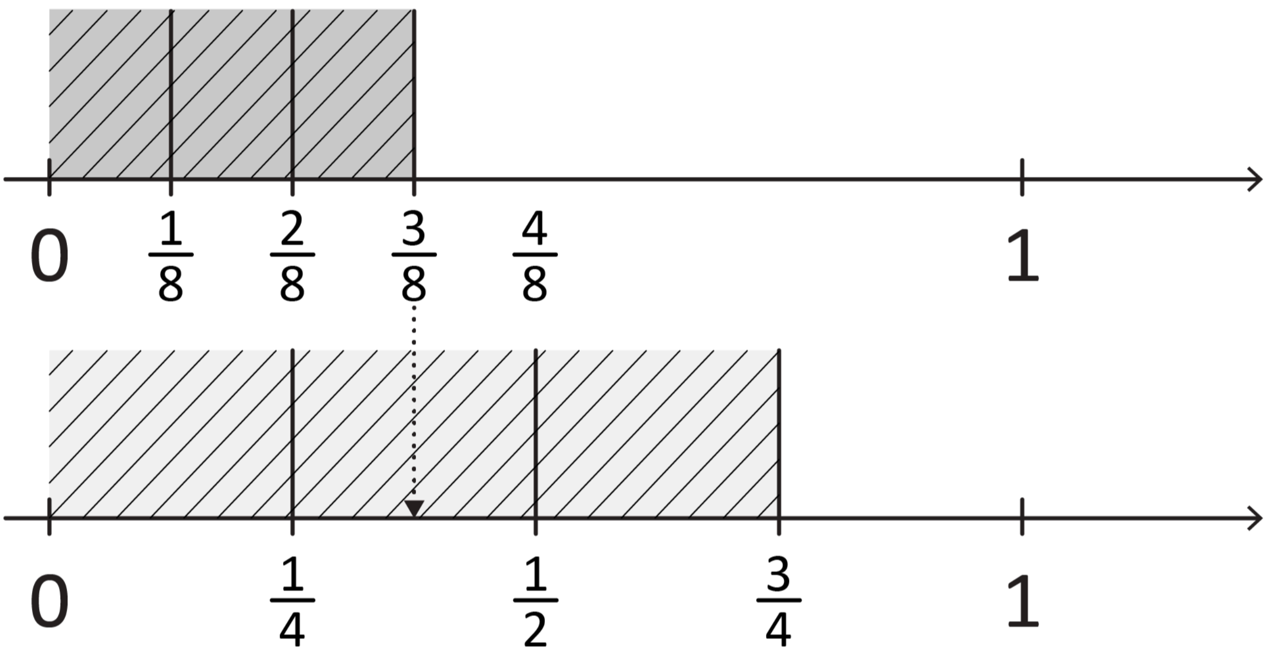
WA6MNAC2

Add and subtract fractions with related denominators using flexible and efficient strategies based on knowledge of equivalence

For example:

- representing fractional quantities with the same or related denominators to add and subtract fractions, such as ‘How much more is $\frac{3}{4}$ than $\frac{3}{8}$?'



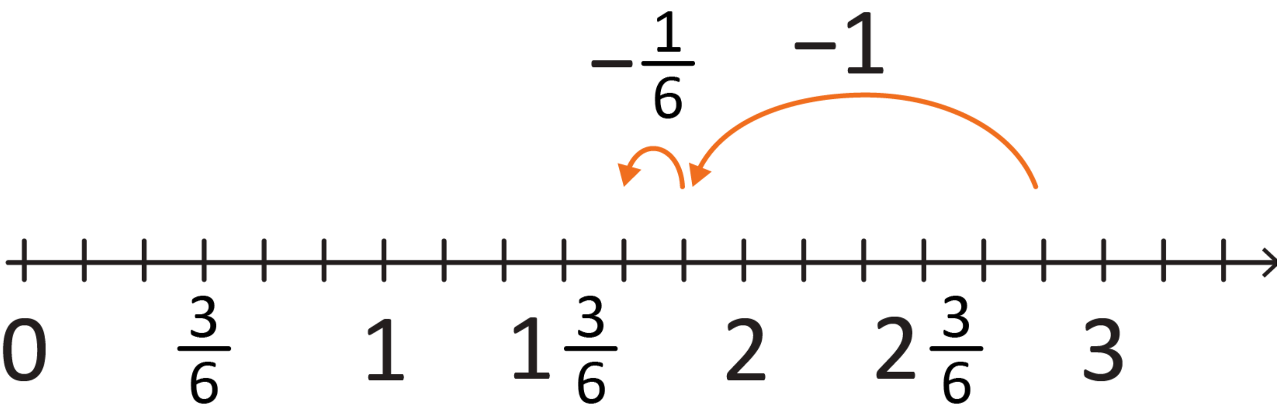
- calculating $\frac{1}{4} + \frac{1}{8}$, mentally or with jottings and diagrams, recognising that $\frac{1}{4} = \frac{2}{8}$, so

  $\frac{1}{4} + \frac{1}{8} = \frac{2}{8} + \frac{1}{8} = \frac{3}{8}$

- adding fractions by simplifying or grouping, such as

  $\frac{7}{8} + \frac{1}{4} = \frac{7}{8} + \frac{2}{8} = 1 + \frac{1}{8}$

- using a number line, such as for $2\frac{5}{6} - 1\frac{1}{6}$



General Capabilities: Numeracy, Critical and creative thinking

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WA6MNAC3

Add and subtract decimals to two decimal places, using flexible and efficient strategies

For example:

- using place value, such as $8.4 + 3.6$

  $= 8 + 3 + 0.4 + 0.6$

  $= 8 + 3 + 1$

  $= 12$

  and

  $5.8 + 7.4$

  $5.8 + 7 = 12.8$

  $12.8 + 0.4 = 13.2$

- multiplying decimals by powers of 10 to solve $2.15 + 2.35$

  $215 + 235 = 450$

  $450 \div 100 = 4.5$

- applying known strategies, such as levelling or constant difference levelling

  $(2.94 + 5.16 \rightarrow 3 + 5.1)$

  or constant difference

  $(2.57 - 1.03 \rightarrow 2.60 - 1.06)$

General Capabilities: Numeracy, Critical and creative thinking

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WA6MNAC4

Multiply decimals by whole numbers and multiply and divide decimals by powers of 10, using flexible and efficient strategies

> **For example:**
>
> - exploring multiplying whole numbers by decimals larger and smaller than one to identify misconceptions that multiplying always makes numbers bigger, such as 120×1.5 and 120×0.5
> - using mental strategies to multiply decimals, such as 3.5×2
> - using place value to mentally multiply and divide decimals by powers of 10, such as
>
> $5.75 \times 10 = 57.5$ $57.5 \div 10 = 5.75$
> $5.75 \times 100 = 575$ $57.5 \div 100 = 0.575$
> $5.75 \times 1000 = 5750$ $57.5 \div 1000 = 0.0575$

General Capabilities: Numeracy, Critical and creative thinking

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WA6MNAC5

Determine a familiar fraction, decimal or percentage of a whole number

> **For example:**
>
> - calculating fractions of whole numbers, such as $\frac{2}{5} \times 30$
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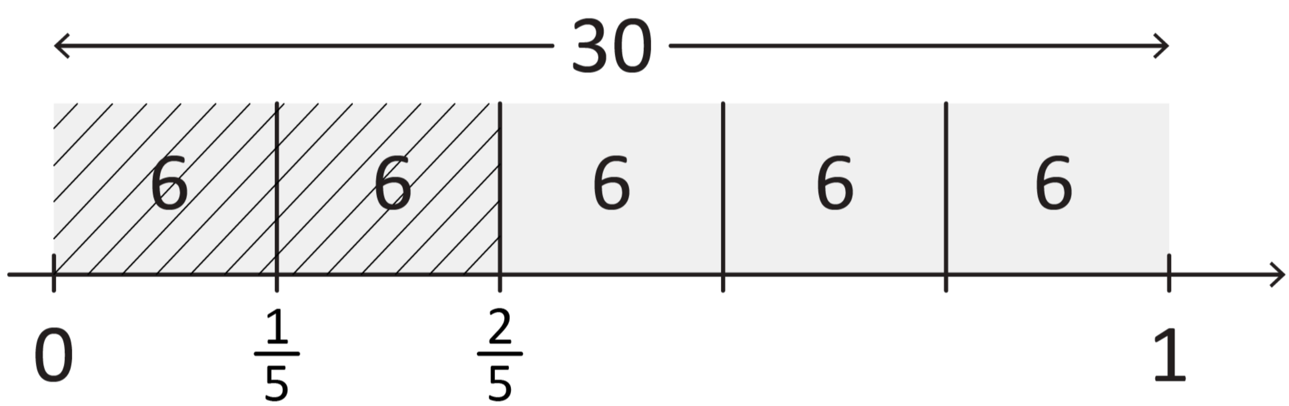
> 
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>
> - explaining how $\frac{1}{3}$ of a quantity can be achieved by dividing by three, and how knowledge of $\frac{1}{3}$ of a quantity can be used to find $\frac{2}{3}$ or $\frac{4}{3}$ of the same quantity
> - finding 0.2 of a whole number, recognising that $0.2 = \frac{1}{5}$ so finding 0.2 of a number means dividing by five
> - finding 10%, 25% and 50% of a whole number, equating 10% to dividing by 10, 50% to finding half by dividing by two and 25% to finding a quarter by dividing by four or by finding half and half again
> 10% of $\$60 = 60 \div 10 = \$6$
> 50% of $105 = 105 \div 2 = 52.5$
> 25% of $1200 = 1200 \div 4 = 300$

General Capabilities: Numeracy

WA6MNAC6

Use estimation and rounding to make reasonable evaluations and justify results

> **For example:**
>
> - using rounding to simplify divisions, such as approximating $997 \div 4$ to $1000 \div 4$, which gives a result of approximately 250
> - estimating fractions, such as $\frac{5}{12}$ is a little less than a half
> - rounding a decimal to a nearby whole number, such as for 25.14×3.5, recognising the answer will be between 3×25 and 4×25, which is between 75 and 100
> - rounding up when determining money required for a purchase, and not rounding for digital financial transactions

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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Financial mathematics

WA6MNAF1

Develop plans for savings goals, predict expenses and identify that saving money with a bank attracts interest

> **For example:**
>
> - choosing a savings goal, such as an end-of-year school camp
> - using a spreadsheet to keep track of money being raised in a bake sale for the end-of-year school camp. Keeping track of income (money received for the cakes purchased) and expenses (money spent on ingredients), recognising the need to prioritise certain expenses (e.g. ingredients over decorations)
> - identifying the importance of checking receipts, including ensuring any % discounts have been accurately applied and using estimation strategies to check the total amount
> - discussing the benefits of using a bank for saving

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Modelling with number

WA6MNAM1

In real-world situations involving whole numbers, order of operations and fractions with the same denominator, use the following process

I. analyse the situation and identify relevant information
II. mathematically represent the situation, including using equations to reach a solution
III. interpret and communicate findings in the context, exploring and justifying decisions

> **For example:**
>
> - modelling situations, such as determining the total cost of Leith hiring a bike for nine hours, analysing the given information, identifying the relevant information to translate the situation into an equation. Deciding on the operation or combination of operations and applying the order of operations, e.g.

$15 + (9 - 4) \times 3 + 2 = \square$ Using efficient strategies to solve, interpreting the answer to be the total cost, explaining findings, such as ‘It would have cost $17 to hire the bike for four hours, and it was an extra $15 for five more hours’

Bike Hire

4 hours for $15!

$3 for each additional hour

Additional items (daily cost):
Helmets $2
Baskets $5

A helmet must be hired by law.

Late returns incur a penalty of $3 per 15 minutes.

General Capabilities: Literacy, Numeracy, Critical and creative thinking

Measurement and geometry

Two-dimensional space and structures

WA6MMGTW1

Explore, visualise and describe translations, reflections or rotations of two-dimensional shapes

> **For example:**
>
> - recognising that translations, reflections or rotations change the position and orientation but not the size of shapes
> - transforming shapes to create tessellating patterns, including using digital tools. Describing the transformations used and discussing why these shapes tessellate, including identifying shapes or combinations of shapes that will not tessellate

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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WA6MMGTW2

Convert between units of length by connecting metric units to the decimal system and extend to units of mass and capacity

> **For example:**
>
> - recognising the significance of the prefixes in units of measurement
> - converting between units, including millimetres, centimetres, metres, kilometres; milligrams, grams, kilograms, tonnes; millilitres, litres, kilolitres and megalitres
> - using place value, such as converting 2600 m to 2.6 km, recognising that 1000 m equals 1 km
> - explaining and using the relationship between the size of a unit and the number of units needed, such as more grams than kilograms will be needed to measure the same object, and so to convert from kilograms to grams you need to multiply by 1000

General Capabilities: Numeracy

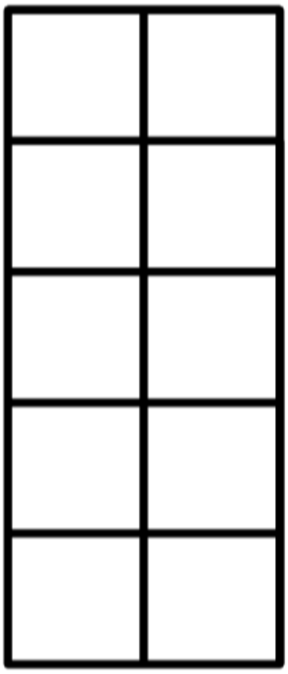
WA6MMGTW3

Describe and test a sequence of steps to determine the area of rectangles based on dimensions

> **For example:**
>
> - drawing a variety of rectangles on grid paper to investigate patterns in the relationship between the length of the base and the height of rectangles, and their area

Base	Height	Area
5 cm	4 cm	20 cm^2
3 cm	6 cm	18 cm^2

- using grid paper to investigate and compare shapes that have the same perimeter but different areas, such as both rectangles have a perimeter of 14 cm but different areas



General Capabilities: Literacy, Numeracy, Critical and creative thinking

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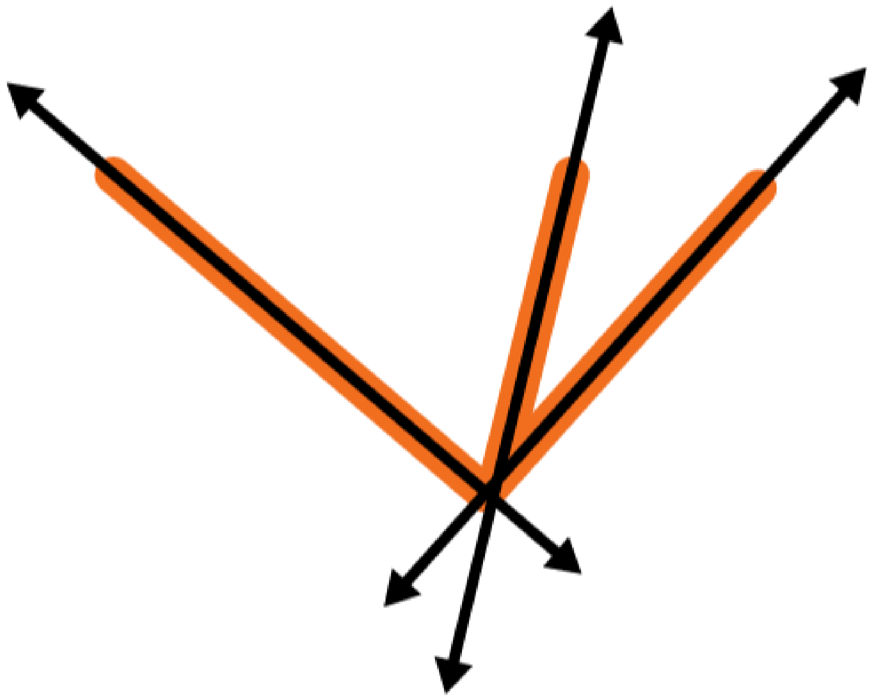
WA6MMGTW4

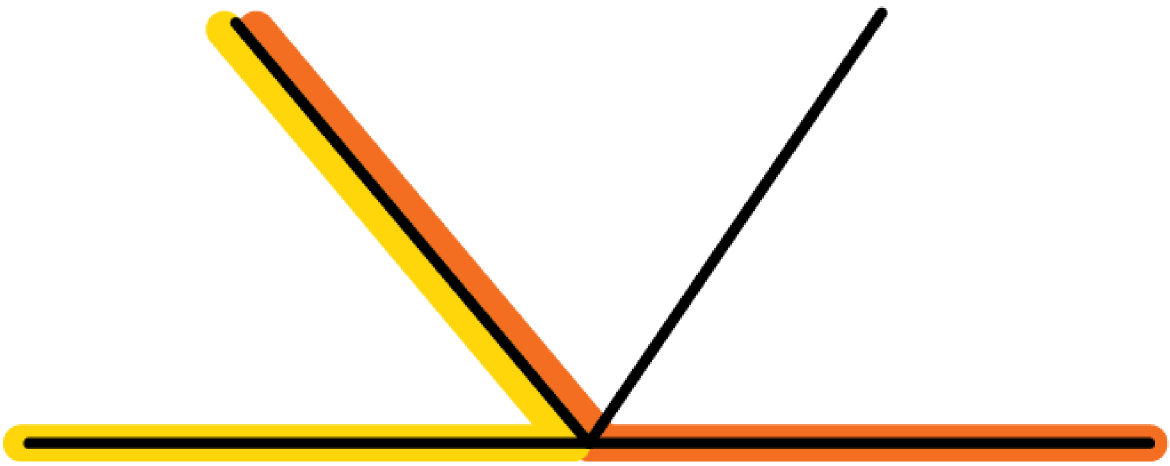
Investigate angles in a right angle, on a straight line, angles at a point and vertically opposite angles, to determine unknown angles and explain reasoning

For example:

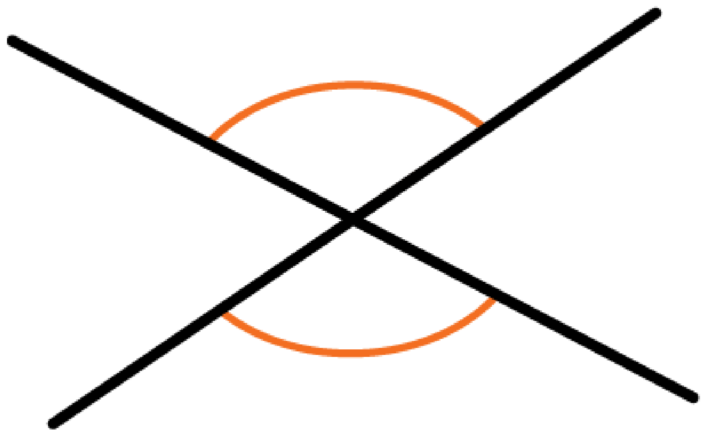
- investigating adjacent angles that form a right angle and establish that they add to 90° (complementary angles)



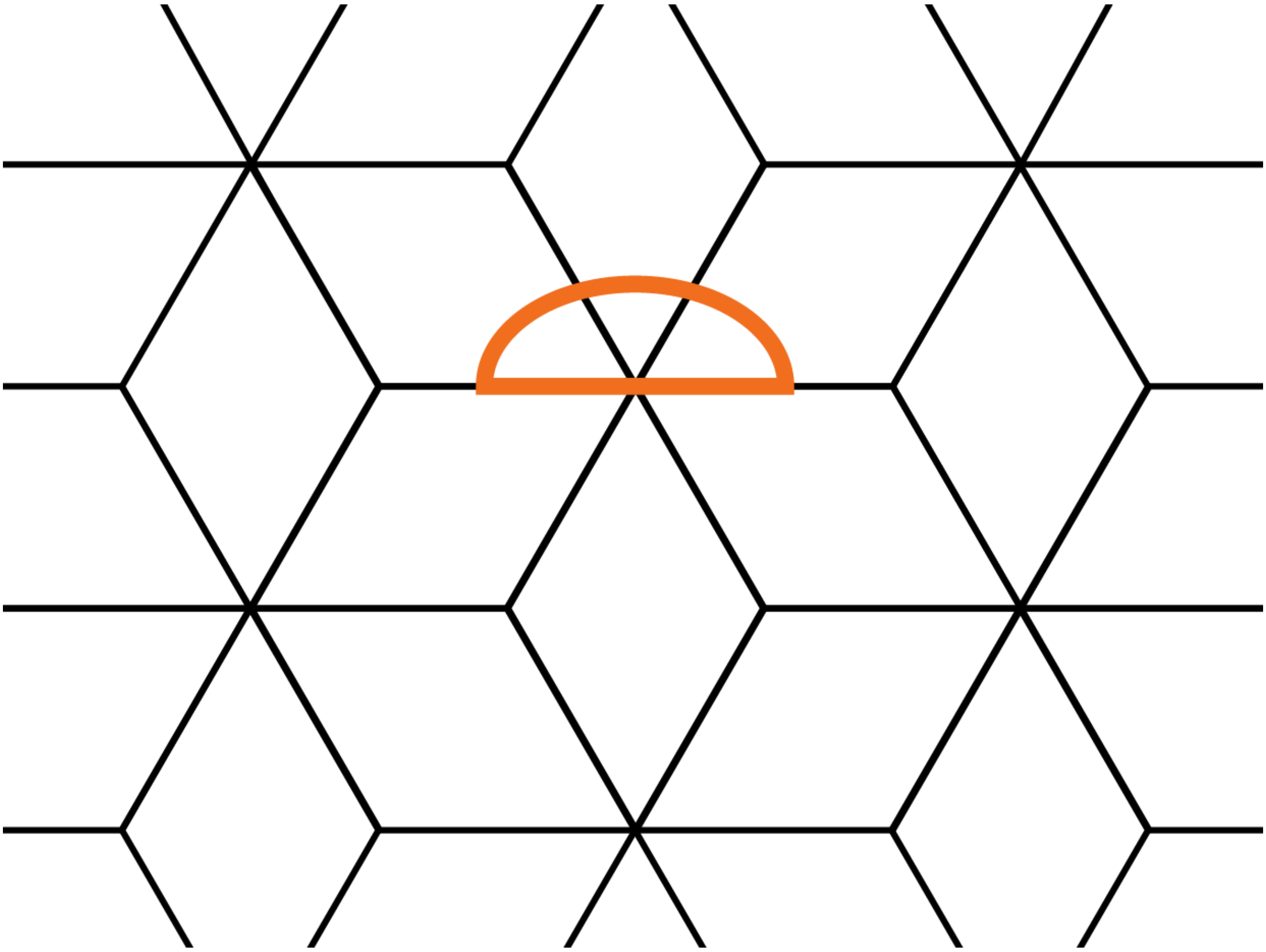
- investigating adjacent angles that form a straight angle and establish that they add to 180°, such as an angle of 120° and 60° (supplementary angles)

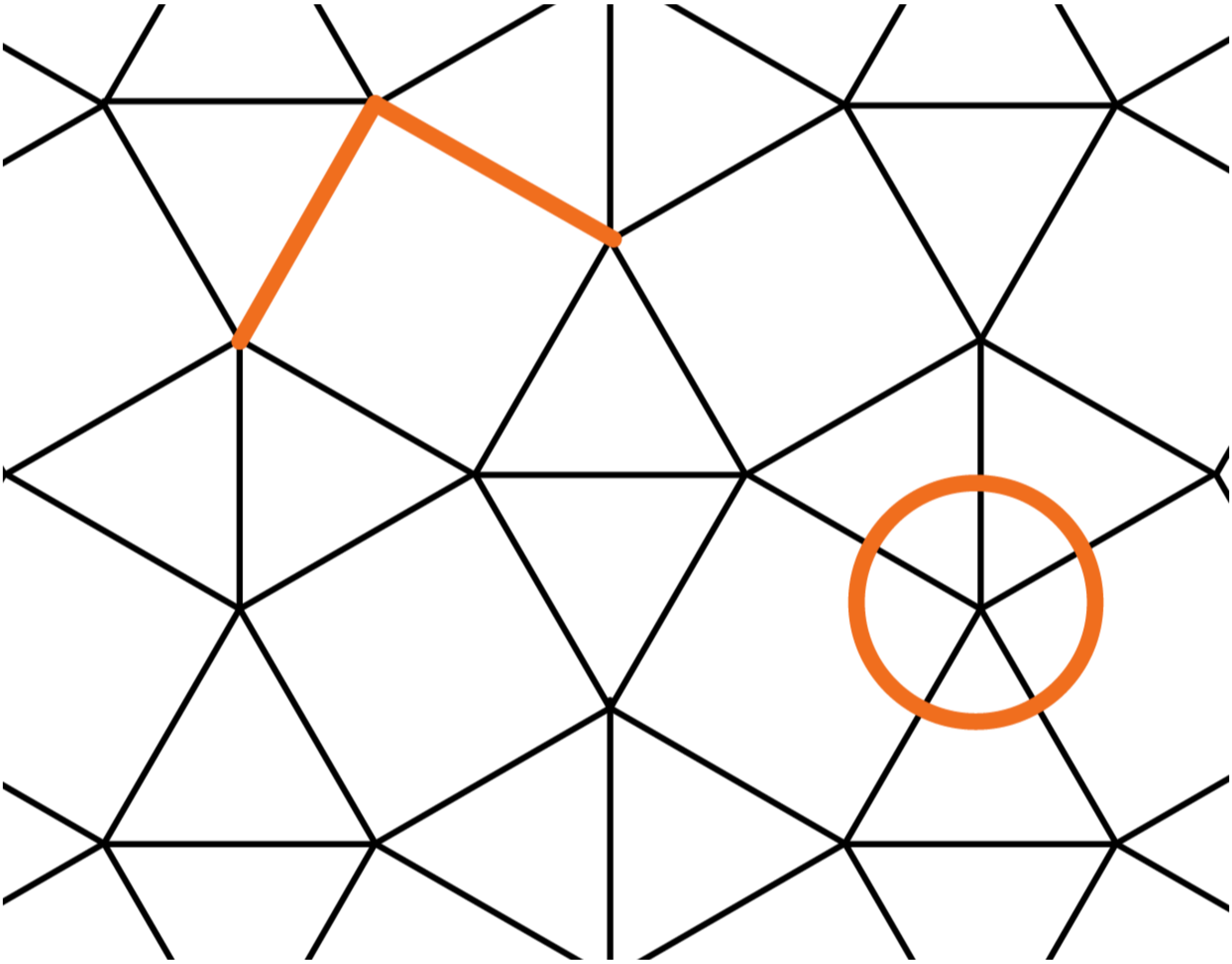


- recognising vertically opposite angles as angles that are directly opposite each other and equal in size, formed when two lines intersect



- recognising and describing angle relationships embedded in diagrams, such as





General Capabilities: Literacy, Numeracy, Critical and creative thinking

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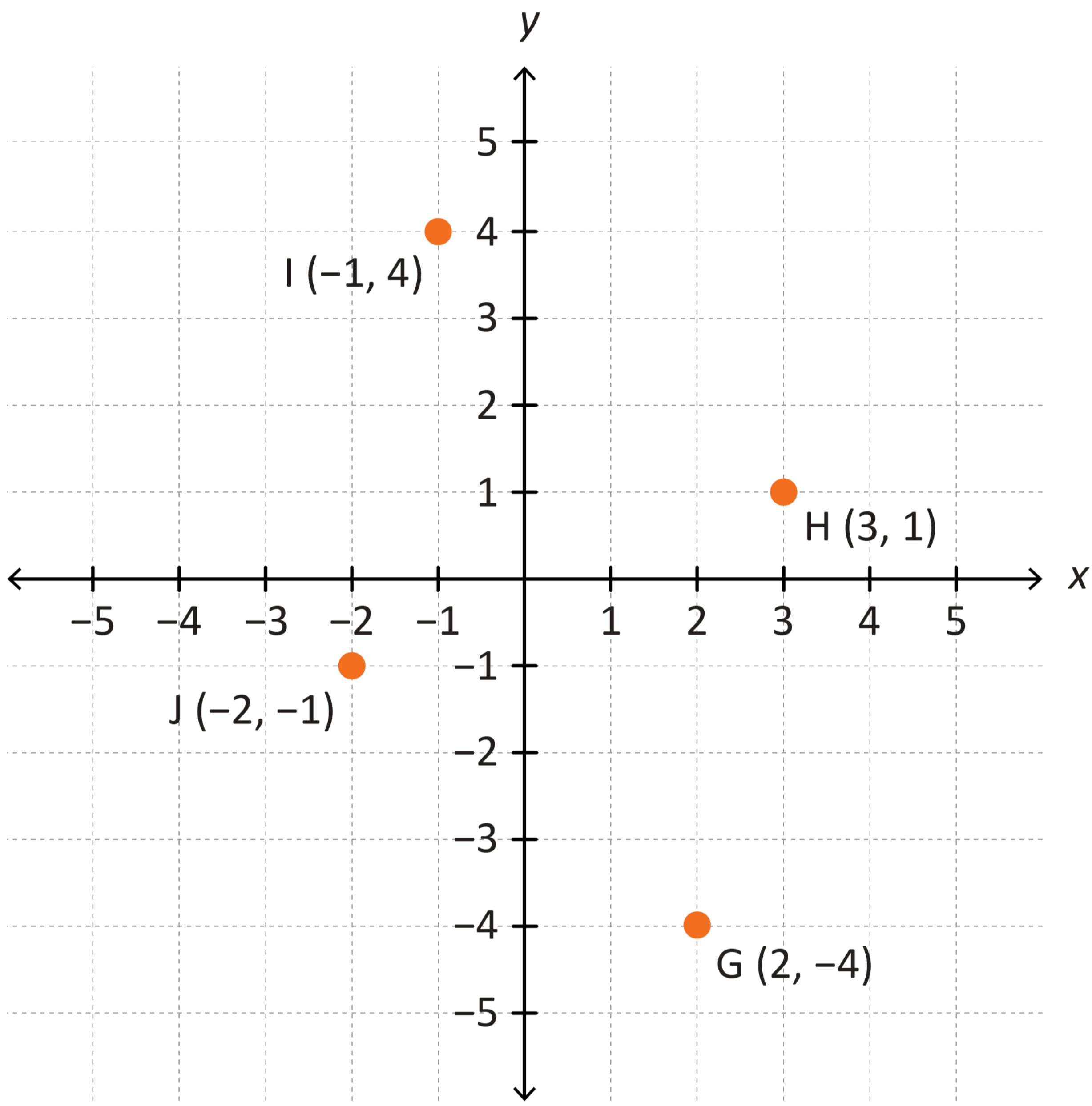
WA6MMGTW5

Explore the Cartesian plane as the intersection of two number lines at zero, using the coordinate system to locate points in all four quadrants

For example:

- recognising positive and negative integers on the number lines when plotting points in all four quadrants on the Cartesian plane



General Capabilities: Numeracy

Three-dimensional space and structures

WA6MMGTH1

Visualise, sketch and construct three-dimensional objects, including prisms and pyramids

For example:

- visualising features to create sketches of 3D objects, showing depth and from different views
- constructing 3D models of prisms and pyramids, given drawings of different views
- creating skeletal 3D models of prisms and pyramids using materials, such as toothpicks, clay, straws and tape

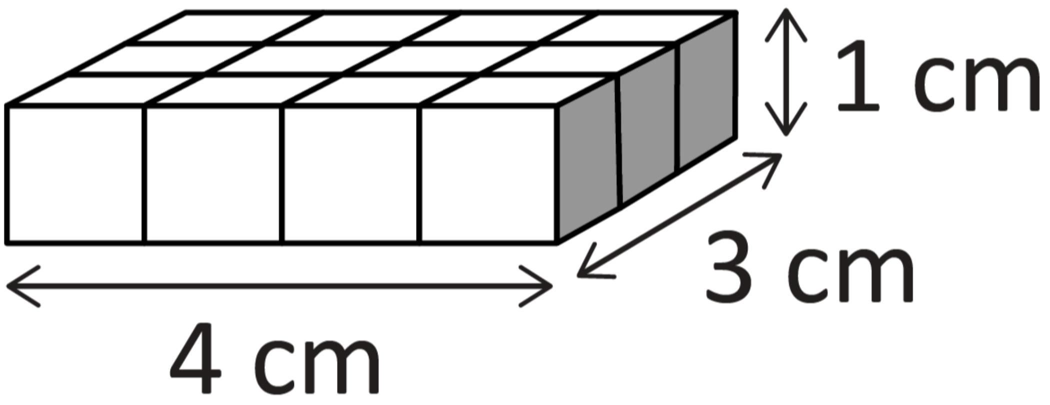
General Capabilities: Numeracy, Critical and creative thinking

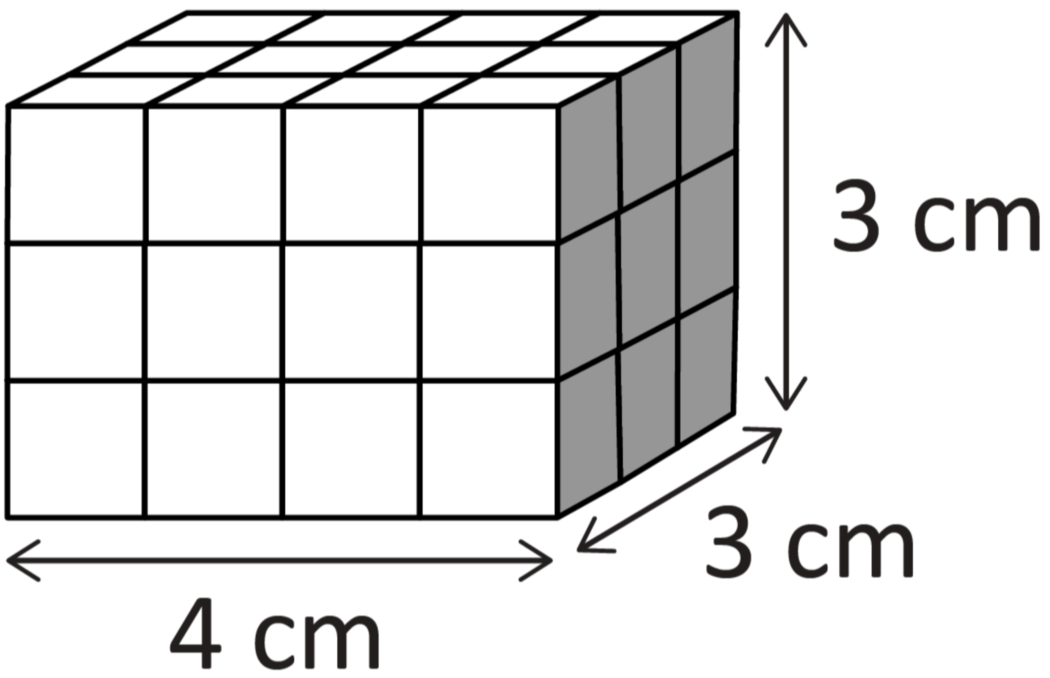
WA6MMGTH2

Describe and test a sequence of steps to determine the volume of rectangular prisms based on dimensions

For example:

- exploring and generalising multiplicative strategies to find volume using diagrams, numbers and words, focusing on the relationship between the number of cubes in a layer and the number of layers in the prism, such as recognising that one layer is $12\ \text{cm}^3$ and multiplying by the number of layers to determine volume



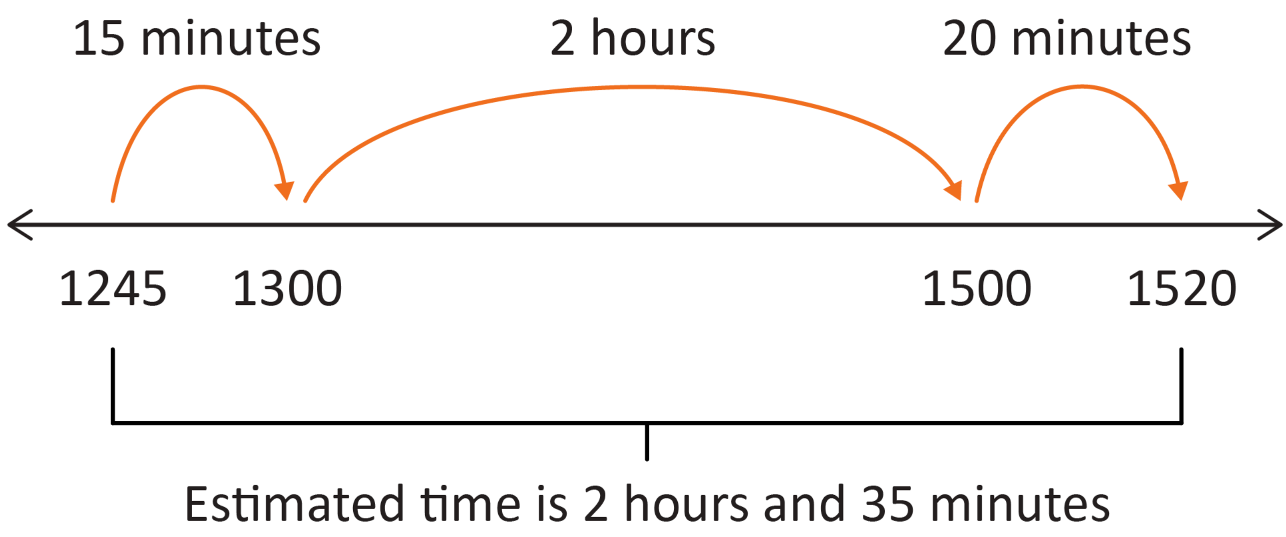


General Capabilities: Literacy, Numeracy, Critical and creative thinking

Non-spatial measurement

WA6MMGN1

Use timetables and itineraries in 12- and 24-hour time systems to determine the duration of events and journeys

> **For example:**
>
> - applying concrete and mental strategies to determine elapsed time, considering the starting and finishing time
>
> 
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>
> - interpreting a variety of timetables and itineraries from real-life situations, such as a school, movie theatre or public transport authority

General Capabilities: Literacy, Numeracy

Modelling with measurement and geometry

WA6MMGM1

In real-world situations involving transformation of two-dimensional shapes, rectangular prisms, pyramids, Cartesian planes, measuring and converting metric units for length, mass and capacity, determining volume and area in metric units or determining the duration of events and journeys, use the following process

I. analyse the situation and identify relevant information
II. mathematically represent the situation to reach a solution
III. interpret and communicate findings in the context, exploring and justifying decisions

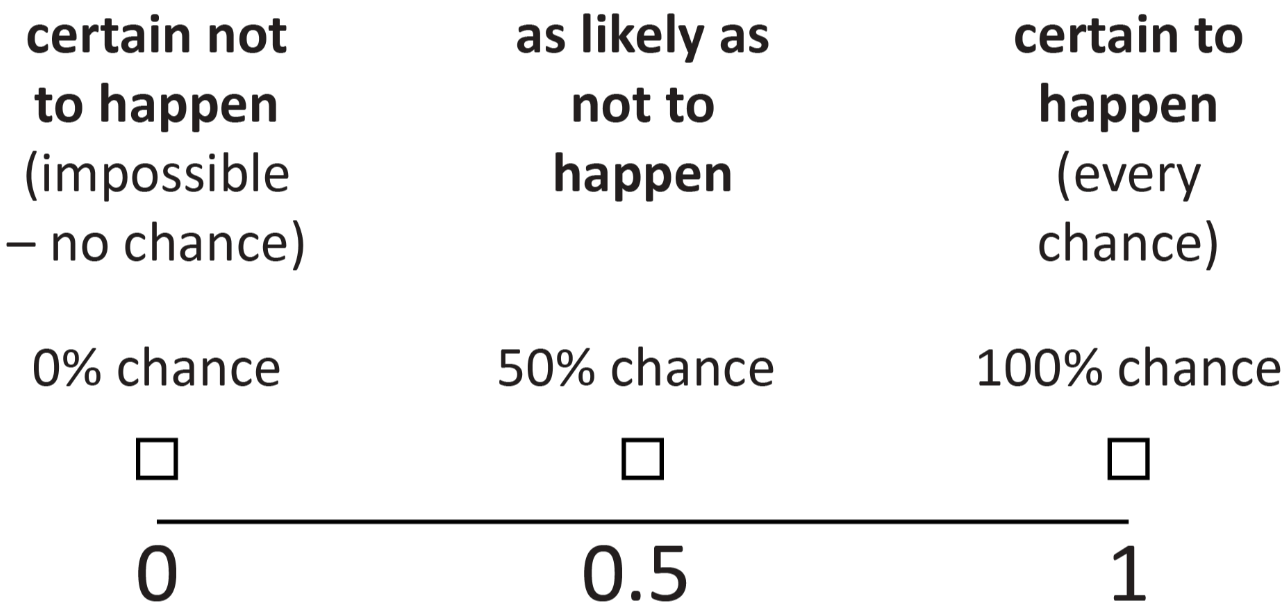
> **For example:**
>
> - planning a class trip to the WA Museum Boola Bardip utilising Transperth timetables and a map of the museum to identify a route and schedule
> - designing a monument incorporating both prisms and pyramids to honour an important person. Investigating how the symbolic meaning of prisms and pyramids could reflect the qualities of the important person

Probability and statistics

Probability

WA6MPSP1

Order everyday chance events and phrases on a scale from 0 to 1, where 0 represents an event that is certain not to happen (impossible) and 1 represents an event that is certain to happen

> **For example:**
>
> - recognising that all measures of probability fall between 0 and 1 or 0% and 100%
> - positioning events and phrases on a scale and justifying placements
>
> 
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General Capabilities: Literacy, Numeracy, Critical and creative thinking

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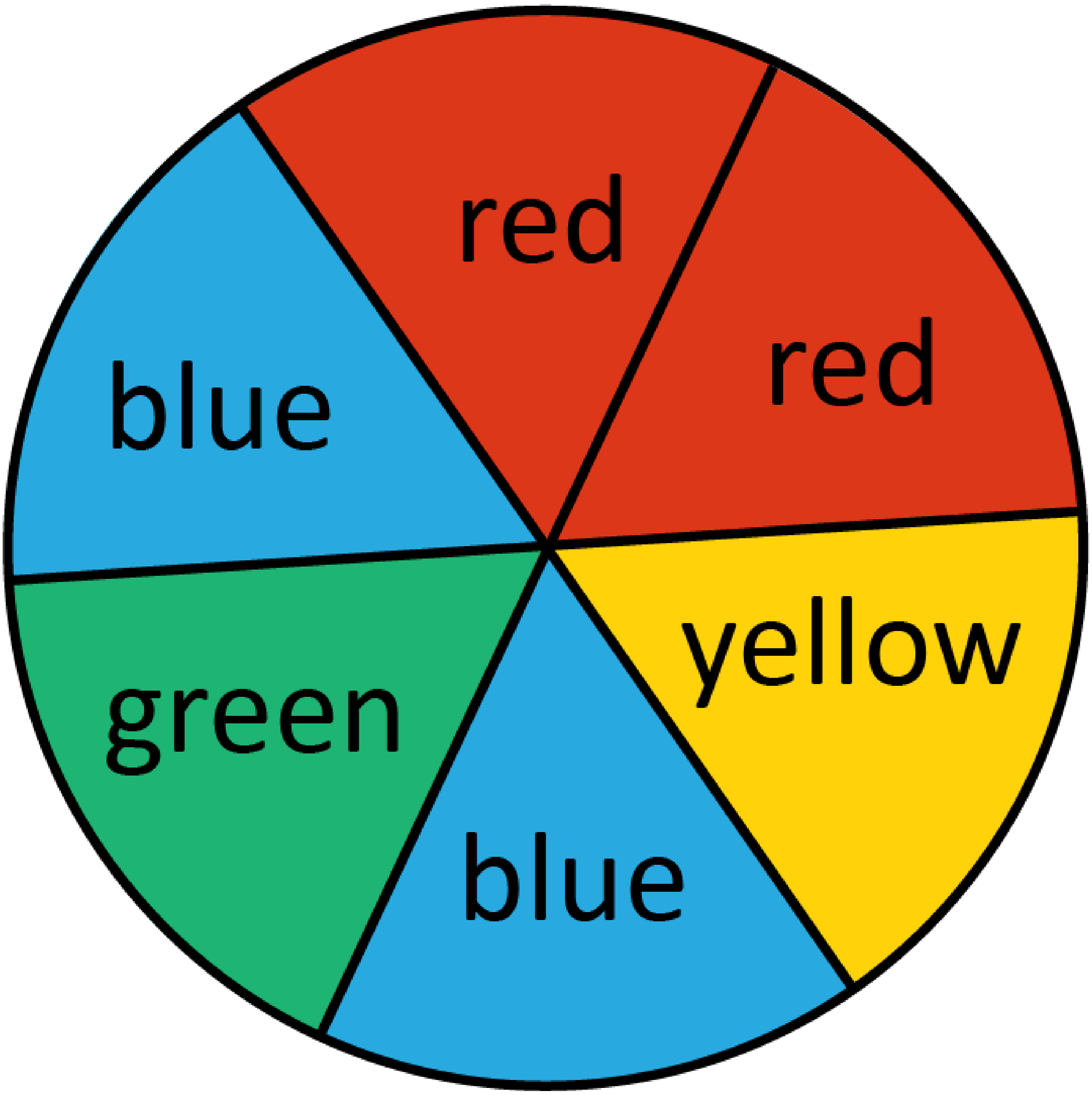
WA6MPSP2

Conduct repeated chance experiments and simulations with equally likely or unequally likely outcomes, including with the use of digital tools, for an increasing number of trials. Compare expected and observed frequencies in terms of variation as the number of trials increase

> **For example:**
>
> - recording the frequency of each outcome for 20, 30 and 40 trials, and comparing the observed frequency with the expected frequency



- combining results for groups of students and the whole class, identifying that the results trend closer to the prediction as the number of trials increases

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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Statistics

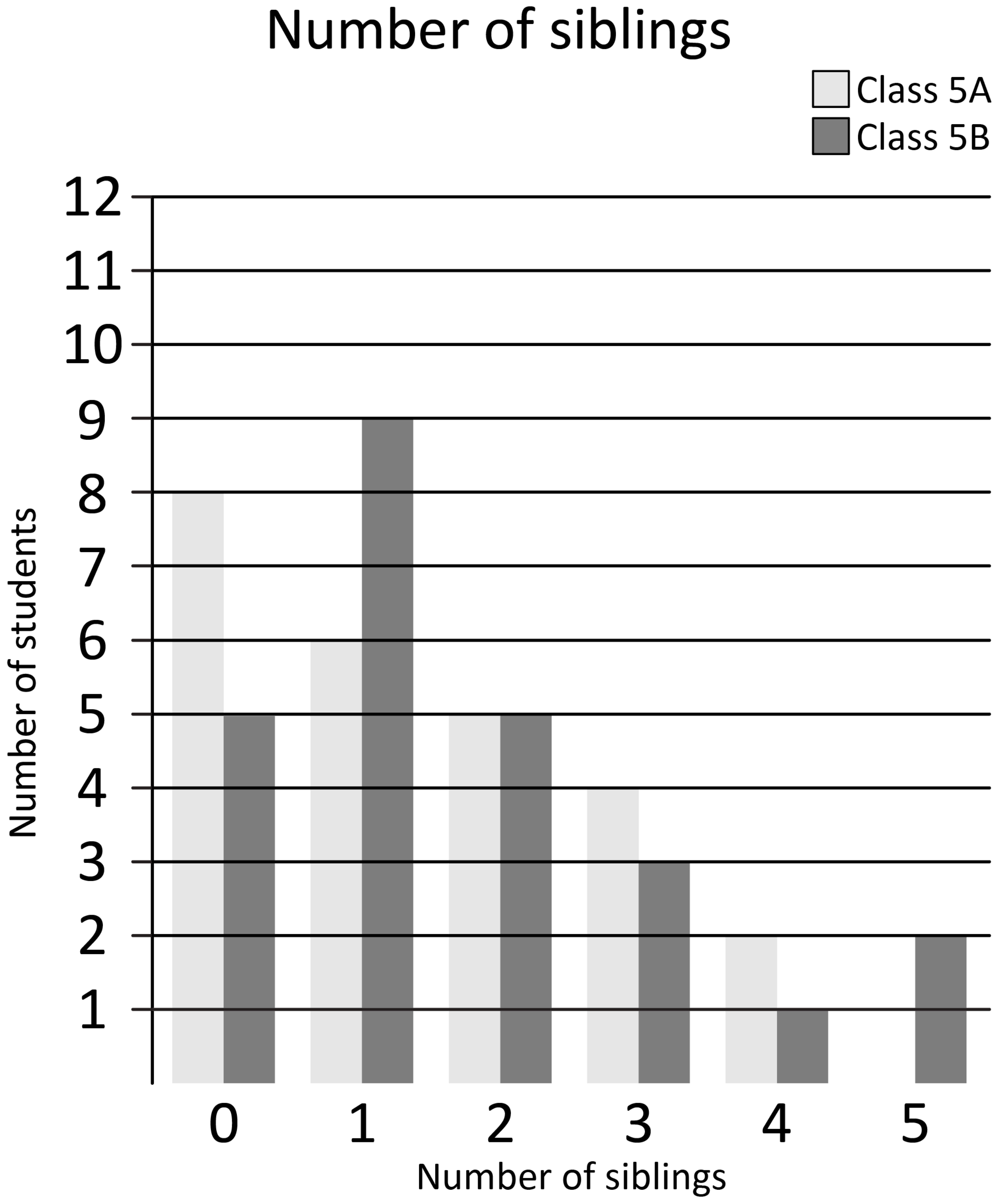
WA6MPSS1

Describe and interpret a range of displays for real-life numerical data, including side-by-side column graphs, using mode, range and shape

For example:

- recognising that shape describes how data is spread on a graph, for instance whether the responses are grouped, spread out, have more high or low values, or show gaps
- identifying the mode as the most frequently occurring response and the range as the difference between the highest and lowest data values recorded

- interpreting data using mode, range and shape, such as Class 5A has a range of 4, a mode of 0 and the number of students decreases as the number of siblings increases. Comparing this to Class 5B



General Capabilities: Literacy, Numeracy, Critical and creative thinking

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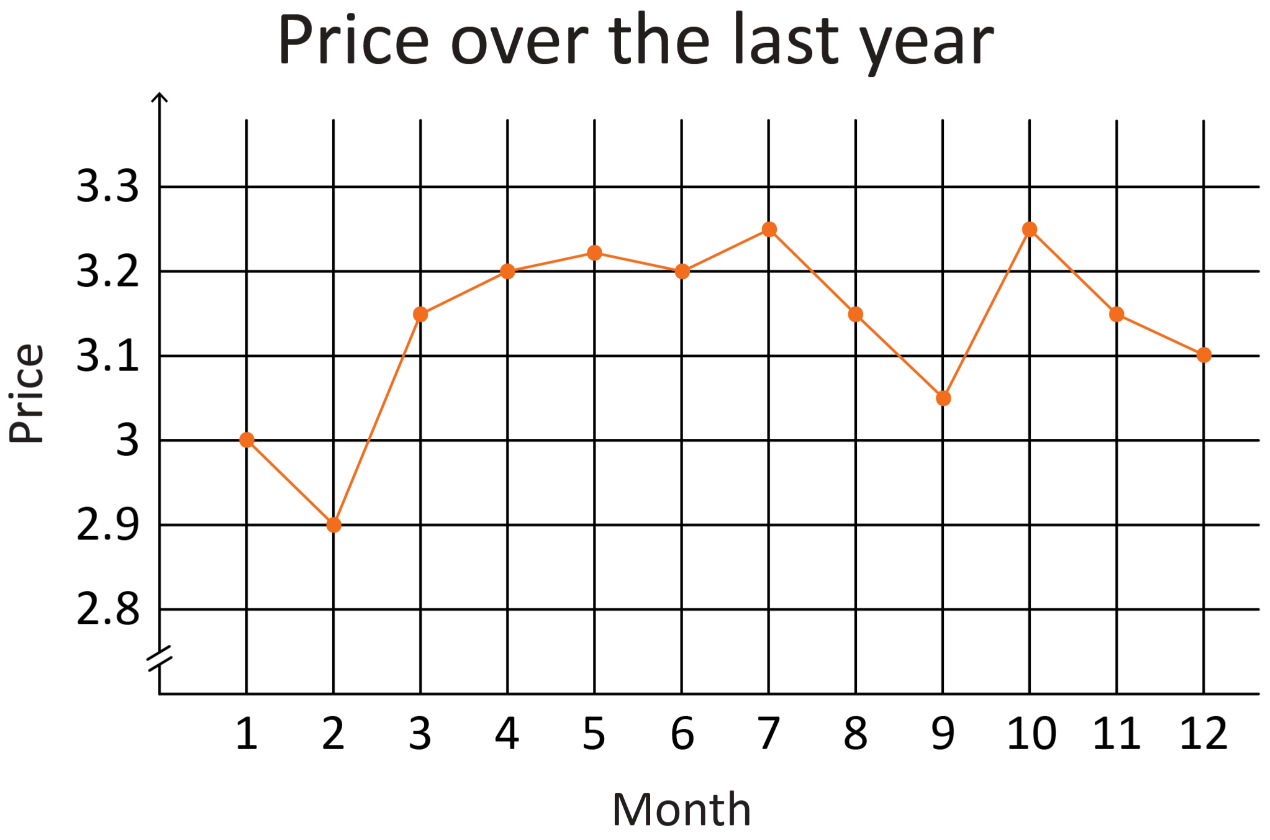
WA6MPSS2

Describe how the features of real-life data displays may influence an audience

For example:

- considering how the broken axis may affect the interpretation of a display, such as choosing the second graph to convey that bread prices have fluctuated dramatically in the last 12 months



- investigating how sizing, colour, labels or scale may have been used in the construction of a display to influence a particular audience or 'tell a particular story'

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding, Digital literacy

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WA6MPSS3

In a real-world context involving continuous and discrete numerical data, use the following process

I. analyse the situation to pose a question
II. choose the most appropriate way to collect data to ensure accuracy and consistency, and make choices to represent data, including line graphs and side-by-side column graphs
III. interpret and communicate findings in terms of the context and describe reasons for variation

For example:

- considering a situation, such as growing plants in different areas of the classroom to formulate a question, such as 'How does the amount of sunlight affect plant growth?' Recording and analysing data, including using a spreadsheet. Choosing a line graph to show plant growth over time, using digital tools as appropriate. Interpreting the data to answer the question, considering any reasons for variation in the data, such as distance from light source

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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Year 7

Year level descriptions

In the early adolescence phase of schooling, students align with their peer group and begin to question established conventions, practices and values. Learning and teaching programs assist students to develop a broader and more comprehensive understanding of the contexts of their lives and the world in which they live.

Mathematics provides opportunities for students to engage with concrete materials, extending to abstract thinking in a range of approaches to learning through the proficiencies of understanding, fluency, problem-solving and reasoning. These reinforce the significance of working mathematically with the content and describe how the content is explored or developed. Students draw on the behaviours of the proficiencies when selecting and using year level content to apply the complete modelling process, leading to an increased understanding of the complexity of the natural environment, society and technology.

In Year 7, students explore and investigate to understand, calculate flexibly and efficiently, and model with a broadening range of numbers, including adding and subtracting with integers. As they transition into efficient multiplicative thinkers, students interpret proportional situations. They develop their abstract thinking through the introduction of algebra.

Students begin to formalise the language of Mathematics by applying mathematical notation, conventions and naming principles in both geometric and measurement situations. They reason with parallel lines, perform transformations of points on the Cartesian plane, classify triangles and explore time zones within Australia. Students generalise their understanding of perimeter, area and volume into formulas for efficient calculation.

Students formalise their understanding of probability through definitions and the construction of simple sample spaces. They connect probability and statistics by engaging in single-stage chance experiments and simulations, use summary statistics, and represent, analyse and critique data.

Achievement standard

By the end of the year:

Students demonstrate the behaviours of the proficiencies of understanding, fluency, problem-solving and reasoning in conjunction with year level content in routine situations. As part of this, they select from and use year level content along with the modelling process, in which they analyse the situation mathematically, represent the problem and interpret and communicate their findings, to solve straightforward, real-world problems in familiar contexts across all strands.

Students make connections between equivalent fractions, equivalent ratios and between fractions, decimals and percentages. They demonstrate flexible and efficient strategies to compare, add and subtract fractions and integers, multiply and divide positive fractions and decimals, and perform percentage calculations. They round to a specified number of decimal places. Students make connections between whole numbers and index notation, and square numbers and square roots. They use variables to represent numbers and link numerical laws and properties to working with variables. They evaluate expressions using numerical substitution and solve linear equations involving one operation. Students describe and represent real-world linear patterns in a table of values and plot points on a graph. Students identify features of transactional money statements.

Students demonstrate use of appropriate language, conventions and notation in measurement and geometry. They apply formulas to determine the perimeter and area of rectangles and volume of rectangular prisms. Students classify triangles according to their properties, find unknown angles and calculate perimeter and area of triangles. They name angle relationships and find unknown angles in parallel lines and plot transformations of points on the Cartesian plane. Students represent different views of rectangular and composite rectangular prisms.

For single-stage chance experiments, students list sample spaces and assign probabilities to outcomes. They describe variation between the estimated and theoretical probability of single-stage chance experiments and simulations. They determine mean, mode, median and range, and identify which measure best reflects the dataset. Students represent and describe data in a stem-and-leaf plot and identify outliers. They make critical comments regarding statements made in the media relating to averages.

Content descriptions

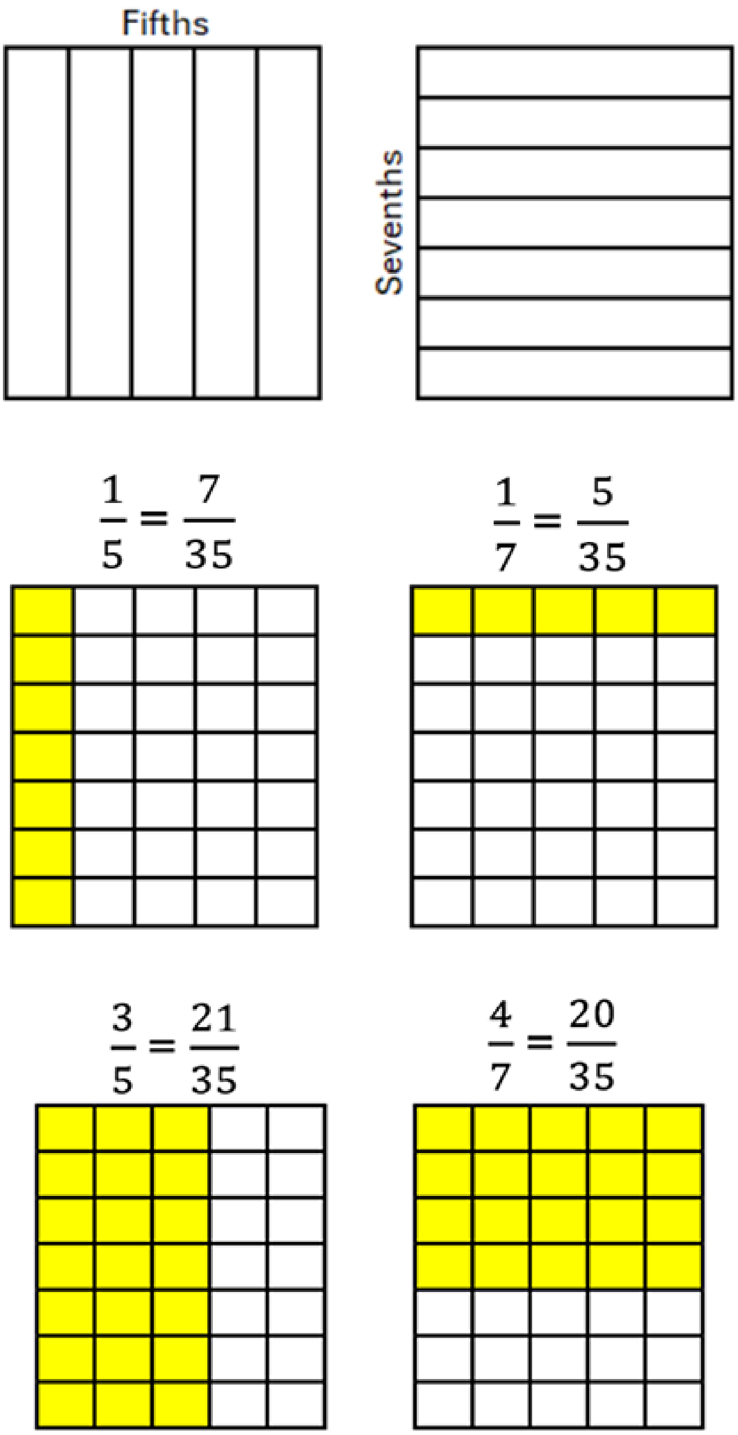
Number and algebra

Understanding number

WA7MNAUN1

Explore and represent equivalent fractions with related and unrelated denominators, visually and numerically

For example:



- using the multiplicative relationships between equivalent fractions to justify that $\frac{2}{5} = \frac{6}{15}$ because if the parts are three times smaller, the number of parts needs to be three times greater

- using knowledge of equivalence to explore the truth of statements, such as $\frac{3}{4} > \frac{2}{3}$ or $\frac{3}{4} + \frac{1}{3} \neq \frac{4}{7}$

General Capabilities: Numeracy, Critical and creative thinking

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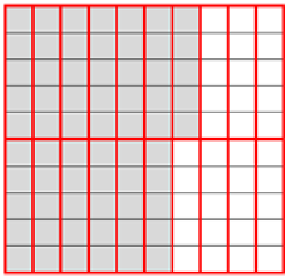
WA7MNAUN2

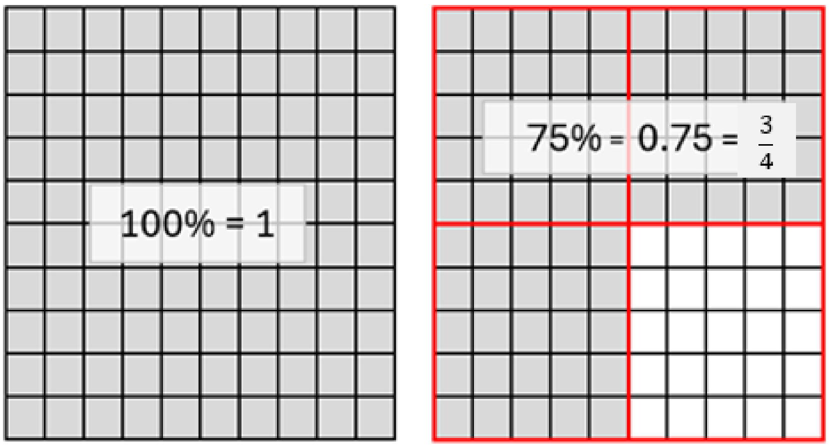
Explore and explain relationships between fractions, decimals and percentages

For example:

- drawing diagrams to explain $65\% = 0.65 = \frac{65}{100} = \frac{13}{20}$



- explaining that as $100\% = 1$, 175% can be represented using a diagram, such as



$175\% = 1.75 = 1\frac{3}{4}$

- investigating, comparing, generalising and validating calculator strategies to move between percentages, fractions and decimals

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA7MNAUN3

Draw and label, or use a given number line, to locate, order and compare with equality and inequality symbols, fractions, terminating decimals, percentages and integers

For example:

- locating $\frac{2}{5}$, −2, −4, 95%, 0.75 and $2\frac{1}{4}$ on a number line and using this to assist in ordering and comparing numbers using inequalities $-4 < -2 < \frac{2}{5} < 0.75 < 95\% < 2\frac{1}{4}$
- comparing $\frac{3}{5}$ and $\frac{6}{10}$ using equality and inequality symbols

  $\frac{3}{5} = \frac{6}{10}$ or $\frac{3}{5} \leq \frac{6}{10}$ or $\frac{3}{5} \geq \frac{6}{10}$

- comparing $\frac{1}{4}$ and $-\frac{1}{3}$ using inequality symbols

General Capabilities: Numeracy

WA7MNAUN4

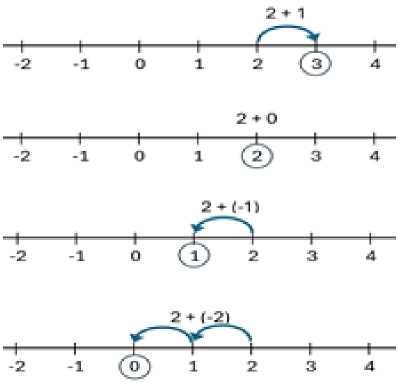
Explore to extend addition and subtraction of positive integers to include negative integers

For example:

- using patterning to explore

$2+2=4$	$5-1=4$	$-3+2=-1$
$2+1=3$	$5-0=5$	$-3+1=-2$
$2+0=2$	$5-(-1)=6$	$-3+0=-3$
$2+(-1)=1$	$5-(-2)=7$	$-3+(-1)=-4$
$2+(-2)=0$	$5-(-3)=8$	$-3+(-2)=-5$
$2+(-3)=-1$	$5-(-4)=9$	$-3+(-3)=-6$

- using number lines to explore



- extending and exploring addition and subtraction as inverse operations of each other by constructing equivalent equations that include negative integers

  If $3 + (-5) = -2$, then

  $-5 + 3 = -2$

  $-2 - 3 = -5$

  $-2 - (-5) = 3$

General Capabilities: Numeracy, Critical and creative thinking

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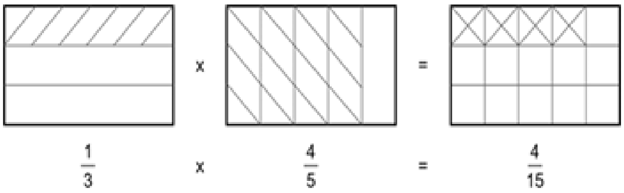
WA7MNAUN5

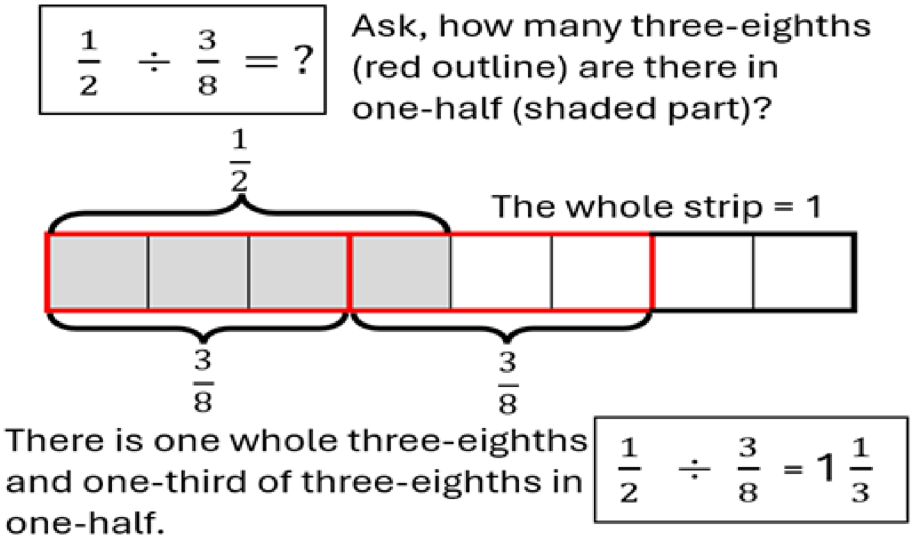
Explore and interpret multiplication and division of positive fractions, visually and numerically

For example:

- representing $\frac{1}{3} \times \frac{4}{5}$



and using to explain $\frac{1}{3} \times \frac{4}{5} = \frac{4}{15}$



General Capabilities: Numeracy, Critical and creative thinking

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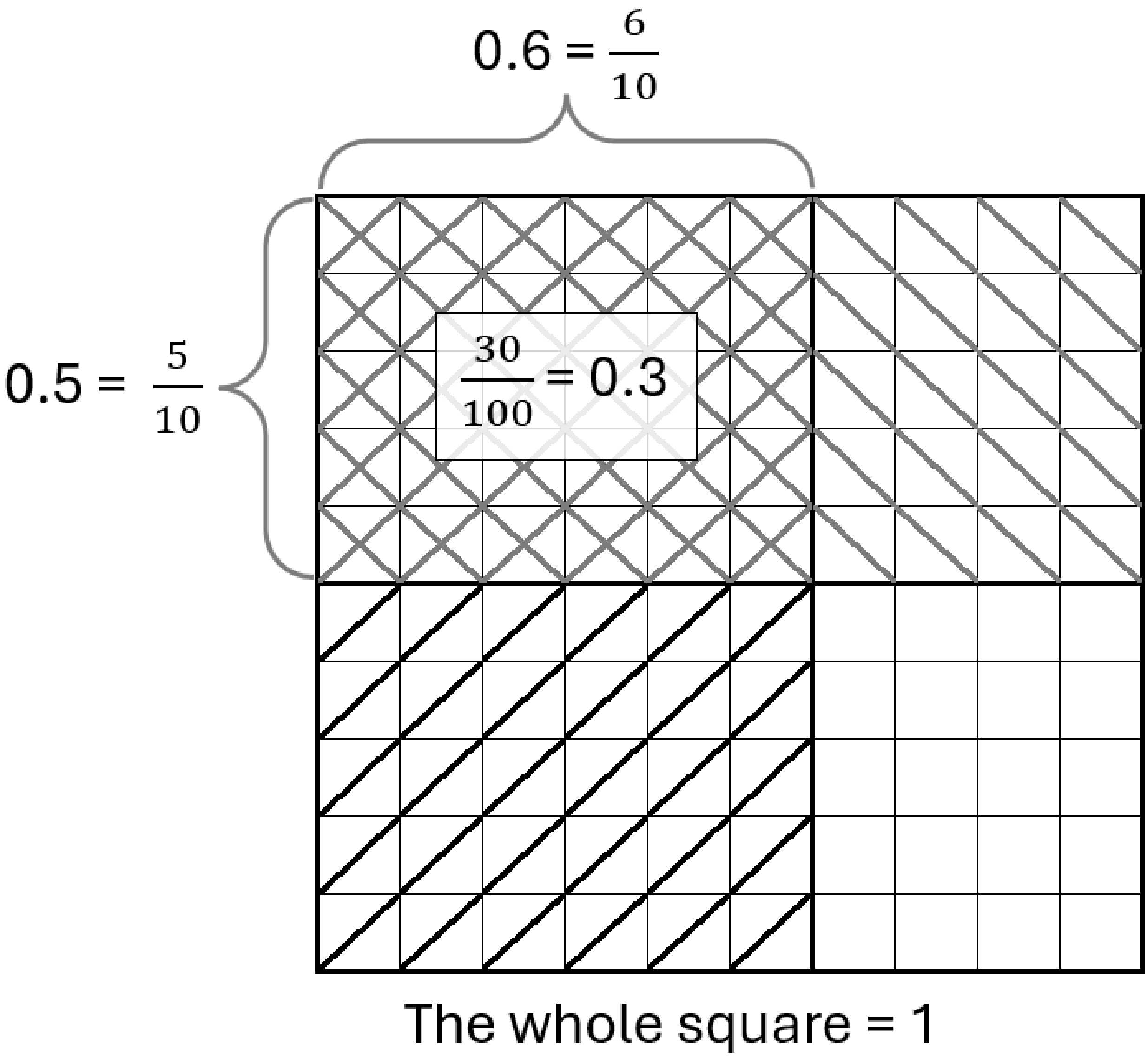
WA7MNAUN6

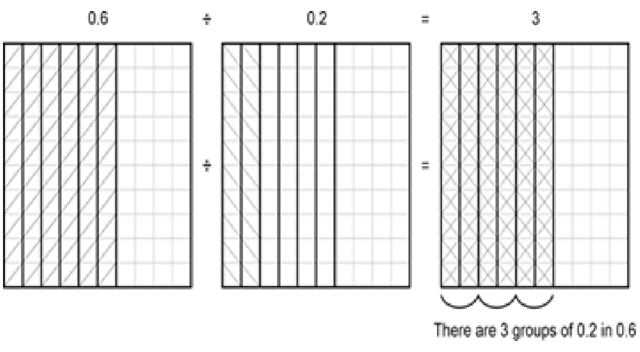
Explore and interpret multiplication and division of positive decimals, visually and numerically

For example:

- representing 0.6×0.5 in a diagram and using to explain and make connections to $\frac{6}{10} \times \frac{5}{10} = \frac{30}{100}$



- representing $0.6 \div 0.2$ in a diagram and using the diagram to explain and make connections to $\frac{6}{10} \div \frac{2}{10} = 3$



General Capabilities: Numeracy, Critical and creative thinking

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WA7MNAUN7

Use place value understanding to explore rounding decimals to a specified number of decimal places

For example:

- using place value to describe 3.64 as between 3.6 and 3.7, $\frac{4}{100}$ greater than 3.6 and $\frac{6}{100}$ smaller than 3.7
- when rounding to three decimal places
  1.6248 rounds to 1.625
  1.6273 rounds to 1.627
  1.6235 rounds to 1.624

General Capabilities: Numeracy, Critical and creative thinking

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WA7MNAUN8

Extend the use of associative, commutative and distributive laws, additive and multiplicative partitioning, inverse operations, order of operations, equality and inequality to validate a range of mental and written strategies involving the four operations on whole numbers, positive fractions and decimals, and addition and subtraction of integers

For example:

- exploring efficiency of strategic use of partitioning, commutativity and order of operations to assist calculation of

  $16 \times 75 = 4 \times 4 \times 3 \times 25$

  $= 4 \times 25 \times 3 \times 4$

  $4 \times 2\frac{3}{4} = (4 \times 2) + (4 \times \frac{3}{4})$

  $-27 + 35 = -27 + (27 + 8)$

  $15.3 - 4.8 = 15.3 + (-5) + 0.2$

- extending the use of inverse operations to assist calculation of an unknown fraction or decimal in an equation with one operation, such as

  $\frac{5}{8} - ? = \frac{1}{4}$

  $\frac{5}{8} - \frac{1}{4} = ?$

  $? \times 3.5 = 16.1$

  $? = 16.1 \div 3.5$

- without calculating, justifying statements as true (T) or false (F), such as

  $48 \times 2.5 = 12 \times 10$

  $8 \div 1.8 \div 2 < 8 \div 2 \div 1.8$

  $15 \times 6 < 15 \times 12 \div 2$

  $-12 + 3 \geq -9 + (-3) + 3$

$\frac{3}{8} + \frac{5}{8} > \frac{5}{8} \times (\frac{1}{5} + \frac{3}{5})$

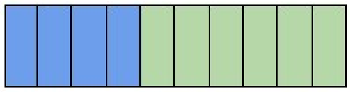
General Capabilities: Numeracy

WA7MNAUN9

Explore and explain the use of ratios and fractions to compare numbers and quantities. Make connections between equivalent fractions and between equivalent ratios

For example:

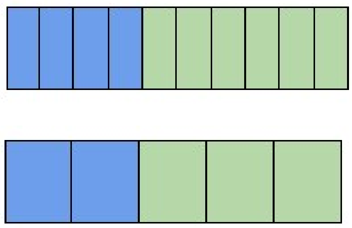
- the bar, or the whole, can be considered as 10 equal parts or 2 unequal parts made up of the 'blue part' and the 'green part'

  exploring ways to compare the sizes of

  - the 'blue part' **and** the 'green part', such as 4:6 or 2:3, 6:4 or 3:2, $\frac{4}{6}$ or $\frac{2}{3}$, $\frac{6}{4}$ or $1\frac{1}{2}$
  - the 'blue part' **and** the 'whole', such as 4:10 or 2:5, 10:4 or 5:2, $\frac{4}{10}$ or $\frac{2}{5}$, $\frac{10}{4}$ or $\frac{5}{2}$
  - the 'green part' **to** the 'whole', such as 6:10 or $\frac{6}{10}$ or $\frac{3}{5}$

- using diagrams to explain equivalent ratios and simplification of ratios,

  such as

  $4:6 = 2:3$ (blue part to green part) or
  $4:10 = 2:5$ (blue part to whole)

- comparing measurements, such as 2 g **to** 0.8 kg
  2:800, 1:400 or $\frac{1}{400}$

General Capabilities: Numeracy, Critical and creative thinking

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Calculating with number

WA7MNAC1

Convert between fractions, decimals and percentages using flexible and efficient strategies

For example:

- mentally converting, with jottings if needed, 12.5% to a fraction using thinking, such as 12.5% is half of 25% ($\frac{1}{4}$) and half of $\frac{1}{4}$ is $\frac{1}{8}$ and converting $\frac{3}{20}$ to a percentage using thinking, such as $\frac{3}{20}$ is equivalent to

$\frac{15}{100}$ and is therefore 15%

- converting 37.5% to a fraction, communicating thinking using a sequence of equations, such as $\frac{37.5}{100} = \frac{375}{1000} = \frac{3}{8}$
- converting $\frac{97}{22}$ to a decimal (2dp), using a calculator

General Capabilities: Numeracy, Digital literacy

WA7MNAC2

Determine percentages of quantities and express one quantity as a percentage of another using flexible and efficient strategies

For example:

- mentally calculating, with jottings if needed, 10% of 570, using the result to calculate 5% and 30% of 570 and finding 25% of 60 using thinking, such as $\frac{1}{4} \times 60$ is 15
- determining the percentage that 0.48 is of 2, communicating thinking using a sequence of equations, such as
  $\frac{0.48}{2}$
  $= \frac{48}{200}$
  $= \frac{24}{100}$
  $= 24\%$
- using a calculator to find 17% of $9.20

General Capabilities: Numeracy, Digital literacy

WA7MNAC3

Add and subtract integers using flexible and efficient strategies

For example:

- mentally calculating, with jottings if needed,
  $-8 + 20$ using $20 - 8$
  $7 - 13$ using $7 - 7 - 6$
  $3 - (-12)$ using $3 + 12$
  $-6 + (-4)$ using $-6 - 4$
- communicating thinking to the solution of
  $-35 + 48 - (-15) + (-11)$
  using a sequence of equations, such as
  $= -35 + (35 + 13) + 15 - 11$
  $= (-35 + 35) + 13 + 4$
  $= 17$

- using a calculator to evaluate $-54 + (-723) - 94 - (-6873)$

General Capabilities: Numeracy, Digital literacy

WA7MNAC4

Add and subtract positive fractions with related and unrelated denominators using flexible and efficient strategies

For example:

- mentally calculating, with jottings if needed, $1\frac{7}{8} + 2\frac{1}{4}$ using thinking, such as $3\frac{7}{8} + \frac{1}{4}$ which is $4\frac{1}{8}$
- communicating thinking using a sequence of equations for $\frac{2}{7} + \frac{2}{3}$ and $2\frac{1}{3} - 1\frac{1}{2}$, simplifying answers where possible
- using a calculator to evaluate $6\frac{3}{7} + 18\frac{5}{9}$

General Capabilities: Numeracy, Digital literacy

WA7MNAC5

Multiply and divide positive fractions using flexible and efficient strategies

For example:

- mentally calculating, with jottings if needed, $5\frac{1}{2} \div \frac{1}{4}$ using thinking, such as there are 20 quarters in 5 and 2 quarters in $\frac{1}{2}$, so the answer is 22 and calculating $\frac{2}{3} \times 6\frac{1}{2}$ using thinking, such as $\frac{2}{3} \times 6$ is 4 and $\frac{2}{3} \times \frac{1}{2}$ is the same as $\frac{1}{2} \times \frac{2}{3}$ or $\frac{1}{3}$ so the answer is $4\frac{1}{3}$
- communicating thinking using a sequence of equations, for calculations, such as

  $\frac{2}{5} \times 2\frac{2}{3}$

  $\frac{2}{7} \div \frac{2}{3}$

  $\frac{1}{2} \times (\frac{2}{5} + \frac{3}{10})$

  simplifying answers where possible
- using a calculator to evaluate $11\frac{2}{7} \div (2\frac{2}{3} \times \frac{1}{2})$

General Capabilities: Numeracy, Digital literacy

WA7MNAC6

Multiply and divide positive decimals using flexible and efficient strategies

For example:

- mentally calculating, with jottings if needed, 0.25×1.2 by using thinking, such as $\frac{1}{4}$ of 1.2 which is 0.3

- communicating thinking using a sequence of equations, for calculations, such as

  $6.4 \div 0.8$

  $= 64 \div 8$

  $= 8$

  7.8×9

  $= 7.8 \times 10 - 7.8$

  $= 78 - 7.8$

  $= 70.2$

  $0.36 \div 8$

  $0.36 \div 2$

  $0.18 \div 2$

  $0.09 \div 2$

  $= 0.045$

- using a calculator to evaluate $27.4 \div 1.05$

General Capabilities: Numeracy, Digital literacy

WA7MNAC7

Use appropriate rounding, estimation strategies and context to check reasonableness of solutions

For example:

- recognising that $6\frac{3}{7} + 18\frac{5}{9}$ could be rounded to $6\frac{1}{2} + 18\frac{1}{2}$ and applying appropriate strategies to approximate (≈) the solution, using this estimation as a tool to check if the solution derived from the actual values is reasonable

- recognising that $695 \div 0.527$ could be rounded to $700 \div 0.5$ and that dividing a number by a number less than one, makes it bigger. Applying appropriate strategies to approximate the solution, using this estimation as a tool to check that the solution derived from the actual values is reasonable

- recognising that an interest of 4.95% of $500 as being approximately 5% of $500 and that multiplying by a number less than one makes the number smaller. Applying appropriate strategies to approximate the solution, using this estimation and the context as a tool to check that the solution derived from the actual values is reasonable

General Capabilities: Numeracy, Critical and creative thinking

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Algebraic techniques

WA7MNAA1

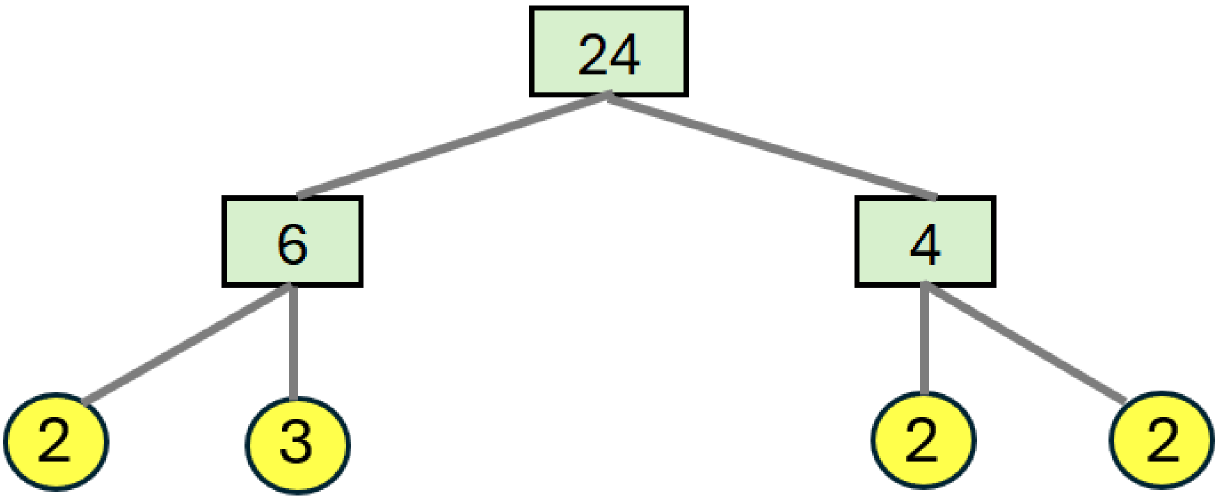
Represent in expanded form, evaluate, and compare numbers expressed in index notation, including powers of 10

> **For example:**
>
> - identifying that 7^3 has a base of 7 and an index or power of 3 and can be written as $7 \times 7 \times 7$ which can be evaluated as 343
> - recognising that $5^4 = 5 \times 5 \times 5 \times 5 = 625$ and that 625 is 25 times as great as 25 and 125 times as great as 5
> - identifying and recognising patterns within powers of 10, such as 1000 is the same as $10 \times 10 \times 10$ which can be written as 10^3 which has a base of 10 and an index or power of 3
> - making connections between powers of 10 and place value, such as $4000 (4 \times 10^3)$ is 100 times as great as $40 (4 \times 10^1)$ and 1000 times as small as four million (4×10^6)

General Capabilities: Numeracy

WA7MNAA2

Extend knowledge of factors to represent natural numbers as products of prime factors using index notation as appropriate

> **For example:**
>
> - making connections between a series of equations and a tree diagram to express 24 as a product of primes
>
> 
>
>
> $24 = 6 \times 4$
> $= 2 \times 3 \times 2 \times 2$
> $= 3 \times 2 \times 2 \times 2$
> We can write this in index form as 3×2^3
> - investigating the use of different pairs of initial factors, such as 8×3 or 2×12 when determining the prime factors of 24

General Capabilities: Numeracy

WA7MNAA3

Explore and explain connections between square numbers and square roots, cube numbers and cube roots, as products of repeated factors

> **For example:**
>
> - deducing that if 9×9 is 9^2 , or 81, then $\sqrt{81} = 9$ and if $6 \times 6 \times 6$ is 6^3 or 216, then $\sqrt[3]{216} = 6$
> - estimating the square root or cube root of a whole number, such as $\sqrt{34}$ by reasoning that it lies between $\sqrt{25}$ and $\sqrt{36}$ and is closer to $\sqrt{36}$ so could be estimated to be 5.8. Using a calculator to confirm a solution of 5.83 (2dp)

General Capabilities: Literacy, Numeracy, Critical and creative thinking

WA7MNAA4

Use real-world contexts or concrete materials to introduce the concept of a variable to represent a number using a letter. Create simple algebraic expressions and evaluate by substituting a given value for the variable/s

> **For example:**
>
> - introducing a variable using unknown numbers of chocolates in two different sized bags where $m = \text{number of chocolates in one small bag}$, $g = \text{number of chocolates in one large bag}$, creating expressions, such as
>   - the number of chocolates in three large bags and one small bag as being $3 \times g + m = 3g + m$
>   - the difference between the number of chocolates in the bags as being $g - m$
>   - combining the number of chocolates in five large bags and two small bags and sharing them between four people as being
>
> $\frac{5g + 2m}{4}$
>
>   and evaluating each expression if the numbers of chocolates in each bag is known
> - using concrete materials, such as algebra tiles, to visually represent numbers as variables
> - using expressions of everyday formulas, such as carry-on bag size for an aeroplane flight being determined by the length + width + depth $(l + w + d)$ and using substitution to determine if a bag is of an appropriate size

General Capabilities: Numeracy, Critical and creative thinking

WA7MNAA5

Extend and apply the associative and commutative laws and properties of numbers to include variables

For example:

- considering the accuracy of the following equations and explaining thinking
  $m + 0 = m$
  $y \times 0 = y$
  $x + x + x = 3x$
  $5 + (m + 3) = (5 + m) + 3$
  $3p + 4 = 3(p + 4)$
  $e \div 3 = \frac{e}{3}$
  $1 \times b = b$
  $g \times 3 = 3 \times g$

General Capabilities: Numeracy, Critical and creative thinking

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Linear and non-linear equations and inequalities

WA7MNALE1

Solve simple linear equations involving up to two operations and verify the solution by substitution

For example:

- using a context, such as 'Three bags with equal numbers of chocolates were tipped into a container that already held five chocolates. There were 62 chocolates altogether. How many were in each bag?'

- exploring and comparing a variety of strategies, such as the balance model, backtracking or inspection, to solve an equation, showing a sequence of steps to communicate thinking. Checking the solution by substitution using jottings or calculator for equations, such as
  $5m - 3 = 12$
  $-3 + m = 11$
  $e - 3 = -2$
  $48 = 14 + 2p$
  $\frac{e}{2} - 1 = 5$
  $0.1v = 12$
  $\frac{1}{4} + 2k = \frac{3}{8}$

  Note: equations involving multiplication and division of negative integers are introduced in Year 8; however, equations, such as $6 - w = 5$, that can be solved by inspection are in the Year 7 curriculum

General Capabilities: Numeracy

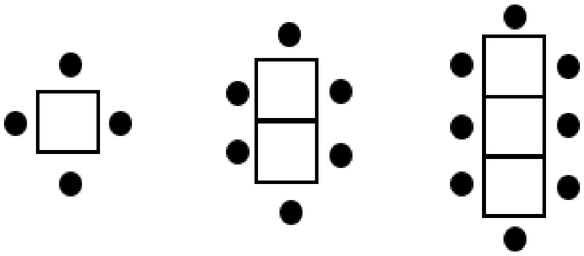
Linear and non-linear patterns and relationships

WA7MNALP1

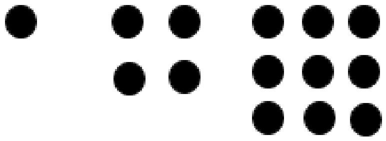
Explore, describe and represent concrete and real-world, linear and non-linear growing patterns using a table of values and a graph. Determine unknown values in the pattern

For example:

- using diagrams or concrete materials, such as matchsticks and counters, to continue growing patterns and represent as number sequences in situations, such as chairs around connecting square tables



or growing dots



  - using words to generalise rules
  - using variables meaningfully to express rules as algebraic equations
  - producing a table of values
  - plotting points in the first quadrant on the Cartesian plane and describing the shape of the relationship

and using the sequence rule, table of values or graph to determine unknown values

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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Financial mathematics

WA7MNAF1

Identify the features of transactional statements and verify transactions. Explain reasons for checking and keeping financial records

For example:

- identifying features of transactional statements (e.g. bank, subscription, canteen, transport, prepaid debit card), including account information, statement period, transaction details (credits and debits), opening and closing balances, fees and interest earned (if appropriate)
- explaining the importance of checking and keeping receipts and invoices, and tracking and verifying transactions for the purpose of budgeting and identifying suspicious transactions, such as 'Alice paid for a meal at a cafe electronically. She had bought a coffee for $5.50 and a wrap for $12.75. When she checked her bank statement on her bank app, she realised she had been charged $31. Explain using calculations, how the error could have occurred and what Alice could do to address the mistake.'

General Capabilities: Numeracy, Digital literacy

Modelling with number and algebra

WA7MNAM1

In real-world situations involving whole numbers, positive fractions, decimals and percentages, addition and subtraction of integers, numbers in index form, linear equations with up to two operations, simple number patterns and/or transactional money statements

I. analyse the situation, decide if an exact or approximate solution is required and determine assumptions and constraints
II. represent the situation mathematically in order to reach a solution
III. interpret and communicate findings in terms of the context and any assumptions or constraints

> **For example:**
>
> - discussing assumptions, if any, choosing a representation, such as a number line or numerical equation, and communicating a solution using correct mathematical notation and use of context, given a problem, such as ‘In Nepal, the temperature overnight was reported to be between $-11°$ and $-13°$. It rose by around $20°$ to $25°$ to reach its daytime peak before falling slightly less than its rise. What was the final temperature?’
> - discussing assumptions, their impact on solutions and communication of solutions in problem situations, such as ‘A restaurant serves between 60 and 80 customers for lunch each day. Many customers order the scallop dish which requires around 0.15 kg of scallops per person. What weight of scallops should the chef order each week?’ While an exact number of kilograms is required to place a weekly order, this is an estimate based on assumptions, such as number of days per week the restaurant is open, number of people choosing the dish, importance of minimising wastage and customer expectation the dish will be available. Considering the range of possible solutions based on different assumptions and, therefore, when communicating a solution, the consequent need to describe and qualify the assumptions and constraints on which it is based
> - people often exercise at $60–80\%$ of their maximum heart rate. A formula determining the maximum heart rate in beats per minute a person should reach when exercising can be calculated using: $220 - \text{age (in years)}$. What would be the target heart rate range for exercise for you at different stages in your life? Discussing assumptions, such as the formula not being suitable to use for very young or very old people. Choosing to represent the situation graphically or by using a linear equation. Interpreting and communicating the solution in terms of age and the initial assumptions

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

Measurement and geometry

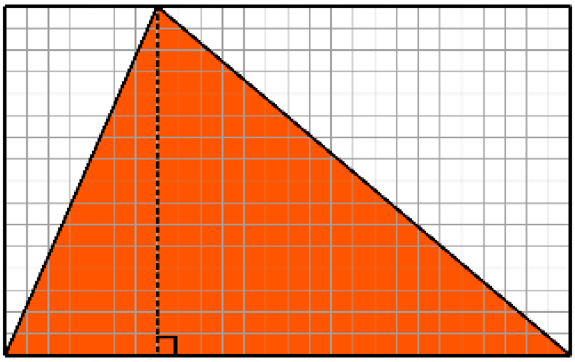
Two-dimensional space and structures

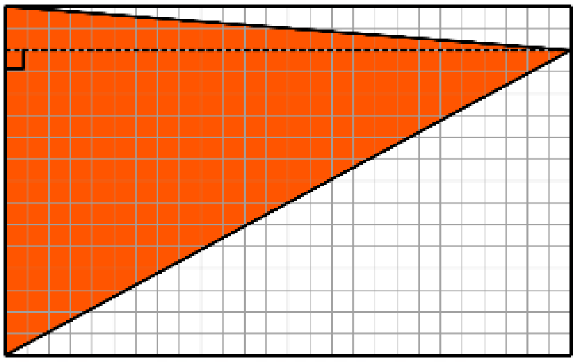
WA7MMGTW1

Establish and apply relationships between lengths of sides, perimeter and area for squares, rectangles and triangles. Generalise and apply formulas, using appropriate units

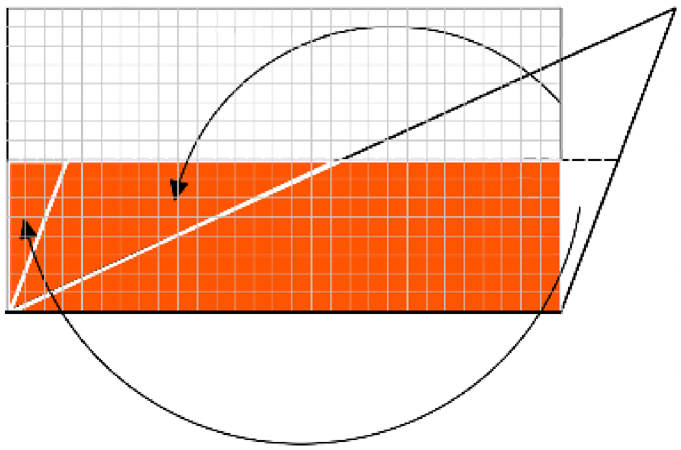
For example:

- establishing the perimeter of an equilateral triangle is three times as long as the length of one of its sides and the length of one side is $\frac{1}{3}$ of the length of the perimeter, generalising as $P = 3l$ where l is the number of length units in a side and P is the number of length units in the perimeter of an equilateral triangle
- using grid paper or dynamic geometry software to demonstrate that for squares and rectangles, the number of length units in the length multiplied by the number of length units in the width provides the number of square units of area, generalising this to $A = l^2$ and $A = l \times w$ respectively. Extending understanding to part-units by comparing areas determined using formula with diagrams showing whole and part-units of length and area
- using grid paper or dynamic geometry software to draw a rectangle, length 5 units and width 1 unit, determining its area and repeating for subsequent rectangles with a constant length of 5 units and widths increasing by 1 unit of length. Graphing the width compared to the area of the rectangles, connecting the graph to the constant increasing area in the sequence of drawn rectangles
- using grid paper to draw a rectangle around any right-angled or acute triangle, comparing the area inside the triangle to that outside the triangle,





explaining how the area of the rectangle can be used to determine the area of a triangle. Generalising to a formula in symbols, such as $= \frac{1}{2} \times b \times h$ and demonstrating how an obtuse triangle can be cut and rearranged to confirm the formula works for any triangle



- using diagrams and reasoning to determine whether it is possible or not to calculate the following and if so, determining
    - the length of a rectangle, given that it has an area of 20 square centimetres
    - the perimeter of a square, given that it has an area of 8100 square metres
    - the area of an isosceles triangle, given that the length of its base is 8 cm, which is the same as the length of its perpendicular height
    - the area of a rectangle of length 5 cm and width 3.4 mm

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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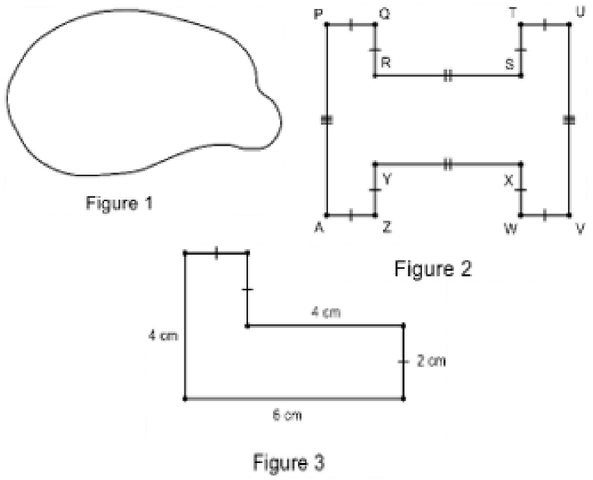
WA7MMGTW2

Explore and explain efficient strategies to determine the perimeter and area of irregular or composite shapes composed of squares and rectangles

For example:



- using rulers, grid paper, scissors, string, geometric reasoning or dynamic geometry software to investigate different ways of determining the perimeter and area of irregular and composite rectangular shapes as in Figures 1, 2 and 3 including exploring
    - the minimum number of lengths requiring measurement in order to reason other lengths and hence calculate perimeter in Figure 2

- different reasoning to find the length of unlabelled sides and hence calculate perimeter and area of Figure 3
- ways of decomposing the shape by cutting or dividing all shapes into squares or rectangles to find area of the parts and hence total area or superimposing a rectangle around each shape determining the area of the rectangle and using subtraction of the area of space within the rectangle but outside the shape, in order to estimate or determine exact area comparing strategies with others to determine and explain the most efficient and accurate method

- drawing different composite square and rectangular shapes that have equal areas but different perimeters and those that have equal perimeters but different areas, justifying using application of formula

General Capabilities: Numeracy, Critical and creative thinking

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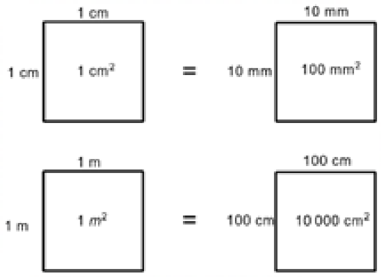
WA7MMGTW3

Explore and establish connections and conversions between units of area

For example:

- using diagrams to represent and explain connections between numbers of different units of equal areas, such as



Recognising that 1 m is 100 times bigger than 1 cm but 1 square metre is 10 000 times as large as 1 square centimetre

- applying knowledge of place value and relative sizes of units to explain conversions between units of area, such as $0.72\ \text{cm}^2$ to mm^2

- extending thinking to include

$1\ \text{ha} = 10\,000\ \text{m}^2$ and

$1\ \text{km}^2 = 1\,000\,000\ \text{m}^2 = 100\ \text{ha}$

General Capabilities: Numeracy, Critical and creative thinking

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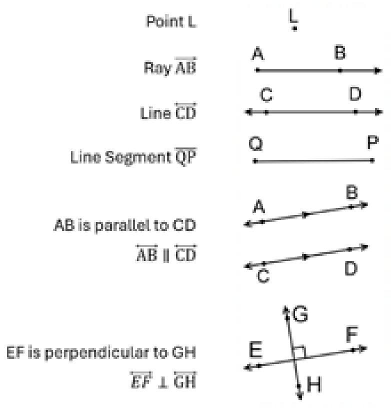
WA7MMGTW4

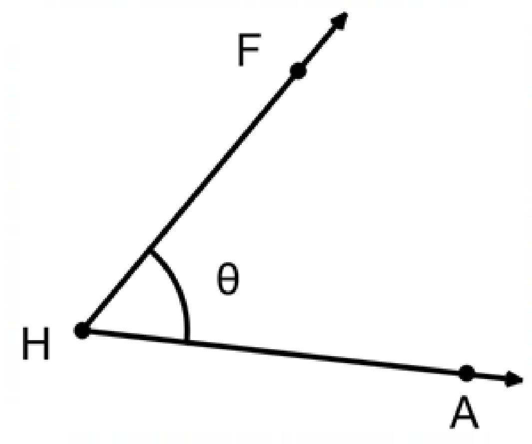
Explore, identify, define, name, label and apply the language, notation and conventions of geometry for points, lines, angles and polygons

For example:

- using conventions, such as the use of capitals and appropriate symbols, to label, name and define points, rays, line segments, lines, parallel lines and perpendicular lines

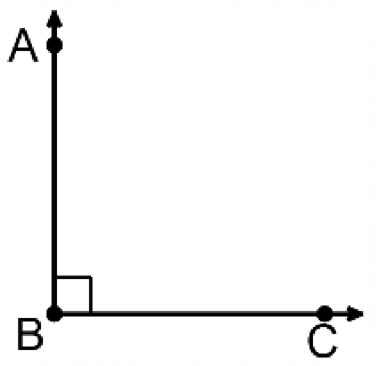


- using conventions, such as capitals, Greek letters and appropriate symbols, to label, name and define acute or obtuse angles



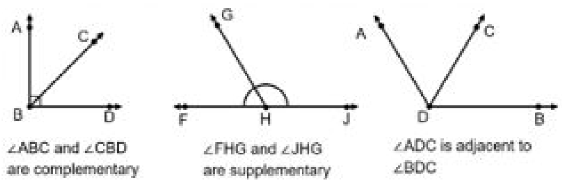
as $\angle H$ or $\angle \mathrm{FHA}$ or $\angle \mathrm{AHF}$ or θ ,

right or perpendicular angles,

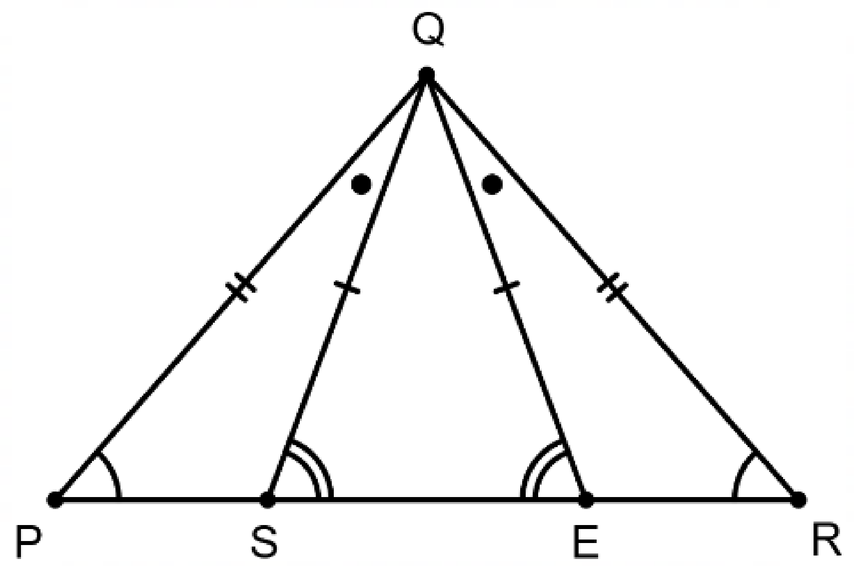


as $\angle \mathrm{ABC}$ or $\angle \mathrm{CBA}$ or $\angle B$,

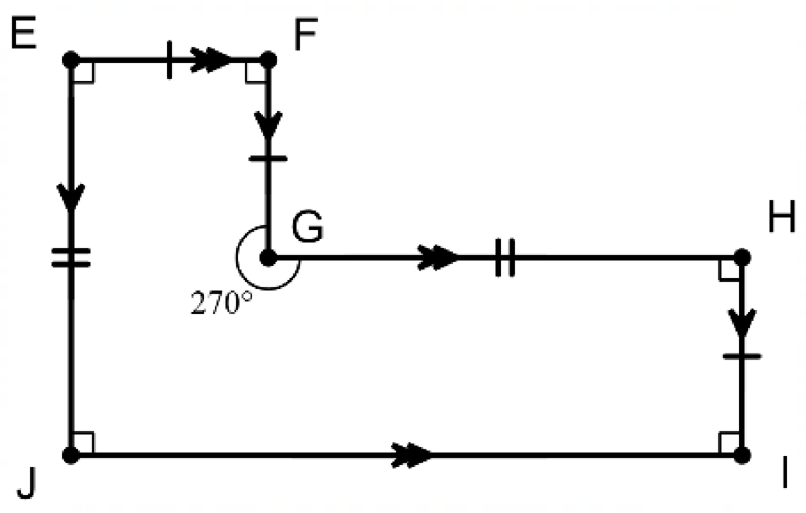
and angles at a point as complementary, supplementary and adjacent angles



- applying correct notation and labelling conventions to triangles and polygons to indicate equal angles, equal sides and parallel sides, such as in Δ PQR and Δ SQE



and in the irregular, concave hexagon EFGHIJ



justifying that EFGHIJ is the same as HIJEFG but is not the same as EIJHGF

- exploring geometric terminology and naming conventions used in dynamic geometry software

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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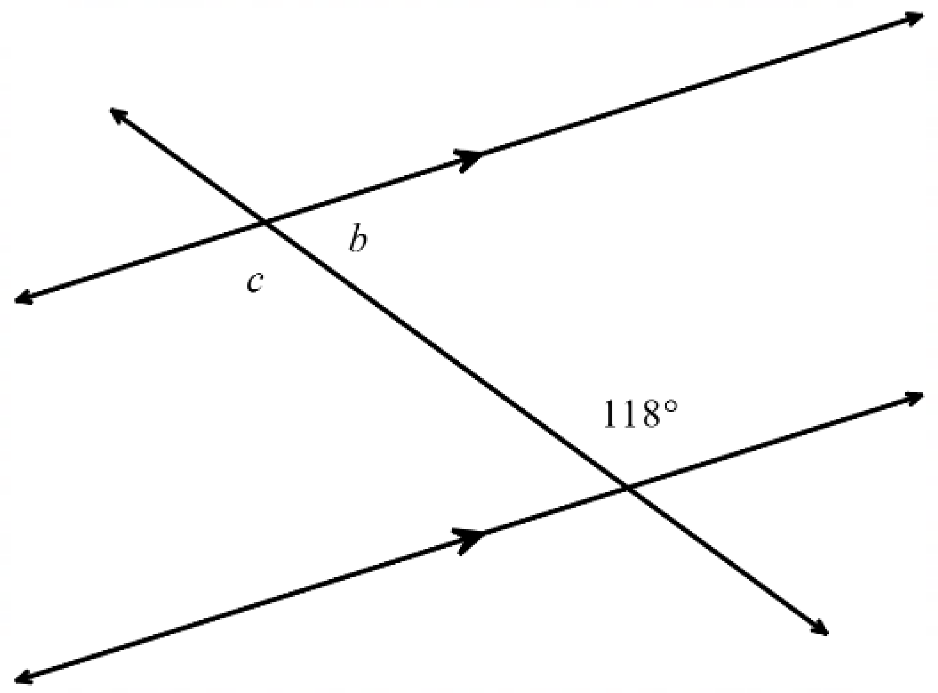
WA7MMGTW5

Investigate, identify and describe corresponding, alternate and co-interior angles formed when two parallel lines are crossed by a transversal. Use relationships to determine unknown angles and explain reasoning

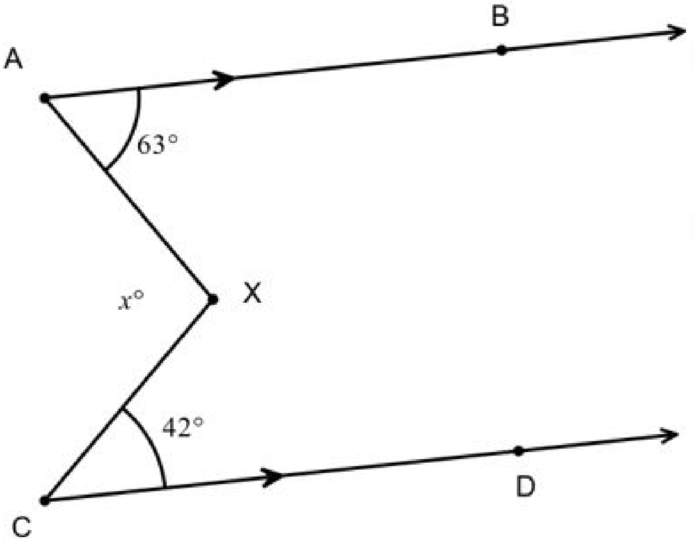
For example:

- using paper folding or dynamic geometry software to create a pair of parallel lines crossed by a transversal, identifying congruent and supplementary angles and labelling pairs of angles as corresponding, alternate and co-interior
- finding unknown angles within pairs of parallel lines. Explaining reasoning in situations, such as



$c = 118°$ (alternate to given angle)
$b = 62°$ (co-interior with given angle)

- finding the value of *x* in the diagram below and giving reasons



constructing a line XY through X parallel to AB and CD

$63° + \angle AXY = 180°$ (co-interior angles, AB || XY)

$\therefore \quad \angle AXY = 117°$ $42° + \angle CXY = 180°$ (co-interior angles CD || XY)

$\therefore \quad \angle CXY = 138°$

$x + \angle AXY + \angle CXY = 360°$ (angles at a point)

$x = 105$

comparing geometric reasoning with others, noting that there are alternative ways of determining and explaining the size of angles

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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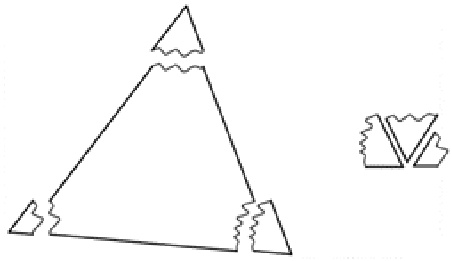
WA7MMGTW6

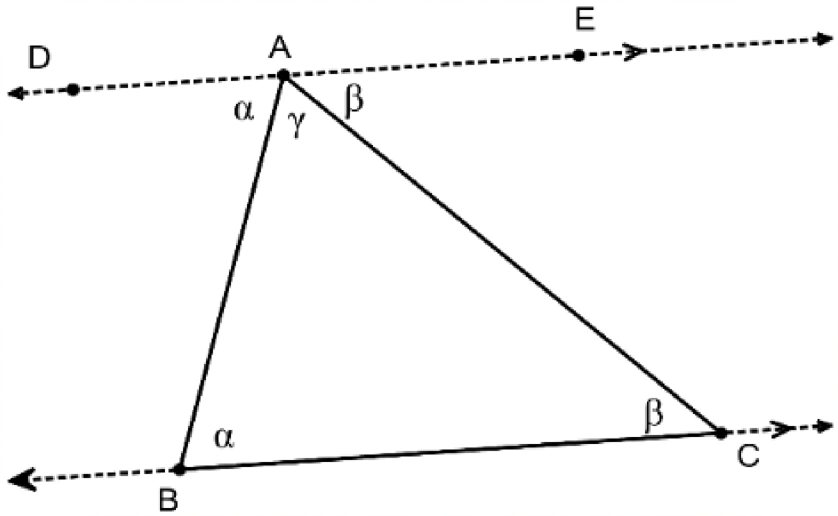
Demonstrate that the interior angle sum of a triangle is 180°

For example:

- using a variety of paper triangles to mark corners, ripping and placing the corners (angles) in a straight line to generalise that the sum of the angles in Δ ABC can be determined by $\angle A + \angle B + \angle C = 180°$



- using knowledge of alternate angles in parallel lines and supplementary angles to explain why the sum of the angles in a triangle is $180°$ using a sequence of logically connected statements



General Capabilities: Literacy, Numeracy, Critical and creative thinking

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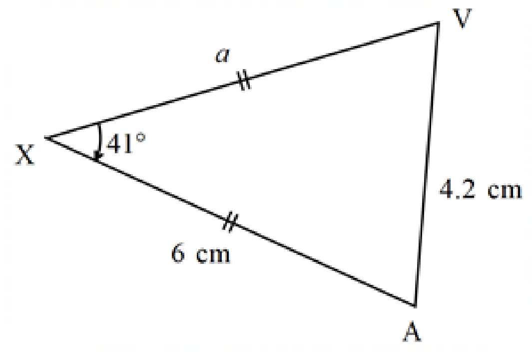
WA7MMGTW7

Explore to classify and name triangles according to their side and angle properties. Use the properties to determine unknown angles in triangles and explain reasoning

For example:

- creating a classification scheme for triangles based on sides and angles, representing it as a sequence of decisions in a flow chart

- using reasoning and/or construction to decide and explain whether the following triangles are possible or impossible, naming and labelling possible triangles and comparing the construction and naming with others
    - a triangle with sides of $4\ \text{cm}$, $10\ \text{cm}$ and $5\ \text{cm}$
    - a triangle with all angles $60°$
    - a triangle with one side of $8\ \text{cm}$ and a $90°$ angle
    - a triangle with two equal sides of $7\ \text{cm}$ and the included angle $45°$
    - a triangle with two equal angles of $95°$
- determining unknown sides and angles in triangles, explaining and comparing reasons with others



$a = 6\ \text{cm}$ (equal sides in an isosceles triangle) $\angle\text{XAV}$ and $\angle\text{XVA} = 69.5°$ (equal base angles of an isosceles triangle)

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA7MMGTW8

Plot coordinates on the Cartesian plane and explore, visualise, predict and determine image coordinates after translation or reflection across the axes, or rotation about the origin

For example:

- plotting the following points $D(-1, 6)$, $E(3, -2)$, $(2, 4)$ onto paper or dynamic geometry software. Using counters for coordinates, paper folding and tracing, digitally moving points and/or reasoning and visualisation to determine the image coordinates (D', E', F') after a transformation, such as
    - 2 units left and 3 units down
    - a reflection across the x-axis
    - a rotation of $180°$ anticlockwise about the origin
    - a rotation of $90°$ clockwise about the origin
- generalising the effect of the transformation on the coordinate (e.g. in a reflection in the x-axis, the y part of the coordinate changes sign from a positive to a negative and vice versa) and investigating whether there are different ways of translating, reflecting or rotating the points that produce the same transformation

General Capabilities: Numeracy, Digital literacy

Three-dimensional space and structures

WA7MMGTH1

Move flexibly between building and drawing rectangular and composite rectangular prisms from different views

> **For example:**
>
> - building rectangular prisms or composite rectangular prisms using at most 12 cubes, visualising and drawing manually, the front, top, side and isometric views. Using dynamic geometry software and the three-dimensional coordinate system to draw representations of rectangular and composite rectangular prisms
> - using provided isometric or perspective drawings of rectangular and composite rectangular prisms to build the prism from blocks
> - exploring the views of rectangular and composite rectangular buildings depicted in map software, highlighting what is the same and what is different between each view

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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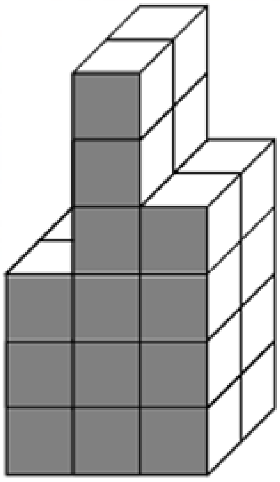
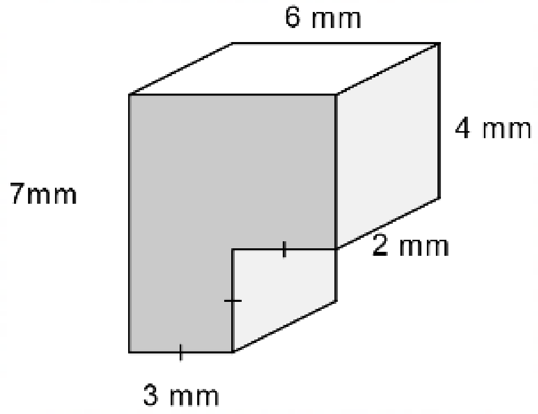
WA7MMGTH2

Establish and apply relationships between the number of identical layers of cubic units, the number of cubic units in each identical layer and volume for rectangular prisms and composite rectangular prisms. Generalise and apply the formula, using appropriate units

> **For example:**
>
> - using layers of interlocking blocks or dynamic geometry software to reinforce that volume is measured in cubic units and that the volume of a rectangular prism is determined by the number of cubic units in any outer layer, multiplied by the number of identical layers in the prism, generalising to $V = l^3$ or $V = l \times w \times h$ for cubes and rectangular prisms respectively. Extending understanding to part-units by comparing volumes determined using the formula with diagrams showing whole and part-units of length and volume
> - using blocks, grid paper or dynamic geometry software to construct a rectangular prism with a length of 5 units, a width of 1 unit and a depth of 1 unit, determining its volume and repeating for subsequent rectangular prisms, with a constant length of 5 units, a constant width of 1 unit and increasing units of depth. Graphing the depth compared to the volume of the prism, connecting the graph to the constant increasing volume in the sequence of prisms and using the graph to determine relationships between the quantities, including finding the depth of the prism if the volume is $18.5\ \text{cm}^3$
> - determining the unknown depth of a rectangular prism given it has a length of $23.4\ \text{cm}$, a width of $13.75\ \text{cm}$ and a volume of $4872.81\ \text{cm}^3$ using a formula
> - using composite rectangular prisms made of blocks or depicted as isometric or oblique drawings to visualise, determine and compare methods for finding volume





General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

Non-spatial measurement

WA7MMGN1

Explore and interpret representations of time zones within Australia using 12- and 24-hour time and determine the local time at different locations considering different times of the year

For example:

- using maps, number lines or digital tools to interpret and explore the language and thinking used in determining and communicating current times across Australia (AEST, ACST, AWST and DST)
- determining current times in Sydney during Daylight Saving Time (DST) if it is 2245 hours in Perth or a quarter past 3 in the afternoon in the Northern Territory or 7.35 pm in Tasmania

General Capabilities: Literacy, Numeracy

Modelling with measurement and geometry

WA7MMGM1

In real-world situations involving perimeter and area of squares, rectangles, triangles and rectangular composite shapes, parallel lines, properties of triangles, transformations of points, views of rectangular prisms and rectangular composite objects, volume and/or Australian time zones

I. analyse the situation, decide if an exact or approximate solution is required and determine assumptions and constraints

II. represent the situation mathematically in order to reach a solution
III. interpret and communicate findings in terms of the context and any assumptions or constraints

> **For example:**
>
> - 'Maya is a Year 7 student who lives in Albany, WA and wants to phone her grandmother who lives in Adelaide, SA. Maya would like to call her grandmother after school and also knows her grandmother goes to bed at 7.30 pm. Determine a suitable time range in which Maya could ring her grandmother.' Realising that an approximate range of times is required, that Maya would need to ring her grandmother after school finishes at approximately 4 pm in Albany and before 7 pm in Adelaide, to allow time for the conversation. Maya would also need to consider if it is daylight saving time or not. Choosing to represent the situation using a number line, communicating the solution as an appropriate local time range for Maya and checking against the initial assumptions
> - 'Charlie is planning to knit a floor rug. She already has 5 m of braiding that she will use to go around the outside edge of the rug. She wants to know if she should knit a square or a rectangular rug to maximise the floor space that the rug will cover. What shape and dimensions should Charlie choose?' Deciding that an accurate answer is required, that the perimeter length needs to be less than 5 m to allow the braid ends to be connected, so a small reduction in the length of braid should be determined, and that the room dimensions and visual appeal of the rug will be constraining factors. Representing the situation using square and rectangle diagrams, choosing the attribute of area to determine the largest floor space and reasoning using patterning, to find possible dimensions that satisfy the braid length (perimeter) of 4.92 m (assuming 8 cm to connect the braid). Determining the area of each possibility in square units. Identifying and communicating that the square of side length 1.23 m provides the maximum possible area of $1.51\ \mathrm{m}^2$, that it satisfies the constraint of 4.92 m and would be a suitable, visually appealing shape

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Probability and statistics

Probability and statistics

WA7MPSP1

Construct sample spaces for single-stage chance experiments, assign probabilities to the outcomes and predict frequencies for different numbers of trials

> **For example:**
>
> - discussing probability terminology, such as probability, favourable outcome, event, trial, sample space, experiment.
>
>   Producing a sample space for a standard six-sided dice $S = \{1, 2, 3, 4, 5, 6\}$ and determining the probability of rolling a 4 using
>
> $P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$ so $P(4) = \frac{1}{6}$, 0.17, 16.67% or a one-in-six chance. In 100 trials, it would be expected that the number of times 4 would occur would be approximately $\frac{1}{6} \times 100 = 16.67$ which would be rounded to 17
>
> - assigning probabilities to events involving coins, dice, spinners, cards, lucky dips etc.

General Capabilities: Numeracy, Critical and creative thinking

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WA7MPSP2

Conduct repeated single-stage chance experiments and simulations to produce datasets, including through the use of digital tools, for an increasingly large number of trials. Discuss and describe variation and estimated probabilities for outcomes, and compare to predictions and theoretical probability, where appropriate

> **For example:**
>
> - conducting an experiment to find the estimated probability of a drawing pin falling point up or point down when tossed. Completing 20 trials, determining relative frequencies (fraction of those that landed point up or point down), comparing with others, describing variation in results and estimating probabilities of each outcome. Repeating over a large number of trials, comparing to others again and using the data to estimate the probability that a pin will land point down when tossed. Comparing this to former estimates of probabilities and discussing the law of large numbers
>
> - conducting a simulation to determine the chance of randomly choosing a person who supports the Dockers from a family comprising three Eagles supporters, two Dockers supporters and one Geelong supporter. Predicting the expected probability before using a random number generator from 1–6, with 1, 2, 3 representing the Eagles supporters, 4 and 5 representing the Dockers supporters and 6 representing the Geelong supporter, over 20 trials. Determining the relative frequency that a Dockers supporter is chosen, comparing results with others, describing variation and estimating the probability. Repeating over an increasingly large number of trials. Comparing results of estimated probability to theoretical probability and discussing variation

- conducting an experiment to predict the unknown distribution of red, green and blue counters in a bag of 12. Randomly choosing a counter, noting its colour, replacing and recording the frequencies of red, green and blue over 50 trials. Determining the relative frequencies for each colour, noting that they add to one. Comparing results with others, describing variation and predicting colour distribution. Repeating over many trials, and again comparing to others, calculating appropriate averages, predicting colour distribution, then confirming the actual number of each colour in the bag. Describing comparisons of predicted and actual colour distribution. Relating to the theoretical probability and discussing variation in the experimental probability as the number of trials increases (law of large numbers)

General Capabilities: Literacy, Numeracy, Digital literacy

WA7MPSP3

Explore and determine the mean, mode, median and range for sets of data and justify, using the context, which measure best reflects the dataset

> **For example:**
>
> - using concrete materials, such as a small collection of
>   - beakers of different amounts of water
>   - 'fake' $100 notes allocated to students ($500, $200, $500, $1400, $600, $0)
>
>   to physically show the mode and the range, reordering to demonstrate the median and redistributing evenly between each data point to show the mean (generalising to a numerical calculation of the mean). Discussing which measure of average best reflects each physical context
> - given the mean number of pets in seven households is 4.25, working backwards to produce and explain seven reasonable data points
> - using data sources, such as primary, discrete data collected from chance experiments, or secondary, continuous data, such as the maximum temperature/rainfall in the class town or city for each month of the year, to determine mean, mode, median and range using written jottings or digital tools depending on the complexity of the data. Explaining and justifying which is the most reflective of the dataset

General Capabilities: Literacy, Numeracy, Digital literacy

WA7MPSP4

Represent primary categorical and numerical data in a Venn diagram, calculate related relative frequencies and interpret results

> **For example:**
>
> - collecting data from a Year 7 maths class on different options for an end of term activity, such as those who would prefer to go to a movie, those who would prefer tenpin bowling, those who are happy with either option and those who like none of the options. Representing the results in a Venn diagram and determining the proportion (relative frequency) of students who chose the movies and not tenpin

bowling (e.g. $\frac{4}{5}$). Analysing results, such as most of our class would like to go to a movie because the film playing at that time is very popular. Acknowledging that the data would vary if recorded on a different day or from a different Year 7 class, impacting conclusions. Discussing whether it would be valid to use the data from this maths class to make decisions for the whole population of Year 7 students

General Capabilities: Literacy, Numeracy

WA7MPSP5

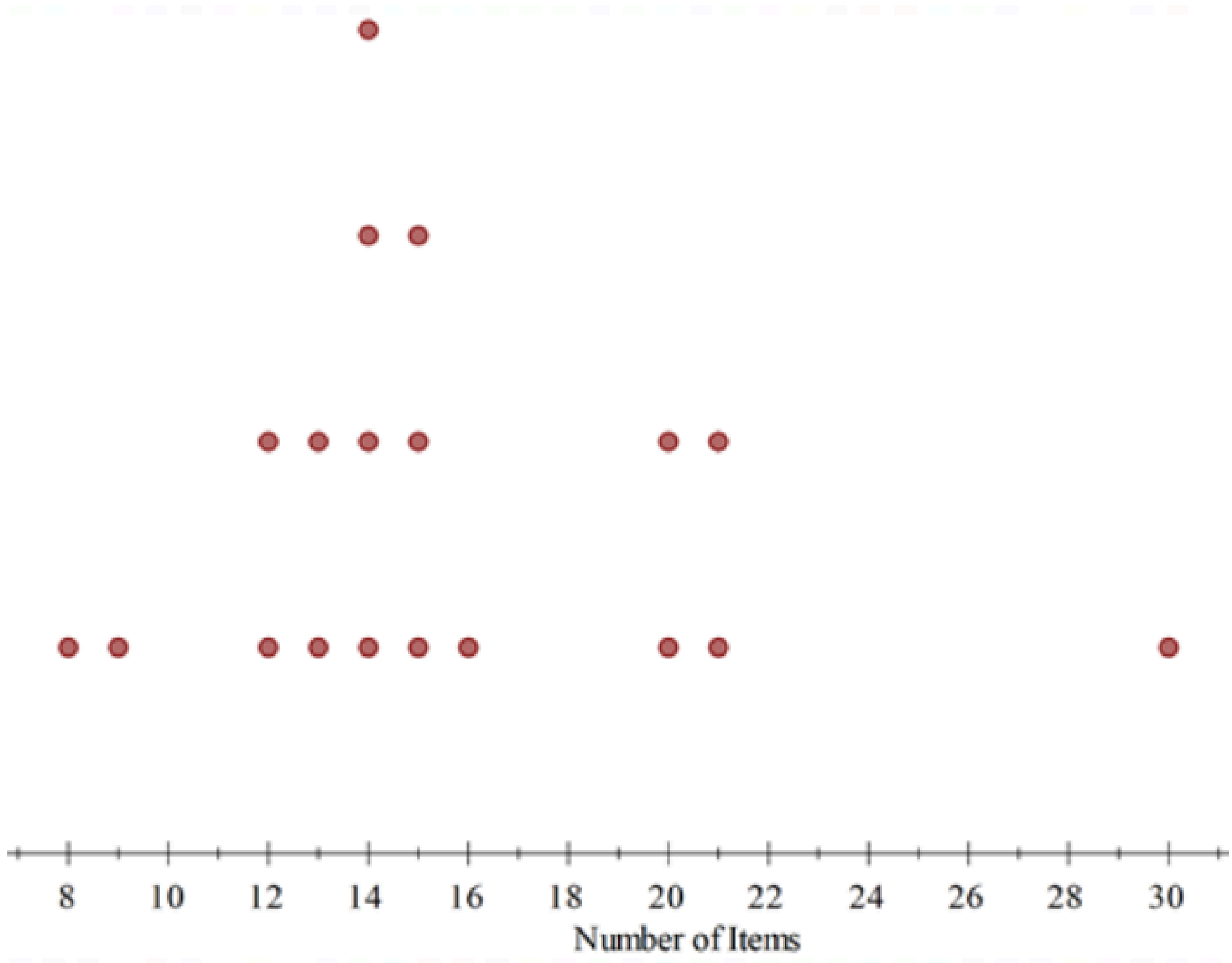
Represent collected data in a stem-and-leaf plot, describe the shape and spread including outliers, and compare to dot plots or column graphs. Use the data to estimate probabilities of specific outcomes

For example:

- collecting data, such as the number of items of stationery in pencil cases, with one-half of the class representing results on a stem-and-leaf plot and the other on a dot plot, manually or by using digital tools.
  Stem-and-leaf plot displaying number of items of stationery in pencil cases

0	8 9
1	2 2 3 3 4 4 4 4 5 5 5 6
2	0 0 1 1
3	0

  Dot plot displaying number of items of stationery in pencil cases



Each half of the class to comment on the shape of the data from their distribution. Identifying clustering of most items between 12–16, non-symmetrical, a spread in items of 22 (30–8), gaps between 9–12 and 16–20 and an outlier of 30, recognising that 30 items could be the result of an error or that one person has a lot of stationery items. Comparing displays and recognising that a stem-and-leaf plot provides detailed information about individual data points and the dot plot (or column/bar graph) provides a clear picture of the mode, shape, spread and outliers of the distribution. Determining the probability that a person chosen at random from the class would be someone from the 'cluster', that is, has 12–16 items of stationery in their pencil case

General Capabilities: Numeracy, Digital literacy

WA7MPSP6

Critically analyse statistical statements made in the media and other real-life situations, that relate to the averages of mean, mode and median. Investigate the impact of chance variation on the dataset from which the averages were determined

For example:

- critically commenting on the terminology and meaning of the averages quoted and the impact of chance variation in data, on statements made in the media, such as
  - the average Australian family has 2.1 children
  - the Fever netball team scored 42 goals in a game on three occasions in the season. This number of goals occurred more than any other score that season

- half of Australia’s population earn less than $75 000 per year and the average wage is $98 000 per year
- a popular confectionery company stating that the typical price of a 180 g chocolate bar is $4.25

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding, Digital literacy

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Modelling with probability and statistics

WA7MPSM1

In real-world situations involving assigning a probability to single-stage chance experiments or simulations, statistical measures, stem-and-leaf plots, dot plots, column graphs and/or Venn diagrams

I. analyse the situation, pose questions as required and determine assumptions and constraints
II. determine appropriate production of a valid and reliable dataset, statistical measures, data representations and analyses, including examination of distributions, to effectively investigate the situation
III. interpret, draw inferences and communicate findings in terms of the context, assumptions, constraints, chance variation and knowledge or insights gained

For example:

- 'What are the chances that the next song on a shuffled playlist is 3.5 minutes long?' Analysing the situation by considering the number of songs on a playlist, where to find the length of a song on the playlist and the time units used to show the length of a song. Posing questions, such as 'How long is a song?' Determining experimental constraints, such as choosing a sample of class members' playlists to represent the group and only using the first 20 songs played. Discussing how data from the sample could be collated and recorded to ensure validity and reliability, including rounding. Representing the data as a stem-and-leaf plot, determining the relative frequencies for lengths (e.g. fraction of songs that were 3.2 minutes long) and the mean, mode, median and range. Analysing the shape and distribution of the data, commenting on variation. Interpreting the results and drawing inferences to predict the probability of the next song being 3.5 minutes long. Communicating findings in terms of the playlist, qualifying that variation would exist due to reasons, such as the order songs were shuffled, the chosen class members and the type of music they prefer, and commenting on whether and why the dataset would be reflective of a broader population of songs

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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Year 8

Year level descriptions

In the early adolescence phase of schooling, students align with their peer group and begin to question established conventions, practices and values. Learning and teaching programs assist students to develop a broader and more comprehensive understanding of the contexts of their lives and the world in which they live.

Mathematics provides opportunities for students to engage in a range of approaches to learning through the proficiencies of understanding, fluency, problem-solving and reasoning. These reinforce the significance of working mathematically with the content and describe how the content is explored or developed. Students draw on the behaviours of the proficiencies when selecting and using year level content to apply the complete modelling process, leading to an increased understanding of the complexity of the natural environment, society and technology.

In Year 8, students explore and investigate to understand, calculate flexibly and efficiently, and model with a broadening range of numbers that includes rational and irrational numbers. As they develop efficiency in multiplicative thinking, they apply and interpret proportional reasoning in fractions, percentages, ratios and rates. Students' abstract thinking develops further as they apply their understanding of the laws and properties of operations with number to algebra.

Students use the language of Mathematics in both geometric and measurement situations, applying reasoning to establish congruency of figures, interpret representations of international time zones, explore circles and use Pythagoras' theorem. They apply their understanding of the properties of quadrilaterals to find the perimeter and area, and generalise volume of right prisms into formulas for efficient calculation.

Students extend their understanding of probability to recognise complementary events and construct sample spaces for two events. They connect probability and statistics by engaging in chance experiments and simulations for simple and compound events. Students critically analyse graphs and tables, and extract data from these representations to determine summary statistics. They investigate and explain techniques for data collection and compare variation between samples of data.

Achievement standard

By the end of the year:

Students demonstrate the behaviours of the proficiencies of understanding, fluency, problem-solving and reasoning in conjunction with year level content in routine situations. As part of this, they select from and use year level content along with the modelling process, in which they analyse the situation mathematically, represent the problem and interpret and communicate their findings, to solve straightforward, real-world problems in familiar contexts across all strands.

Students identify and order rational and irrational numbers and terminating or recurring decimals. They demonstrate flexible and efficient strategies to carry out calculations involving the four operations with integers, positive rational numbers and percentage increase and decrease. They use proportional reasoning to find unknown numbers in equivalent ratios and fractions and use familiar rates. Students apply the index laws to numbers with positive-integer indices. They simplify algebraic expressions involving the four operations and make connections between expanding and factorising algebraic expressions with numerical factors. They solve linear equations involving two operations and graph linear relationships on the Cartesian plane from a table of values. Students identify advantages and disadvantages and determine penalties of various forms of payment.

Students classify quadrilaterals according to their properties, find unknown angles and calculate perimeter and area of quadrilaterals. They use Pythagoras' theorem to find an unknown length in a right-angled triangle. Students apply formulas to determine the circumference and area of circles and the volume of right prisms. They identify and name corresponding sides and angles of congruent figures. They determine the duration of events, involving 12- and 24-hour time systems, across multiple time zones.

Students construct sample spaces, assign probabilities and conduct chance experiments and simulations for two simple or compound events and identify variation in estimated probabilities. They determine the probability of complementary events. Students determine the mean, mode, median and range from graphs and tables, and describe the effect of an outlier on these statistical measures. They use secondary data in Venn diagrams and two-way tables to determine related probabilities. Students explain issues related to the collection of data and compare variation in same-sized random samples. They make critical comments on graphical representations and tables in the media.

Content descriptions

Number and algebra

Understanding number

WA8MNAUN1

Investigate, define, identify and use correct notation for rational and irrational numbers, including terminating, recurring and rounded decimals

> **For example:**
>
> - investigating to explain that 3, 37% and 0.6666… ($0.\dot{6}$ or 0.67 rounded to 2dp) are rational numbers as they can be written as the fractions $\frac{3}{1}$, $\frac{37}{100}$ and $\frac{2}{3}$
> - investigating to explain that $\sqrt{2}$ and π are irrational numbers as they cannot be written as fractions with an integer numerator and denominator and that $\frac{22}{7}$ and 3.14 are approximations (≈) for π
> - investigating terminating, recurring and rounded decimals resulting from dividing 1 by 1, 2, 3, … to 15, comparing to dividing 10 by 1, 2, 3, … to 15, using correct notation

General Capabilities: Numeracy

WA8MNAUN2

Draw and label, or use a given number line, to locate, order and compare with equality and inequality symbols, rational and irrational numbers, including numbers written in index form and percentages

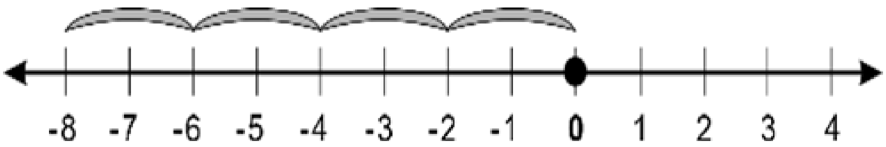
> **For example:**
>
> - locating $-1\frac{3}{4}$, 3, − 0.5, $\sqrt{2}$, 25%, 2^1 , 140% and $\frac{1}{3}$ on a number line. Writing the numbers in ascending order. Comparing numbers using <, >, ≤, ≥ or = e.g. $\frac{1}{3} < \sqrt{2}$
> - justifying why the following statement is true
> $-4\frac{1}{20} > -4.1$

General Capabilities: Numeracy

WA8MNAUN3

Explore to extend multiplicative thinking with positive integers to include multiplication and division of negative integers

> **For example:**
>
> - exploring using patterning
> $$\begin{array}{ll} 3 \times 2 = 6 & -3 \times 2 = -6 \\ 3 \times 1 = 3 & -3 \times 1 = -3 \\ 3 \times 0 = 0 & -3 \times 0 = 0 \\ 3 \times (-1) = -3 & -3 \times (-1) = 3 \\ 3 \times (-2) = -6 & -3 \times (-2) = 6 \end{array}$$
> - representing 4×-2 as repeated addition on a number line



- considering 4×-3 as

  $(-3)+(-3)+(-3)+(-3)$ and -4×-3 as $-[(-3)+(-3)+(-3)+(-3)]$

- extending and exploring multiplication and division as inverse operations of each other, by constructing equivalent equations that include negative integers

  $4 \times (-7) = -28$

  $-28 \div 4 = -7$

  $-28 \div -7 = 4$

- investigating

  $(-5)^2$ and -5^2

  $(-5)^3$ and -5^3

  and justifying that there are two numbers that can be squared to give a result of 25 and that $\sqrt[3]{-125} = -5$

General Capabilities: Numeracy, Critical and creative thinking

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WA8MNAUN4

Extend the use of associative, commutative and distributive laws, additive and multiplicative partitioning, inverse operations, order of operations, equality and inequality to validate a range of mental and written strategies involving the four operations on any rational number

For example:

- selecting, using and communicating partitioning, commutativity, and order of operations to evaluate

  -74×4

  $= 4 \times (-70) + 4 \times (-4)$

  $-63 \div 150$

  $= \frac{-63}{150}$

  $= \frac{3 \times (-21)}{3 \times 50}$

  $= \frac{-21}{50}$

  $-\frac{1}{2} + \frac{3}{5} \times \frac{1}{2}$

  $= -\frac{1}{2} + \frac{3}{10}$

$= \frac{3}{10} - \frac{1}{2}$

- flexibly using associativity and inverse operations to create equivalent equations and using this to solve for unknown fractions or decimals in equations with one operation, such as

  $4\frac{1}{8} - \ ? \ = -2\frac{1}{4}$

  $4\frac{1}{8} - (-2\frac{1}{4}) = \ ?$

  $4\frac{1}{8} + 2\frac{1}{4} = \ ?$

  $4 + 2 + \frac{1}{8} + \frac{1}{4} = \ ?$

  $? \times 3.5 = -16.1$

  $? = -16.1 \div 3.5$

- without calculating, justifying statements as true (T) or false (F)

  $9.6 + 8 \div 4 > 9.6 \div 8 + 4$

  $9 \div (0.4 + 3) = (9 \div 0.4) + (9 \div 3)$

General Capabilities: Numeracy, Critical and creative thinking

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WA8MNAUN5

Explore and apply proportional reasoning to determine unknown numbers in equivalent ratios and fractions

For example:

- giving reasons why n must be 2 if $\frac{n}{5} = \frac{6}{15}$
- moving flexibly between ratios and fractions in order to calculate an unknown, such as $7:4 = a:6$

  $\frac{7}{4} = \frac{a}{6}$

  $4a = 42$

- dividing a quantity in a given ratio, such as sharing $570 in the ratio 2:1

General Capabilities: Numeracy, Critical and creative thinking

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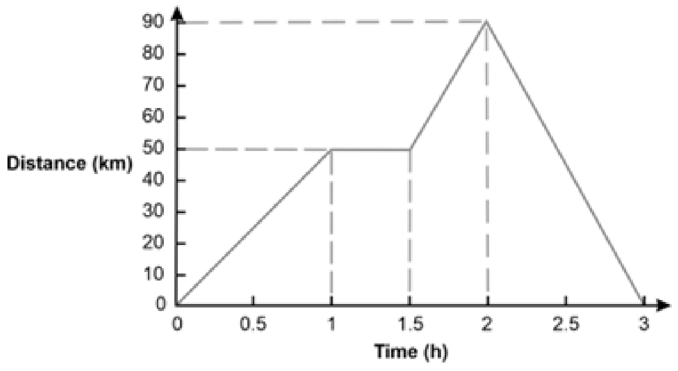
WA8MNAUN6

Identify, interpret, compare and use familiar rates, including those represented as graphs that show a quantity varying over time

For example:

- identifying 30 km/h as a rate, interpreting 30 km as the distance travelled in every one hour, reasoning that this is equivalent to 60 km travelled in two hours, 15 km in 30 minutes or 0.5 km in one minute

- determining that purchasing cheese at $3.75/100 grams is a better buy than purchasing the same product at $42/kilogram
- given the following distance-time graph of a car travelling to visit a friend,



  explaining the significance of the horizontal line segments and calculating the average speed of the trip in the first two hours

General Capabilities: Numeracy, Critical and creative thinking

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Calculating with number

WA8MNAC1

Calculate percentage increases and decreases, using knowledge of fractions and decimals to improve efficiency

For example:

- mentally, with jottings if needed, increasing 360 by 10% using thinking, such as $360 + 10\%$ of 360

  $= 360 + \frac{1}{10} \times 360$

  or by using strategies, such as 110% of 360 = 1.1 of 360

- mentally, with jottings if needed, decreasing 360 by 25% using $360 - 25\%$ of 360

  $= 360 - \frac{1}{4} \times 360$

  or by using strategies, such as

  75% of 360

  $= \frac{3}{4}$ of 360

- using a calculator to increase a population of 5560 by 107% using

  5560 + 1.07 of 5560

  or

  2.07 of 5560

WA8MNAC2

Multiply and divide integers using flexible and efficient strategies

> **For example:**
>
> - mentally calculating, with jottings if needed, $\frac{-35}{5}$ by using thinking, such as $5 \times \boxed{} = -35$ or calculating $(2)(-6)(-45)$ using thinking, such as $2 \times (-45) \times (-6)$ which is the same as $-90 \times (-6)$ or 540
> - communicating thinking to the solution of 24 × (−13) using a sequence of equations, such as
>
> $= 20 \times (-13) + 4 \times (-13)$
>
> $= -260 + (-52)$
>
> $= -260 - 52$
>
> $= -260 - 50 - 2$
>
> $= -312$
> - using a calculator to evaluate
>
> $-817 \div (-35)$ to two decimal places (2dp)

General Capabilities: Numeracy, Digital literacy

WA8MNAC3

Use flexible and efficient strategies for calculations involving the four operations with rational numbers, including those written in index form, using rounding, estimation or the context to check reasonableness of results

> **For example:**
>
> - mentally, with jottings if needed, calculating
>
> $5 - 6^2 \div (-4)$
>
> $(2^4 \times 10^2 \div 4^2)^2$
>
> $(-8^2) \div (2^3)$
> - communicating thinking using a sequence of equations for calculations, such as
>
> $-5.1 + 8.3 - 0.6$
>
> $1\frac{1}{2} + 5\frac{3}{5}$
>
>   and using rounding and properties of numbers to check reasonableness of results

- using a calculator to efficiently determine a solution to the hypotenuse of a right-angled triangle (e.g. $\sqrt{3.2^2 + 7.85^2}$ cm) checking that the solution is reasonable, using an estimation of $\sqrt{3^2 + 8^2}$ and knowing that the hypotenuse is the longest side

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

Algebraic techniques

WA8MNAA1

Develop and apply the index laws for numbers in index form with positive-integer and zero indices

For example:

- exploring expressions, such as

  $3^2 \times 3^4$

  $= (3 \times 3) \times (3 \times 3 \times 3 \times 3) = 3^6$

  or $3^{(2 + 4)}$

  and

  $3^5 \div 3^2$

  $= \frac{3 \times 3 \times 3 \times 3 \times 3}{3 \times 3}$

  $= 3^3$

  or $3^{(5 - 2)}$

  and

  $(3^3)^2$

  $= 3^3 \times 3^3$

  $= 3 \times 3 \times 3 \times 3 \times 3 \times 3$

  $= 3^6$

  or $3^{(3 \times 2)}$

  and

  $(2 \times 3)^2$

  $= (2 \times 3) \times (2 \times 3)$

  $= 2^2 \times 3^2$

  and

$\left(\frac{2}{3}\right)^2$

$= \left(\frac{2}{3}\right) \times \left(\frac{2}{3}\right)$

$= \frac{2^2}{3^2}$

and using words to generalise each of these rules

- exploring the meaning of the zero-index using examples, such as

$\frac{5^2}{5^2} = \frac{25}{25} = 1$

and

$\frac{5^2}{5^2} = 5^{2-2}$

$= 5^0$

$\therefore \quad 5^0 = 1$

and by patterning in

$2^4, 2^3, 2^2, 2^1, 2^0$

- applying the index laws to simplify and evaluate, mentally or with a calculator, depending on the complexity of the numbers, communicating thinking using a sequence of equations for expressions, such as

$(2^2)^3 \times 2 \times 2^3$

$((3^8 \div 3^6) \times 3^2) \div 3^0$

$\left(\frac{4}{3}\right)^2$

General Capabilities: Numeracy, Critical and creative thinking

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WA8MNAA2

Extend and apply knowledge of additive and multiplicative partitioning, order of operations and the associative and commutative laws of numbers, to create or simplify algebraic expressions involving the four operations

For example:

- simplifying the following algebraic expressions

$-m + 2 - 5m - 8 + 2m$

$3p \times 2e \times (-5)$

$15x \div 3x$

$\frac{3p}{6}$

- considering the accuracy of the following equations and explaining thinking

$a \div 6b = a \div 2 \div 3b$

$-12h \times 10a = 3h \times (-8 \times 5a)$

$3a \times (-5b) \times 2g = 5a \times (-2b) \times 3g$

$m \div (-3) = 5m \div (-15)$

$20 \div (a + b) = (20 \div a) + (20 \div b)$

- creating and comparing algebraic expressions to represent the total number of chocolates for a group of four people, each holding a bag that has the same unknown number of chocolates, if each person is given an extra three chocolates

General Capabilities: Numeracy, Critical and creative thinking

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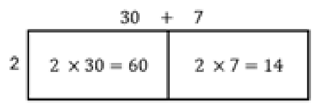
WA8MNAA3

Extend and apply knowledge of the distributive law with numbers to algebraically expand and factorise expressions with a common numerical factor

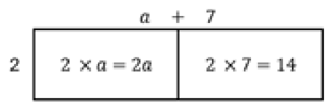
For example:

- connecting the visual representation of 2 × 37 using an area model



$= 60 + 14$

to $2(a + 7)$



$= 2a + 14$,

factorising using reverse thinking

$2a + 14$

$= 2 \times a + 2 \times 7$

$= 2(a + 7)$

and verifying the factorisation using the distributive law

- expanding and simplifying expressions, such as

$5(2g - h + 6)$

$-4(-3p + (-2))$

$-7b + 5(b - 3)$

communicating thinking using a sequence of equations

- factorising expressions, such as

  $-4 - 12k$

  $2m - 6m^3 + 18 - 4de$

General Capabilities: Numeracy, Critical and creative thinking

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Linear and non-linear equations and inequalities

WA8MNALE1

Solve linear equations involving up to three operations, including those with negative coefficients or requiring collection of like terms, and verify the solution by substitution

For example:

- using a strategy, such as the balance model, backtracking or inspection, to solve an equation, showing a sequence of steps to communicate thinking. Checking the solution by substitution using jottings or calculator for equations, such as

  $1 - 2p = -10$

  $3 - 4h + 6h = 7$

  $4.3 = \frac{5y - 1.2}{2}$

  $\frac{8p}{3} = \frac{3}{5}$

- communicating thinking to solve

  $3p - 5q = 5$ when $p = -4$

General Capabilities: Numeracy

WA8MNALE2

Determine and explain why there are two solutions to a quadratic equation of the form $x^2 = k$ if $k > 0$

For example:

- reasoning that for $x^2 = 9$ there are two values of x that satisfy this quadratic equation given that 3^2 and $(-3)^2$ both equal 9
- solving $b^2 + 7 = 71$ and verifying solutions
- using knowledge of quadratic equations in situations involving Pythagoras' theorem, recognising that the solution must be positive because of the context

General Capabilities: Numeracy, Critical and creative thinking

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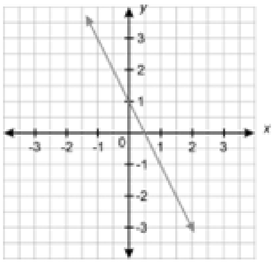
Linear and non-linear patterns and relationships

WA8MNALP1

Use a table of values to move flexibly between the equation of a line represented by $y = mx + c$ and its graph and make connections between the algebraic and graphical solution of the equation. Explore and explain similarities and differences between multiple lines on the same axes

> **For example:**
>
> - using an equation, such as $y = 5x - 1$, generate a table of values, plot the coordinates on the Cartesian plane and create a line, recognising that there is an infinite number of ordered pairs in any given linear relationship. Connecting the graph of $y = 5x - 1$ to solving equations, such as $9 = 5x - 1$ and $1.5 = 5x - 1$, graphically and algebraically
> - extracting a table of values from a linear graph and using the table to determine the equation of the line
>
>   For the following graph
>
>   
>
>
>   a corresponding table of values can be produced
>
> | | | | | |
> |---|---|---|---|---|
> | x | -1 | 0 | 1 | 2 |
> | y | 3 | 1 | -1 | -3 |
>
>   The equation of the linear relationship can be determined as $y = -2x + 1$ by recognising the values of m and c as the constant difference between values of y (i.e. -2) and the value of y when $x = 0$ respectively
>
> - investigating and describing similarities and differences between multiple lines, using terms such as steepness, direction, parallel, non-parallel, pass through the same point, increasing and decreasing

General Capabilities: Numeracy, Critical and creative thinking

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Financial mathematics

WA8MNAF1

Identify the advantages and disadvantages of various forms of payment for goods and services, and determine penalties, such as interest charged and fees, inherent in these payments

For example:

- considering payment forms, such as cash, EFTPOS, debit and credit payments, online payments, gift cards and transfers, including potential savings, costs and risks
- identifying and calculating surcharges and total amounts paid when using a debit card for purchases (0.75% transaction fee on top of the $25 purchase)
- determining penalties for not adhering to the terms of credit, such as ‘buy now, pay later’ schemes, in situations such as ‘Bill made an $80 purchase, paying four $20 instalments. He was late to pay each monthly instalment which incurred a $10 late fee each month. What percentage of the purchase price did Bill spend on late fees?’

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Modelling with number and algebra

WA8MNAM1

In real-world situations involving rational and irrational numbers, ratios, rates, percentage increases and decreases, numbers in index form, the distributive law, factorisation, linear equations with up to three operations, linear or simple quadratic relationships and/or penalties involved in different forms of goods and services payment

I. analyse the situation, decide if an exact or approximate solution is required and determine assumptions and constraints
II. represent the situation mathematically in order to reach a solution
III. interpret and communicate findings in terms of the context and any assumptions or constraints

For example:

- using a supermarket catalogue to create problems, such as
  ‘Which is the better deal, and by how much, when buying two orange juices costing $4.15 each – a discount of 30%, buying one and getting one for half price or three for $10?’
  While it is clear an exact solution is required, it is necessary to discuss assumptions, such as only wanting to buy two juices, even if buying three could be cheaper per juice, and rounding all calculations to the nearest 5c. Such assumptions would impact the representation, solution and communication of the solution
- ‘A childcare centre has 26 children, but some children need extra supervision. Using a ratio of one adult per three children for most children and one adult per two for the rest, determine how many adults should be caring for the children?’
  Discussing that an exact solution is required and that it needs to be assumed that the number of adults would need to be rounded up to ensure compliance. Representing the situation with equivalent ratios and communicating the solution using whole numbers and the context of the child care
- using measurement knowledge to solve problems, such as ‘The equal sides of a tile in the shape of an isosceles triangle are 2 cm longer than the shorter side. If the perimeter of the tile is 22.5 cm, estimate the number of tiles that will fit along a 1 m line.’ Discussing assumptions, if any, choosing to represent the situation as an algebraic equation, communicating the solution in terms of the triangle and assumptions and constraints made

Measurement and geometry

Two-dimensional space and structures

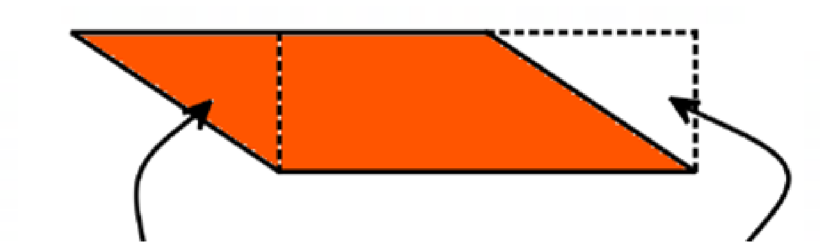
WA8MMGTW1

Establish and apply relationships between lengths of sides, perpendicular lengths, lengths of diagonals, perimeter and area for parallelograms, trapeziums, rhombuses and kites. Generalise and apply formulas, using appropriate units

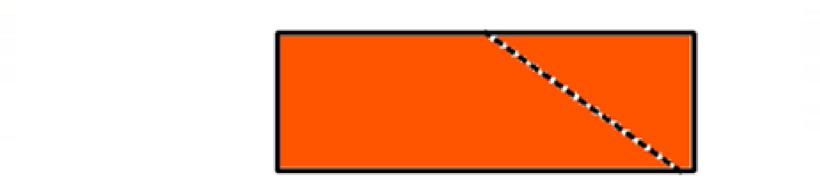
For example:

- using properties of quadrilaterals to establish relationships between lengths of sides and perimeters, generalising as formulas in symbols, such as the perimeter of a rhombus is $P = 4l$ where l is the number of length units in a side and P is the number of length units in the perimeter
- using scissors or dynamic geometry software to strategically cut parallelograms, trapeziums, rhombuses and kites, translating, reflecting or rotating the cut piece/s to form a rectangle and derive formulas for area, including
  - generalising that the area of a parallelogram is equal to the length of one of the parallel sides multiplied by perpendicular distance between the parallel sides, leading to a formula, such as

    $\text{Area parallelogram} = b \times h$

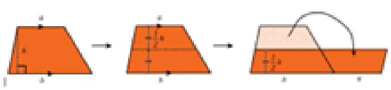
    

    cutting the triangle from one end and moving it to fit on the other end

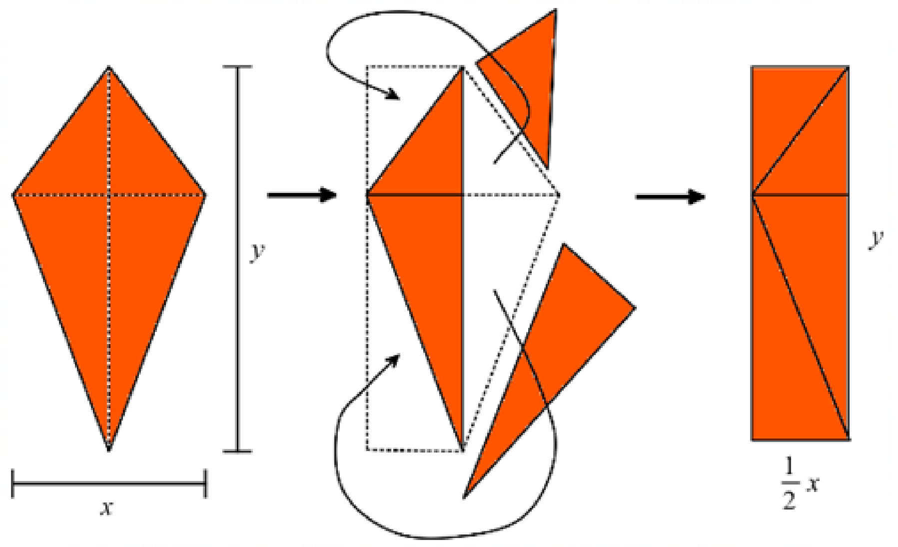
  - generalising that the area in a trapezium is the same as half the length of the perpendicular height multiplied by the total length of the parallel sides, leading to a formula, such as

    $\text{Area trapezium} = \frac{1}{2}h \times (a + b)$

  - generalising that the area of a kite is the same as half the product of the diagonals, leading to a formula, such as

    $\text{Area kite} = \frac{1}{2} \times x \times y$



and discussing what is the same before and after each quadrilateral is cut and transformed

- drawing different parallelograms, trapeziums, rhombuses or kites, each with an area of $15\ \text{cm}^2$
- investigating different ways of finding the area of a hexagon using the formulas of a rectangle, parallelogram, trapezium or triangle

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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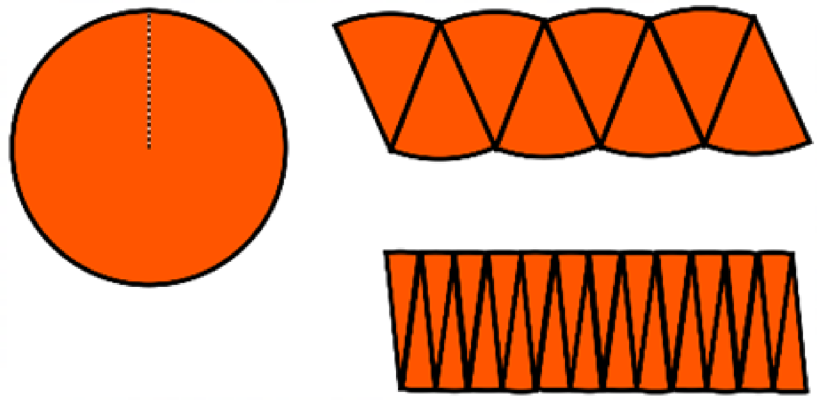
WA8MMGTW2

Identify, describe and explore the relationship between the radius, diameter and circumference of a circle and use this to establish and apply formulas to determine perimeter and area, using appropriate units

For example:

- using concrete examples of circles to establish relationships between the lengths of the diameter and the circumference by measuring using string; creating and interpreting a graph of diameter and circumference and entering results into a spreadsheet to show that circumference ÷ diameter is approximately 3.1 (π). Generalising and applying relationships, such as the number of length units in the perimeter or circumference of a circle is approximately 3.1 (π) times greater than the number of length units in the diameter and the length of the radius is approximately $\frac{1}{6}$ of the circumference
- establishing the relationship between the number of length units in the radius and number of square units in the area of a circle by dissecting a cut out circle into many congruent sectors, arranging them to form an approximate parallelogram with a height of r and a base of πr and generalising the area of a parallelogram determined to $\text{Area circle} = \pi r^2$



- determining the circumference of a circle given that its area is $37\ \text{cm}^2$ and the area of a circle given that its circumference is 171 mm

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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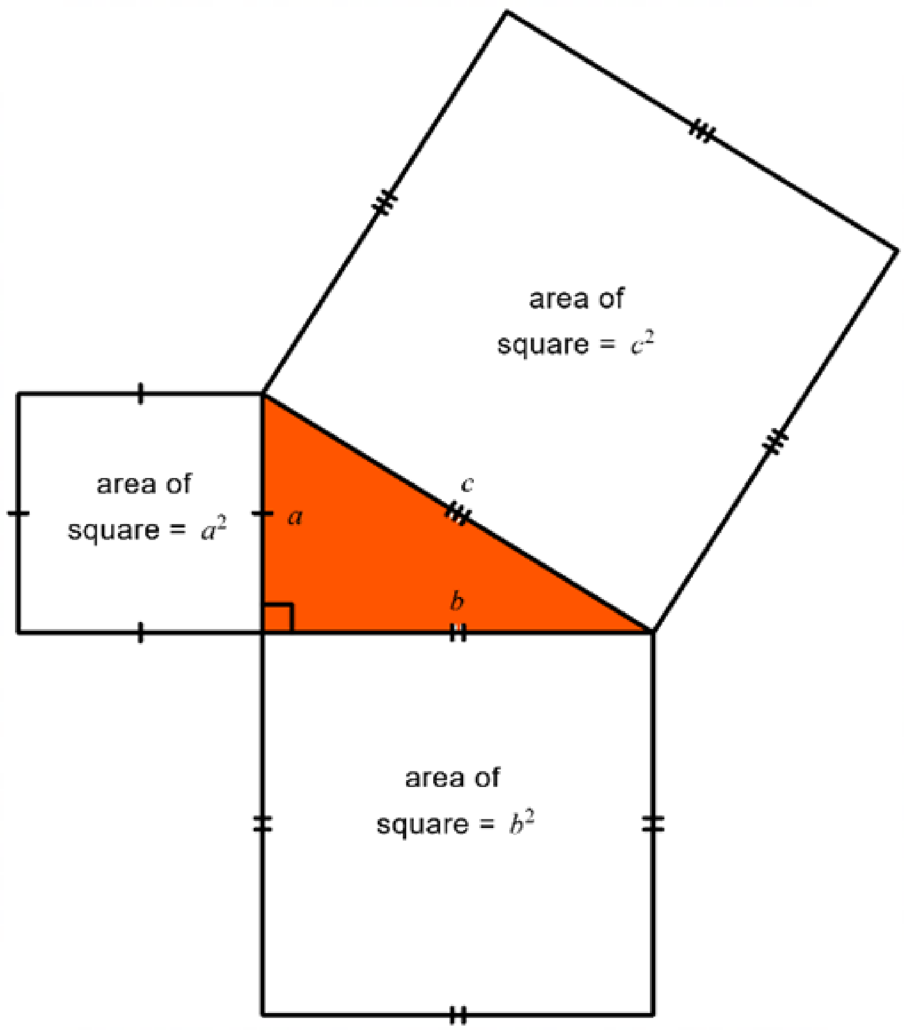
WA8MMGTW3

Investigate in order to establish, define and use Pythagoras' theorem to determine the length of an unknown side in a right-angled triangle

For example:



- using grid paper or dynamic geometry software to draw a right-angled triangle, identifying the shorter sides and the hypotenuse, and exploring the relationship between the area of the squares on the two shorter sides to the area of the square on the hypotenuse. Testing, predicting and generalising the relationship to the formula $a^2 + b^2 = c^2$
- applying Pythagoras' theorem to determine the length of the diagonal of a rectangle with width 7 cm and length 9 cm, in centimetre units, communicating the solution through a sequence of equations and demonstrating that the solution would be positive due to the context

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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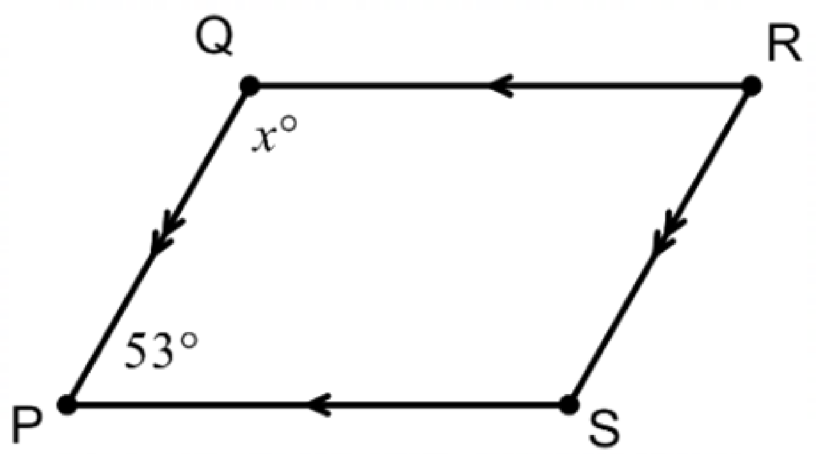
WA8MMGTW4

Explore, identify, classify and establish properties of quadrilaterals, including the interior angle sum. Use this to determine unknown sides and angles in quadrilaterals and explain reasoning

For example:

- creating a classification scheme for quadrilaterals based on sides, angles and diagonals, representing it as a sequence of decisions in a flow chart
- exploring strategies, such as cutting a quadrilateral into two triangles or marking the corners of a quadrilateral, ripping and placing the corners (angles) in a revolution, to generalise to the sum of the interior angles in any quadrilateral being 360°
- determining unknown angles in a quadrilateral, explaining reasons, such as finding the value of x given PQRS is a parallelogram



$\text{QR} \parallel \text{PS}$ (opposite sides of a parallelogram are parallel)

$x + 53° = 180°$ (co-interior angles are supplementary),

$\therefore x = 127$

- extending interior angle sum of triangles and quadrilaterals to include other polygons

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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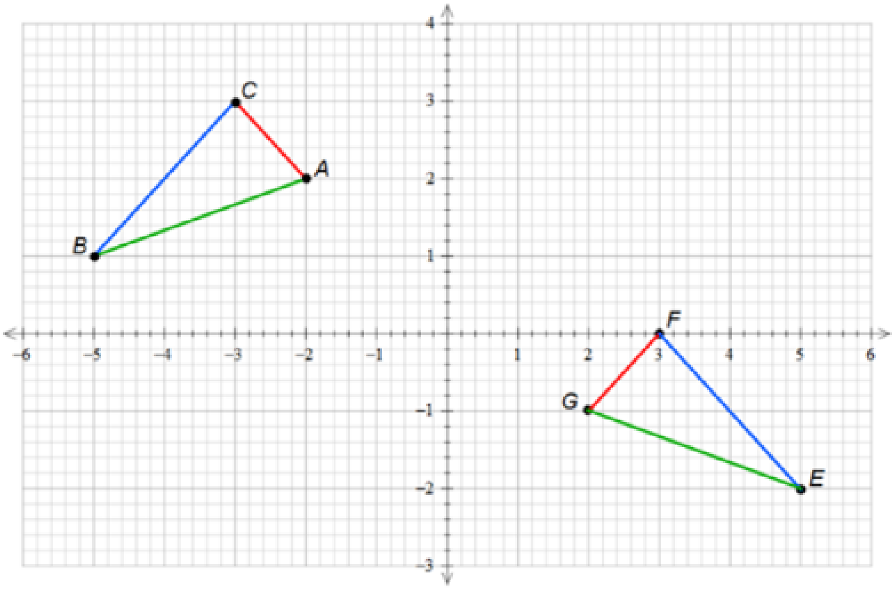
WA8MMGTW5

Recognise and identify equal corresponding sides and equal corresponding angles of congruent figures. Explore, visualise, predict and determine the translation, reflection, rotation, or combination of these transformations, to match one congruent figure to another

For example:



- identifying and expressing in both words and symbols that between the figure and its image, corresponding sides are the same length, corresponding angles are the same size, the triangles are the same shape and size and are therefore congruent (ΔABC ≅ ΔGEF)
- using cardboard triangles, paper folding and tracing, dynamic geometry software and/or reasoning to predict, visualise and investigate different combinations and patterns of transformations that exactly match the figure of ΔABC to its image ΔGEF. Investigating to determine whether a sequence of transformations is commutative or not
- investigating, exploring and creating designs in textile patterns involving the translation, reflection and rotation of congruent figures using dynamic geometry software or A3 paper and shape templates

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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Three-dimensional space and structures

WA8MMGTH1

Explore in order to visualise and draw cross-sections of different solids and use this to identify prisms

For example:

- using concrete materials, such as a block of cheese, round cake, chocolate moulded into a triangular prism, ice-cream cone, chocolate ball or playdough square-based pyramid, to visualise, predict, draw then cut different cross-sections, identifying that any cross-section taken parallel to the base in a right prism is always congruent
- exploring how many different polygons can be formed from the cut faces of a cube

General Capabilities: Numeracy, Critical and creative thinking

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WA8MMGTH2

Establish and apply relationships between the area of a uniform cross-section, the length perpendicular to that uniform cross-section and the volume of right prisms. Generalise, apply formulas and use this to connect to capacity if required, using appropriate units

> **For example:**
>
> - recognising that right prisms have uniform cross-sections that match the base of the prism and that the volume is determined by calculating the area of the cross-sectional base, multiplying it by the number of units of height, in relation to that base, generalising to $V = \text{base area} \times \text{height}$ and using the formula to determine the volume in cubic units, converting to capacity units if required
> - using knowledge of volume of prism formulas and capacity conversions to find
>   - the capacity of a 16 cm tall trapezoidal prism that has trapezoidal sides with parallel lengths of 17.2 cm and 12.3 cm, and a perpendicular height of 11.5 cm
>   - the side length of a cube with capacity 32 L (using $\sqrt[3]{32}$)
>   - possible dimensions of a triangular prism that holds $40\ \text{cm}^3$

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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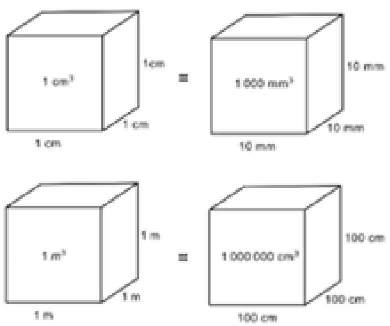
WA8MMGTH3

Explore and establish connections and conversions between units of volume and between units of volume and capacity

> **For example:**
>
> - using models and diagrams of cubes or rectangular prisms to represent and explain connections between numbers of different units of equal volume, such as
>
>   
>
>
>   recognising that 1 cm is 10 times as large as 1 mm but 1 cubic cm is 1000 times as large as 1 cubic mm
>
> - applying knowledge of place value and relative sizes of units to explain conversions between units of volume, such as $0.25\ \text{m}^3$ to cm^3
>
> - using materials, such as a milk carton, a medicine syringe or graduated beakers, to show that $1\ \text{cm}^3 = 1\ \text{mL}$, $1000\ \text{cm}^3 = 1\ \text{L}$, and extending thinking to $1\ \text{m}^3 = 1000\ \text{L}$ or 1 kL, $1000\ \text{m}^3 = 1\ \text{megalitre (ML)}$ and $1\,000\,000\ \text{m}^3 = 1\ \text{gigalitre (GL)}$

- choosing an object with a known volume in cubic centimetres, immersing it in water to show the millilitres it displaces, and connecting to conversion between units, such as $95 \ \mathrm{cm}^3$ is the same as 95 mL
- using knowledge of conversions, place value and relative sizes of units to assist in explaining the capacity in millilitres of a $0.73 \ \mathrm{m}^3$ storage container

General Capabilities: Numeracy, Critical and creative thinking

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Non-spatial measurement

WA8MMGN1

Explore and interpret representations of national and international time zones using 12- and 24-hour time, and determine the duration of events across multiple time zones

For example:

- using maps, number lines or digital tools to interpret and explore the language and thinking used in determining international time zones and comparing times in Beijing, China to Suva, Fiji and Toronto, Canada
- recognising the challenges of planning regular three-hour virtual meeting times for a company that has staff in both Vancouver, Canada and Perth, Western Australia, and determining suitable starting and ending times for these meetings, including consideration of Coordinated Universal Time (UTC)

General Capabilities: Literacy, Numeracy

Modelling with measurement and geometry

WA8MMGM1

In real-world situations involving perimeter and area of quadrilaterals and circles, properties of quadrilaterals, transformations of figures, Pythagoras' theorem, congruency, cross-sections, volume or capacity of prisms and/or international time zones

I. analyse the situation, decide if an exact or approximate solution is required and determine assumptions and constraints
II. represent the situation mathematically in order to reach a solution
III. interpret and communicate findings in terms of the context and any assumptions or constraints

For example:

- 'Barri has designed a disposable sugar sachet, the shape of a triangular prism. The length of the prism is 52 mm, and the end face has a base length of 0.9 cm and perpendicular height of 8 mm. Barri is asked to redesign the sachet to have a volume of $2 \ \mathrm{cm}^3$ by altering just one dimension. Determine the new dimensions of the sugar sachet.' Assuming that an exact solution would be required given the degree of accuracy of the measurements and the size of the object. Choosing to represent the situation as an oblique sketch and with the formula for volume of a triangular prism. Showing calculations using the same units and solving equations to determine new dimensions. Communicating the solution with the chosen units and explaining that the solution could differ, depending on the dimension chosen to alter

Probability and statistics

Probability and statistics

WA8MPSP1

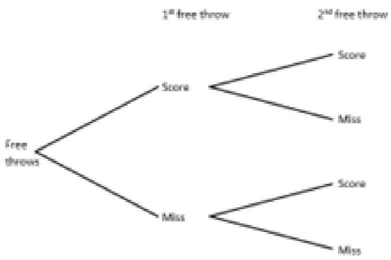
Construct sample spaces, such as lists, simple tree diagrams, tables or arrays, to show all possible outcomes for two events. Assign probabilities to outcomes and events including those involving ‘and’, ‘not’, ‘at least’, exclusive ‘or’ and inclusive ‘or’

For example:

- constructing sample spaces to show all outcomes for two free throws at a basketball game, where a score or miss is equally likely, such as
  S = {SS, SM, MS, MM}
  or

		Second free throw	
		Score	Miss
First free throw	Score	S, S	S, M
	Miss	M, S	M, M

  or



  and using to assign probabilities to events, such as
  P(a miss first and a score second)
  P(a miss first or a score second)
  P(at least one score)
  P(not having any misses)
- creating a sample space to show all outcomes for rolling two standard dice

Dice 1 \ Dice 2	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

  and using to assign probabilities to events, such as
  P(rolling a 1 first)
  P(rolling at least one 5)
  P(not rolling a double)

> P(rolling a 4 first and a 4 second)
> P(sum of 6)

General Capabilities: Numeracy, Critical and creative thinking

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WA8MPSP2

Recognise that complementary events have a combined probability of one and use this relationship to calculate probabilities

> **For example:**
>
> - exploring the language of complementary events, such as rolling a 3 on a dice and not rolling a 3 on a dice. Verifying that as the probability of rolling a 3 on a dice is $\frac{1}{6}$ and the probability of not rolling a 3 on a dice is $\frac{5}{6}$, then the sum of the event and its complement is 1
> - using complementary events to assign the probability of not choosing a red 3 from a standard pack of cards by
>
>   P(not choosing a red 3)
>
> $= 1 - \mathrm{P}$ (choosing a red 3)
>
> $= 1 - \frac{2}{52}$
>
> $= \frac{50}{52}$ or $\frac{25}{26}$
>
>   or 96.2% or 0.962
>
>   or a 25 in 26 chance
> - using complementary events to assign a probability to rolling at most a 9 from a 10-sided dice roll by identifying that P(at most a 9) and P(10) are complementary events.
>
>   Using $1 - \frac{1}{10}$ to determine the expected probability

General Capabilities: Numeracy, Critical and creative thinking

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WA8MPSP3

Conduct repeated chance experiments and simulations for two events to produce datasets, including through the use of digital tools, for a large number of trials. Discuss, explain and compare variation and estimated probabilities for simple and compound events

> **For example:**
>
> - conducting a simulation to determine the estimated probability of not getting two tails when tossing two coins. Predicting the probability before using a virtual coin toss or random number generator with 1 representing 'heads' and 2 representing 'tails', to flip two coins for a large number of trials. Comparing results with others, and with predicted and theoretical probability results, discussing and explaining any variation and making connections to complementary events

- conducting an experiment to find the estimated probability of a difference of 6 between the sum of the numbers shown on two white dice and the number shown on a green dice. Completing an actual or virtual dice experiment to produce a large set of data, representing results in a column graph and using to estimate the probability of the difference being 6. Comparing the shape of the graph in terms of clusters, gaps, outliers and symmetry with others and noting variation. Explaining and justifying conclusions made regarding the probability of a difference of 6 between the sum of the numbers shown on two white dice and the number shown on a green dice

General Capabilities: Literacy, Numeracy, Digital literacy

WA8MPSP4

Analyse data represented in stem-and-leaf plots, column graphs and frequency tables to determine the mean, mode/s, median and range. Describe the effect of any outliers on the statistical measures

For example:

- using information recorded in a table, on the number of cups of coffee consumed by people in an office on a given day

Number of cups	0	1	2	3	9
Frequency	12	27	16	5	1

determining the mean, mode, median and range of the data with and without the outlier, manually or with digital tools. Comparing and describing differences in the two sets of statistical measures (with and without outlier), recognising the impact of the outlier on mean and range and understanding why it is often more appropriate to use the mode or median to describe data when an outlier is included in the calculations

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA8MPSP5

Use secondary data represented in two-way tables and Venn diagrams to describe events, including those that are mutually exclusive. Estimate related probabilities and make predictions as appropriate

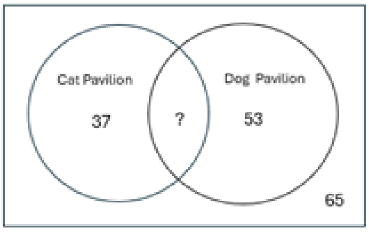
For example:

- using data represented in completed or partially completed two-way tables, such as information concerning jackets ordered by members of a netball club and whether the jacket had a hood or a name printed on the back

		Name printed	No name printed	Total
Hood	Hood	68	112	
	No hood		26	
	Total	82		

Determining and describing the jackets of 14 students as having a name printed but not having a hood, 68 students as having a hood and a name printed and 194 students having a hood, or a name printed. Determining related probabilities, such as a student being chosen at random who does not have a name printed on their jacket. Identifying and describing that the outcomes of having a hood and having a name printed are not mutually exclusive, as they can occur at the same time. Using the data to predict that in a future group of 300 netball club members $\frac{112}{220} \times 300 = 152.72$, or approximately half the members, would order a jacket with a hood but no name, acknowledging that future outcomes will be affected by chance variation, such as fashion trends

- using data represented in a completed or partially completed Venn diagram concerning a sample of 180 people at the Perth Royal Show being asked by the organisers to complete a questionnaire as to whether they had visited the cat pavilion only, the dog pavilion only, both or neither.



Reasoning that the number of people who had visited both pavilions was 25. Determining related probabilities of a person being chosen at random for a follow-up survey, who had visited at least one of the pavilions. Recognising that visiting the cat pavilion and the dog pavilion are not mutually exclusive events, as both events could occur. Understanding that for mutually exclusive events, such as buying a Royal Show entrance ticket online or buying it at the gate, a Venn diagram depicting how visitors purchased tickets would appear as two non-interconnecting circles. Determining that approximately 36% of visitors to the Royal Show do not go to either pavilion and using this to inform possible action for future Royal Shows, acknowledging that future predictions will be affected by chance variation, such as weather

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA8MPSP6

Investigate and explain techniques for data collection, including census, survey, experiment and observation, and explain the practicalities and implications of obtaining data through these techniques

For example:

- defining and identifying data collection techniques and explaining the practicalities and implications of each in situations, such as
  - Australian Bureau of Statistics, five-yearly surveys
  - polling voters
  - testing a new medicine
  - studying animal behaviour

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding, Digital literacy

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WA8MPSP7

Explore, analyse and compare variation between results from same size random samples drawn from the same population. Identify and explain how chance variation impacts on data validity, reliability and conclusions drawn

> **For example:**
>
> - defining sample, random sample and population in the context of producing a dataset of the heights of a sample of five students, randomly chosen from the Year 8 school population using a random number generator. Determining mean, mode, median, range and proportions (e.g. fraction or percentage of students between 150 and 155 cm tall) manually or with digital tools from this sample. Repeating for different samples of five students, comparing and analysing variation between the samples, including the effect of outliers. Repeating the process with samples of 15 students, recognising the effect of the increased sample size on variation. Commenting on the impact of the sample size and the validity and reliability of the datasets as indicators of the height of the total population of Year 8 students

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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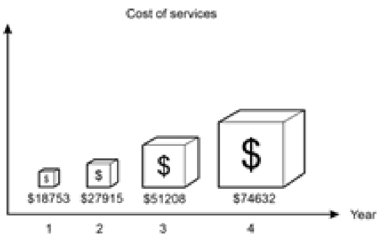
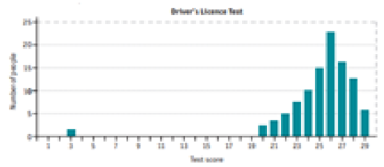
WA8MPSP8

Critically analyse visual representations and tables in the media and other real-life situations to identify misleading or inaccurate features and interpretations. Recognise the impact of the validity and reliability of the data used

> **For example:**
>
> - using the following graphical display showing the cost of services for a day care business, to describe misleading features
>
> 
>
>
> Recognising the title and axes labels do not provide clarity as to the purpose of the graph, the scale is distorted or non-existent, the use of selective data from only four years and no acknowledgement of the source presenting the display, thus meaning the data that the graph is depicting may not be valid and reliable. There is an inappropriate use of size for each of the 'blocks', creating the impression that the fourth year had at least four times the cost of services than the second. The graph is a biased representation and influences the audience into thinking that the cost of services has increased more than it actually has
> - the results of people completing the written learner driver's licence test in a town in WA for one month were represented in a column graph for the State Licensing report. The report stated: 'There was great variation in results; however, the average score was 25.3 and most people scored above the pass mark of 26'



Critically looking at the validity of including the outlier in the comments and calculations, the mean being the chosen 'average' and the accuracy of the calculations

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding

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Modelling with probability and statistics

WA8MPSM1

In real-world situations involving two-stage chance experiments or simulations, complementary events, data collection methods, same-sized random sampling and/or analysis of graphs, tables and data

I. analyse the situation, pose questions as required and determine assumptions and constraints
II. determine appropriate production of a valid and reliable dataset, statistical measures, data representations and analyses, including examination of distributions, to effectively investigate the situation
III. interpret, draw inferences and communicate findings in terms of the context, assumptions, constraints, chance variation and knowledge or insights gained

For example:

- 'Who has the best chance of winning in a game of throwing two 10-sided dice where Player One wins if the sum of the dice is 9, 10, 11 or 12, Player Two wins if the sum is 13, 14, 15 or 16 and Player Three wins for all other sums of dice?' Analysing the situation by considering the sums of dice Player Three could win with. Posing questions, such as 'What are all the possible sums when two 10-sided dice are rolled?' and 'What is the probability of each sum occurring?'. Making assumptions, such as the dice are fair and that the game is based on a single roll of the dice. Deciding to represent the problem by producing an array or by conducting a simulation over a large number of trials. Using representation, data or complementary events to determine the chance of each player winning and hence, the winner. Communicating the results in the context of the game and with a consideration to variation
- 'The health department has noticed an increase in back problems in 13–15-year-old adolescents. The recommended weight of backpacks for this age group is 8 kg or less. Investigate if this backpack weight is typical for adolescents in this age group.' Analysing the situation by considering an appropriate sample size of backpacks to be weighed, the units of weight to be used and what is meant by 'typical'. Posing questions, such as 'How much does a backpack weigh?' From the analysis, determining constraints, such as a sample of backpacks from Year 8 students should be weighed rather than the population of Year 8 students and the backpacks should be weighed to the nearest half kilogram. Making assumptions, such as the measuring scale is accurate and that the weight reflects what students carry on a daily basis. Randomly choosing a student sample using a random number generator, weighing their backpacks and recording the results in a frequency table. Determining the mean, mode, median and range, representing in an appropriate graph and commenting on variation and the effect of outliers. Repeating the task with a different same-sized sample to consider variation. Communicating the findings in terms of the average weights of the sample backpacks compared to the

> recommended weight, and commenting if and why this result would be reflective of the broader population

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding, Digital literacy

Year 9

Year level descriptions

In the middle adolescence phase of schooling, teaching and learning programs encourage students to develop an open and questioning view of themselves as active participants in their society and the world.

Mathematics provides opportunities for students to engage in a range of approaches to learning through the proficiencies of understanding, fluency, problem-solving and reasoning. These reinforce the significance of working mathematically with the content and describe how the content is explored or developed. Students draw on the behaviours of the proficiencies when selecting and using year level content to apply the complete modelling process, leading to an increased understanding of the complexity of the natural environment, society and technology.

In Year 9, students explore and investigate to understand, calculate flexibly and efficiently, and model with real numbers, writing solutions in exact or approximated form. They engage with financial mathematics by calculating simple interest and exploring ways in which people earn money. They work flexibly, both algebraically and graphically with linear equations, developing an understanding of gradient.

Students explain and determine perimeter and area of composite figures. They apply Pythagoras' theorem to solve perimeter and area problems. Through construction, drawing and geometric reasoning, students establish conditions for congruent triangles, explore properties of similar figures and develop the trigonometric ratios. Students extend their use of formula to include volume, capacity and surface area of right prisms and cylinders.

Students connect probability and statistics by collecting data from experiments and simulations related to two-stage chance experiments, both with and without replacement. They analyse comparative graphs in context using statistical language and critically analyse statistical processes and claims made in the media that relate to data sampling.

Note: the optional content in Year 9 is intended to build and extend students' year level knowledge according to areas of interest, understanding of content and preparation for subsequent study. The content descriptions are optional. Teachers may choose from optional content according to the needs of the student/s.

Achievement standard

By the end of the year:

Students demonstrate the behaviours of the proficiencies of understanding, fluency, problem-solving and reasoning in conjunction with year level content in routine situations. As part of this, they select from and use year level content along with the modelling process, in which they analyse the situation mathematically, represent the problem and interpret and communicate their findings, to solve straightforward, real-world problems in familiar contexts across all strands.

Students compare and order real numbers, including those expressed in scientific notation. They demonstrate flexible and efficient strategies to carry out the four operations with real numbers, expressing solutions in exact or approximated form. They express real numbers in scientific notation. Students apply the index laws to variable bases with positive-integer and zero indices, expand and factorise expressions with an algebraic factor and expand binomial products. Students solve linear equations involving brackets. They use the Cartesian plane to find the distance between two points and the gradient and midpoint of a line segment. They graph straight lines using the gradient and y-intercept, and identify and represent direct proportion algebraically and graphically. Students solve simple quadratic equations algebraically and use tables of values to graph the function. They determine interest using the simple interest formula and perform calculations that relate to earning income.

Students determine the perimeter and area of composite shapes, including those involving right-angled triangles. They use triangle and angle properties to show reasoning as to why triangles are congruent and find unknown sides and angles. They use properties of similar figures and scale to determine real-life lengths from scale drawings and apply a trigonometric ratio to find unknown sides and angles in right-angled triangles. Students apply formulas to determine the volume, capacity and surface area of right prisms and cylinders.

Students construct sample spaces, assign probabilities and conduct experiments and simulations for two-stage events, both with and without replacement, and identify variation in estimated probabilities. They describe data represented in tables using statistical measures and relative frequencies to make inferences. They compare back-to-back stem-and-leaf plots and comparative histograms using shape and spread. Students describe different sampling methods and make critical comments on statistics relating to data sampling in the media.

Content descriptions

Number and algebra

Understanding number

WA9MNAUN1

Investigate very large and very small numbers and move flexibly between their exact and approximated scientific notation

> **For example:**
>
> - exploring contexts for very large numbers (population count, national debt, mass of celestial bodies, data storage) and very small numbers (nanoparticles, sizes of cells, microseconds, microtransactions, tolerances in engineering, winning times in Olympic events) and comparing these quantities or measurements to those in more familiar contexts
> - exploring instruments used to measure very large and very small measures, such as the capacity of liquid in a scientific pipette being 5.72×10^{-4} mL, which is 0.000572 mL
> - choosing to write the size of an atom given as 2.45×10^{-7} mm exactly as 0.000000245 mm, and the speed of light given as 299 792 458 ms as approximately 2.998×10^{8} ms for accuracy, efficiency or context

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNAUN2

Investigate, define, compare and order real numbers, with equality and inequality symbols, including those expressed in scientific notation

> **For example:**
>
> - explaining that the real number system includes both rational and irrational numbers
> - using efficient strategies to write the real numbers below in descending order and comparing the numbers using <, >, ≤, ≥ or =
> $\sqrt{\pi}, -1\frac{1}{2}, 3.25 \times 10^{-1}, 2^{0}, \sqrt[3]{-8.1}, 0.15, 1.4^{0}, 5.5 \times 10^{-2}, \sqrt{3}$
> - without calculating, justifying why the following are true:
> $3.25 \times 10^{-1} > 5.5 \times 10^{-2}$
> $\sqrt{\pi} < \pi$
> $\sqrt{2} > \frac{1}{2}$
> $\sqrt[3]{-64} > \sqrt[3]{-125}$

General Capabilities: Numeracy

WA9MNAOpt1

Year 9 optional
Explore to develop a sequence of steps to flexibly move between recurring decimals and fractions

General Capabilities: Numeracy, Critical and creative thinking

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Calculating with number

WA9MNAC1

Use flexible and efficient strategies for calculations involving the four operations with real numbers and express solutions in exact form or as an approximation

For example:

- determining the unknown side of a right-angled triangle, with a side length of 1 cm and a hypotenuse of 2 cm, as being exactly $\sqrt{3}$ cm and approximately as 1.73 cm
- recognising that solutions can be written exactly or as approximations, such as in a semicircle with a radius of 6 units, the area can be written exactly as 18π square units and approximately as 56.55 square units (2dp)
- determining the average speed of light if the distance between the Earth and the Sun is 1.496×10^{11} metres and the time taken for light to travel from the Earth to the Sun is 4.987×10^{2} seconds, discussing differences between rounding decimal numbers before calculating and rounding the solution

General Capabilities: Numeracy

Algebraic techniques

WA9MNAA1

Extend and apply index laws with positive-integer indices and the zero index to variable bases, and simplify where appropriate

For example:

- generalising numerical expressions, such as
  $2^2 \times 2^3 = 2^{2+3} = 2^5$
  to $a^m \times a^n = a^{m+n}$
  and hence simplifying
  $b^3 \times b^7 \times b$ or $3e^2 \times 5e^4$

  $2^5 \div 2^3 = 2^{5-3} = 2^2$
  to $a^m \div a^n = a^{m-n}$

  and hence simplifying
  $\frac{h^8}{h^3}$ or $3ay^5 \div 6ay^3$

  $(2^2)^3 = 2^{2\times 3} = 2^6$
  to $(a^m)^n = a^{mn}$

  and hence simplifying
  $(7e^2)^3$ or $(2cd^2)^4$

  $(\frac{2}{3})^2 = (\frac{2^{1\times 2}}{3^{1\times 2}}) = \frac{2^2}{3^2}$

to $(\frac{a}{b})^n = \frac{a^n}{b^n}$

and hence simplifying
$(\frac{2b}{3e})^3$ or $(\frac{5}{4x^2})^2$

$2^3 \div 2^3 = 2^{3-3} = 2^0 = 1$
to $a^0 = 1$

and hence simplifying
$-5k^0 - 7$

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNAA2

Extend and apply index laws with numerical expressions of base-10 to include negative-integer indices. Develop the relationship between these negative indices and equivalent fractions and decimals

For example:

- applying patterns, index notation and laws to negative integer numerical indices

  Patterns

10^1	10^0	10^{-1}	10^{-2}	10^{-3}
10	1	$\frac{1}{10^1}$ or $\frac{1}{10}$	$\frac{1}{10^2}$ or $\frac{1}{100}$	$\frac{1}{10^3}$ or $\frac{1}{1000}$
10	1	0.1	0.01	0.001

- using index laws
  $10^2 \div 10^6$

  $= \frac{10^2}{10^6}$

  $= \frac{10 \times 10}{10 \times 10 \times 10 \times 10 \times 10 \times 10}$

  $= \frac{1}{10^4}$

  and making connections to
  $10^2 \div 10^6$
  $= 10^{2-6}$
  $= 10^{-4}$
  $\therefore 10^{-4} = \frac{1}{10^4}$ or $1 \div 10\,000$

- explaining when using scientific notation that $3.2 \times 10^{-4} = 3.2 \times \frac{1}{10\,000} = \frac{3.2}{10\,000}$ or 0.00032

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNAA3

Explore and apply the distributive law to expand and factorise algebraic expressions with a common algebraic factor, including collecting like terms where appropriate

For example:

- extending expansion and factorisation using the distributive law with numerical factors, to expansion and factorisation with algebraic factors

  $2b(m + 3c)$
  $= 2b \times m + 2b \times 3c$
  $= 2bm + 6bc$

  and factorising using reverse thinking

  $2bm + 6bc$
  $= 2b \times m + 2b \times 3c$
  $= 2b(m + 3c)$
  verifying factorisation using the distributive law

- expanding and simplifying expressions, such as
  $3b(2b - c + 1)$
  $-2k(- 3k + (- 2))$
  $-2b(b - 3) - 7b + 5$
  $x(x + 2) - 6(x - 3)$
  communicating thinking using a sequence of equations

- factorising expressions, such as
  $9mp - 3m + 6me$
  $-2d^2 - 3d^3$

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNAA4

Explore and apply the distributive law to expand binomial products, collecting like terms where appropriate

For example:

- connecting the visual representation of expansion with numbers, such as
  $24 \times 37 = (20 + 4)(30 + 7)$

$$\begin{array}{c|c|c|} & \multicolumn{2}{c}{30+7} \\ \hline 20 & 20\times 30 & 20\times 7 \\ + & & \\ 4 & 4\times 30 & 4\times 7 \\ \hline \end{array}$$

to

$(a + b)(c + d)$

$= ac + ad + bc + bd$

$$\begin{array}{c|c|c|} & c & + \; d \\ \hline a & ac & ad \\ + & & \\ b & bc & bd \\ \hline \end{array}$$

and using to expand and simplify

$(x + 3)(x + 8)$

$$\begin{array}{c|c|c|} & x & + \; 8 \\ \hline x & x^2 & 8x \\ + & & \\ 3 & 3x & 24 \\ \hline \end{array}$$

$= x^2 + 8x + 3x + 24$

$= x^2 + 11x + 24$

- communicating thinking of expansion and simplification of equations, such as

  $(y - 4)^2$

  $(c + 8)(c - 8)$

  $(2y - 3p)(- y - 7p)$

  $(b + 2)^2 - 5$

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNAOpt2

Year 9 optional

Explore efficient strategies to simplify expressions that involve addition, subtraction, multiplication or division of algebraic fractions with an algebraic term in the numerator and a whole number denominator

General Capabilities: Numeracy, Critical and creative thinking

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Linear and non-linear equations and inequalities

WA9MNALE1

Solve linear equations involving brackets and/or a variable on each side of the equation, and verify the solution by substitution

For example:

- using a strategy, such as the balance model, backtracking or inspection, to solve an equation, showing a sequence of steps to communicate thinking. Checking the solution by substitution, using jottings or calculator for equations, such as
  $3(y-5) = -6y$
  $9 = 7(-4r-1)$
  $3d - 7 = 5d + 8$
  $\frac{2-e}{5} = -4$
  $4(\frac{3}{4}c - 3) = 2c + 2$
- communicating thinking to solve
  $5e + 2f = 7.5 - 3f$ when $e = 0$ or when $f = 0$

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNALE2

Determine and explain why there are up to two solutions to a quadratic equation of the form $ax^2 = k$ and verify the possible solution/s by substitution

For example:

- reasoning and appropriately communicating that for $x^2 = k$ there could be
  - two solutions ($k > 0$)
  - one solution ($k = 0$)
  - no real solutions ($k < 0$)
- considering $2x^2 = 10$, explaining that x can be expressed in exact form as
  $x = \pm\sqrt{5}$ or as a decimal approximation of
  $x \approx \pm 2.24$ (2dp)
  and checking by substitution $2(\sqrt{5})^2$ and $2(-\sqrt{5})^2$ both equal 10

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNAOpt3

Year 9 optional
Solve linear equations that involve simple algebraic fractions with numerical denominators and verify the solution by substitution

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNAOpt4

Year 9 optional
Solve quadratic equations in factorised form using the null factor theorem and verify the solution/s by substitution

General Capabilities: Numeracy, Critical and creative thinking

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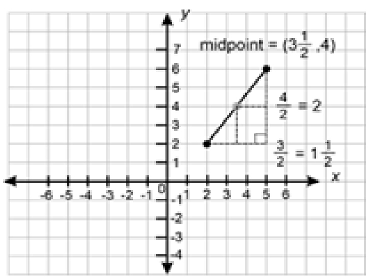
Linear and non-linear patterns and relationships

WA9MNALP1

Use the Cartesian plane to explore finding the distance, gradient and midpoint between two points

> **For example:**
>
> - plotting and joining two points on a Cartesian plane to form a line segment. Investigating, explaining and validating strategies, such as
>   - using vertical and horizontal units between points to form sides of a right-angled triangle and applying Pythagoras' theorem to determine the distance between the two points
>   - counting the vertical difference and horizontal difference between the points and dividing the vertical rise by the horizontal run to determine gradient (m). Connecting gradient to ratio, such as 3 units to the right and 6 units up, is the same as 5 units to the right and 10 units up. Distinguishing between line segments with positive and negative gradients and those with the same gradient. Exploring gradients of horizontal and vertical line segments
>   - using the halfway points of the vertical and horizontal units to determine the midpoint
>
> 
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General Capabilities: Numeracy, Critical and creative thinking

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WA9MNALP2

Move flexibly between the equation of a line, represented by $y = mx + c$, and its graph using the gradient and y -intercept. Graph the equation of a line represented in $ax + by = c$ form

> **For example:**
>
> - interpreting the coefficient of $x(m)$ in $y = mx + c$ form as the gradient/slope/rate of change and the constant (c) as the vertical intercept
> - finding the equation of a line given the gradient (m) and the vertical intercept or by extracting the gradient and vertical intercept from a graph and using the equation to determine whether a given point lies on the line
> - graphing an equation of a line using the gradient and vertical intercept in $y = mx + c$ form, verifying the accuracy of the drawn graph using another point
> - graphing an equation of a line written in $ax + by = c$ form using the vertical and horizontal intercepts (i.e. when $x = 0$ and when $y = 0$), verifying the accuracy of the drawn graph using another point
> - investigating to compare and contrast graphs and equations of horizontal, vertical and parallel lines

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNALP3

Identify rates as direct proportion, represent algebraically and graphically, and use both forms to predict unknown values and interpret in the context of the situation

> **For example:**
>
> - identifying that the relationship between the rate at which water drips from a tap into a bucket in each minute (d) and the amount of water in the bucket (w) in millilitres are directly proportional. Recognising that if there are 600 mL in the bucket after 40 minutes, then this can be represented as $w = 15d$. Representing the equation graphically, recognising that $m = 15$ and that the y-intercept is (0, 0), predicting unknown values and interpreting in context

General Capabilities: Numeracy, Critical and creative thinking

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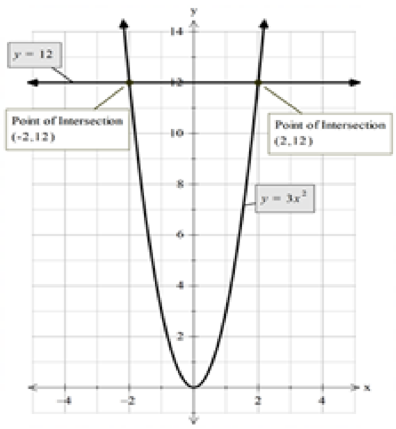
WA9MNALP4

Use a table of values to plot points and graph quadratic functions of the form $y = ax^2$, describe key features and make connections to the algebraic solution/s of $ax^2 = k$

> **For example:**
>
> - graphing $y = 3x^2$ manually or digitally, recognising that it is a parabola with an axis of symmetry at $x = 0$ and a turning point at (0, 0) and using it to solve $3x^2 = 12$, verifying the solution algebraically
> $3x^2 = 12$
> $x^2 = 4$
> $x = \pm 2$
>
>   
>
>
>   and connecting to the graphical solutions of $(-2, 12)$ and $(2, 12)$

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA9MNAOpt5

Year 9 optional
Develop and use the algebraic formulas for finding the distance, midpoint and gradient between two points

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNAOpt6

Year 9 optional

Rearrange formulas, including $ax + by = c$, to change the subject of the formula

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNAOpt7

Year 9 optional

Investigate indirect proportion, represent algebraically and graphically, use both forms to predict unknown values and interpret in the context of the situation

General Capabilities: Numeracy, Critical and creative thinking

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Financial mathematics

WA9MNAF1

Explore, explain and perform calculations that relate to earning income. Identify the elements of an income statement/pay slip, including employer superannuation contributions and income tax as a deduction from gross income

> **For example:**
>
> - explaining that individuals earn an income in various ways, including salaries, wages (penalty rates), self-employment, commission or piecework
> - determining the total wage earned for a casual employee on $23.50 per hour who works fixed hours on the weekend and earns time and a half for all hours worked on a Saturday
> - determining the exact commission on the sale of a $755 painting given the commission rate of $4\frac{2}{3}\%$
> - explaining the features and calculations within pay slips relating to pay period, payment details, deductions, tax paid, superannuation, gross and net pay, overtime and bonuses, year-to-date totals, employer information and payment method

General Capabilities: Numeracy, Critical and creative thinking

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WA9MNAF2

Develop and use the simple interest formula to solve problems relating to saving and borrowing

> **For example:**
>
> - a car loan of $14 500 is charged with 9.25% p.a. simple interest over a three-year period. How much interest will be charged for each month of the loan?

General Capabilities: Numeracy, Digital literacy

WA9MNAOpt8

Year 9 optional

Use authenticated websites to explore and compare different savings account options based on their characteristics (interest rates, fees, withdrawal policy)
or

Compare price, quality, terms and conditions of goods and services, such as phone plans and digital subscriptions

General Capabilities: Numeracy, Critical and creative thinking, Ethical understanding

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Modelling with number and algebra

WA9MNAM1
In real-world situations involving scientific notation, real numbers, linear equations with variables and/or brackets on either side of the equation, quadratic graphs and equations, direct proportion and/or simple interest, earning income or income statements

I. analyse the situation, decide if an exact or approximate solution is required and determine assumptions and constraints
II. represent the situation mathematically in order to reach a solution
III. interpret and communicate findings in terms of the context and any assumptions or constraints

> **For example:**
>
> - suppose ‘Moogle Millions’ made a profit of approximately 4.15×10^7 dollars in 2025. The company hopes profits will increase significantly over the next few years. Explore what would happen if profits doubled, tripled or increased by a fixed amount each year. Which pattern best explains how profits would become 100 times higher by 2029? Discussing that as the original profit would have been rounded in its representation in scientific notation, it follows that the solution will also be rounded. Exploring assumptions, such as no market changes over the stated years, representing the solution using scientific notation and applying index laws ($4.15 \times 10^7 \times 10^2$), communicating the answer using scientific notation and the context
> - using a newspaper bank advertisement on rates paid for investments over different time periods to investigate and compare the simple interest Suzie would earn if she invested $2500 at the given rates for the time periods advertised
> - ‘A fruit grower packs small and medium bags of oranges. The medium bags have two more oranges than the small bags. If the grower packs 500 oranges into 40 bags, how many oranges should be put in each medium bag?’
>   Discussing assumptions, such as all 500 oranges are packed, choosing to represent the situation as an algebraic equation and communicating the solution in terms of the fruit growing context and assumptions and constraints made

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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Measurement and geometry

Two-dimensional space and structures

WA9MMGTW1

Explore, explain and use efficient strategies to determine the perimeter and area of composite shapes involving triangles, quadrilaterals and/or circles, (including sectors), using appropriate units

For example:

Figure 1

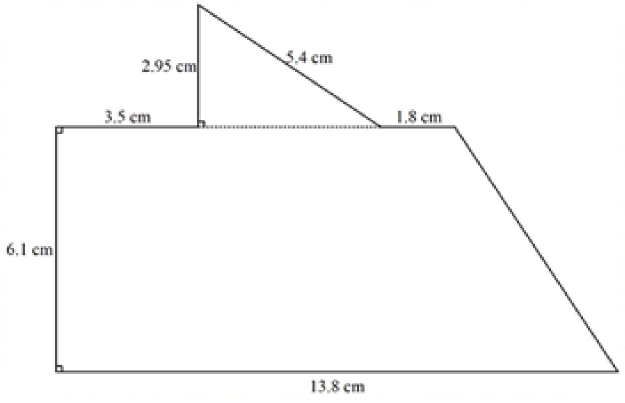


Figure 2

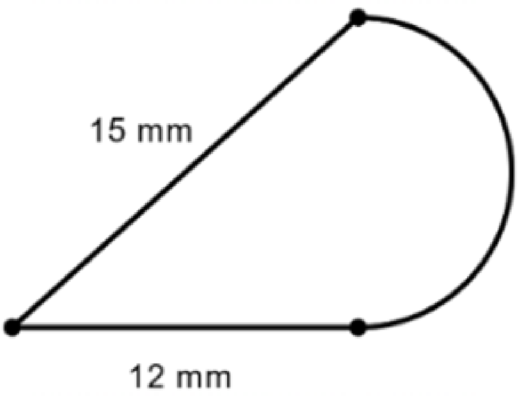
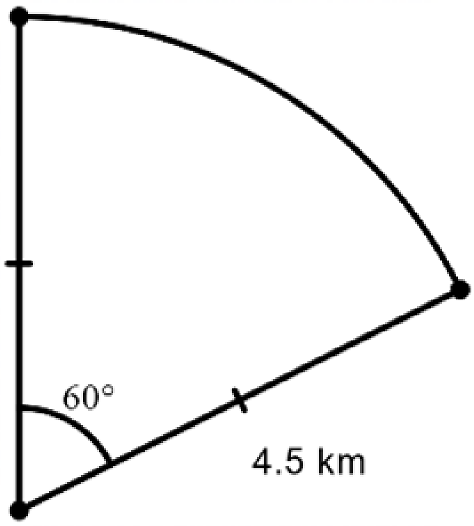


Figure 3



- exploring ways to determine perimeter using strategies appropriate to the shape, such as direct measurement, calculations of circumference and Pythagoras' theorem, knowledge of transformations or symmetrical parts. Applying addition of the length parts to determine an estimation, approximation or exact value of perimeter in appropriate length units
- exploring ways to determine area using strategies appropriate to the shape, such as decomposition into familiar polygons and circles or parts thereof and applying formulas, superimposing a rectangle around a shape and subtracting the area of space within the rectangle but outside the shape, from the area of the rectangle, knowledge of transformations or symmetrical parts. Applying strategies to determine an estimation, approximation or exact value of area in appropriate square units and comparing strategies with others to determine and explain the most efficient and accurate method

General Capabilities: Numeracy, Critical and creative thinking

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WA9MMGTW2

Use Pythagoras' theorem to determine the perimeter and area of shapes involving right-angled triangles, in both exact and decimal approximation form. Investigate and apply the converse of Pythagoras' theorem to establish whether a triangle is right-angled

For example:

- determining the height, area and perimeter of a right-angled triangle, with base length of 12 cm and hypotenuse 16 cm, in exact $(\text{height} = \sqrt{112}\text{ cm, P} = 28 + \sqrt{112}A = 6\sqrt{112}\text{ cm}^2)$ or approximate units $(\text{height} = 10.583\ \text{ cm}, = 38.583\ \text{ cm}, A = 63.498\ \text{ cm}^2)$
- exploring and using Pythagoras' theorem and geometric reasoning to determine the area of a rhombus with a side length of 8 cm and one diagonal of 15 cm or a regular hexagon with side length 5 cm
- using vertical and horizontal units between points plotted on a Cartesian plane and applying Pythagoras' theorem to determine the distance between the two points
- justifying whether 5 cm, 12 cm and 13 cm is a Pythagorean triad (lengths that form the sides of a right-angled triangle)

General Capabilities: Numeracy, Critical and creative thinking

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WA9MMGTW3

Explore to identify and describe conditions for triangles to be congruent. Use this to determine unknown sides or angles in pairs of congruent triangles and explain reasoning

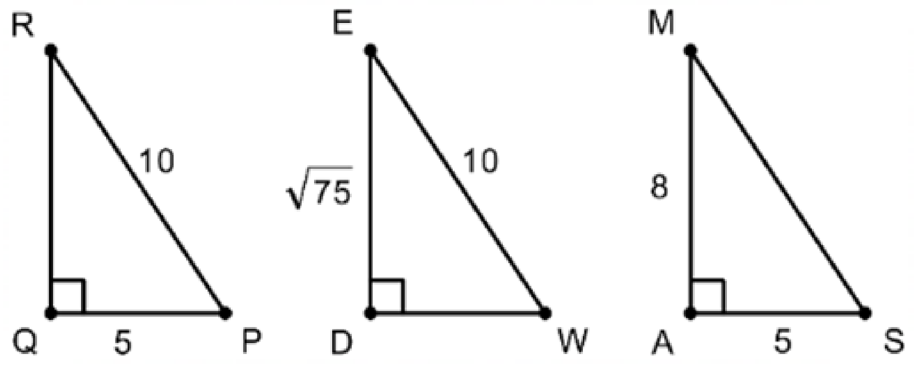
For example:

- constructing manually or by using dynamic geometry software, the following triangles with the given conditions
    - three sides of 5 cm, 6 cm and 4 cm
    - two sides of 7 cm, 5 cm and an included angle of 27°
    - one side of 5 cm and two angles, 32° and 54°
    - one right angle, a hypotenuse of 11 cm and one side of 4.5 cm
    - two sides of 5.5 cm and 4 cm and a non-included angle of 63°

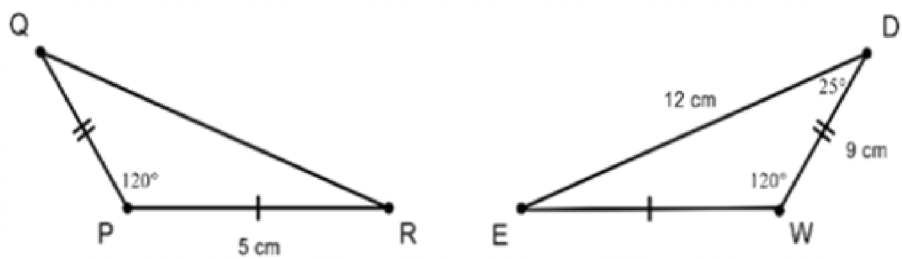
    - three angles of 51°, 24° and 105°

  comparing with others, reasoning and generalising which conditions confirm congruent triangles (SSS, SAS, RHS, AAS) and which do not

- identifying and justifying two triangles being congruent in words and symbols, communicating that $\triangle \mathrm{PQR} \cong \triangle \mathrm{WDE}$ because all three pairs of corresponding sides are equal to each other (SSS)



- determining the test that confirms that two triangles are congruent, the congruency statement and unknown sides or angles, giving reasons in situations such as



  the two sides and included angle of ΔPQR is equal to the two sides and included angle of ΔWDE, therefore $\triangle \mathrm{PQR} \cong \triangle \mathrm{WDE}$ (SAS)

  $\overline{QR} = 12$ cm as it corresponds to $\overline{DE}$ and $\angle Q = 25°$ as it corresponds to $\angle D$

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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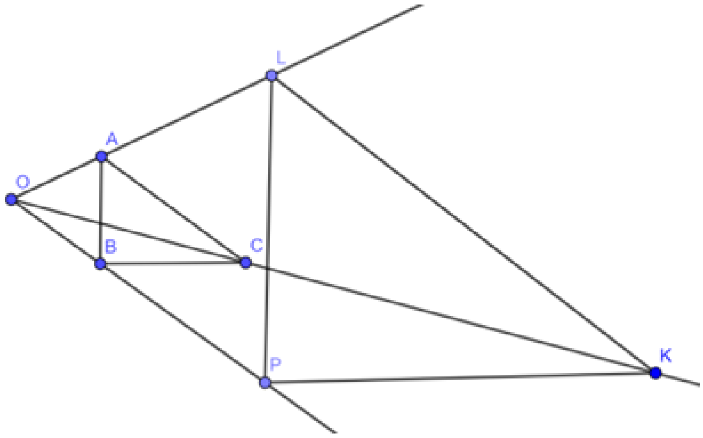
WA9MMGTW4

Construct similar figures by enlargement and reduction, and use this to establish, explain and apply properties of similar figures

For example:

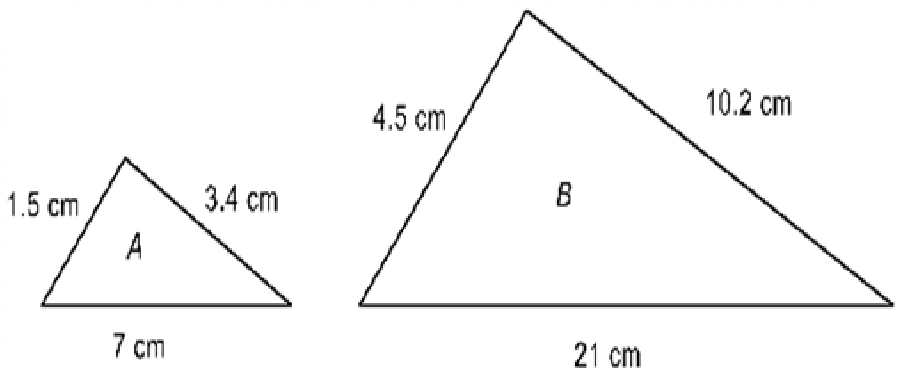
- constructing similar figures using techniques such as dynamic geometry software, grids, compasses and centres of enlargement or reduction. Design, test and refine a sequence of steps for producing similar figures



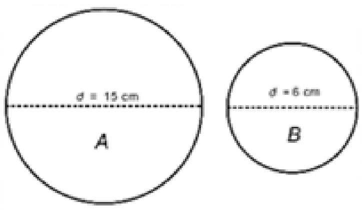
- comparing what is the same and what is different between the original figure and its image, noting that they are the same shape but not necessarily the same size. Measuring to generalise the properties of similar figures
  - corresponding angles are the same size
  - corresponding sides are in proportion
  ($\frac{\overline{AB}}{\overline{LP}} = \frac{\overline{AC}}{\overline{LK}} = \frac{\overline{BC}}{\overline{PK}}$)

  and recognising that this proportion is the scale or scale factor. Writing a similarity statement between pairs of similar figures, such as $\triangle \mathrm{ABC} \sim \triangle \mathrm{LPK}$

- determining the scale factor between similar figures using corresponding lengths in situations, such as



$\frac{4.5}{1.5} = \frac{10.2}{3.4} = \frac{21}{7} =$ scale factor of 3 and



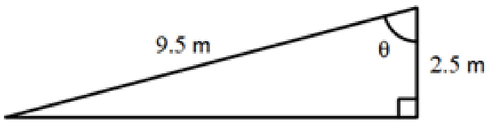
$\frac{6}{15} =$ scale factor of $\frac{2}{5}$,

recognising that an original figure will be enlarged if the scale factor is greater than one but will be reduced if the scale factor is greater than zero but less than one

General Capabilities: Literacy, Numeracy, Critical and creative thinking

WA9MMGTW5

Use similarity to investigate and explain the constancy of the sine, cosine and tangent ratios for a given angle in right-angled triangles. Choose and use a trigonometric ratio to determine the length of an unknown side or the size of an unknown angle

> **For example:**
>
> - identifying and labelling the hypotenuse, adjacent and opposite sides with respect to a given angle in a right-angled triangle in any orientation
> - verifying the constancy of each trigonometric ratio for a given angle by creating a series of similar triangles, measuring corresponding side lengths and identifying and determining the constancy of the different ratios between sides using the trigonometric notation of $\sin\theta$, $\cos\theta$, $\tan\theta$
> - choosing and using a trigonometric ratio to determine the length of a wheelchair ramp in the following situation
>
> 
>
>
> - choosing and using a trigonometric ratio to determine the size of the missing angle θ in the following situation
>
> 
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General Capabilities: Numeracy, Critical and creative thinking

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WA9MMGTW6

Apply the properties of similarity to determine scales, lengths and angles of real-life figures from scale drawings, maps, plans and photographs

> **For example:**
>
> - exploring and interpreting maps, plans and scale drawings, recognising that between the scale drawing and actual figure, angles are equal, and sides are in proportion
> - determining the scale used between a floor plan and actual house, the width of a bookcase using the scale or ratio on its construction plan and the actual radius of a cell, given a scale and its biological drawing
> - exploring scale by mapping a 30 000 hectare bushfire onto a map
> - exploring scale in photographs, such as taking a picture of a person standing next to a tree, using their height and the tree's height in the image and the actual height of the person to determine a scale factor and hence estimating the actual height of the tree, discussing and finding ways to improve the estimate

General Capabilities: Numeracy

WA9MMGOpt1

Year 9 optional

Explore and apply Pythagoras’ theorem and trigonometry to simple situations involving right-angled triangles in three-dimensional contexts projected to two dimensions

General Capabilities: Numeracy, Critical and creative thinking

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WA9MMGOpt2

Year 9 optional

Explore the relationship between sine and cosine ratios and the unit circle, determine their approximate values for angles from 0° to 360°, and identify pairs of angles that share the same ratio value

General Capabilities: Numeracy, Critical and creative thinking

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WA9MMGOpt3

Year 9 optional

Apply deductive reasoning and use a sequence of logically connected statements to produce proofs of congruent triangles

General Capabilities: Numeracy, Critical and creative thinking

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Three-dimensional space and structures

WA9MMGTH1

Establish, explain and apply formulas to determine the volume, capacity and surface area of cylinders, using appropriate units

> **For example:**
>
> - recognising that a cylinder has uniform cross-sections of a circle and multiplying the area of this circle by the height, generalising to $Volume\ cylinder = \pi r^2 \times h$ to determine the volume in cubic units, converting to capacity units if required
> - decomposing a cylinder into a sketched net or visualising its net, as comprising two equal circles and a rectangle, showing the addition of the area of the faces to generalise to a formula $SA\ cylinder = 2\pi r^2 + 2\pi rh$ and applying to determine the surface area in square units
> - investigating and designing the net of the largest possible cylindrical closed container on an A3 sheet of cardboard by considering which attribute best determines ‘largest’, applying formulas, calculations and conversions before constructing, comparing and ordering cylinders with others

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA9MMGTH2

Explore and explain efficient strategies to determine the surface area of right prisms using appropriate units

> **For example:**

- using strategies, such as unfolding a cardboard prism, visualising or drawing the net or different two-dimensional views of the prism, recognising congruent faces and showing related area calculations of the different faces, in a sequence of steps, adding to determine the surface area in square units
- determining the depth of a rectangular prism, given that it has a surface area of 726 cm^2, length of 15 cm and width of 5 cm
- determining the surface area of a cube given that its volume is 15.625 m^3

General Capabilities: Literacy, Numeracy

WA9MMGTH3

Use dynamic geometry software to explore and construct familiar objects in three dimensions using transformations of two-dimensional figures

For example:

- using dynamic geometry software to show transformations of two-dimensional figures in the plane through a third dimension to create prisms and cylinders, such as constructing a right cylinder by rotating a rectangle around a line in the plane of the rectangle or rotating a triangle around an axis to create a cone. Describing different views of the object, altering the dimensions and highlighting what is the same and what is different about the original object and its alteration
- drawing a two-dimensional shape and describing how to transform images of the shape to represent a three-dimensional object, using at least two different transformations, such as translating, rotating or reflecting a square to form a cube

General Capabilities: Literacy, Numeracy, Digital literacy

Modelling with measurement and geometry

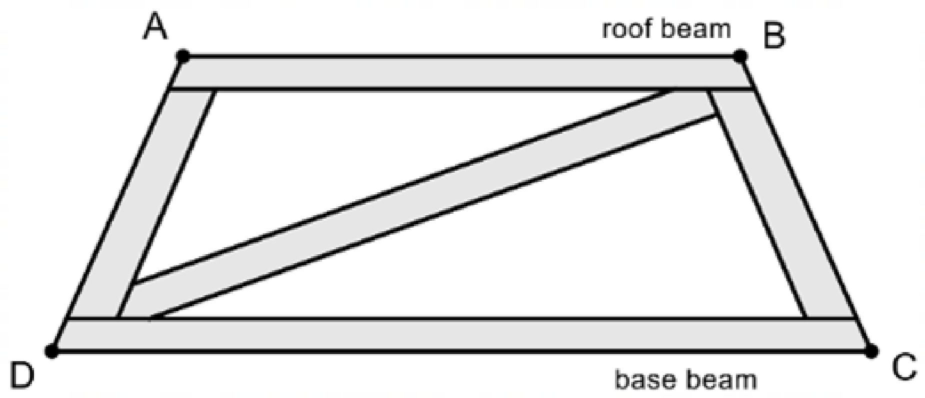
WA9MMGM1

In real-world situations involving the perimeter and area of composite shapes and right-angled triangles, enlargement and reduction of similar figures, finding unknown side lengths and angles using trigonometric ratios, scale in similar figures and/or volume, capacity and surface area of right prisms and cylinders

I. analyse the situation, decide if an exact or approximate solution is required and determine assumptions and constraints
II. represent the situation mathematically in order to reach a solution
III. interpret and communicate findings in terms of the context and any assumptions or constraints

For example:

- ‘A building firm is constructing a roof beam with an 11.4 m base beam parallel to a 7.6 m roof beam and 1.5 m apart. The firm wants to insert a truss to connect from one top corner to the diagonal bottom corner to ensure that the beam is rigid. What is the total length of roof beam needed to build the whole roof beam?



Deciding that an exact answer to the nearest millimetre is required for building purposes and that the truss is straight. Choosing to use properties of an isosceles trapezium and Pythagoras’ theorem to determine the length of the truss and the length of the equal, non-parallel sides and the perimeter of a trapezium to determine the total length of beam. Communicating the solution in the context of the roof beam and to the nearest millimetre

General Capabilities: Literacy, Numeracy, Critical and creative thinking

Probability and statistics

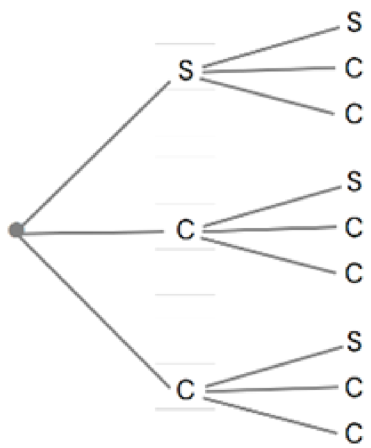
Probability and statistics

WA9MPSP1

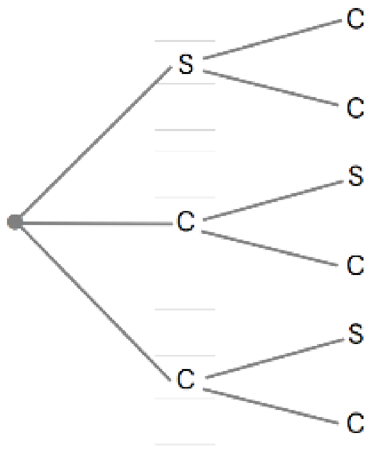
Construct sample spaces to show outcomes for two-stage chance experiments both with and without replacement. Assign probabilities to outcomes and make informal connections to independent and dependent events

For example:

- the digits 1, 2, 3, 4 and 5 are written on ping-pong balls and placed in a bag. Two balls are drawn consecutively without replacement to make a two-digit number. Choosing from a list, an array or a tree diagram to show all possible outcomes and use this to find the probability that the number formed is even. Repeating with replacement. Noting that the events of drawing one ball after another with replacement are independent as the outcome of the first event does not impact the outcome of the second event
- choosing to construct a tree diagram sample space to represent a box of three chocolates, containing one strawberry and two caramel chocolates, with two chocolates being randomly selected, one after the other. Determining the probability of selecting two caramels in each of the following scenarios
  - with replacement:
    the three chocolates have coloured wrappers to indicate the flavour. A chocolate is randomly chosen, the flavour noted and the chocolate replaced (independent events)

    P(caramel, caramel) $= \frac{4}{9}$
  - without replacement: three unwrapped chocolates, needing to be eaten to identify flavours (dependent events)



P(caramel, caramel) = $\frac{2}{6} = \frac{1}{3}$

General Capabilities: Numeracy, Critical and creative thinking

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WA9MPSP2

Conduct repeated two-stage chance experiments and simulations, both with and without replacement, to produce datasets, including through the use of digital tools. Discuss, compare and interpret variation and estimated probabilities for compound events

For example:

- conducting an experiment to find the estimated probability of choosing two hazelnut chocolates, one after the other, from a box containing seven strawberry, 11 hazelnut and two caramel chocolates for the following scenarios
  - with replacement:
    the chocolates have coloured wrappers to indicate the flavour. A chocolate is randomly chosen, the flavour noted, the chocolate replaced, and a second chocolate is chosen, and the flavour is noted.
  - without replacement:
    the chocolates are unwrapped, needing to be eaten to identify flavours (not replaced), choosing and using an appropriate tool, such as a 1–20 random number generator (1–7 strawberry, 8–18 hazelnut and 19–20 caramel) to simulate and generate large datasets for both scenarios.

  Using the datasets to estimate the probability of drawing a hazelnut followed by another hazelnut in each case, comparing and explaining differences in the estimated probabilities of drawing two hazelnuts between scenarios. Discussing and interpreting the effects of with/without replacement and the chance variation on the datasets and the conclusions drawn.

  For students engaging in Year 9 optional, extending and connecting to the probability formula for independent events

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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WA9MPSP3

Analyse data with multiple variables represented in tables. Describe using statistical measures and relative frequencies to make inferences

> **For example:**
>
> - describing data in tables involving variables, such as the states of Australia, their population, land size and median house price. Determining proportions, such as the percentage of Australians who live in NSW or the fraction of land size that WA is of Australia, and statistical measures, such as the average median house price. Discussing possible inferences that could be made from the data

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding

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WA9MPSP4

Explore, choose and create graphical or visual representations and justify choice with regards to context, purpose, data type and intended audience

> **For example:**
>
> - using digital tools and a variety of data types and data sizes, explore and create standard and non-standard representations, such as histograms, line graphs, Venn diagrams, two-way tables, stem-and-leaf plots, stacked bar charts, pie charts, infographics or 3D graphs
> - discussing determining factors involved in choosing an appropriate representation, such as where the representation will be displayed, whether the data is categorical, numerical, continuous or discrete, the message the representation will be intending to convey, who will be viewing the representation and what level of detail or complexity in the representation is appropriate for the audience
> - using topics of interest involving sports statistics, environmental data over time, social media 'likes' etc., selecting and creating an appropriate display based on the dataset and the determining factors

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA9MPSP5

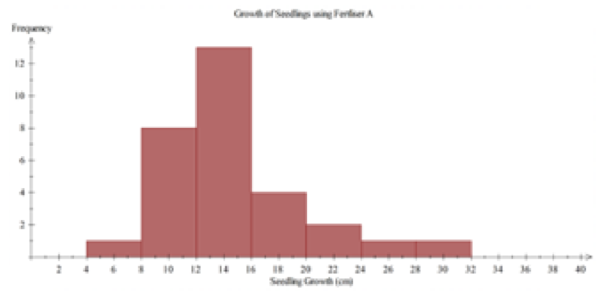
Interpret and compare multiple datasets represented in back-to-back stem-and-leaf plots and histograms with consideration of shape, spread and centre

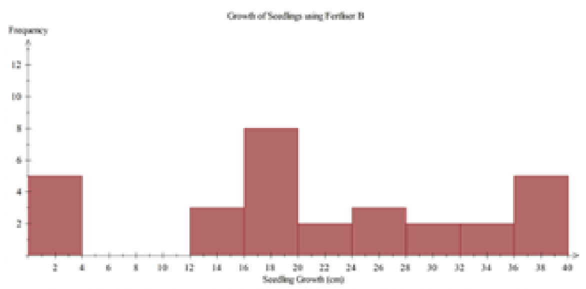
> **For example:**
>
> - given a back-to-back stem-and-leaf plot of pulse rates before and after exercise to show heart beats per minute, describing the shape of ‘before’ and ‘after’ as both being non-symmetrical and positively skewed.

before	pulse rate	after
9 8 8 8	6	
6 6 4 1 1 0	7	
8 8 6 2	8	6 7 8 8
6 0	9	0 2 2 4 5 8 9
4	10	0 4 4
0	11	8
	12	4 4
	13	
	14	6

Comparing the gaps (a significant gap between 124 and the outlier of 146 in the ‘after’ pulses), clusters (tighter clustering around the middle values in the ‘after’ pulses) and range (significantly higher in the ‘after’ pulses due to the outlier). Determining the mean, mode and median of ‘before’ and ‘after’, recognising that the ‘after’ pulses have more than one mode and calculating the differences between the mean and the median between each of the datasets. Generalising to ‘after’ pulse rates have higher central tendencies, a wider spread and tighter clustering compared to the ‘before’ pulse rates

- describing comparative histograms displaying the growth of seedlings after different fertilisers are applied, in terms of shape and spread





- describing the shape, spread and centre of the Australian Bureau of Statistics population pyramid, in terms of the age structure of people in Australia over time

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA9MPSP6

Describe different sampling methods and analyse how the different methods can affect the results of surveys. Identify and explain how chance variation impacts on the data validity, reliability and conclusions drawn from surveys

For example:

- exploring and explaining systematic, stratified, clustered, convenience or capture-recapture sampling methods, and deciding which method would be most representative of a total population (e.g. stratified sampling would be most suitable when surveying employee satisfaction across different departments)
- conducting a survey of fellow students to decide a suitable frozen yoghurt to be included in the school canteen menu. Assigning different sampling methods to groups of students to each produce two different samples. Each group analysing their samples separately to determine and compare findings, discussing chance variation. Comparing results between groups and discussing factors that could explain differences. Suggesting and explaining which of the sampling methods produces a valid and reliable dataset on which a prediction of the total population for this scenario can be made
- explaining how the number of people with Type A blood could be estimated, justifying choices of sampling size and method with consideration to ensuring the collection of valid and reliable data

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding

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WA9MPSP7

Critically analyse statistics in the media and other real-life situations relating to data samples, including the effect of chance variation on sample analyses

For example:

- critically commenting on whether a statistical sample is produced from valid and reliable data by considering factors, such as sample size, random variation, measurement or recording error, inconsistencies or biases in where, when and how the data sample was collected. Recognising that uncertainty in sampling produces uncertainty of conclusions. Commenting on situations such as
  - a newsagent in a small country town advertises that they have had two million-dollar winners in the past week, encouraging customers to buy the winning ticket from them
  - a major national supermarket conducted a survey about brand preferences using an online form. They received 65 responses
  - researchers want to estimate the population of fish in a dam using the capture-recapture method. They capture and tag 100 fish, release them back into the dam and after a month, capture another 100 fish, finding that 10 of them are tagged. They estimate the population of fish in the dam to be 1000
  - a pharmaceutical company conducts a paid clinical trial with 30 volunteers to test the efficacy of a new drug for headaches

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding

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WA9MPSOpt1

Year 9 optional
Identify independent and dependent two-stage chance events using $P(A \text{ and } B) = P(A) \times P(B)$ and sample spaces, such as tree diagrams, to determine the probability of independent events

General Capabilities: Numeracy, Critical and creative thinking

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WA9MPSOpt2

Year 9 optional

Produce and organise accurate and valid, ungrouped continuous data to construct histograms and frequency polygons. Determine summary statistics and analyse the distribution in terms of centre, shape and spread

General Capabilities: Numeracy, Critical and creative thinking

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Modelling with probability and statistics

WA9MPSM1

In real-world situations involving two-stage chance experiments or simulations both with or without replacement, different sampling methods, choosing and creating graphical representations and/or analysis of tables and comparative graphs

I. analyse the situation, pose questions as required and determine assumptions and constraints
II. determine appropriate production of a valid and reliable dataset, statistical measures, data representations and analyses, including examination of distributions, to effectively investigate the situation
III. interpret, draw inferences and communicate findings in terms of the context, assumptions, constraints, chance variation and knowledge or insights gained

> **For example:**
>
> - 'What are the chances of Kevin randomly choosing two consecutive days in April next year to holiday?' Analysing the situation by recognising that days cannot be repeated and that the last day of April cannot be included as the first of the consecutive days. Posing questions, such as 'How many days in April?' and 'What are the possible consecutive dates?' Producing a large set of data using a random calendar date or sequence generator, representing the results in a table and using to estimate the probability of randomly choosing two consecutive April dates. Interpreting and communicating the result, recognising and acknowledging the impact of chance variation
> - 'Oliver is planning his holidays for next year. He wants to spend a week in Esperance for one holiday and a week in Geraldton for another holiday. Oliver wants to determine the best months to holiday according to the weather.' Analysing the situation by determining the important aspects of weather. Posing questions, such as 'What is the maximum temperature in each month for each of the places in the last 10 years?' Making assumptions, such as his employer allowing him to take holidays in two separate weeks. Producing valid and reliable data from a source, such as the Bureau of Meteorology. Choosing and using appropriate comparative representations depending on the data collected. Analysing and interpreting the representations, communicating when Oliver should travel to these destinations in reference to the initial qualifications

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding, Digital literacy

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Year 10

Year level descriptions

In the middle adolescence phase of schooling, teaching and learning programs encourage students to develop an open and questioning view of themselves as active participants in their society and the world.

Mathematics provides opportunities for students to engage in a range of approaches to learning through the proficiencies of understanding, fluency, problem-solving and reasoning. These reinforce the significance of working mathematically with the content and describe how the content is explored or developed. Students draw on the behaviours of the proficiencies when selecting and using year level content to apply the complete modelling process, leading to an increased understanding of the complexity of the natural environment, society and technology.

In Year 10, students further engage in financial mathematics by calculating and interpreting income tax and compound interest. They explore the effect of error when using approximate rather than exact real values, including in measurement situations. They apply the index laws and extend their algebraic and graphical understanding to include inequalities, simultaneous equations and quadratic and exponential functions.

Students make connections between measurement and geometry by considering the effect on perimeter, area, volume, capacity and surface area when similar figures and objects are enlarged or reduced. Students use geometric reasoning to establish conditions for similar triangles. They find unknown sides and angles in right-angled triangles using Pythagoras' theorem or trigonometry and extend their three-dimensional spatial reasoning and use of formula to include volume, capacity and surface area of composite objects.

Students connect probability and statistics by collecting data from two- and three-stage chance experiments and simulations, both with and without replacement, to model conditional events. They interpret, compare key features and analyse multiple boxplots. Students broaden their understanding of analysis to include commenting on association in bivariate and categorical data. Students critically analyse statistical reports in the media and identify potential sources of bias.

Note: the optional content in Year 10 is intended to build and extend students' year level knowledge according to areas of interest, understanding of content and preparation for subsequent study. The content descriptions are optional. Teachers may choose from optional content according to the needs of the student/s.

Achievement standard

By the end of the year:

Students demonstrate the behaviours of the proficiencies of understanding, fluency, problem-solving and reasoning in conjunction with year level content in routine situations. As part of this, they select from and use year level content along with the modelling process, in which they analyse the situation mathematically, represent the problem and interpret and communicate their findings, to solve straightforward, real-world problems in familiar contexts across all strands.

Students compare the difference in using approximate to exact values on final calculations. They substitute values into real-life formulas to find unknowns using digital tools. Students solve one variable linear inequalities, verify the solution and represent on a number line. They solve linear simultaneous equations graphically. Students graph quadratic and exponential functions, and relate key graphical and algebraic features. They calculate income tax and use repeated simple interest to determine compound interest.

Students use Pythagoras' theorem and/or trigonometry to determine unknown sides and angles in right-angled triangles involving angles of elevation and depression. They determine the effect on perimeter, area and volume when shapes and objects are enlarged or reduced. They use triangle and angle properties to show reasoning as to why triangles are similar and find unknown sides and angles in similar triangles. They determine volume, capacity and surface area of composite objects.

Students choose and construct appropriate sample spaces for two- and three-stage chance experiments and assign probabilities to events involving conditional statements. They conduct chance experiments and simulations to model conditional probability. They describe possible association between variables represented in a scatterplot or two-way table. Students represent multiple datasets using boxplots and compare shape, spread and centre. They critically analyse claims, inferences and conclusions in the media and identify potential sources of bias.

Content descriptions

Number and algebra

Understanding number

WA10MNAUN1

Move flexibly between real number inequalities expressed as a worded statement, algebraically or on a number line

> **For example:**
>
> - representing all measurements greater than or equal to two metres as an inequality and on a number line
>
> 
>
>
> - representing a range of temperatures greater than or equal to −1.5° and less than 3° as inequalities and on a number line
>
> 
>
>
> - writing an inequality to match the following number line solution
>
> 
>

General Capabilities: Numeracy

Calculating with number

WA10MNAC1

Use absolute and percentage error to compare the result of using approximate rather than exact real numbers on final calculations

> **For example:**
>
> - investigating the use of absolute error to determine percentage error
>
> $\text{Percentage error} = \frac{\text{Absolute error}}{\text{Exact result}} \times 100\%$
>
>   where absolute error is the positive difference between the approximate and exact results in situations, such as
>
>   - calculating the surface area of cylinders using 3.14 instead of the exact value of π or by using π and then rounding the area of the base circles
>   - applying Pythagoras' theorem or trigonometry to find a missing side in a right-angled triangle and using 1.41 instead of the exact value of $\sqrt{2}$
>   - applying the compound interest formula and using 2.67% interest per annum instead of $2\frac{2}{3}\%$

General Capabilities: Numeracy, Critical and creative thinking

Algebraic techniques

WA10MNAA1

Extend and apply index laws with positive-integer indices and variable bases, to include negative-integer indices

> **For example:**
>
> - applying patterns, index notation and laws to establish $a^{-1} = \frac{1}{a}$, $a^{-2} = \frac{1}{a^2}$, $a^{-3} = \frac{1}{a^3}$ and $a^{-n} = \frac{1}{a^n}$
> - representing expressions involving negative-integer indices as expressions involving positive-integer indices and vice versa
> $x^{-3} = \frac{1}{x^3}$, $2x^{-1} = \frac{2}{x^1} = \frac{2}{x}$,
>
> $\frac{4}{x} = \frac{4}{x^1} = 4x^{-1}$, $\frac{1}{x^2} = x^{-2}$
> - justifying why these statements are true
> $\frac{9x^5}{3x^{-5}} = 3x^{10}$
>
> $9x^{-7} \div 3x^5 \neq 3x^2$
>
> $(3a^2)^2 \times a^3 \neq 6a^7$
>
> $a^5 \times a^{-7} = \frac{1}{a^2}$

General Capabilities: Numeracy, Critical and creative thinking

WA10MNAA2

Substitute values into real-life linear, quadratic or simple exponential formulas to determine unknowns using digital tools

> **For example:**
>
> - using software and a real-life situation; such as the height of a toy rocket launched from the ground being modelled by a quadratic given as $h = -16t^2 + 128$, to find unknown values of h and t
> - using spreadsheets to investigate $A = P(1 + i)^n$ for different values of unknowns

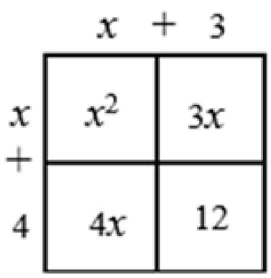
General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

WA10MNAA3

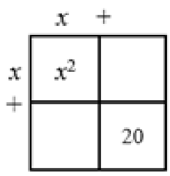
Extend and apply knowledge of the expansion of binomial products to explore the factorisation of monic quadratics

> **For example:**

- extending knowledge of expansion of $(x + 4)(x + 3)$



  to make connections to factorising quadratics using knowledge of multiples and partitioning
  $x^2 + 7x + 12$
  $= x^2 + 4x + 3x + 12$
  $= x(x + 4) + 3(x + 4)$
  $= (x + 4)(x + 3)$
- factorising $x^2 + 12x + 20$ by making connections between the expansion of binomial products and the area model and generalising to a factorising strategy involving factors of 20



- factorising quadratics, such as
  $x^2 + 14x + 24$, $x^2 - 3x - 28$ and $x^2 + 9x - 22$

General Capabilities: Numeracy, Critical and creative thinking

WA10MNAOpt1

Year 10 optional
Simplify algebraic products and quotients involving indices with integer and fractional indices

General Capabilities: Numeracy, Critical and creative thinking

WA10MNAOpt2

Year 10 optional
Establish the connection between fractional indices and surds. Perform the four operations with surds and rationalise the denominator if required

General Capabilities: Numeracy, Critical and creative thinking

WA10MNAOpt3

Year 10 optional
Interpret and use base-10 logarithmic scales on graphs of real-life contexts

General Capabilities: Numeracy, Critical and creative thinking

WA10MNAOpt4

Year 10 optional
Factorise monic and non-monic quadratic expressions using techniques, such as completing the square, perfect squares, difference of squares and grouping in pairs for four-term expressions

General Capabilities: Numeracy, Critical and creative thinking

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WA10MNAOpt5

Year 10 optional
Explore efficient strategies to simplify expressions that involve addition, subtraction, multiplication or division of algebraic fractions with an algebraic expression in the numerator and/or denominator, including the use of factorisation

General Capabilities: Numeracy, Critical and creative thinking

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Linear and non-linear equations and inequalities

WA10MNALE1

Solve one-variable linear inequalities involving brackets and/or a variable on each side. Represent the solution on a number line and verify the solution by substitution

For example:

- using a strategy, such as the balance model, backtracking or inspection, to solve an inequality, showing a sequence of steps to communicate thinking for inequations, such as $-2m - 1 \geq -3m + 2$, representing solutions as

  

  and checking by substituting a value greater than 3, such as $m = 4$, to show that the inequality holds true
  $-2(4) - 1 \geq -3(4) + 2$
  $-9 \geq -10$

General Capabilities: Numeracy, Critical and creative thinking

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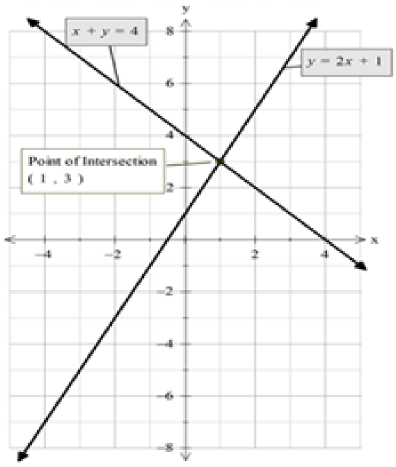
WA10MNALE2

Determine the solution to linear simultaneous equations in the forms $y = mx + c$ or $ax + by = c$ graphically and verify the solution by substitution

For example:

- graphing linear equations, such as $y = 2x + 1$ and $x + y = 4$, manually or digitally



verifying that (1, 3) is the point of intersection of the lines

Note: consider graphs of parallel lines that will have no solution

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MNAOpt6

Year 10 optional
Determine the solution to linear simultaneous equations in the forms $y = mx + c$ or $ax + by = c$ algebraically and verify the solution by substitution or using digital tools

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MNAOpt7

Year 10 optional
Identify the region on the Cartesian plane defined by linear inequalities

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MNAOpt8

Year 10 optional
Solve monic and non-monic quadratic equations graphically and algebraically, including the use of the quadratic formula, factorising techniques and digital tools, and verify the solution/s by substitution

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MNAOpt9

Year 10 optional
Use algebraic techniques to solve exponential equations that involve terms with related bases

General Capabilities: Numeracy, Critical and creative thinking

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WA10MNAOpt10

Year 10 optional

Solve cubic equations in the form $ax^3 = k$ or in factored form, algebraically or using digital tools

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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Linear and non-linear patterns and relationships

WA10MNALP1

Use a table of values to plot points and graph quadratic functions of the form $y = ax^2 + c$. Identify and relate key graphical and algebraic features and make connections to the graphical and algebraic solution/s of $ax^2 + c = k$. Use digital tools to explore the shapes, features and related solutions to more complex quadratic functions

> **For example:**
>
> - making connections between graphical and algebraic representations of quadratics, such as
>   - the shape and concavity of the parabola and the value of the coefficient of x^2
>   - the vertical intercept and the value of *c*
>   - the horizontal intercepts when they exist
>   - approximate solutions to examples, such as $-x^2 + 4 = 3.2$ using the graph of $y = -x^2 + 4$ and checking solutions using digital tools
> - using digital tools to graph easily factorised monic quadratics, such as $y = x^2 + 4x$ and $y = x^2 - 7x + 10$, exploring connections between the graphical and algebraic representations, and approximating and checking solutions to related equations

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MNALP2

Use a table of values to plot points and graph exponential functions of the form $y = a^x$ where $a > 0$. Identify and relate key graphical and algebraic features and use these to determine graphical solutions of related equations. Use digital tools to explore the shapes, features and related solutions to more complex exponential functions

> **For example:**
>
> - making connections between graphical and algebraic representations of exponentials, such as
>   - the vertical intercept
>   - the direction of the graph given the nature of
>   - the shape of the graph given the value of
>   - the asymptote
>   - approximate solutions to examples, such as $2^x = 3$ and $y = 2^{2.5}$ using the graph of $y = 2^x$ and checking solutions using digital tools
> - using digital tools to explore real-life exponential growth and decay situations, making connections between the algebraic and graphical forms

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MNALP3

Identify and distinguish between linear, quadratic and exponential functions represented by equations, tables of values and graphs

> **For example:**
>
> - recognising that in a table of values, if there is a constant difference between consecutive values of the dependent variable, then the function is linear. If there is a constant second difference between consecutive values of the dependent variable, then the function is quadratic, and if the ratio between the consecutive values of the dependent variable is constant, then the function is exponential
> - investigating the links between algebraic and graphical representations of each function type using digital tools

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MNAOpt11

Year 10 optional
Use gradient and/or point/s to graphically and algebraically determine equations of parallel and perpendicular lines

General Capabilities: Numeracy, Critical and creative thinking

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WA10MNAOpt12

Year 10 optional
Graph monic and non-monic quadratics of the form $y = ax^2 + bx + c$ or $y = a(x - p)^2 + q$ and their transformations, manually and using digital tools. Identify and connect key graphical and algebraic features and make connections to the algebraic solution/s

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MNAOpt13

Year 10 optional
Explore and use strategies, including digital tools, to model the equation of a quadratic function from a table of values or graph

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MNAOpt14

Year 10 optional
Explore, describe and interpret circles and rectangular hyperbolas where the asymptotes are parallel to the axes, and their transformations, using digital tools, and make connections between the algebraic and graphical representations

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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Financial mathematics

WA10MNAF1

Explore, explain and calculate income tax, including the use of tax tables

> **For example:**
>
> - investigating differences in tax paid for incomes $10 either side of one of the intervals in a current financial year tax bracket

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MNAF2

Apply repeated simple interest to develop the compound interest formula and solve problems that relate to saving and borrowing

> **For example:**
>
> - using a loan of $5000 over three years at 4% p.a., show repeated yearly simple interest calculations and demonstrate connections to the compounded interest formula
> - using the compound interest formula to calculate the exact interest earned for an investment of $12 000 at $3\frac{1}{3}\%$ over four years, compounded yearly
> - comparing different finance options, such as ‘Is it better to borrow $2000 from a store finance program that offers 8.55% p.a. simple interest over 18 months or from a bank that offers 7.2% p.a., compounded yearly, over two years?’

General Capabilities: Numeracy, Digital literacy

WA10MNAOpt15

Year 10 optional
Use authenticated websites to investigate how changes to the principal, rate of return, voluntary contributions and time can affect superannuation balances
or
Compare characteristics of insurance, such as young driver car insurance or holiday insurance, and recognise that the cost is higher when the risk is higher

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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Modelling with number and algebra

WA10MNAM1
In real-world situations involving real numbers, absolute and percentage error, linear inequalities, simultaneous equations, real-world formulas, quadratic or exponential functions, taxation and/or compound interest

I. analyse the situation, decide if an exact or approximate solution is required and determine assumptions and constraints
II. represent the situation mathematically in order to reach a solution

III. interpret and communicate findings in terms of the context and any assumptions or constraints

For example:

- using situations such as ‘Jackson invests $12 500 in an account, paying $3\frac{1}{3}\%$ compounded fortnightly for $2\frac{1}{2}$ years. Compare the amount earned if the interest is calculated using 3.33% rather than its exact $3\frac{1}{3}\%$.’ Discussing that an exact solution is required, with assumptions, such as there are 26 fortnights in a year. Choosing to represent the situation using the compound interest formula and communicating the comparison between rates using the exact difference, percentage difference and language of the problem
- the cross-sectional shape of a tunnel can be represented by the quadratic equation $y = -0.75x^2 + 12$. Determine, using digital tools, (i) the maximum height of the tunnel and (ii) the maximum height a truck entering a tunnel could be, if its width is 4.8 m. Discussing assumptions, such as the x-axis is level ground and the tunnel is one way, choosing to represent graphically using digital tools, interpreting the turning point of the graph as its maximum height and communicating the solution in terms of the tunnel and assumptions and constraints made
- ‘The growth of a bacteria changes according to the function $y = 800 \times 1.26^t$, where t is the time in hours after bacteria has been placed in a petri dish for experimental analysis. Using digital tools, determine when the number of bacteria will double?’

  Discussing assumptions, such as the petri dish is closed so no further bacteria can enter, and the function remains consistent over time. Choosing to represent the function as an equation or graph to find an approximate answer, acknowledging the limitations of accuracy in the graph, and using digital tools to refine the solution. Communicating the appropriately rounded solution in the context of the problem, with assumptions and constraints made

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

Measurement and geometry

Two-dimensional space and structures

WA10MMGTW1

Use Pythagoras' theorem and/or trigonometry to determine unknown sides and angles in right-angled triangles involving angles of elevation and depression

> **For example:**
>
> - identifying, describing and defining angles of elevation and depression
> - determining unknown sides and/or angles of elevation or depression from given right-angled triangle diagrams by choosing an appropriate trigonometric ratio, applying and communicating solutions in appropriate units through a sequence of equations
> - sketching and labelling right-angled triangle diagrams and choosing and using appropriate techniques to solve problems in situations, such as
>   - determining the angle of depression from the top of an 820 m hill to a bushfire 5.8 km away
>   - calculating the height of a flagpole erected on top of a Perth building given that at a distance of 300 m from the base of the building, the angles of elevation to the bottom and top of the flagpole are 37° and 39° respectively

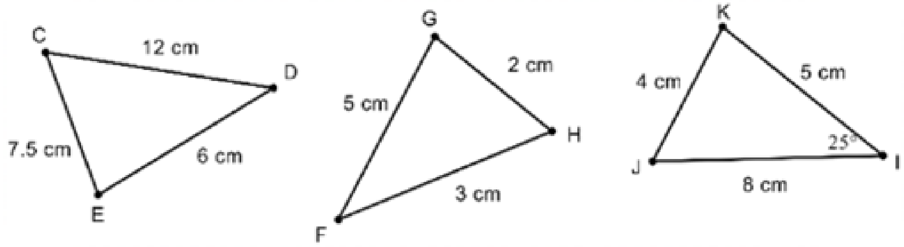
General Capabilities: Numeracy

WA10MMGTW2

Explore to identify and describe conditions for triangles to be similar. Use this to determine unknown sides and angles in pairs of similar triangles and explain reasoning

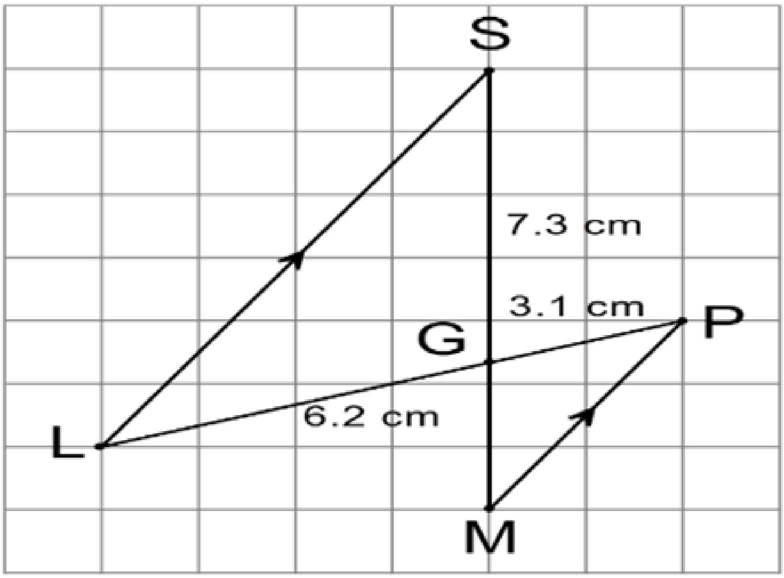
> **For example:**
>
> - constructing manually or by using dynamic geometry software, the following pairs of triangles with the given conditions
>   - triangle 1 with sides of 3 cm, 5 cm and 4.5 cm and triangle 2 with sides of 6 cm, 10 cm and 9 cm
>   - triangle 1 with an angle of 35° and a side of 8 cm and triangle 2 with an angle of 35° and a side of 4 cm
>   - triangle 1 with sides of 4 cm and 6 cm, with an included angle of 25° and triangle 2 with sides of 6 cm and 9 cm, with an included angle of 25°
>   - triangle 1 with angles of 127° and 21° and triangle 2 with angles of 32° and 21°
>   - triangle 1 with sides of 4 cm, 7.5 cm and 4.5 cm, and triangle 2 with sides of 3 cm, 15 cm and 9 cm
>   - triangle 1 with hypotenuse of 13 cm, a 90° angle and the shortest side being 10 cm, and triangle 2 with hypotenuse of 6.5 cm, a 90° angle and the shortest side being 5 cm
>
>   comparing with others to see if similar triangles have been produced, reasoning and generalising conditions to confirm similar triangles (SSS, SAS, RHS, AA)

- identifying and justifying two triangles being similar in words and symbols



recognising ΔCDE ~ ΔIJK because all three corresponding sides are in proportion (SSS) (scale = $\frac{2}{3}$)

- determining the test that confirms two triangles are similar, the similarity statement and finding unknown sides or angles, giving reasons in situations such as



communicating in both words and symbols that ∠SGL = ∠MGP because they are vertically opposite and ∠GSL = ∠GMP as they are alternate angles, in parallel lines
∴ ΔSGL ~ ΔMGP because two angles of ΔSGL are equal to two angles of ΔMGP (AA)

$\overline{\text{GM}} = 3.65$ cm as corresponding sides in similar triangles are in proportion

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MMGTW3

Investigate, explore and determine the effect on the perimeter and area of shapes when they are enlarged or reduced by a scale factor

For example:

- determining the perimeter and area of figures, such as triangles, quadrilaterals, circles and composite figures, that have been drawn using grid paper, a compass or dynamic geometry software. Making a prediction as to what the perimeter and area of their similar figures would be when side lengths or diameters have been doubled, tripled or halved. Testing this prediction by constructing the scaled figures manually or by using dynamic geometry software to determine the new perimeter and area.

Using results to explain and make generalisations as to the effect on perimeter ($\times\ k$) and area ($\times\ k^2$) when a figure has been scaled with a given scale factor (k)
- determining the scale factor when a semicircle with an area of $42.47\ \text{cm}^2$ is enlarged to have an area of $679.59\ \text{cm}^2$

General Capabilities: Numeracy, Critical and creative thinking

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WA10MMGOpt1

Year 10 optional

Apply right-angled trigonometry to two-dimensional situations involving navigational bearings

General Capabilities: Numeracy

WA10MMGOpt2

Year 10 optional

Explore to establish and use the sine, cosine and area rule to determine unknown sides and angles for any triangle

General Capabilities: Numeracy, Critical and creative thinking

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WA10MMGOpt3

Year 10 optional

Use the unit circle and dynamic geometry software to explore and represent trigonometric functions graphically

General Capabilities: Numeracy, Critical and creative thinking

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WA10MMGOpt4

Year 10 optional

Solve simple trigonometric equations graphically, algebraically or using the unit circle and verify solution/s by substitution

General Capabilities: Numeracy, Critical and creative thinking

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WA10MMGOpt5

Year 10 optional

Explore geometric relationships and apply deductive reasoning and a sequence of logically connected statements, to produce proofs of similar triangles and angle/chord/radius/tangent properties in circles

General Capabilities: Numeracy, Critical and creative thinking

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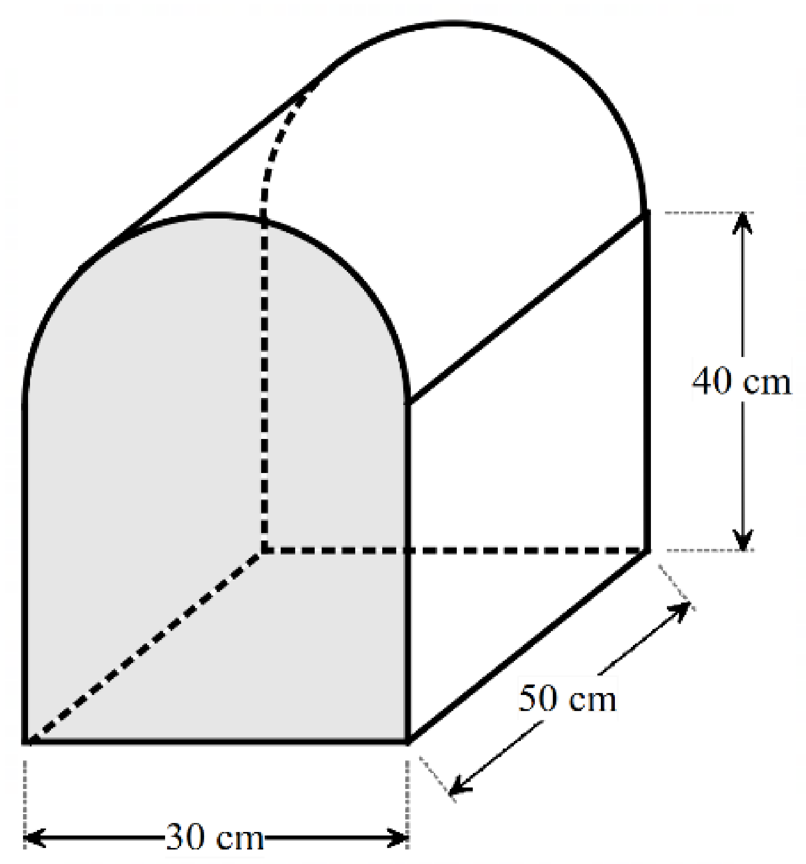
Three-dimensional space and structures

WA10MMGTH1

Use efficient strategies and apply formulas to determine the volume, capacity and surface area of composite solids, using appropriate units

For example:



- exploring the orientation of the object to identify the base end and the height. Use strategies to calculate the volume, such as dividing the base end into familiar shapes to determine the area of the cross-sectional base and multiplying by the height or dividing the composite solid into familiar prisms, applying appropriate formulas and adding to find total volume. Using the volume to determine capacity and comparing strategies with others, deciding which approach is the most efficient
- visualising or drawing the net or different two-dimensional views of a composite object, recognising congruent faces and showing area calculations of the different faces in a sequence of steps, adding to determine the surface area in square units

General Capabilities: Numeracy

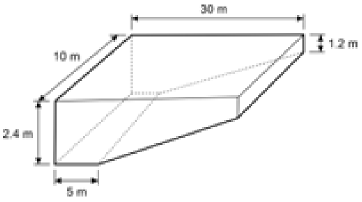
WA10MMGTH2

Investigate, calculate and identify the impact of errors on the accuracy and outcome of results in measurement situations

For example:

- determining the length of a wooden beam leaning against a wall, using a clinometer or large protractor to measure an angle of elevation and applying trigonometry and/or Pythagoras' theorem. Comparing calculated and actual measurements of the wooden beam and discussing accuracy of results in terms of absolute and percentage error, the measuring instruments, the size of the measurement unit used and human error

- determining the planned height of an access ramp being built from a carpark to the entrance of a leisure centre, given that the plan is for it to be 3.5 m long, on an inclination of 12°. Discussing any implications of the miscalculation or error of 1° on the access ramp build
- investigating the effect on capacity of the following fibreglass swimming pool if the 2.4 m height was accidentally read as 2.6 m, determining percentage error and discussing the impact of the misreading



General Capabilities: Numeracy, Critical and creative thinking

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WA10MMGTH3

Investigate in order to determine the effect on surface area and volume when objects are enlarged or reduced by a scale factor

For example:

- constructing similar objects with a given scale using dynamic geometry software, cardboard, grid or isometric paper and discussing what is the same and what is different about the object and its enlarged/reduced image. Calculating and comparing the surface area and volume of the object and its image. Determining that the surface area of the object is enlarged or reduced by the square of the scale factor ($\times\ k^2$) and that the volume of the object is enlarged or reduced by the cube of the scale factor ($\times\ k^3$)
- determining the scale factor when a rectangular prism with a volume of 44.8 cm^3 is enlarged to have a volume of 1209.6 cm^3
- exploring the impact of changes in scale on the surface area and volume in situations involving 3D printing or production prototypes

General Capabilities: Numeracy, Critical and creative thinking

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WA10MMGOpt6

Year 10 optional
Explore, explain and apply efficient strategies and formulas to determine the surface area and volume of right pyramids, right cones, spheres and related composite solids

General Capabilities: Numeracy, Critical and creative thinking

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Modelling with measurement and geometry

WA10MMGM1
In real-world situations involving Pythagoras' theorem, trigonometry and angles of elevation and depression in right-angled triangles, volume, capacity and surface area of composite objects, the impact of errors in measurement and/or the effect of enlargement and reduction on perimeter, area and volume of similar figures and objects

I. analyse the situation, decide if an exact or approximate solution is required and determine assumptions and constraints
II. represent the situation mathematically in order to reach a solution
III. interpret and communicate findings in terms of the context and any assumptions or constraints

> **For example:**
>
> - ‘A drone is hovering at a height of 455 m. The angles of depression from the drone to a boat in trouble are at first 9° and then 14°. How far has the troubled boat travelled between the two observations?’ Making assumptions, such as the sea is relatively flat, the drone’s height is constant and that it does not change position. Using representations, such as a diagram, choosing a trigonometric function, such as tangent and determining the difference between the horizontal distance from the drone at 9° to 14°. Communicating the solution in terms of the drone and the boat and with reference to the assumptions
> - ‘A construction firm needs to build a bespoke hollow cylindrical 1.25 m cement pipe with an outer diameter of 32.5 cm and a pipe thickness of 6.5 cm. Determine the volume of cement required to make the pipe and the maximum capacity of the pipe at any given time.’ Making assumptions, such as an exact volume and capacity is required and that the thickness of the pipe is uniform with the pipe cross-section congruent throughout the length of the pipe. Choosing to use consistent, appropriate units, such as metres. Selecting efficient strategies and appropriate formulas to determine the volume of the pipe in cubic metres, rounding up appropriately to ensure enough cement. Determining the volume of the hollow pipe and converting to capacity in litres. Communicating the solution to the total volume of the pipe in cubic metres and the capacity of the pipe in litres, making reference to the assumptions made

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

Probability and statistics

Probability and statistics

WA10MPSP1

Choose and construct appropriate sample spaces to show outcomes for two- and three-stage chance experiments both with and without replacement. Assign probabilities to events involving conditional statements, such as ‘if ... then’, ‘given’, ‘of’, ‘knowing that’

> **For example:**
>
> - recognising and interpreting the notation and language of conditional probability, such as from a dice roll and a coin toss, determining
>
>   P(3 and a tail | dice shows a prime number)
>
>   i.e. the probability of flipping a 3 and a tail, given that the dice roll was a prime number. Choosing from a list, array or tree diagram to show all outcomes and assign a probability
> - choosing and representing a suitable sample space, such as an array, for rolling two eight-sided dice to assist in determining the probability of a sum of 14, given that both dice show an even number. Use knowledge and reasoning of conditional events to restrict the sample space and assign conditional probabilities
> - choosing to represent the sample space of drawing three marbles from a bag containing two red, one green and two blue marbles, one after the other without replacement, in a tree diagram. Determining the probability of events, such as 'drawing a blue marble third, given the first was red' and 'drawing a green marble first, knowing that the second was blue'

General Capabilities: Numeracy, Critical and creative thinking

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WA10MPSP2

Conduct repeated chance experiments and simulations to model conditional probability and produce datasets using digital tools. Discuss, compare and analyse variation and estimated probabilities for conditional events

> **For example:**
>
> - conducting an experiment to find the estimated probability of drawing a red marble third, given the first was green, when drawing three marbles consecutively, without replacement, from a bag containing 19 red, eight blue and 13 green marbles. Choosing and using an appropriate tool, such as a 1–40 random number generator, to simulate the situation. Producing a large number of trials to estimate the probability of drawing a red marble third, given the first was green. Comparing results with others, recognising the effect of ‘without replacement’, conditional events and chance variation on the datasets produced and the estimated probability. Comparing and analysing differences between the initial prediction and the estimated probability.
>   For students engaging in Year 10 optional, extending and connecting to the probability formula for conditional events

- conducting a simulation of an amusement arcade game where the prizes are three toys, with the chances of winning a toy dinosaur being 20%, a fluffy dice 50% and a stuffed toy 30%. Making predictions of the average number of played games required to win at least one of each of all three prizes. Choosing and using an appropriate tool, such as a virtual spinner or a random 1–10 number generator, to simulate the situation and to produce a large dataset. Using data to estimate the average number of games played to collect at least one of each of all three prizes. Comparing results with others, describing and explaining variation of results. Discussing, comparing and analysing differences in the average number of games needed to win at least one of each of all three prizes, in situations where it was given that the first prize drawn was
    - a stuffed toy
    - a toy dinosaur
    - a fluffy dice
- conducting simulations to produce datasets for situations involving chance, such as the ‘birthday problem’ or ‘three door problem'

General Capabilities: Numeracy, Critical and creative thinking, Digital literacy

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WA10MPSP3

Analyse bivariate data represented in a two-way table using proportions and comment on possible association between categorical variables

For example:

- forty people in the suburb of Gosnells were randomly surveyed about dog adoption from the local dog shelter to produce data to guide an adoption advertising campaign. Using the data given in a two-way table

Living near a dog shelter and dog adoption					
		Live ≤ 5km from shelter	Live >5 km from shelter	Total
Dog adoption	Have adopted a dog	15	4	19
	Have not adopted a dog	16	5	21
	Total	31	9	40

  determining proportions, such as $\frac{15}{31}$ of the population that live within 5 km of the dog shelter have adopted a dog, which is similar to $\frac{4}{9}$ of the population that live further than 5 km from the dog shelter and have adopted a dog. Commenting that proximity to a dog shelter does not appear to have an association with dog adoption. Discussing how the information could be used to guide the dog adoption advertising campaign

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding

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WA10MPSP4

Represent secondary data in two-way tables or Venn diagrams and assign probabilities to outcomes involving conditional statements

> **For example:**
>
> - choosing from a Venn diagram or two-way table to represent information, such as a 200-people survey about which social media platform they use. 115 use 'Quickgram' only, 13 use 'Z' only and 45 use neither. Determine the probability that a person chosen at random uses 'Z' given that they also use 'Quickgram'.
>   Note: situations involving mutually exclusive events should also be included

General Capabilities: Numeracy, Critical and creative thinking

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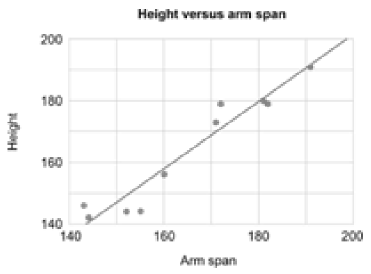
WA10MPSP5

Represent the relationship between bivariate data in a scatter plot and draw a trend line by eye if appropriate. Use the graph and context to describe any association in terms of strength, direction, linearity and outliers. Make predictions and recognise and explain any limitations of the model

> **For example:**
>
> - representing data of the arm span and height, in centimetres, of a group of students, manually or with digital tools, depending on the amount or complexity of the data, in a scatter plot.
>
> 
>
>
>   Describing the data as having a strong, positive, linear relationship between arm span and height, with arm span increasing as height increases and with no outliers. Using the trend line to make predictions between known data values, such as predicting the height of a student with an arm span of 165 cm (interpolation), and beyond known data values, such as predicting the arm span of a student who is 210 cm tall (extrapolation), discussing the validity of each prediction and explaining the predictions limitations in terms of chance variation

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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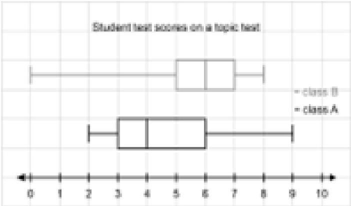
WA10MPSP6

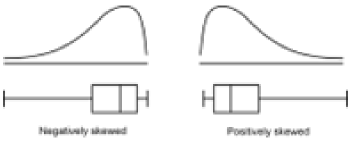
Represent and analyse boxplots. Explain differences between multiple boxplot datasets in terms of shape, spread and centre. Compare or match the shapes of boxplots to distributions depicting the same data

> **For example:**
>
> - using a five-number summary calculated from a dataset to assist in producing a boxplot manually or with digital tools, depending on the size and complexity of the data, and conversely, using a given boxplot to extract a five-number summary. Analysing the distribution in terms of shape (symmetry, skewness and clustering), spread (range and IQR) and centre (median)

- representing multiple datasets, such as test results for Class A and Class B, manually or with digital tools, on the same scale and comparing differences between the shape, spread and centre of the distributions



- analysing a negatively skewed distribution, where the mean is smaller than the median, and comparing it with a positively skewed distribution, where the mean is larger than the median



- describing the shape of boxplots and matching the distributions to histograms and dot plots

General Capabilities: Literacy, Numeracy, Critical and creative thinking

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WA10MPSP7

Critically analyse the claims, inferences and conclusions of statistical reports in the media and other real-life situations, and identify potential sources of bias

For example:

- critically analysing the source, sampling method, validity and reliability of the dataset, accuracy and appropriateness of statistical measures and representations, interpretation and extrapolation of results, including conclusions regarding association, and overall implications and recommendations outlined, to identify possible bias in reports involving
  - the effectiveness of a new exercise regimen in a magazine
  - climate change trends and the impact on a local ecosystem reported in the community newspaper
  - a post on social media concerning crime rates in your local area
  - satisfaction ratings and reports on social media concerning restaurants or other places to visit

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding

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WA10MPSOpt1

Year 10 optional
Use weighted tree diagrams and/or formulas to assign probabilities to two- and three-stage chance events including situations involving conditional probability

General Capabilities: Numeracy, Critical and creative thinking

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WA10MPSOpt2

Year 10 optional
Produce, organise and represent accurate and valid data in a cumulative frequency graph, and use this to analyse quartiles and percentiles

General Capabilities: Numeracy, Critical and creative thinking

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WA10MPSOpt3

Year 10 optional
Determine the mean and standard deviation of a dataset. Investigate, analyse and interpret the effect of individual data values, including outliers, on the standard deviation

General Capabilities: Numeracy, Critical and creative thinking

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Modelling with probability and statistics

WA10MPSM1
In real-world situations involving two- and three-stage chance experiments both with and without replacement, conditional probability or statements, boxplots, bivariate data and/or two-way tables

I. analyse the situation, pose questions as required and determine assumptions and constraints
II. determine appropriate production of a valid and reliable dataset, statistical measures, data representations and analyses, including examination of distributions, to effectively investigate the situation
III. interpret, draw inferences and communicate findings, in terms of the context, assumptions, constraints, chance variation and knowledge or insights gained

> **For example:**
>
> Using the modelling process to investigate topics, such as
>
> - analysing the number of kicks made by two Western Australian football teams in the previous football season, comparing to the number of games won and providing advice to coaching staff for the upcoming season
> - investigating the validity of a statement in a health magazine that read, 'The average adolescent has a resting heart rate of approximately 80 beats per minute which is reduced depending on the number of hours of exercise they do per week'

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Ethical understanding, Digital literacy

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WA10MPSOpt4

Year 10 optional
Using the modelling process to design and conduct a chance experiment, simulation or statistics experiment on a topic of interest

General Capabilities: Literacy, Numeracy, Critical and creative thinking, Digital literacy

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